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Executive Summary

This project covered two areas: use of INFINIUM Pure Oxygen Anodes™ for production of pure aluminum, and low-cost production of aluminum-scandium master alloy for high-performance alloys.

Pure Oxygen Anodes experiments and literature research identified key obstacles to long-term operation, and mitigated several of them. At this point, low-temperature operation in potassium cryolite with thin tubes and porous silver anodes seems the best route to long-term operation.

At this point, there is not a clear path to 4000 hours at 0.5-1 A/cm² performance, as would be required for large-scale commodity aluminum production with cost similar to carbon anodes. But there is likely a route to 250 hours at 0.25-0.5 A/cm², which could make this technology feasible for niche products *e.g.* aluminum-scandium master alloy.

The project did identify an alternative Pure Oxygen Anodes commercialization pathway in the form of recycling dross waste from aluminum casting operations. These anodes can potentially use molten salt bath with higher density than liquid aluminum, allowing long dross residence time for dissolution. INFINIUM is looking to pursue this pathway further with a large aluminum company.

Regarding production of Al-2%Sc commercial master alloy composition for Al-Sc alloys, this project reached several findings around two production methods.

First, the team produced 10 kg of approximately 2 wt% Sc in Al, which is the standard commercial master alloy composition. This confirmed existing literature showing that ScF_3 reacts with liquid Al metal to produce Al-Sc alloys. However, the high cost of ScF_3 (60% higher than Sc_2O_3, Sc basis), the low maximum observed yield of 27% ScF_3 conversion, lack of novelty, and a prospective commercialization partner's desire to avoid producing ScF_3 are challenges to moving forward with this method.

Second, this project demonstrated a novel process for direct electrolytic reduction of scandium oxide with a liquid aluminum cathode, with potentially the lowest possible raw material costs. This required the formulation of a low-density scandium-rich bath chemistry with sufficient oxide solubility for electrolysis, and understanding parallel metallothermic and electrolytic reactions. Here challenges to commercialization are all technical: carbon dust formation reduces current efficiency; using air to remove this carbon dust would damage cell materials of construction; harvesting alloy and replenishing the liquid aluminum cathode could remove bath and lead to scandium yield loss; and maintaining ideal bath composition will require control of Sc_2O_3 feed rate, current density, and idle time, such that frequent chemistry measurements may be necessary.

This report describes the high-temperature chemistry fundamentals involved in all of the above, plus about 70 of the experiments which contributed to further understanding of these phenomena, detailed narratives and schematic designs for industrial use of these processes, and preliminary cost estimates for these processes.

Comparison of Accomplishments with Goals and Objectives

As described in the Executive Summary above, this project began with the goal of demonstrating feasibility of using INFINIUM Pure Oxygen Anodes™ to replace carbon anodes in aluminum production. This goal posed both technical and commercial challenges to scale-up of higher difficulty than were anticipated at the start of the project. These challenges, and a new opportunity in scandium, led the team to pivot the project goal to developing low-cost production of aluminum-scandium alloys. Progress toward both of those goals is described below.

Goal 1: Zero-Emissions Primary Production of Aluminum

Pure Oxygen Anodes were tested as potential replacements for carbon anodes used in the Hall-Héroult cell which produces nearly 100% of today's aluminum. These anodes use a solid electrolyte tube, typically made of stabilized zirconia, between the molten salt electrolyte and anode, resulting in high-purity oxygen by-product.

Toward this end, the original project goal was to run at 200 A for 1000 hours, *i.e.* pass 200,000 A·h of charge, using a vessel with 10 tubes of 3.2 cm diameter and 25 cm immersion depth. This would be a stepping stone to 1000 A for 1000 hours, which is an important aluminum industry commercialization benchmark. The project's best experiment achieved 680 A·h charge passed in a small 10 cm depth vessel with a single 1.9 cm diameter anode tube, that is, 42 times lower anode active area. One could thus say the project fell short of its original charge-per-area goal by a factor of 200,000 ÷ 680 ÷ 42 = 7. This somewhat understates the additional effort needed to reach 200,000 A·h, in that all ten tubes would need to reach seven times higher performance (current density × tube lifetime) over 4.2x larger area than the best single-tube experiment here.

An intermediate goal was to reach 200 A for 250 hours, a performance benchmark which would justify use of these anodes for Al-Sc production. The Al-Sc alloy production bath composition selection occurred too late in the project to test Pure Oxygen Anodes for this application.

Goal 2: Low-Cost Production of Aluminum-Scandium Master Alloys

A prospective partner is developing a new scandium oxide mine/production facility, and is interested in low-cost use of Sc_2O_3 to make Al-Sc alloys. Due to low Sc solubility in Al, the standard commercial master alloy composition is Al-2 wt% Sc. INFINIUM undertook two activities to make this alloy: metallothermic reduction and electrolysis. The goal in this project was to produce 10 kg of Al-2%Sc alloy using Sc_2O_3 as the primary source of scandium.

Metallothermic reduction used the reaction between ScF_3 and Al metal to produce 10 kg of approximately 2 wt% Sc in Al, with AlF_3 reaction by-product. This confirmed existing literature showing that ScF_3 reacts with liquid Al metal to produce Al-Sc alloys. There are several obstacles to commercialization of this pathway:

- Today's ScF_3 price is $1300/kg, *i.e.* $2947/kg Sc basis. This is 60% higher than

$1200/kg for Sc_2O_3, *i.e.* $1840/kg Sc basis. Sc_2O_3 is the most stable oxide on the periodic table, so making ScF_3 requires reaction with HF at 700°C.

- The best experiment achieved yield of 27% ScF_3 conversion. Most of the ScF_3 volatilized and is difficult to recover, though with further work it may be possible to use filters to recover some of it from the vacuum system.
- Based on the work to date, INFINIUM has no patentable inventions associated with this process at this point.
- Our current commercialization partner expressed a desire to avoid production of ScF_3, so pursuing this would require INFINIUM to find an additional partner or develop ScF_3 production capability in-house.

The electrolysis activity developed a novel method for direct reduction of Sc_2O_3 into a reactive aluminum cathode, and experiments produced 200 g of Al-Sc alloy with >2 wt% Sc using this method. Commercialization of this method will require overcoming several technical obstacles:

- Experiments produced carbon dust on the molten salt surface, which provides an electronic conduction pathway and reduces current efficiency.
- Using air to remove this carbon dust would damage the current cell materials of construction which are TiB_2 and graphite.
- Harvesting Al-Sc alloy and replenishing the liquid aluminum cathode require a siphon or ladle to be immersed through the molten salt bath. This could remove some bath and lead to Sc_2O_3 yield loss. The ratio of metal withdrawal to minimum required Sc_2O_3 feed is 100:3, and the ratio of metal batch size to bath could be as high as 10:1, so a small amount of bath loss relative to metal would be a large loss relative to the bath and oxide.
- Ideal bath composition is characterized by ratios of Sc:Al cations, O:F anions, and Na:(Al+Sc) bath solute. Experiments show the window of operation is not terribly narrow. But keeping all of these in specification will require control of Sc_2O_3 feed rate relative to current (controls oxide content), current density (controls Sc:Al reduction ratio), idle time (controls extent of bath-metal metallothermic reaction vs. electrolysis), and some amount of NaF addition (makes up for losses *e.g.* evaporation). Such control could be challenging, and frequent chemistry measurements may be necessary, which would increase cost.

In short, this project approximately reached Technology Readiness Level (TRL) 5 with the ScF_3 metallothermic reduction process known in literature, and TRL 4 with the novel electrolysis process. Further technical work could potentially lead to increased metallothermic yield and solutions to the electrolysis challenges.

Summary of Project Activities

As described above, project activities focused around two main goals: zero-emissions primary production of aluminum, and low-cost production of aluminum-scandium alloys.

Zero-Emissions Primary Production of Aluminum

The bulk of the project (9 of 12 quarters) was spent developing Pure Oxygen Anodes™ for molten salt electrolysis to produce aluminum metal from its oxide with no direct carbon emissions. This approach uses solid electrolyte tubes, usually made of zirconia, to protect an inert anode from the molten salt environment. Others had tried this before, from Swiss Aluminum in the 1960s, to Robert Rapp in the 1990s. Also in the 1990s, Uday Pal at Boston University established a foundation of theory and experimental practice for the use of this method to produce metals such as magnesium. Because this is the most carbon-intensive step in the aluminum value chain, this would pave the way toward substantial emissions reductions in the aluminum industry.

If successful, Pure Oxygen Anodes could:

- Eliminate the carbon anode plant, with all of its volatile organic emissions;
- Eliminate the anode effect which results in perfluorocarbon (PFC) production;
- Eliminate fluoride fume and HF emissions in the high flow rate anode gas stream;
- Eliminate many emissions control and permitting requirements;
- Reduce Al product impurities, most of which come from the carbon anode;
- Create high-purity oxygen gas by-product;
- Enable use of natural gas to provide chemical energy in place of carbon;
- Create a reducing environment, which enables use of a TiB_2-lined vessel;
- Eliminate the need for a solid fluoride bath “frozen sidewall”, and enable insulation around the vessel to dramatically reduce thermal energy losses;
- Eliminate anode-gas back-reaction, potentially increasing current efficiency;
- Reduce cell footprint by using vertical electrodes.

This report section through p. 204 documents work toward this goal as follows:

- Introduction, p. 8: background chemistry and other fundamentals involved in materials and engineering approaches to improving anode system lifetime beyond prior attempts.
- Industrial cell design narrative, p. 27: rough design and principles of operation of an industrial aluminum electrolysis cell using this technology.
- Techno-economic energy, emissions, and cost modeling, p. 46: detailed models of industrial cell energy use, emissions particularly CO_2 and equivalents, and capital and operating cost.
- Experiments, p. 63: electrolysis runs and other experiments for development of materials and subsystems required for implementation of this technology.

Introduction

INFINIUM and Boston University worked to develop Pure Oxygen Anodes™ technology for aluminum production. The first 6 quarters of the project established a baseline of chemically stable material in the molten salt bath but degradation during electrolysis. It also ruled out the use of gadolinia-doped ceria (GDC) solid electrolyte. There have since been two approaches to improving membrane lifetime during electrolysis: the materials approach, including optimizing the bath and solid electrolyte chemistry; and balance of system engineering, particularly switching current to promote intermittent "healing" of the solid electrolyte.

Materials Approach to Solid Electrolyte Tube Lifetime

The materials approach to improving tube lifetime, product purity, overpotential and current density consists of three strategies:

- Change the bath chemistry to reduce zirconia solubility;
- Change membrane material to one whose reduced form has lower affinity for aluminum (*i.e.* cerium metal has lower aluminum affinity than zirconium), and one with higher ionic conductivity;
- Reduce solid electrolyte thickness to reduce overpotential and enable higher current density.

Details on the first two approaches follow.

Bath-Membrane Interactions

There are two candidate bath chemistries, which we have denoted the "BU flux" based on CaF_2-MgF_2, and the INFINIUM or potassium cryolite bath based on AlF_3-KF. Characteristics of the CaF_2-MgF_2 bath are:

- Low zirconia solubility (p. 77), which is consistent with electrolysis experiments showing negligible zirconium in aluminum product using this bath;
- Very low volatility: negligible evaporation over 120 hours at 1050°C (Quad Cell Experiment 1, p. 143).
- High eutectic temperature: 970°C;
- High alumina solubility when CaO is added: around 10 wt% at 1200°C (p. 75), resulting in reasonable Al^{3+} ion fraction;
- High density, leading to floating liquid aluminum product;
- Magnesium impurity in aluminum product: 4 wt% in an early experiment, about 12% (Quad Cell Experiment 1, p. 143).

Characteristics of the AlF_3-KF bath are:

- Very low eutectic temperature: 558°C (*cf.* 698°C for AlF_3-NaF);[1]

- High alumina solubility: at cryolite ratio of 1.3 ($KF:AlF_3$ mole ratio) solubility is over 8 wt% at 850°C (p. 74, *cf.* 5 wt% for AlF_3-NaF);
- Very high Al^{3+} ion fraction to support good cathode current density;
- Much lower static zirconia corrosion rate than AlF_3-NaF;
- High volatility: significant losses at 850°C, clouding the viewport;
- Moderate zirconia solubility (p. 82).

Based on these characteristics, in the near term we would like to pursue a hybrid of these approaches, avoiding MgF_2 to prevent Mg co-reduction, but adding CaF_2 to AlF_3-KF in order to reduce volatility by dilution, and increase density to create a floating aluminum pad – which further reduces bath evaporation, along with other "Configuration 3" benefits described on page 29.

Going further, big breakthroughs in membrane materials and bath interactions require that we understand the fundamental chemistry and thermodynamics of molten salts and their electrochemical reactions. For this reason, this section concludes with a deep dive into molten salt-oxide solutions. A discussion of potassium cryolite starts on page 11, and "BU flux" starts on page

Membrane Material

Regarding membrane material, characteristics of zirconia (stabilized by yttria) include:

- Mechanical robustness: transformation toughening leads to outstanding high-temperature strength;
- Ease of processing: slip casts easily;
- Commercial availability of thin-electrolyte tubes such as those from Acumentrics with strong high-conductivity Ni-YSZ cermet support;
- Low cost today, prospects for $35-60/kg;
- High metal affinity for aluminum, as mentioned above.

Characteristics of ceria (doped by gadolinia or samaria) include:

- Not terribly hard to slip cast, though slip viscosity is higher than that of zirconia;
- Likely lower solubility than zirconia in AlF_3-KF;
- Lower metal affinity for aluminum than zirconium, leading to lower co-reduction and presence in product (the one experiment we ran had just 0.47% Ce in Al product);
- Potential low future cost, as ceria is a high-volume by-product of rare earth mining – some even treat it as a tailing;
- Much higher ionic conductivity than zirconia at low temperatures below 1000°C;
- Higher electronic conductivity than zirconia, which is good for this application;
- Very low toughness, to the point where we were not able to use a monolithic

ceria tube in an experiment.

Based on these favorable ceria characteristics as a membrane but with poor mechanical robustness, a best-case solution would use ceria as a coating exposed to the bath, with another structural support material behind it. The support material must conduct away negative charges and allow oxygen to pass through. It must therefore either be another oxide ion conductor such as zirconia, or a porous conducting material such as porous strontium-doped lanthanum manganite (LSM) or nickel-ceria or nickel-zirconia cermet. The cermets would require presence of a reducing fuel such as hydrogen or natural gas in order to prevent oxidation of the metal phase.

Membrane electronic conductivity deserves further explanation. Part of the corrosion could be membrane reduction due to electronic conduction in the bath, with low electronic conduction of the membrane, as described in INFINIUM PCT Patent Application PCT/US14/63743. That is, electronic conductivity of the bath can effectively turn the bath into an extended cathode, and reduce the membrane material. If the membrane has high electronic conductivity, then electrons passing through the bath continue harmlessly through the membrane, without reducing it.

Materials Strategies Moving Forward

With the above considerations in mind, INFINIUM is pursuing the following cell bath and membrane materials strategies:

1. Bath with low zirconia solubility. Running with CaF_2-based bath at 1000-1100°C should result in long membrane lifetime, particularly if there is an option for replacing magnesium fluoride. High temperature leads to good zirconia conductivity.
2. Thin ceria coating on zirconia support. Dip coat, co-cast, plasma spray, or otherwise make a 20-200 μm coating of ceria on a zirconia support tube. This will likely need to operate above 1000°C to achieve reasonable zirconia conductivity and system overpotential. But it will require developing detailed sintering curves for ceria and zirconia, and will be hard to match shrinkage throughout the sintering process.
3. Thin ceria coating on porous LSM or other oxide conductor support. Although dip coating may work, the lower sintering temperature of LSM and related conductors than ceria leads to plasma spray as most likely technique to succeed.
4. Thin ceria coating on porous nickel-ceria cermet support. Dip coat or co-cast 20-200 μm ceria onto NiO-ceria support tube, co-fire them together, then reduce the NiO-ceria in hydrogen or natural gas to make a nickel-ceria cermet. This has been done in literature, and results in a strong, highly conductive tube.

Each of these materials strategies is under investigation at INFINIUM, as described in the M1.1-M1.3 sections below. In addition to these materials strategies, INFINIUM is also pursuing engineering changes which could lead to significant materials performance improvements, as described on page 26.

Potassium Cryolite Molten Salt Bath Chemistry and Thermodynamics

Bath chemistry plays a significant role in determining overall electrolysis performance, as small variations in bath composition can result in drastically different performance and failure modes.[2] The conventional Hall-Héroult process uses a sodium cryolite based bath, but electrolysis with inert anodes introduces an additional compatibility constraint, which requires optimizing new bath chemistries. We have focused on two candidate chemistries, the "BU flux" based on CaF_2–MgF_2 and the "INFINIUM bath" based on potassium cryolite AlF_3–KF.

The INFINIUM bath chemistry has also been widely studied in the scientific literature in recent years, and much progress has been made in characterizing the bath makeup at the atomic-scale, giving insight into the mechanisms that ultimately drive performance. Consequently, INFINIUM has developed a working understanding of the relevant atomic processes in the bath: **(1) before addition of Al_2O_3** (cryolite thermodynamics), **(2) after Al_2O_3 addition** (dissolution reactions and formation of oxyfluoro-anion complexes), and **(3) during electrolysis** (anodic and cathodic reactions). This report will present the first two of these, and discuss cathodic reactions. Anodic reactions and membrane interactions will follow in the next report.

Before addition of Al_2O_3: Cryolite Thermo

Benefits of using potassium based cryolite as opposed to the conventional sodium cryolite are membrane compatibility, alumina solubility, and operation at lower temperatures where parasitic and corrosive reactions are more easily mitigated and an inert anode can be maintained. As seen in the KF–AlF_3 phase diagram shown in Figure 1,[3] the eutectic temperature is low, 558 ºC at cryolite ratio (CR) ~ 1.22 (KF:AlF_3 mole ratio), compared to 698 ºC for the NaF-AlF_3 system (at CR ~ 1.17, NaF:AlF_3 mole ratio).[4] The liquidus curve represents an upper bound on the CR, upon which precipitation of solid K_3AlF_6 (CR=3) occurs. At 800ºC, for example, CR<1.5 is required to avoid solid precipitation. As the AlF_3 content increases (CR reduced toward 1) the vapor pressure of $KAlF_4$ gas also increases, resulting in increased volatility of the bath.

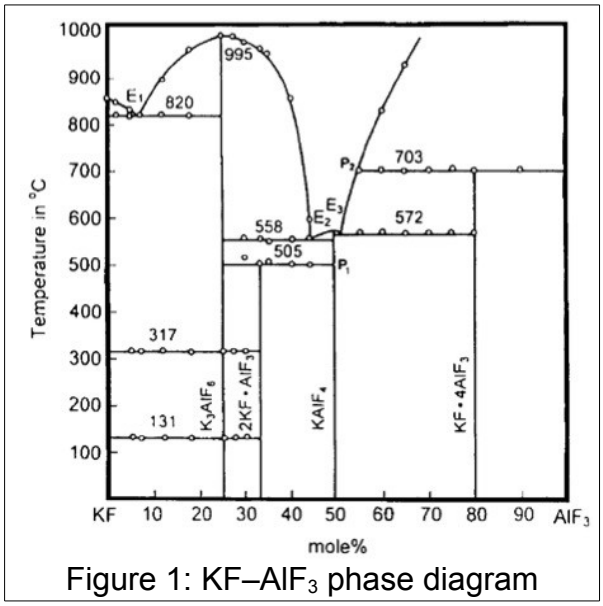


Figure 1: KF–AlF_3 phase diagram

At the atomic scale, the cryolite bath is not comprised of isolated free ions of Al^{3+}, F^-, and K^+, but rather complexes consisting of Al^{3+} occupying the geometric centers coordinated by F^- ligands, which are separated from each other by mobile K^+ ions (that carry the majority of the current).[5] At the temperatures of interest (~ 600 – 900ºC) the relevant composition range is roughly 1 < CR < 2, and the majority anion complexes are

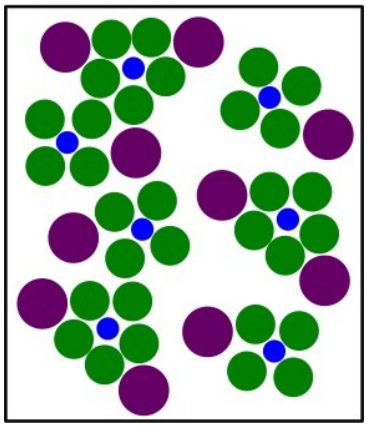


Figure 2: Schematic depiction of AlF_5^{2-} and AlF_4^- complexes (blue Al^{3+} and green F^-) coordinated by mobile K^+ ions (purple) in cryolite melt.

AlF_4^- and AlF_5^{2-} (drawn schematically in Figure 14).[6,7,8] The higher the CR, the higher the relative concentration of AlF_5^{2-} to AlF_4^- complexes in the bath.[9]

These two anionic complexes are involved in the ensuing Al_2O_3 dissolution reactions, as well as the charge-transfer reactions at the anode and cathode. Therefore, understanding their role and evolving concentration in the bath is important to optimizing the bath chemistry.

After Al_2O_3 addition: Dissolution and Oxyfluoro-Anion Complex Formation

Al_2O_3 solubility in the cryolite melt is a crucial operating parameter– there must be some minimum solubility for electrolysis to be possible at all. Fortunately, another beneficial property of potassium cryolite melts is the high solubility of Al_2O_3 (in comparison to Na cryolite) at relatively low temperature as seen in Figure 3.[4,10,11] For example, at 800ºC and CR = 1.3, there is ~6 wt% Al_2O_3 solubility in the potassium cryolite flux.

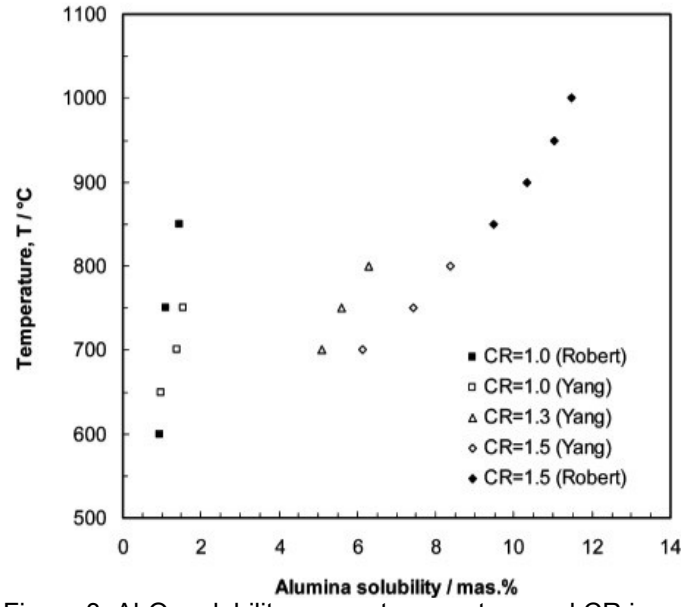


Figure 3: Al_2O_3 solubility versus temperature and CR in $KF–AlF_3$.

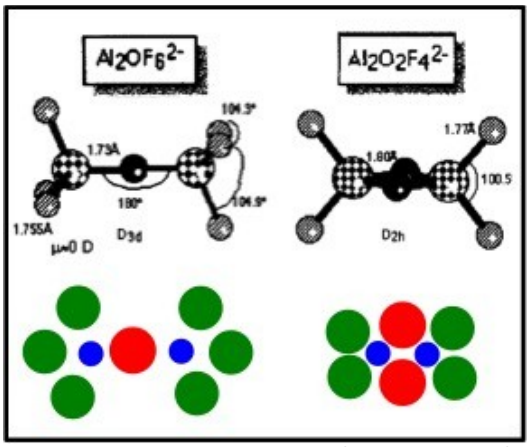


Figure 4: Structure of $Al_2OF_6^{2-}$ and $Al_2O_2F_4^{2-}$ complexes formed by dissolving Al_2O_3 in $KF\text{-}AlF^3$ (Al^{3+}, F^-, and O^{2-} in blue, green, and red, respectively). K^+ ions are not shown for simplicity.

The atomic structure of $KF–AlF_3–Al_2O_3$ melts has been studied in detail in the literature through a variety of methods (Raman spectroscopy, first-principles calculations, and others), concluding that Al_2O_3 is dissolved by forming complexed oxyfluoroaluminate anions.[5,11-15] Raman spectroscopy investigations confirm that the dissolved species' are Al dimers with bridging Al–O–Al bonds, and for baths with CR < 3, the majority species are $Al_2OF_6^{2-}$ and $Al_2O_2F_4^{2-}$ as shown in 3d and depicted

schematically in 2d in Figure 4.[5,13] There are then two possible Al_2O_3 dissolution reactions, which form each of these species, respectively:

$$2\ Al_2O_3 + 4\ AlF_5^{2-} \rightarrow 3\ Al_2O_2F_4^{2-} + 2\ AlF_4^- \quad (1)$$

$$Al_2O_3 + 2\ AlF_5^{2-} + 2\ AlF_4^- \rightarrow 3\ Al_2O_6^{2-} \quad (2)$$

As seen in both reactions 1 and 2, AlF_5^{2-} complexes are required for Al_2O_3 dissolution, which rationalizes the low Al_2O_3 solubility with CR=1 seen in Figure 3. Also, reaction 1 is favored over reaction 2 as the cryolite ratio is increased. As Al_2O_3 content is increased, again reaction 1 is favored over reaction 2. Additionally, in the bulk, the equilibrium between both oxyfluoro-anion species ($Al_2O_2F_4^{2-}$ and $Al_2OF_6^{2-}$) is maintained by reaction 3:

$$2\ Al_2OF_6^{2-} \leftrightarrow Al_2OF_6^{2-} + 2\ AlF_4^- \quad (3)$$

The relationships and interaction in composition space highlighting reactions 1, 2, and 3 can be summarized by constructing the KF-Al_2O_3-AlF_3 ternary shown in Figure 6. To this point, the relevant atomic species and complexes in the bath have been identified, namely AlF_5^{2-} and AlF_4^- in cryolite, and the resulting $Al_2O_2F_4^{2-}$ and $Al_2OF_6^{2-}$ produced from adding and dissolving Al_2O_3. As a potential is now applied across the cell to initiate electrolysis, these same species are also involved in the charge-transfer reactions that produce oxygen and metallic aluminum at the anode and cathode, respectively.

During Electrolysis: Cathodic Reactions

At the cathode, Al metal is produced from one of the Al-containing complexes, likely either AlF_4^- or $Al_2OF_6^{2-}$,[42,53] by reactions 4 and 5:

$$5\ AlF_4^- + 3\ e^- \rightarrow 4\ AlF_5^{2-} + Al_{liq} \quad (4)$$

$$2\ Al_2OF_6^{2-} + 8\ AlF_4^- + 6\ e^- \rightarrow Al_2O_2F_4^{2-} + 8\ AlF_5^{2-} + 2\ Al_{liq} \quad (5)$$

Both cathode reactions 4 and 5 consume AlF_4^- and produce AlF_5^{2-}, where 4 involves the disintegration of an AlF_4^- and 5 involves the disintegration of $Al_2OF_6^{2-}$ at the cathode shown in Figure 5 and Figure 6, respectively.

In the literature, some suggest that reaction 5 is more likely because of the strength of the AlF_5^{2-} and AlF_4^- complexes,[42] and also the observed ease of Al deposition when oxygen-containing species are introduced into the bath.[53]

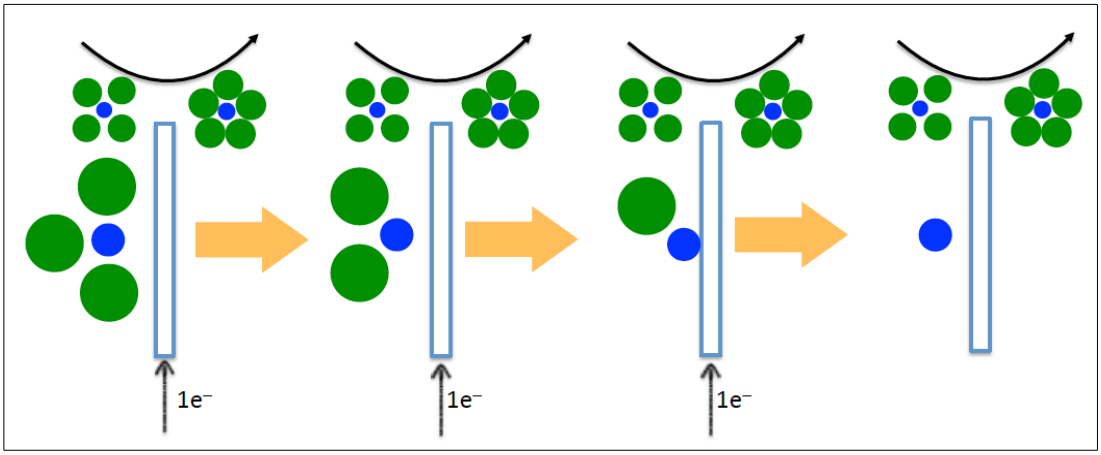


Figure 5: Schematic depiction of reaction process 4 at the cathode

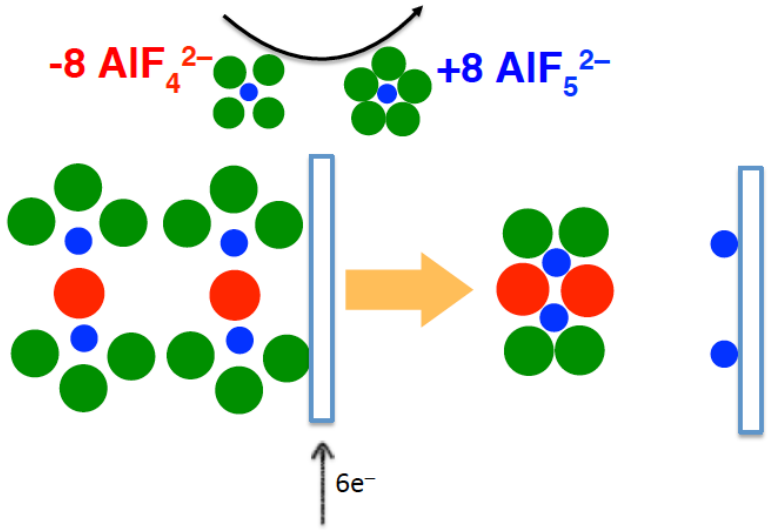


Figure 6: Schematic depiction of reaction process 5 at the cathode

BU Flux Molten Salt Bath Chemistry and Thermodynamics

Unlike potassium cryolite, which has been specifically investigated in significant detail in the literature, insight into the BU-flux must come from the combined investigation of its underlying subsystems, which have been investigated in the literature: **(1) fluoride (CaF_2–MgF_2)**, **(2) calcium (CaF_2–CaO)**, and **(3) oxide (CaO–Al_2O_3) binary subsystems**, and finally, the **(4) CaF_2–CaO–Al_2O_3 ternary system**. We navigate through this composition space to gain insight into the bath chemistry that best supports high-efficiency Al metal production. At the atomic scale, this translates to optimizing the bath composition to maximize the formation of Al-containing (anionic) complexes that can easily participate in the anodic and cathodic charge-transfer reactions. In cryolite melts, these complexes are dimerized aluminum oxy-fluorides.

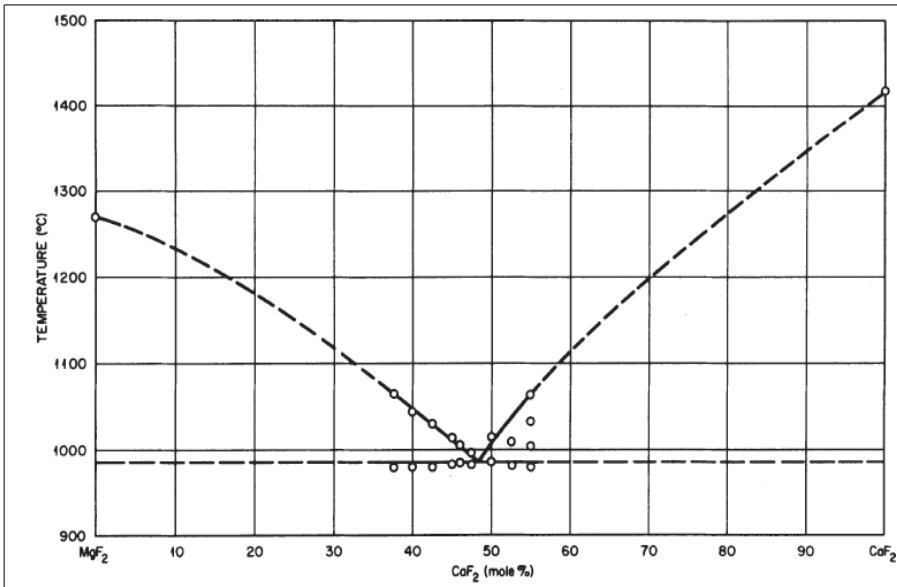


Figure 7: *MgF_2 – CaF_2 Phase Diagram*

Cesium(I)	0.60	1.7	0.72	
Potassium(I)	0.73	1.4	0.60	
Sodium(I)	0.87	1.15	0.50	
Lithium(I)	1.0	1.0		1.30
Barium(II)	0.87	1.15	0.50	1.40
Strontium(II)	0.91	1.1	0.48	1.35
Calcium(II)	1.00	1.00	0.43	1.30
Iron(II)	1.0	1.0		
Manganese(II)	1.0	1.0		
Magnesium(II)	1.3	0.78	0.34	1.25
Aluminium(III)	1.65	0.60	0.26	
Silicon(IV)	2.1	0.48	0.21	
Boron(III)	2.36	0.42	0.18	
Hydrogen(I)	2.5	0.40		
Phosphorus(V)	3.0	0.33		
Sulfur(VI)	4.0	0.25		
$(CH_3CO)_2O$		0.47		

Figure 8: *Optical basicity (Λ) values of notable oxides, fluorides, and sulfides*

(1) Fluoride Subsystem: CaF_2 – MgF_2

The fluoride subsystem plays an important role in the bath, as it is responsible for forming the main solvent species in the bath and lowers the operating bath temperature. In comparison with the other chemical sub-systems, CaF_2 – MgF_2 remains in liquid form at the lowest temperature. As seen in **Figure 10**, adding MgF_2 to CaF_2 brings the melting point from ~ 1400°C to nearly 1000°C at the eutectic point at near 50% (mole%) MgF_2 composition.[16] At the atomistic level, some level of insight into the complexation behavior can be gained

by looking at the optical basicity of the end-members. A table of measured optical basicity is shown in **Figure 8** for some notable oxides and fluorides.[17] MgF_2 (Λ=0.34) is more acidic than CaF_2 (Λ=0.43), consistent with forming MgF_4^{2-} complexes solvated by Ca^{2+} cations and depressing the liquidus temperature consistent with **Figure 10**.

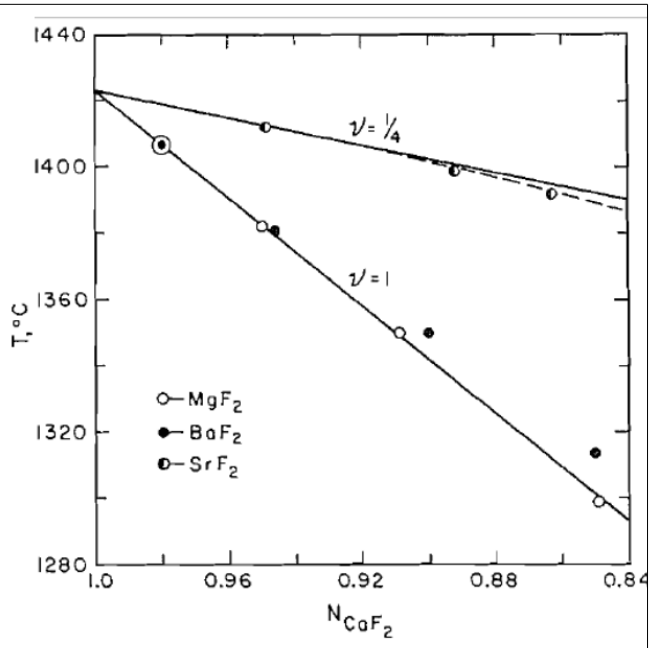


Figure 9: *Depression of the freezing point of CaF_2 by MgF_2, BaF_2, and SrF_2*

The presence of a high concentration of MgF_4^{2-} which are likely electrochemically active at the cathode may also explain the Mg impurity found in Al production experiments performed with the BU bath. The number of MgF_4^{2-} present in the bath may be able to be reduced while still depressing the liquidus temperature of the bath sufficiently, through the addition of an alternative basic fluoride additive, such as BaF_2 (Λ=0.5) or SrF_2 (Λ=0.48). In such cases, the depression in the liquidus temperatures comes predominantly from the increase in entropy of mixing rather than complexation (with an acidic additive). This is also consistent with an increase in molar volume upon the addition of SrF_2 and BaF_2 to CaF_2, compared to the addition of MgF_2 to CaF_2 which results in a decrease in molar volume.[18] The freezing point depression of CaF_2 with addition of MgF_2, BaF_2, and SrF_2 are highlighted in **Figure 9**.[19]

(2) Ca Subsystem: CaF_2 – CaO – (MgF_2)

The CaF_2–CaO binary system has been well characterized as a simple eutectic, with the eutectic point at 15.3 mol% CaO at 1362°C shown in **Figure 17**.[20] In the liquid phase, the system behaves as an ideal solution,[20] meaning that there

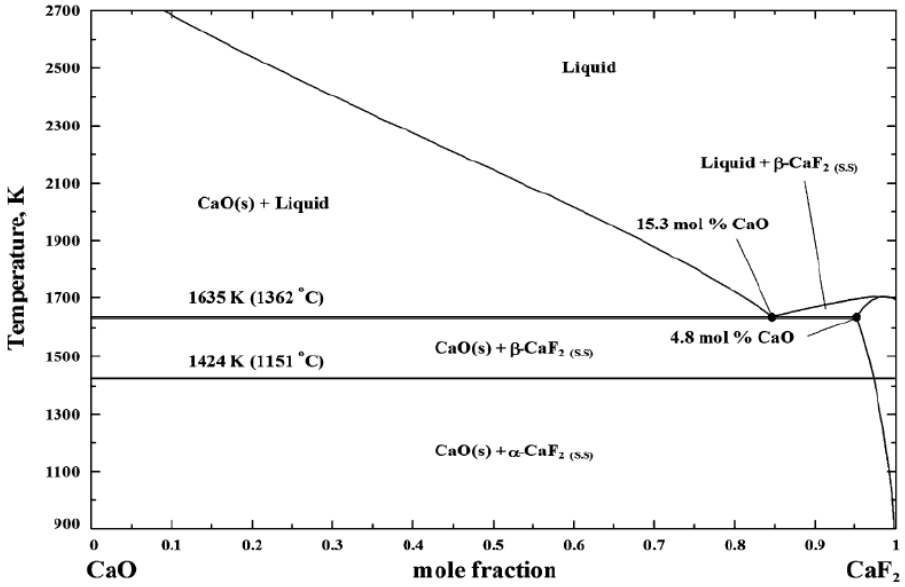


Figure 10: *CaO–CaF_2 Phase Diagram*

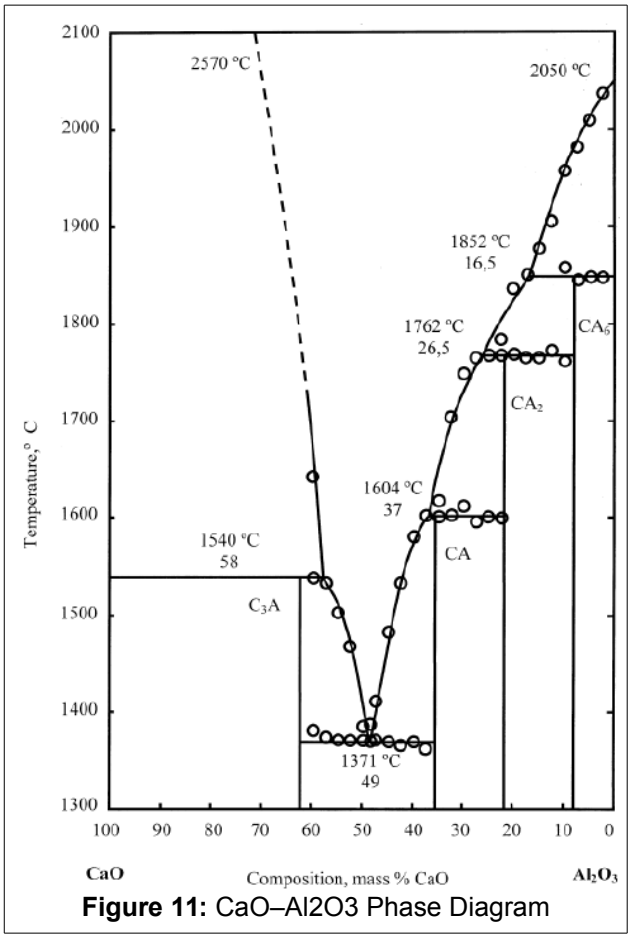


Figure 11: CaO–Al2O3 Phase Diagram

is little interaction (such as complexation) between CaO and CaF_2 components when mixed together. Gibbs free energy calculations[21] for the replacement of 1 mol of CaF_2 in $(CaO)_x$–$(CaF_2)_{1-x}$ by 1 mol of MF_2 (M = Mg, Fe, Sr, Ba) give some additional insight into the complexation behavior relevant to the BU-flux. For Mg and Fe replacement, the free energy of mixing (ΔG_s) is negative ($\Delta G_s < 0$) for all *x* (all CaO–CaF_2 compositions), and for Sr and Ba replacement, $\Delta G_s > 0$ for $x > 0.2$. In more CaO-rich compositions, ΔG_s decreases significantly for Mg and Fe, but increases significantly for Sr and Ba.[21] This suggests that with more acidic fluoride addition (MgF_2 and FeF_2) not only is there complexation with CaF_2, but there is also complexation with CaO resulting in Mg– or Fe– oxyfluoride anions. It seems that Sr and Ba (basic) additions result in little interaction with CaF_2 and possibly repulsive interaction with CaO.

(3) Oxide Subsystem: CaO – Al_2O_3

The CaO–Al_2O_3 binary is a well studied glass-forming system, with extensive characterization of Al and Ca anion coordination at high temperature (shown in **Figure 11**).[22] As a liquid (above 2000°C), Al_2O_3 has short-range order dominated by AlO_4 tetrahedra.[23] The glass-forming range is centered at ~64 mol% CaO, roughly coinciding with the deepest eutectic in the Al_2O_3–CaO phase diagram at ~49 mol% CaO (1371°C). 50 mol% CaO corresponds to a fully polymerized glassy network, and by 60–70 mol% CaO, the network is depolymerized.

This structural behavior can be understood in the context of the acid/base behavior of oxides and

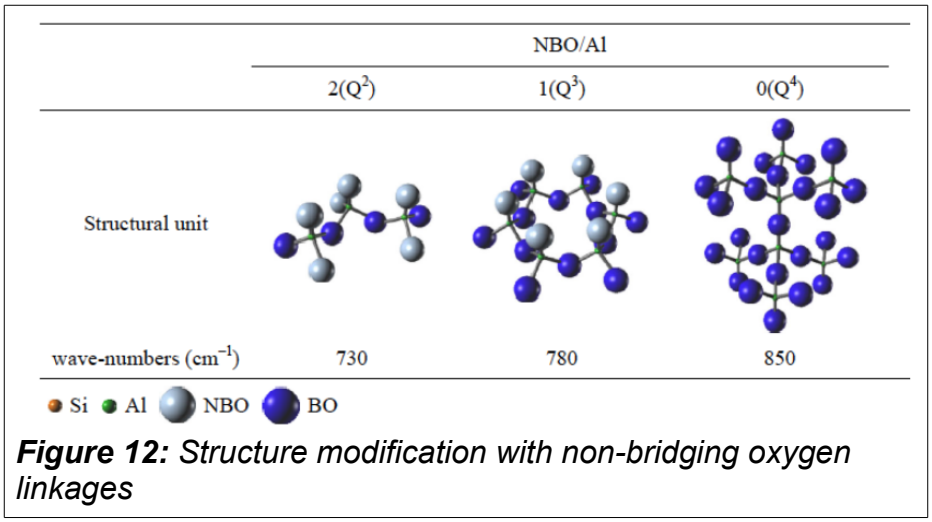


Figure 12: *Structure modification with non-bridging oxygen linkages*

the specific role of non-bridging oxygen bonds illustrated in **Figure 12**.[24] Beginning with pure Al_2O_3, each oxygen anion is located directly adjacent to multiple Al cations, forming Al-O-Al polyhedral linkages. If every O in the system is bonded locally to 4 Al, as is the case in liquid Al_2O_3, then the 3D SiO_2 structure is formed containing only bridging oxygens. As CaO (Λ=1.0) is added to Al_2O_3 (Λ=0.6), additional O from CaO can be incorporated into Al_2O_3 and modify its structure.

As such, in the glass-forming literature CaO is a network modifier and Al_2O_3 is the network former, reducing the number of non-bridging oxygens per Al. Oxygen connected to only 3 Al (as opposed to 4 in pure Al_2O_3) can arrange together to form polymeric rings, and oxygen connected to only 2 Al can form smaller chains and eventually dimers. In the context of Al_2O_3 dissolution to form anionic complexes that easily produce Al and O_2 gas at the cathode and anode respectively, the role of CaO seems to be instrumental in weakening strong Al–O bonds present in Al_2O_3. This is consistent with INFINIUM's recent results showing higher Al_2O_3 dissolution in baths containing higher CaO concentration, indicating some kind of network-modifying behavior of the Al_2O_3 crystal structure in the bath. At first glance, it appears that both MgF_2 and Al_2O_3 components compete to react with CaO, but ideally Al_2O_3–CaO interactions are maximized.

(4) CaF_2–CaO–Al_2O_3 Ternary

The CaF_2-CaO-Al_2O_3 is the closest system studied in detail in the literature that resembles the BU flux composition, its relevance owing to its utility as a slag-refining system. Of note, at high temperature there is a region with 2 immiscible liquids occurring at higher CaF_2 content (which coincides with the concentration regime where we would like to operate for electrolysis, rich in fluoride). This is seen in the ternary phase diagram shown in **Figure 13**. One is likely a glassy-polymerized CaO-Al_2O_3 melt with high viscosity, and the other is a less viscous fluoride-rich molten salt. With guidance from cryolitic melts which are known to support facile Al metal production, the target bath composition should try to capture the low-viscous fluoride-rich phase but with as high a concentration of dimerized aluminum oxyfluoro-anions. Understanding the role of CaF_2, CaO, and MgF_2 in light of their complexation behavior is crucial towards optimizing the bath composition.

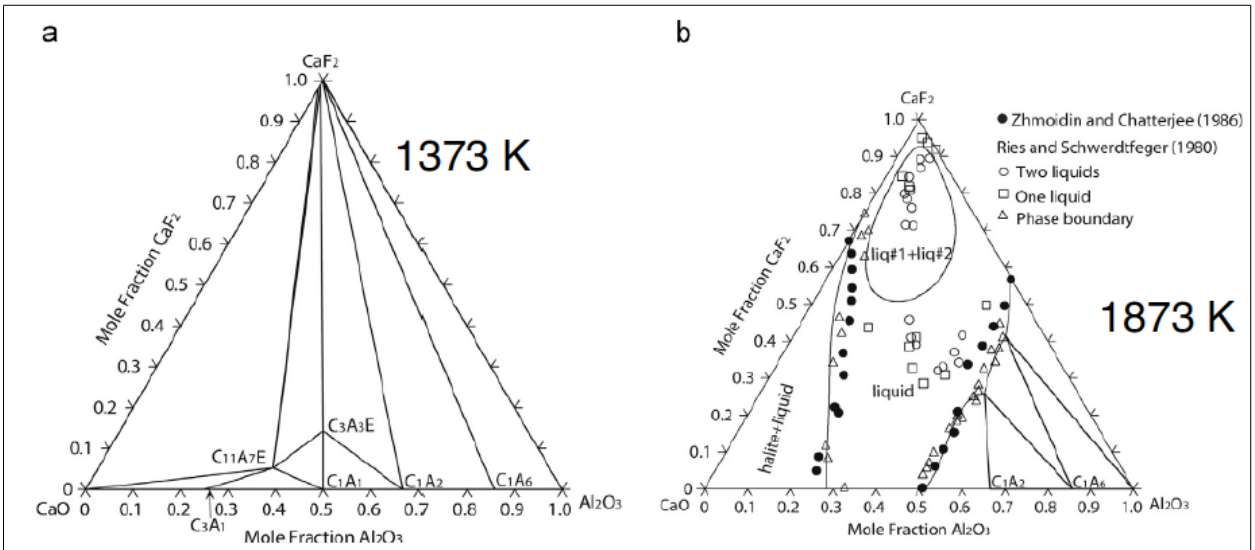


Figure 13: *$CaO–CaF_2–Al_2O_3$ Ternary Phase Diagram*

Molten Salt Decision

BU flux has the advantages of lower zirconia solubility and corrosion rate, lower volatility and high density – leading to floating aluminum product, whose advantages and disadvantages are described on p. 29. The main disadvantages of BU flux are magnesium impurity in the product and high minimum temperature. Magnesium is present in most aluminum alloys, so that is not a big problem.

The high minimum temperature for the BU flux (at least 1000°C, generally 1100°) requires us to use a liquid silver anode. And to date, whether producing aluminum or magnesium, liquid silver ejection by rapid oxygen flow has led to clogging and/or thermal stress fracture of YSZ in every high-current experiment. There are proposed methods to deal with this as follows:

- Use a porous alumina or other oxygen transport membrane, see p. 49.
- Develop a new solid-state high-temperature anode based on lanthanum chromite or similar sintering-resistant oxide conductor.

The second is beyond the scope of this project, and we did not have the time or resources to complete the first. For these reasons, in order to have a working high-current system in the timeframe of this project, **INFINIUM selected $KF\text{-}AlF_3$ based bath for use in the pilot cell**, and will keep the BU flux in mind for future development.

Because of the low density of this bath composition, this decision **commited INFINIUM to using Configuration 2** (p. 27) for our Pilot Cell, and tentatively for near-term commercial aluminum alloy and/or powder production.

Membrane Degradation Overview

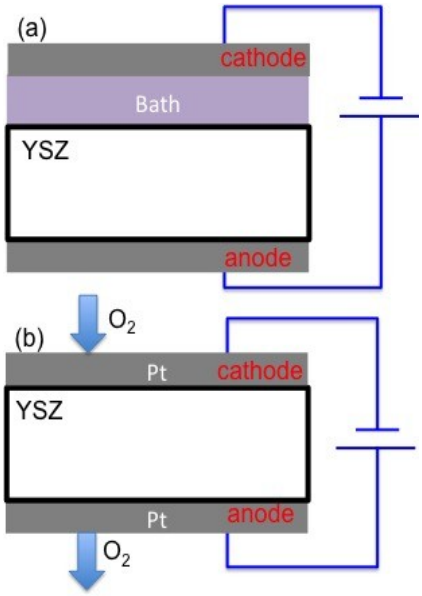


Figure 14: Schematic diagram illustrating SOM assembly in (a) aluminum electrolysis compared to (b) direct Pt contacts as studied in the literature.

Membrane degradation is the key challenge impeding reaching current immediate performance goals in efficiency (90%), lifetime (50–100 hrs), and current density (0.5 A/cm^2), and as such resolving this issue paves the path toward scale-up and commercial deployment. After accomplishing proof-of-concept and producing metal with the desired purity, INFINIUM's attention has been devoted to tuning the set of experimental operating parameters with respect to applied voltage, temperature, electrode configuration (specifically membrane thickness), bath chemistry, and solid oxide membrane (SOM) processing to minimize membrane degradation.

To systematically navigate through this parameter space, we review here the observed failure mechanisms in our experiments and compare under the context of the relevant solid oxide conductor literature to clarify understanding, optimize our design, and iteratively improve upon performance. In our experiments, SOM degradation is encountered through accumulated resistance– observed either through higher overpotential to drive the same amount of current, or through lower current with constant applied overpotential. There are two qualitative failure modes that we observe repeatedly: **(1) blackening/darkening of the SOM** and **(2) leaching of Y** from exposed surfaces and grain boundaries of the SOM into the bath.

At the atomistic level, the ideal SOM functions by incorporating O^{2-} ions from the bath (from $Al_2O_2F_4^{2-}$ oxyfluoranion complexes reacting to form $Al_2OF_6^{2-}$) into the outer membrane surface and simultaneously evolving ½ O_2 gas while extracting 2 electrons at the inner membrane to complete the external circuit, as illustrated in Figure 14a. This process, ideally, leaves the SOM physically and chemically unchanged. Membrane

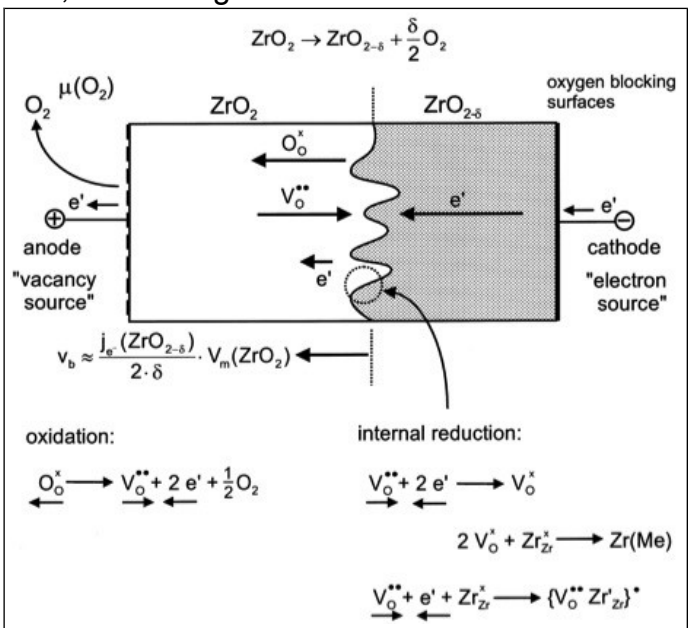


Figure 15: Depiction of electrochemical blackening of ZrO_2 due to internal reduction of membrane material.[26]

degradation therefore occurs when these inner and outer reactions (rates and chemistry) are unbalanced and/or if there are other undesired side reactions.

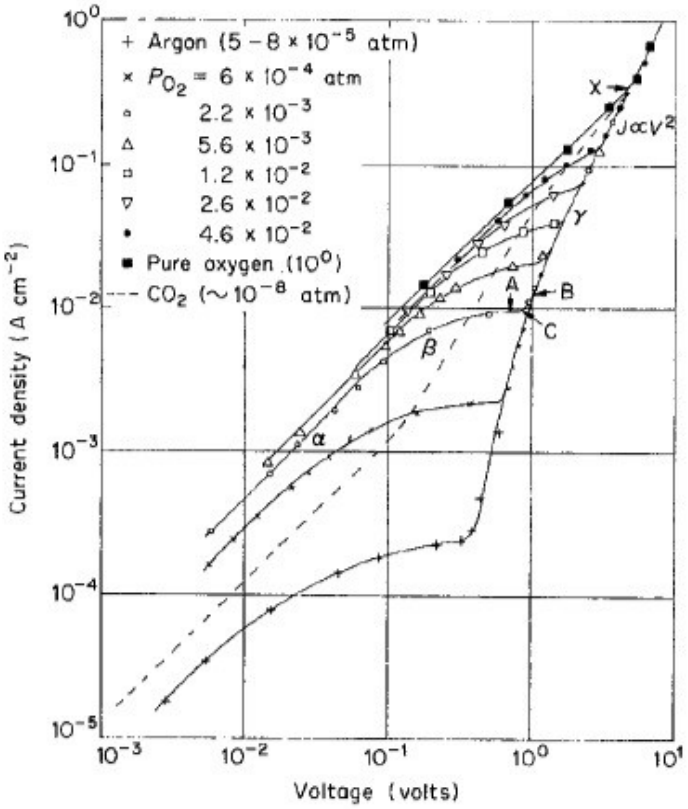


Figure 16: Dependence of ZrO_2 blackening on current density, voltage, and oxygen pressure.[25]

(1) YSZ Blackening. So-called 'blackening' of ZrO_2 and YSZ has been observed emanating from the cathode side when large currents (high overpotential) are passed through YSZ (with metal anodic and cathodic contacts as illustrated schematically in Figure 14b) leading to net reduction of the material.[25,26] With a O^{2-} blocking cathode (e.g. dense Pt film contact), net electrons (in lieu of O^{2-} ions) injected from the cathode (e.g. reduction) couple with oxygen vacancies ($V_O^{\cdot\cdot}$) in YSZ and lower the net charge from 2+ to 1+ ($V_O^{\cdot}$) or even neutral (V_o^x, so-called 'F-centers'), or they can reduce Zr^{4+} to Zr^{3+} (T-centers), as shown in Figure 15. These new defect states alter the optical properties and can cause YSZ to yellow or blacken. Forming F-centers (over T-centers) is reported to be energetically favorable in YSZ,[27] and the corresponding darkened regions of YSZ show mixed-conducting behavior (e.g. higher electronic conductivity) and evidence of a metal-insulator transition.[28] At the same time, blackening may be accompanied by a decrease in fracture strength, but also a blackened sample may 'revert to its original state' upon annealing in air or oxygen,[29] which supports the experimental evidence we have obtained highlighting membrane repair through reversible current pulses.

Observation of darkened bands in the SOM highlight that the flux of O^{2-} ions entering the membrane is not identically equivalent to the flux of O^{2-} leaving the SOM (as O_2 gas), and consequently the unbalanced current is supported by other conducting species, likely electrons in our case. The literature, fortunately, provides a guide toward minimizing membrane darkening by controlling the oxygen chemical potential as seen in Figure 16. In conductivity measurements performed at high temperature (1380°C) in experimental apparatus corresponding to Figure 14b, there are two clear conduction regimes– at low voltage the current density is sustained by O^{2-} conduction (as desired) and at higher voltage and current density, blackening occurs accompanied with a characteristic elbow in the curve. As the system's O_2 pressure is *increased*, the transition to SOM blackening occurs at higher voltage and current, which offers us

practical guidance. At higher O_2 pressure, electronic conduction in ZrO_2 is mainly p-type (as opposed to n-type at lower O_2 pressure)[26] which should make it more difficult for electronic carriers (F-centers and T-centers) to contribute to darkening. From Figure 16, the effect of O_2 partial pressure is significant as the onset of darkening can occur at orders of magnitude higher current density and applied overpotential, as desired in our application. Also, the (intrinsic) n-type conductivity is more sensitive to change in temperature compared to O^{2-} ion conductivity, which is mostly constant due to extrinsic introduction of charge carriers, and suggests that lower temperature operation corresponds to the highest O^{2-} conductivity relative to e^- conduction.

To conclude, SOM darkening occurs when the electronic conduction contribution to the current becomes comparable to the O^{2-} ion contribution, and to minimize this effect we target operating regimes where the latter is promoted at the expense of the former. In consultation with the literature, this suggests maximizing the O_2 chemical potential (meaning oxide-saturated bath and high oxygen pressure inside the anode tube), lowering the temperature, and enhancing O^{2-} migration by reducing the diffusion length (e.g. thinner tubes) to reduce membrane degradation and extend operating life.

(2) Yttrium Leaching. By directly interfacing the SOM with a molten bath, there is the added chance for yttrium to leach out of the membrane into the bath (comparing Figure 14a to Figure 14b, where Y leaching to air is very unlikely). For every 2 Y^{3+} incorporated on the Zr^{4+} sublattice (e.g. $2Y_{Zr}'$) there is an accompanying single oxygen vacancy ($V_O^{\cdot\cdot}$), which is required to carry the current across the membrane (from anode to cathode during electrolysis). Therefore, losing the Y^{3+} to the bath under neutral conditions by definition means irreversibly losing charge carriers and degrading the O^{2-} conductivity of the SOM, also possibly initiating the membrane darkening discussed above. There is a clear entropic driving force driving Y leaching into the bath, which can be mitigated by introducing YF_3 (or Y_2O_3) into the bath as an additive and keeping the bath saturated with oxide, as seen in the following proposed chemical reaction:

$$Y_2O_{3\,(ZrO2)} + 3Al_2OF_6^{2-}{}_{(bath)} \rightarrow 2YF_{3\,(bath)} + 3Al_2O_2F_4^{2-}{}_{(bath)}. \quad (6)$$

In this regard, the membrane microstructure likely plays a large role– grain boundaries in YSZ are known to be higher in Y content (e.g. Y segregation)[30] and penetration of the bath flux into YSZ grain boundaries has also been observed.[31] Therefore, one other route toward mitigating yttrium leaching is to reduce the role of grain boundaries by increasing the SOM grain size or even incorporating a single-crystalline YSZ coating. Indeed, single crystalline YSZ also displays greater resistance to blackening compared to polycrystalline samples.[25]

Summary of materials challenge

At the materials level, the current density target (~0.5 A/cm^2) translates to the SOM requiring the ability to sustain a high flux of O^{2-} ions, specifically sufficient diffusivity for diffusion times on the order of 10^0 seconds through the SOM. At the same time, the SOM lifetime target requires equivalently stringent criteria: maintaining stability with the bath by minimizing the reaction-rate/flux involving electrons and non-O^{2-} ions across the

SOM, specifically with diffusion/reaction times on the order of 10^5 seconds or longer. Without the combined judicious selection of bath composition, temperature, SOM thickness/geometry, and operating parameters (current/voltage program), these two requirements can often be at odds with each other.

Degradation during static soak tests

The static soak tests, where the SOM is kept in the bath without any applied electromotive potential, are a useful way to separate sources of SOM failure. Electrochemical blackening/darkening requires a high voltage bias applied across the membrane, but Y-leaching is a net charge-neutral process driven by entropy. Therefore, from static soak experiments we can obtain an idea of the background Y-leaching rate and then later extrapolate the additional effects of running electrolysis.

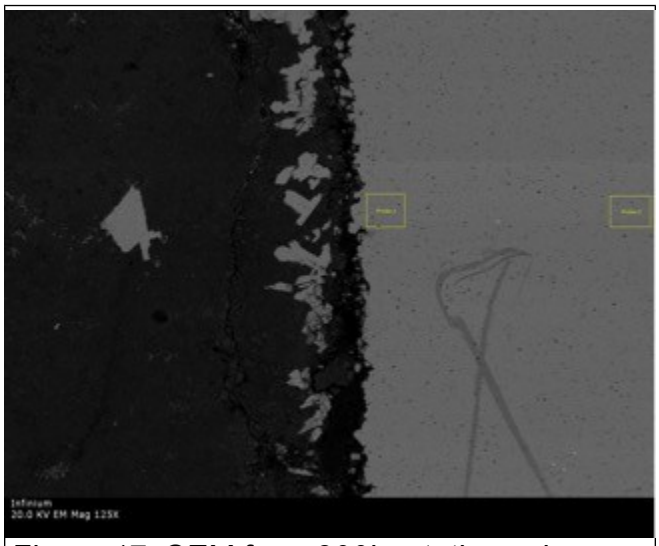


Figure 17: SEM from 200hr static soak experiment. K-Cryolite, CR=3, saturated Al_2O_3, 1.5 wt% YF_3, 965°C

From earlier reports, 200hr and 500hr static corrosion tests (comparing Na- vs. K- cryolite, YF_3 addition, saturated vs. unsaturated with Al_2O_3, and varying the CR ratio) show ~200 μm bath penetration into the SOM but with a characteristic undissolved discontinuous outer layer from the YSZ membrane, with an example shown in **Figure 17**. In comparison to static soak tests with bubble-induced agitation reported below on p. 82 and YSZ membranes that have undergone electrolysis, there is no such preserved outer YSZ layer. In static soak experiments, lack of agitation/stirring ensures the preserved outer layer locally saturates the membrane surface with dissolved YSZ and reduces the driving force for further Y-leaching, and underestimates the background dissolution rate under operating conditions.

New static soak tests show two common phenomena: (1) Y-leaching from the membrane surfaces and (2) elements comprising the bath being detected in the membrane interior. Both of these phenomena were also observed in detail in the literature,[32] specifically examining YSZ membrane degradation in static $CaCl_2$–CaO melts at similar temperature (~1000°C). In this work, Y-leaching is observed and accompanied by a crystallographic phase transition from cubic to monoclinic , leading to a reduction in O^{2-} diffusivity and increase in porosity initiated from volume expansion. Furthermore, a positive kinetic feedback loop is completed as the bath penetrates through the created porosity and reacts further with the membrane, with confirming evidence of Cl being detected in grain boundaries in the membrane interior (**Figure 18**). This explanation also supports the observations (SEM micrographs) described in detail

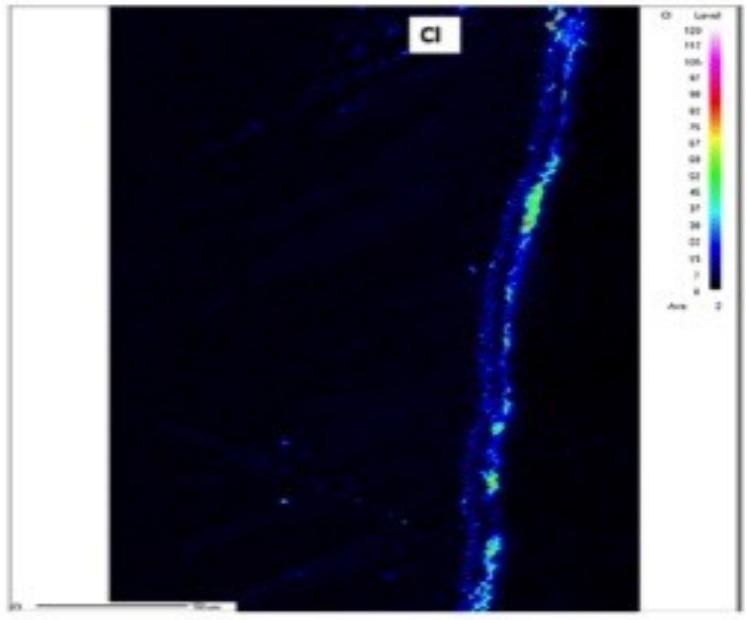


Fig. 6. Chlorine mapping of YSZ- 8 mol% reacted in molten $CaCl_2$ at 1000°C for 16 h.

Figure 18: Chlorine mapping of YSZ-8mol% reacted in molten $CaCl_2$ at 1000°C for 16 hr, from ref 32, showing clear bath penetration into YSZ grain boundaries.

below where components of the bath are detected in the membrane interior.

Degradation during electrolysis

As a significant voltage bias is applied to drive O^{2-} ions through the SOM to maintain our target current density, electrochemical blackening/darkening now comes into play. From the overview of membrane degradation above, the membrane darkening can be thought of as a consequence of the current passing through the SOM being sustained more by conducting electrons than transported O^{2-} ions (from the bath), leading to membrane reduction and O^{2-} blocking. First, as described in the previous section on static soaks, background Y-leaching reduces the number of available O^{2-} vacancies in the membrane, so mitigating Y-leaching should also directly benefit SOM stability during electrolysis and should remain a top priority. Second, reducing the SOM thickness is the most direct way to reduce O^{2-} migration times in the SOM, and by virtue of sustaining more of the total current with O^{2-} migration, also reduces the O^{2-} mass transport limitation which drives SOM blackening. Third, to produce metallic Al at the cathode, we have learned through experiment and data from the literature[33],[34],[35] that a critical current density is required, at the risk of producing so-called 'metal fog' or partially reduced 'black crud' instead. Supported by experiments using thinner membranes in this report (e.g. **Exp035** in **Task 2.1**), higher current density can be sustained resulting in the reliable formation of high-purity metal product (improved initial cathodic current efficiency). The eventual formation and accumulation of electronically conducting products in the bath that migrate back to the anode, however, provide a consistent reservoir of electrons to drive and exacerbate membrane blackening. Unaddressed, this can be thought of as another positive feedback loop that accelerates membrane degradation after several hours of reliable operation.

Recommendation and Future Directions

A clearer picture emerges suggesting that the initiating step toward accelerated SOM degradation is Y-leaching, and suggests that if this issue is sufficiently addressed, the rate of corrosion (even during electrolysis) can be significantly improved by reducing or

avoiding feedback loop driven degradation. Although more precise calibration of bath composition (e.g. tailoring CR ratio, temperature) is still required,

1. YF_3 and Y_2O_3 addition appear effective in reducing the driving force for Y-leaching (reducing bath penetration and qualitative membrane degradation with 10wt% Y_2O_3 addition to the bath in Task 2.1).

The low price of Y_2O_3 ($6/kg on July 5, 2016 according to mineralprices.com) helps with this. Another route to improved lifetime supported consistently by data is:

2. Optimizing the membrane microstructure (coarsened grains) to minimize the role of grain boundaries acting as short-circuit pathways that accelerate SOM degradation driven by bath penetration.

Thinner anodes, supported by initial data in this report, facilitate faster migration of O^{2-} and offset deleterious reactions having to support the applied current.

Engineering Approach to Longer Solid Electrolyte Lifetime

The engineering approach to tube lifetime seeks to improve membrane performance by controlling aspects of the environment around it. For example, the above understanding of material degradation mechanisms leads to three engineering solutions:

- Mechanisms to minimize or separate the cathode reaction products, such as particular complex ions, from back-reacting with the anode.
- Current/voltage pulsing is another actionable strategy that can mitigate reverse membrane blackening (reduction) and extend SOM life.
- Vigorous stirring can increase oxide ion concentration at the membrane-bath interface. This is based on the observation that static tests with bath saturated in alumina perform much better than those with below-saturation alumina concentration, and that oxide ion concentration in bath during electrolysis falls both locally in a boundary layer around the membrane and globally due to removal through the membrane.
- Our experiments also seemed to show that providing excess alumina beyond saturation can address depletion over time, as it does not dissolve but stays in suspension and dissolves as electrolysis removes oxide from the bath. In an operating Hall-Héroult cell, this is not an option because suspended oxide with higher density than the bath could settle and get under the liquid aluminum pad, forming a sludge layer increasing resistance between the cathode and pad. This also may not be an option in our system in Configuration 2 (Figure 81 p. 101).

These represent specific challenges and engineering solutions to achieve both high current density and sufficient SOM anode lifetime.

Based on these observations, our hypothesis is that accelerated membrane corrosion during electrolysis vs. static testing is at least partially due to low oxide concentration in the bath at the membrane interface. The other possible explanation is bath electronic conduction with low membrane electronic conductivity as described on page 10. But the high current efficiency observed in many experiments suggests that bath electronic conductivity is too low for this to be a major problem.

Modeling suggests that stirring would thin the boundary layer in the bath adjacent to the membrane with low alumina concentration, resulting in faster oxide ion (or related complex ion) mass transfer, and higher oxide and alumina concentration at the membrane-bath interface.

Industrial Aluminum Electrolysis Cell Design Narrative

Overview

INFINIUM has developed three cell design options for applying Pure Oxygen Anodes™ technology to aluminum production. They offer increasing techno-economic reward for increasing levels of risk. Configuration 2 uses a traditional graphite block cathode with steel current collector bars for bath chemistry with lower density than liquid aluminum. This document describes the three design configurations and detailed cell design in a narrative, with the design geometries themselves and narratives as separate sections.

Design Principles

Industrial aluminum electrolysis cell design is very different from experimental design. Cost is king, the cell must be made of cheap materials, and should be as energy-efficient as possible. The Hall-Héroult cell is very successful in this regard: it is brutally simple, inexpensive to build, robust, and more efficient than comparable processes.

There are several materials compatibility rules governing the construction of cells:

- Liquid aluminum cannot touch steel, it dissolves steel rapidly.
- Carbon is relatively stable in contact with liquid aluminum, e.g. behind or under a liquid aluminum pad, it rapidly forms a stable aluminum carbide (Al_4C_3) layer.
- Molten salt bath catalyzes rapid oxidation of metals and some compounds such as TiB_2 by dissolving protective oxide layer, exposing metal for further oxidation.
- Molten salts also catalyze rapid corrosion of carbon at the bath-metal interface by dissolving the protective Al_4C_3 layer.
- Steel at cathode potential may not touch bath: aluminum forms at the interface.

In addition, there is considerable concern among industry partners about our ability to make certain features work in an industrial cell. One in particular indicated that the steel shell is “a big red flashing light” in our cell design. The floating liquid aluminum pad configuration described in the proposal is also a risky design. Based on this, INFINIUM has developed three design concepts, including one which can simplify deployment, though which loses some of the benefits of our more sophisticated cell designs.

Design Configurations

Figure 19 shows three INFINIUM industrial aluminum electrolysis cell design concepts as they stand today. Benefits and drawbacks of each design are as follows.

Conventional cell with Pure Oxygen Anodes™ insert

Figure 19a shows a conventional Hall-Héroult cell with liquid aluminum cathode and frozen salt crust side-wall, but with zirconia Pure Oxygen Anode tubes in clusters with the anode bus and oxygen manifold. This minimal risk deployment strategy could bring the following advantages over conventional Hall-Héroult:

- No carbon anode plant;
- No impurities introduced by carbon anodes;
- Clean anode gas with no HF or other bath volatiles;
- Anode flexibility: produce oxygen by-product or use natural gas to reduce energy cost;
- Recover anode gas thermal energy;
- No need to lower the anodes due to consumption (cf. carbon);
- Fixed-geometry leads and lower energy loss;
- Some passive response to runaway heating: zirconia resistance decreases with temperature.

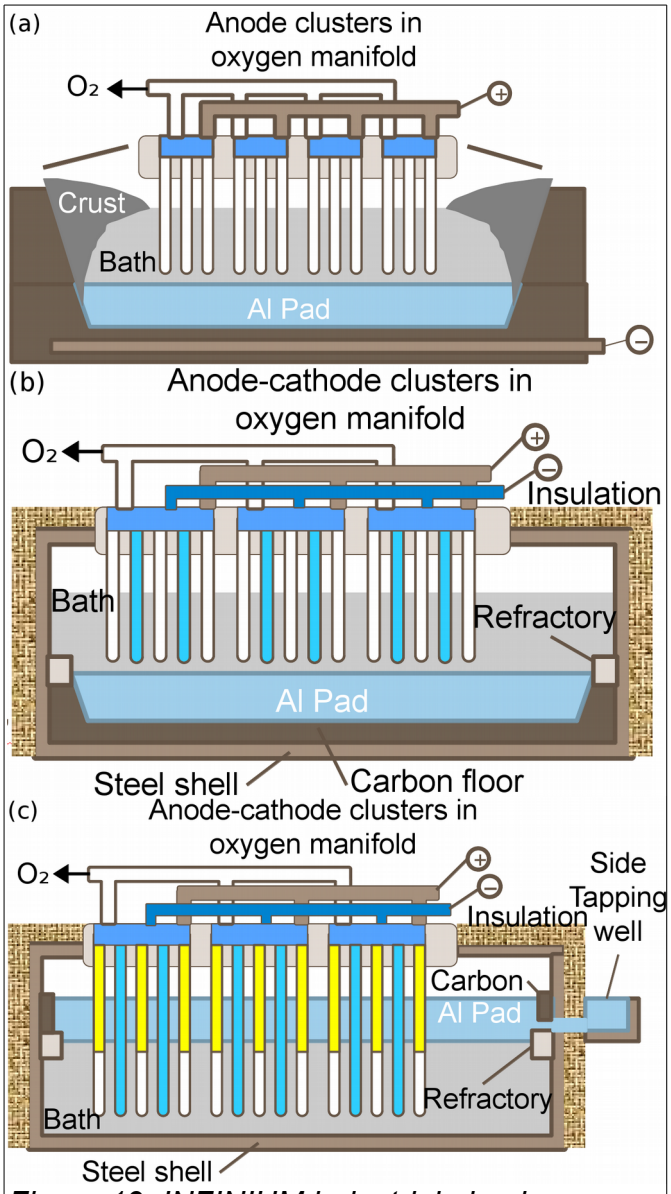


Figure 19: INFINIUM industrial aluminum electrolysis cell design concepts: (a) conventional cell with insert; (b) sealed steel shell with low-density bath; (c) sealed steel shell with floating aluminum pad.

This design meets all of the design criteria by using a steel cathode bus bar in a carbon floor, with the bath-metal interface protected by the frozen side-wall. No metal components protrude from the surface, so catalyzed oxidation is not a problem.

This design does however introduce a potential problem: rough crust breakage during alumina feeding could potentially break one or more tubes. Also, average current density in zirconia tubes is low, and current is concentrated at the tube ends, tube cost is thus high relative to its productivity. And it may require auxiliary bath stirring, as there is no CO_2 anode gas lift.

Sealed steel shell with low-density bath

Figure 19b shows a cell design with sealed steel shell and liquid aluminum pad on the bottom. A carbon floor separates the aluminum from the steel shell. The shell is stable in contact with the bath. A highly-stable refractory material, such as partially-stabilized zirconia, boron nitride, silicon nitride, or a conductor such as titanium diboride (TiB_2) protects the steel and carbon from the bath-metal interface; this can be in the form of bricks or plates, or applied directly to the shell by thermal spray or similar method.

The cathode for this arrangement must be an electrical conductor which is stable at bath-metal interface, for which only TiB_2 and ZrB_2 are known to work well. Those materials are expensive, with TiB_2 costing around $50/kg and ZrB_2 considerably more. Closed-end TiB_2 tubes will have lower cost, and if filled with liquid aluminum, the resulting cathodes will conduct well, breakage should not contaminate the product.

This configuration adds the following benefits beyond that of Figure 19a:

- Higher productivity per unit cell cross section and smaller cell house building;
- A TiB_2-lined steel shell at cathode potential is stable in the inerted sealed cell;
- Insulation around the steel shell reduces thermal losses;
- Cell heat balance management by direct external heating or cooling need not deal with crust management;
- Feeding is simplified without crust to break;
- Higher zirconia anode tube productivity per unit cost due to vertical cathodes;
- No building crane is required.

This also resolves the salt crust breakage issue of the first configuration. However, if the steel shell is cathodic, then aluminum metal will form at the steel and cause problems. This can happen one of two ways: a cathode can short to the shell or to the aluminum pad, or volatile reactive metal such as sodium or calcium can condense on the shell in electrical connection to the bath. We will either over-insulate between the bath and shell in order to be conservative, or investigate methods to electrically determine when a short happens, isolate the offending cathode, and resume operation.

Sealed steel shell with floating aluminum on high-density bath

Figure 19c shows a cell design with floating liquid aluminum pad on a high-density molten bath. This configuration is not possible with a conventional Hall-Héroult type cell, as the aluminum would rapidly oxidize in the presence of air and/or the CO_2 anode product, but is enabled by the withdrawal of anode gas through the zirconia tubes and by the sealed and inerted vessel.

This configuration would require sheathing to separate liquid aluminum from the zirconia tubes in order to avoid shorting and tube damage. This sheath can be made of boron nitride, silicon nitride, or a conducting material such as TiB_2. If it is TiB_2 or highly-

conductive material, it will need spacers between the sheath and zirconia tubes to prevent shorting, and gas flow down the annulus between them to prevent aluminum ingress into the annulus. If BN, Si_3N_4 or low-conductivity zirconia (*e.g.* 3 mol% Y_2O_3 or 6 mol% CaO), then it can be a separate tube with spacers, or a coating applied directly to each tube by plasma spray or similar method.

Additional potential benefits of this configuration are as follows:

- Further simplified alumina feeding: the Hall-Héroult requires alumina with prescribed particle size distribution, as fines sinter easily into large particles which sink. In this cell, particles or agglomerations which sink remain in the bath and dissolve, permitting use of fines, and potentially reducing alumina cost.
- Bath composition: because alumina can sink and remain in the bath, *i.e.* does not form sludge, the cell can operate with alumina at or above saturation. That is, it can include suspended oxide in bath, or large oxide particles such as dross or sintered fines, without concern about forming a sludge layer between the pad and cathode. This can improve raw material flexibility, ionic conductivity, current density and productivity, and may extend zirconia tube lifetime.
- Impurities tolerance: calcium increases bath density; in today's Hall-Héroult cell and Configuration 2 this leads to a density inversion: bath goes from less dense than the pad to more dense. Configuration 3 welcomes high-density impurities, and opens the door to hydrated lime as an upstream precipitating agent.
- Lower bath evaporation: the aluminum pad acts as a barrier to evaporation of $KAlF_4$ or other volatile bath species.
- Less product contamination: if a membrane tube breaks in the configurations of Figure 19a or 19b, the dense liquid metal anode falls into the liquid aluminum product; in this case it falls away from the aluminum, though a small amount might contact aluminum on its way up to the pad.
- Simplified tapping: it may be possible to add a side tapping well. This could facilitate automated tapping by a robotic device moving from cell to cell.
- Enhanced magnetohydrodynamic (MHD) stirring: all of the current buses, anode and cathode, are above the bath, leading to a continuous current sheet and very high magnetic field in the bath, resulting in vigorous stirring for enhanced alumina mass transfer both across the cell and vertically over all electrode surfaces.

Detailed Design Narrative: Configuration 2

Based on the request of a potential aluminum scale-up partner, INFINIUM conducted a design study for a 200 kA scale industrial electrolysis cell. It will use Configuration 2 corresponding to INFINIUM's low-density bath. This also provides a compromise between risk and benefit: this will allow separate anode (top) and cathode (bottom) electrical buses, reducing complexity of the upper manifold and bus connections. However, this configuration will complicate metal removal, and will result in significant

movement of the molten salt bath as the metal pad grows and shrinks under it during a 24-hour cycle of production and tapping. It will also not have as much MHD stirring opportunity.

Overall cell dimensions and performance

Figure 20 shows the overall cell top view and longitudinal cut-away section. The cell uses 600 anode tubes at 333 A each, with 73 cm length 3.4 cm ID 4.0 cm OD (reducing to 3.8 cm OD during its lifetime), immersed with 30 cm active length, giving 320 cm² inner anode area and 1.04 A/cm² maximum current density. The 480 cathode tubes are closed-end TiB_2 tubes 4 cm outer diameter extending 86 cm from the lid downward into the aluminum pad, operating at 0.74 A/cm² as they extend through the full 45 cm bath. They are filled with liquid aluminum for high conductivity, low cost, and no product contamination.

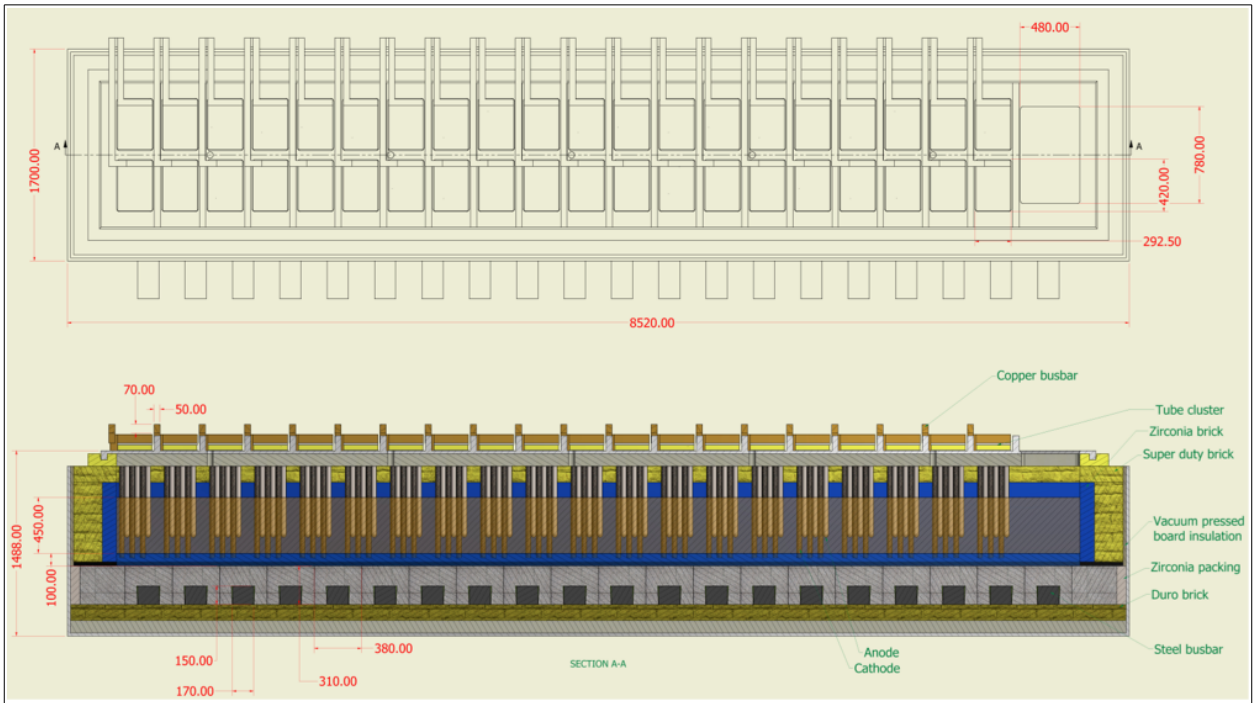


Figure 20: Top view and cut-away section of the INFINIUM 200 kA cell.

Cell production rate at 200 kA is 64 kg/hour at 95% current efficiency. With a density of 2.3 g/cm³ and inner cell dimension above, this leads to metal height rise rate of 4.2 mm/hour, or 10.1 cm over a 24-hour period between collections. If electrodes remain in place as the aluminum level rises and falls below them, then cathodes will need to extend through the depth of the bath plus 10 cm and some clearance into the liquid metal, and zirconia tube immersion depth will change by this much with each tapping.

With the bath moving up and down about 10 cm, its overall thickness is 45 cm: 30 cm for the tube active area (with anode liquid against YSZ inner surface), 10 cm extra to move up and down, plus an extra 4 cm to maintain complete coverage of the anode active area (3 cm below minimum, 2 cm above maximum). The liquid aluminum pad is

10-20 cm thick: 10 cm just after tapping, 20 after 24 hours of production. There is an additional 10 cm of head space above the maximum bath height, and 10 cm of insulation hanging below the lid. Figure 21 shows schematically this effect of liquid aluminum height change on cell geometry.

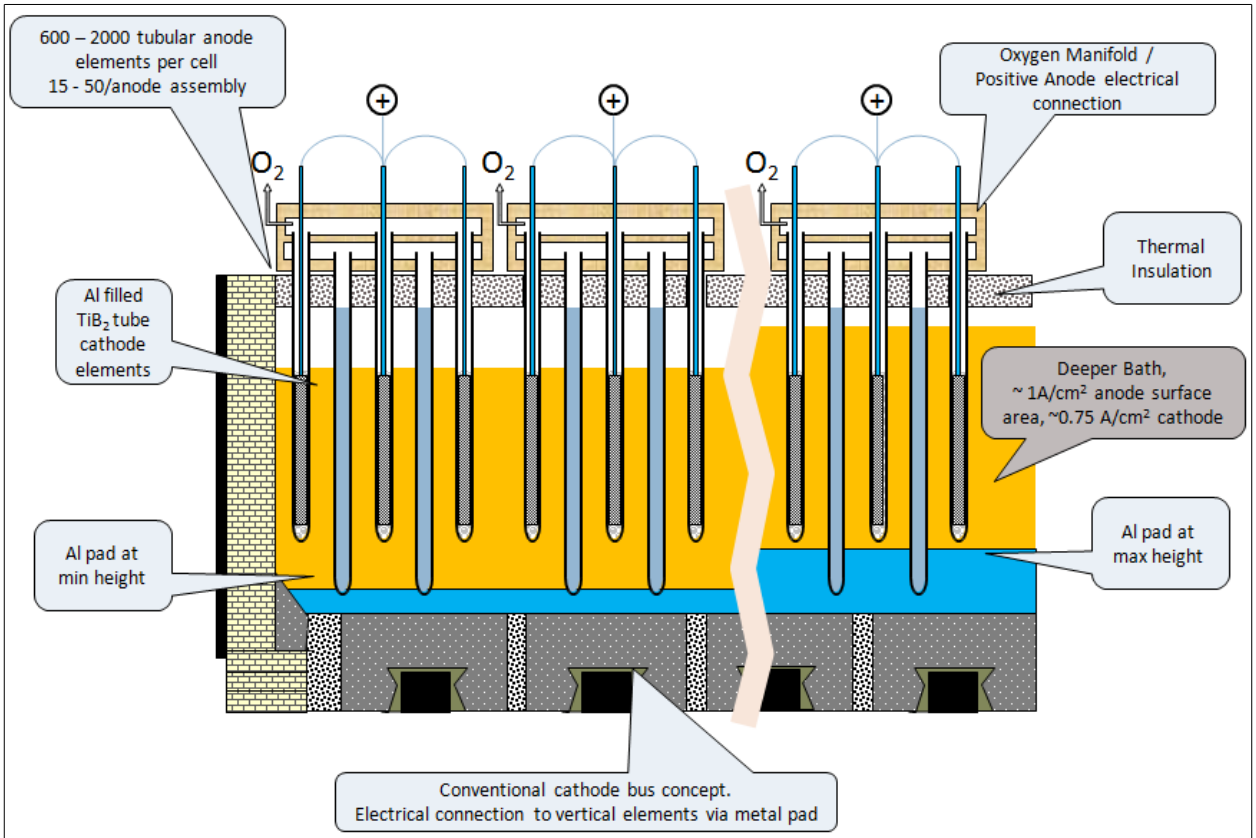


Figure 21: Schematic showing geometry changes during metal production and tapping.

Steel side walls 6 cm thick surround the cell. To be conservative, the design includes 1-3 (currently 3) courses of refractory bricks leading from below the lowest bath-metal interface height up above the highest bath surface height, about 60 cm, in order to prevent aluminum reduction at the steel which would severely damage it.

The present design features a layer of low-density insulating sintered zirconia brick, such as those made by Zircoa, extending from well below the bath-metal interface as a conservative measure to prevent any bath ingress into the steel. It has two additional layers of lower-cost insulating brick, such as Allied Minerals Superduty, in a double Flemish bond 3-course thick arrangement. This provides chemically robust zirconia facing the bath and at the bath-metal interface, and with low conductivity such that the bath melt isotherm is either in the zirconia or as close behind it as possible.

With a high-quality TiB_2 liner electrodeposited inside the steel shell, then brick layers are not necessary. This TiB_2 liner would in turn require an inerted cell; without it bath-catalyzed oxidation would destroy the liner.

Bottom Cathode Current Collector

The cathode current collector consists of twenty extruded graphite blocks running the full width of the cell, 38 cm wide, 31 cm high; these support the aluminum pad. Twenty steel bars 12.5 cm wide 10 cm high conduct current out of the cell with maximum current density of about 80 A/cm² (516 A/in²). Each is placed into the groove of an extruded graphite block, and filled with cast iron between the steel and graphite to form a good electrical connection and prevent graphite swelling. There is thus 20 cm carbon separating steel current collectors from the aluminum pad. Seams are filled with carbon paste. The steel bars protrude from the side of the cell, passing through the shell with insulation to reduce the probability of the shell reaching cathode potential.

Below the carbon blocks are two courses of high-density insulating brick, such as Fizion Technologies Duro, followed by vacuum pressed insulation.

Figure 22: Anode-cathode cluster section showing active anode area.

Electrode clusters and top plate

Electrodes are arranged into clusters shown in Figure 22 each with five rows of three anodes, interspersed with four rows of three cathodes staggered from the anodes, for a total of 15 anodes, 12 cathodes, and 2500 A current. Overall cluster dimensions are approximately 29 cm wide by 42 cm long. (Cathodes have the same outer diameter than anodes, but anode current density is measured at the inside and has a shorter active length, so cathode current density is about 0.7 A/cm².) They are arranged in two rows of 20 clusters with inner width about 91 cm (two cluster lengths plus center feed port area and other structures) and inner length about 773 cm (20 cluster widths plus supports and other structures and tapping well door).

This electrode arrangement leads to an effective anode-cathode distance (ACD) of 1.93 cm as predicted by FEA. At 0.5 A/cm² current density, the bath resistance overpotential is 0.57 V, which meets the project's energy target goals. INFINIUM is exploring design options for further reducing this overpotential, such as increasing electrode length to decrease current density, or changing configuration to reduce ACD.

Passive TiB_2 cathode connection tubes dip into the molten aluminum pad below, and are filled with molten aluminum (charged with solid metal to start). The cluster bottom plate is insulated beneath with an insulation system to be determined. The TiB_2 tubes are fixed by both the insulation and the cluster bottom plate. The zirconia tubes then

emerge upward through this layer and are fixed to the middle cluster lid. The top cluster lid (or external lid) electrically connects to the center current collector conducting rod of each anode, this is a bolt connection. It may be possible to assess anode failure either open or closed by measuring current collector conducting rod top temperature using infra-red imaging.

A sealed interior space between the middle cluster and top/external cluster lids collects oxygen produced at the anodes at a single gas line attachment. This gas phase is separate from the molten aluminum used in the passive cathodes. At 5000 A, oxygen flow rate from each cluster is about 0.3 standard liters/minute (SLPM). The external lid can be removed to reveal the zirconia tubes, allowing for inspection and removal; the cluster bottom plate can be removed (breaking the cell seal) to inspect the TiB_2 cathode tubes. The whole cluster can be removed as a single cartridge and replaced.

The 2.5 cm thick main lid plate has large cut-outs for each electrode cluster bottom plate. It includes cross-ribs 5 cm wide by 15 cm high in order to provide strength and rigidity needed to support the weight of electrode clusters over a hot cell, these ribs are welded to the top/outside surface. A 10 cm thick insulation system hangs beneath the lid plate, to go between the clusters' integrated 10 cm thick insulation blocks.

The lid includes five alumina feeders down the center line, with four pairs of clusters between each feeder, and two cluster pairs on each end. There is a central gap of about 7 cm between the clusters allowing room for alumina feed ports. There is also a large door on one end of the top plate for tapping, as shown on the right of Figure 24 and in more detail on the right of Figure 23.

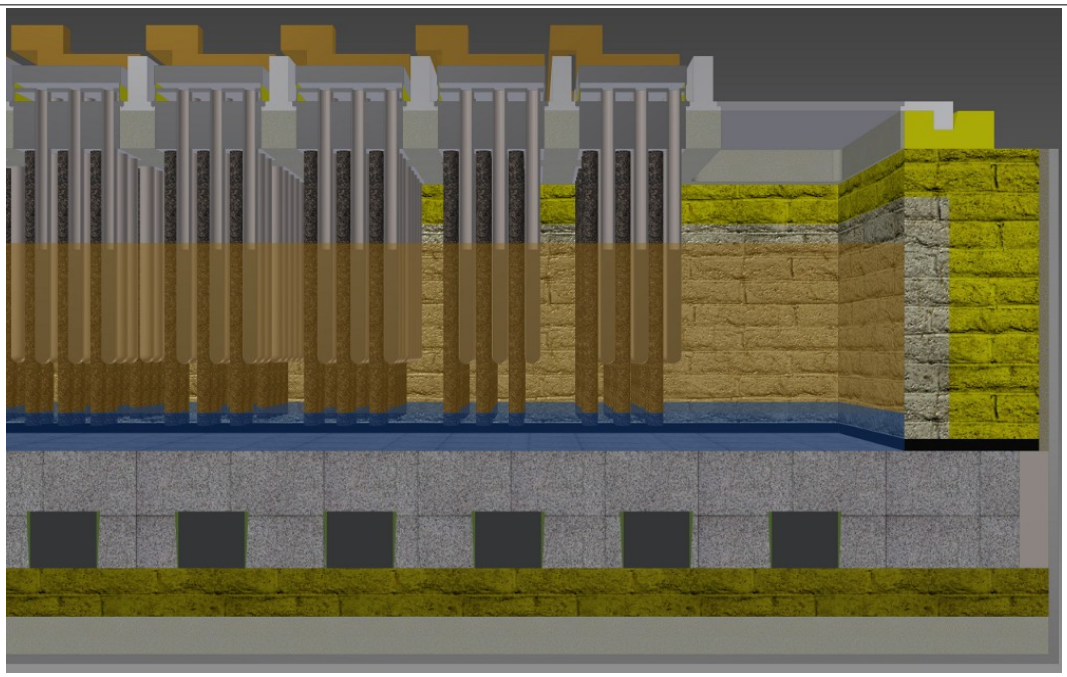


Figure 23: Complete cell cross-section detail showing anode-cathode clusters and door for metal withdrawal.

At the end of each cluster opposite the cathode bus, there is a copper bus bar 5×7 cm in cross-section joined to the end of the aluminum cluster top plate, which serves as the cluster anode bus connection. Its current density is 143 A/cm². These cross the width of the cell above the lid ribs, connect together at the top edge opposite the cathode bus, and run down that side of the cell as a 10×7 cm bus riser, also with 143 A/cm² current. These can either each connect directly to one steel bar at the cathode of the neighboring cell, or join together in a large cell bus.

Detailed Design Narrative: Configuration 3

Based on the request of a potential aluminum scale-up partner, INFINIUM conducted a design study for a 200 kA scale industrial electrolysis cell using configuration 3. Anode and cathode buses will both be at the top, which increases the complexity of the upper manifold and bus connections. There will also be increased MHD stirring. However, this configuration will simplify metal removal, and will eliminate movement of the molten salt bath as the metal pad will grow on top of it.

Overall cell dimensions and performance

Figure 24 shows the overall cell top view and longitudinal cut-away section. The cell uses 600 anode tubes at 333 A each, with 93 cm length 3.4 cm ID 4.0 cm OD (reducing to 3.8 cm OD during its lifetime), immersed with 30 cm active length, giving 320 cm² inner anode area and 1.04 A/cm² maximum current density. The 480 cathodes are TiB_2 coated nickel rods of 4 cm outer diameter extending 87 cm from the lid downward through the aluminum pad, operating at 0.74 A/cm².

As before, anodes are on an 8.5 cm square lattice, with cathodes also on an 8.5 cm square lattice translated to the centers of the anode lattice, leading to effective anode-cathode distance of about 2 cm. Overall cell footprint is 852×170 cm, with inner pad footprint of 773×91 cm. Vertical current density in the aluminum pad is thus just below 2.8 A/cm².

Cell production rate at 200 kA is 67 kg/hour. With a density of about 2.3 g/cm³ and inner cell dimension above, this leads to metal pad height increase of 4.2 mm/hour, or 10.1 cm over a 24-hour period between collections. The liquid aluminum pad is 10-20 cm thick: 10 cm just after tapping, 20 after 24 hours of production. There is an additional 10 cm of head space above the maximum pad height, and 10 cm of insulation hanging below the lid.

Because the growing/sinking pad is on top of the bath, the solid electrolyte-bath interface is stationary. Each solid electrolyte tube is sheathed through the alumina pad by a ceria tube in order to prevent short-circuiting.

The aluminum cathode/pad can connect to the cathode bus in two ways. One is through the vertical TiB_2-coated vertical cathode extenders to the top lid. The top lid then becomes cathodic, with the solild electrolyte tubes providing insulation between the lid and anode current collectors. Due to low zirconia conductivity at the low temperature of the lid, this should be an effective insulator in that part of the cell.

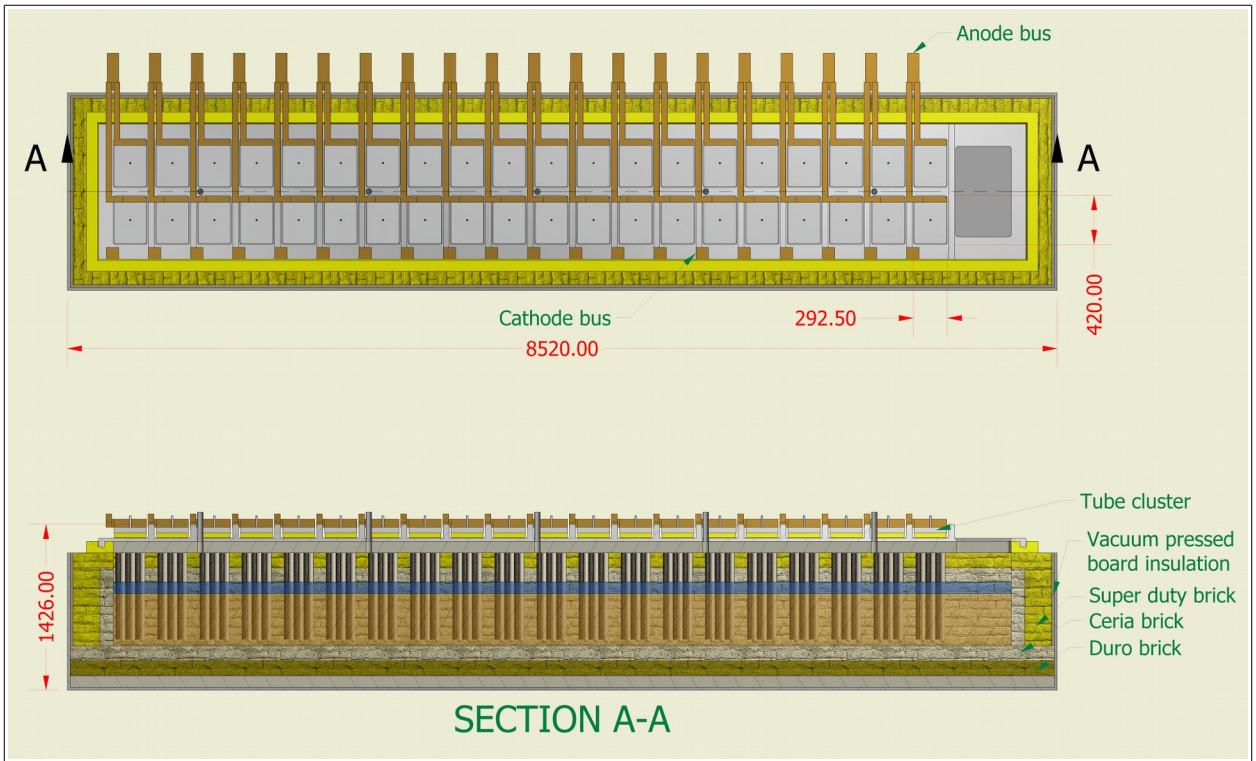


Figure 24: Top view and cut-away section of the floating-pad INFINIUM 200 kA cell.

The second option is to use liquid/solid aluminum cathode connections like those described below (p. 39). Though it exhibits high current density and small cross section, this design would produce significant horizontal current in the pad, with potential MHD implications.

Figure 25 shows the metal withdrawal door. Unlike the bottom pad design, the liquid metal siphon would have to find the best height for liquid metal tapping, instead of simply going to the bottom of the cell.

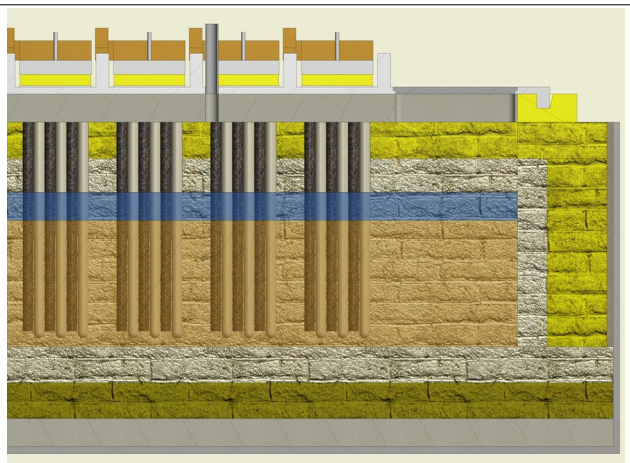


Figure 25: Cell cross-section detail showing anode-cathode clusters and door for metal withdrawal.

Other Cell Design Considerations

Lorentz forces and magnetohydrodynamic (MHD) stirring

The large current buses in the bottom steel bars and top copper bus bars will induce a magnetic field in the cell. At the mid-plane between them, the field from the top and bottom will roughly cancel each other, so any net field will come from asymmetry: greater proximity of most of the bath to the top or bottom. Assuming much of the current passes horizontally through the liquid aluminum due to its high conductivity, that is much closer to the bath, and will dominate the field there.

Approximating the aluminum pad as an infinite current sheet with current density per unit length I/L, the magnetic field due this current is given by:

$$B = \mu \frac{I}{L} \tag{7}$$

In our system, I/L is 27.8 kA/m, and the field is in the direction of the length of the cell, across the anode clusters. In the bath, the main component of the current is between anodes and cathodes, in the direction of the length of each anode cluster and the width of the cell. Minimum current density in the bath is given by half the anode current divided by the cross section area per anode, which is 83A÷(6cm×30cm) = 0.46 A/cm² = 4.6 kA/m². Based on this, Lorentz force acts in the vertical direction, alternating between upward and downward in the channels between rows of anodes and cathodes, leading to alternating vertical convection cells. Force per unit volume *f* is given by:

$$\vec{f} = \vec{J} \times \vec{B}, |\vec{f}| = J \mu \frac{I}{L}. \tag{8}$$

Using the magnetic constant for a vacuum $\mu_0 = 1.26 \times 10^{-6}$ m-kg/s²A² gives a force magnitude of 161 N/m³. Maximum velocity in a channel is given by:

$$u_{max} = |\vec{f}| \frac{L^2}{16 \eta}, \tag{9}$$

where *L* is the channel width and η is the viscosity. In a 2 cm wide molten salt channel with viscosity 0.03 Pa-s, maximum velocity is 13 cm/s. This is a reasonable velocity for alumina distribution, and with density 2300 kg/m³, leads to a Reynolds number close to 206, indicating laminar flow. This result is very sensitive to channel width *L* such that electrode geometry will play a major role in determining actual MHD-driven velocity.

Looked at another way, dividing force by density gives about 6.9 cm/s² acceleration. Over the 30 cm vertical distance, this gives maximum velocity as: $u_{max} = \sqrt{2ad}$ or about 20 cm/s. This is relatively close to the result of the previous calculation.

Note that magnetic field will only be this high in the immediate vicinity of the aluminum pad, considerably smaller above it and on the ends of the cell. The analysis above therefore applies to Configuration 3 (aluminum pad on top) in the top portion of the cell near both the anode and cathode buses, as shown schematically in Figure 26. In

Configuration 2, anode and cathode buses produce magnetic fields which approximately cancel, leading to perhaps one-tenth of this field on average, and one-tenth of this maximum velocity. The field could go one way on top and the other way on the bottom, leading to opposite convection cells; the cathode-induced field will be stronger on one side and the anode-induced field stronger on the other.

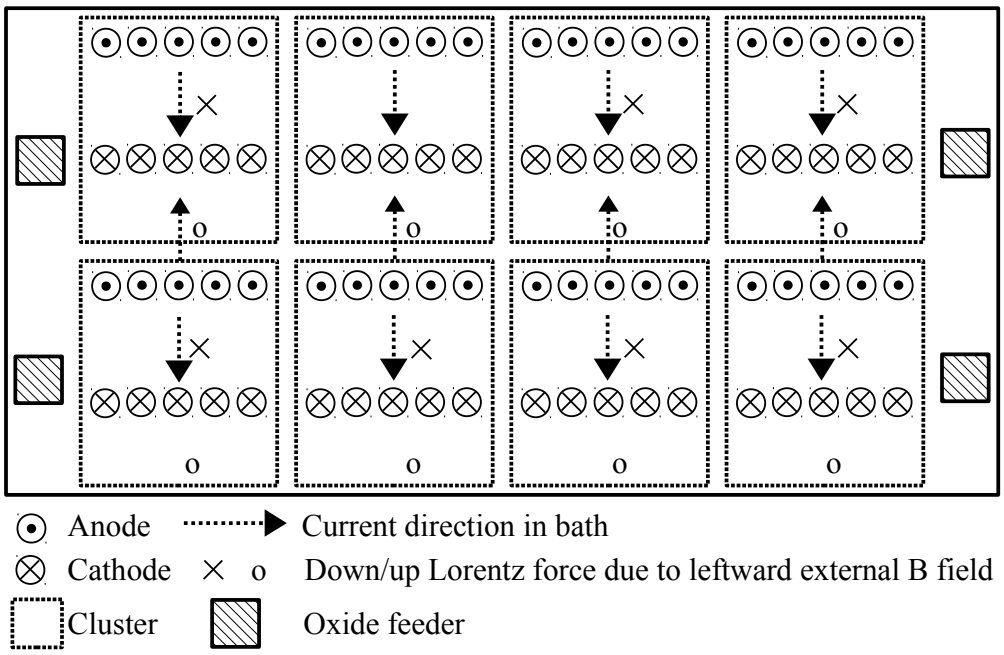


Figure 26: Illustrative cell top view showing anode-cathode arrangement in rows to allow flow due to Lorentz force with a horizontal external magnetic (B) field.

To complicate matters further, current in the vertical electrodes will create additional fields, which will largely cancel each other in Configuration 2, but reinforce each other in Configuration 3. In Configuration 2, getting current from anode bus above to cathode bus below will require vertical bus bars, which will induce their own fields. For all of these reasons, we will need FEA to get a better estimate of velocities. That said, this is likely a good order-of-magnitude approximation, and velocities as low as a couple of cm/s will likely distribute alumina very well, indicating that this system will likely obtain sufficient stirring from MHD to mix alumina well and operate at high current density.

Alumina feeding

As mentioned above, there are five alumina feeders distributed along the center spine, these are shown as small circles in Figure 24 (top view). The feeder will use a sealed valve system in order to maintain the inert/reducing environment in the cell, as feeding will take place much more frequently than tapping. Examples of valve system design include double block and bleed which INFINIUM is currently testing for the 200 A cell, or a rotary or star valve.

As mentioned above, in Configurations 2 and 3 there is no crust to break, making

feeding somewhat simpler. The feed system is therefore similar to the Dow Chemical Company “E Cell” design for magnesium electrolysis.

Note that our selective oxygen ion-conducting anodes can directly measure oxygen activity in the bath, providing an excellent sensor of oxide concentration and distribution to determine when—and where—feeding is necessary.

Cell assembly

The cell is assembled in sections. Since the cathodes are fed from one end, and the next cell is connected in series, one can connect the anode bus directly to the cathode bus of the next cell.

The cell shell is first cleaned, then a layer of ceramic paper is glued to bottom and end, and side parts of the cell. Into the bottom section of the cell vacuum pressed board is then installed, covered by two courses of Duro Superduty brick. On to this layer is installed the cathode bars, these are made by installing a steel connection rod into the machined cathode block and using molten cast iron as the sealant.

After installation of the cathode the free side wall is then installed, the cell is then lined with vacuum pressed board, then bricked. The bottom courses of bricks are carbon blocks or graphite blocks interlocked with a Superduty backing. This is then followed by a course of Superduty bricks with a zirconia face brick (on the hot face side), that is interlocked. The free board of the cell is lined with Superduty brick.

To start the cell, as with a conventional cell, first pour molten bath into an empty shell without any tube clusters installed, wait 8 hours for it to seal in all of the refractories, then pour in liquid metal which settles into the bottom. The electrode clusters can then lower into the cell through the top plate.

Aluminum Cathode Current Collector

As mentioned above, Configuration 3 with a floating aluminum cathode has many advantages. But using a conventional graphite cathode current collector would result in very high current density, very high overpotential, and very high energy consumption. Even Configuration 2 produces 2-4 times the cathode current density of a conventional Hall-Héroult cell. With conventional graphite-steel current collectors this could add considerably to energy consumption by shifting away from the optimal balance between electrical and thermal losses.

Some aluminum producers have experimented with copper core steel bus bars for lower resistance. This would help compensate for the much higher current density through the bottom of our cell than in a conventional cell.

One approach to connecting to a floating pad is to use vertical graphite connectors similar to those of the Hoopes cell[36] in widespread use for refining high-purity aluminum. While they can operate at high current density, the Hoopes cell requires very high voltage and energy consumption in part due to the high overpotential in graphite conductors.

The INFINIUM aluminum current collector creates a non-carbon high-conductivity metallic electrical connection to a floating or submerged liquid aluminum pad in an aluminum reduction cell in one of two ways. First is an aluminum connection from the liquid pad all the way to the external cathode bus near the environment temperature. Because the typical cell operating temperature at 950°C is hundreds of degrees above the aluminum melting point at 660°C, this aluminum connection will be liquid on the end of the aluminum pad in the cell, and solid on the end of the cathode bus connection.

The liquid can be contained in conduit lining materials with low solubility in aluminum, such as but not limited to carbon *e.g.* graphitic carbon, or silicon carbide, or boron nitride, or aluminum oxide, or other aluminum-bearing compounds with elements having low solubility in aluminum. Any of these materials can be used as dense solids, powders, or coatings.

Because these conduit materials are relatively brittle, it is advantageous to provide a strong steel sheath around the conduit, both for mechanical support to minimize stresses on the conduit, and also to limit the amount of metal that can flow out of the conduit in the event of breach of the conduit lining.

Figure 27 shows two such cathode current collector arrangements. In the first, a horizontal conduit or ledge in the cell contains liquid aluminum with a temperature gradient between the cell temperature and aluminum melting point, and solid aluminum plates or shafts connect at the end of that conduit or ledge away from the bath and anodes. In the second, a tube or similar conduit draws liquid aluminum upward from the cell toward the solid aluminum connection. In both cases, the solid aluminum connects directly to the external cathode bus. Both of these arrangements also apply to a cell with low-density bath and submerged aluminum pad, in this case the tube or horizontal conduit would have an end submerged into the aluminum pad below the bath.

It can be advantageous in some circumstances to include a second high-melting conductive material at the environment end of the solid aluminum conductor, such as copper or nickel. With such a connector present, if a strong process transient event heats the conductor rapidly and melts all of the aluminum, the second conductor will remain solid and maintain a connection to the liquid aluminum, avoiding an open connection.

Second, one can connect liquid aluminum from the pad to a second aluminum-saturated liquid metal with low solubility in liquid aluminum, such as tin, bismuth, or lead. As with the first configuration, the liquid aluminum and second liquid metal are contained in materials with low solubility in both metals, such as but not limited to carbon *e.g.* graphitic carbon, or silicon carbide, or boron nitride, or aluminum oxide, or titanium diboride. Any of these materials can be used as dense solids, powders, or coatings. Bismuth and lead have the advantage of minimal interaction with steel, so a break in the liquid metal-facing material would not dissolve and puncture steel the way tin would.

The denser second metal is at a lower temperature than the cell liquid aluminum near it, thus liquid aluminum above the denser metal is stratified (colder second-metal-saturated higher-density liquid aluminum is on the bottom) and resists convection,

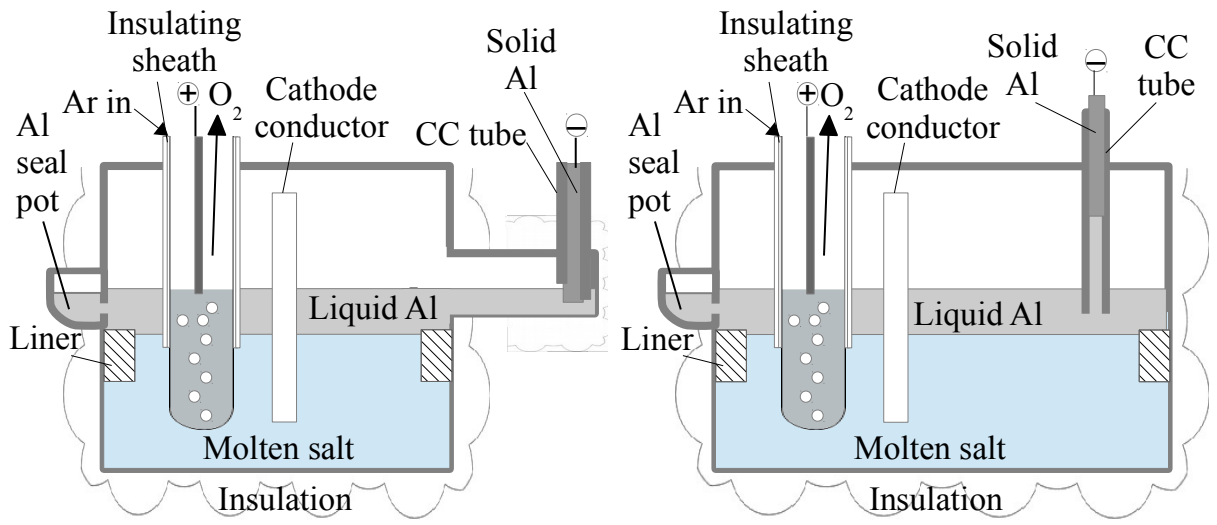


Figure 27: Electrolysis cell configuration with high-density salt and floating aluminum pad (not showing raw material feeder), showing two configurations of aluminum current collectors (CC): liquid aluminum ledge with insulated solid aluminum plate or rod connection (left), and insulating current collector tube or conduit drawing liquid aluminum into contact with solid aluminum (right).

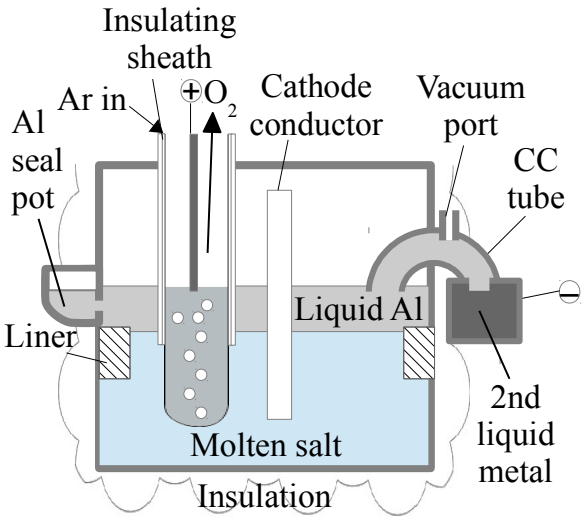


Figure 28: Electrolysis cell configuration with high-density salt and floating aluminum pad (not showing raw material feeder), showing two configurations of aluminum-dense liquid metal current collectors (CC): liquid aluminum ledge with second liquid metal pool (left), and insulating current collector bridge tube to second liquid metal pool (right).

minimizing transport of the second metal through the liquid aluminum and into the aluminum product. Cathode bus connection to the second liquid metal is by immiscible

solid metal, such as a steel connection to bismuth or lead, or carbon connection to tin.

Figure 28 shows two arrangements for this configuration. In the first, a horizontal conduit or ledge contains liquid aluminum with a temperature gradient between the hotter cell temperature and cooler second metal, and the cooler second liquid metal is in a pool below that ledge. In the second, a liquid aluminum bridge tube connects the liquid aluminum pad to the external pool of second liquid metal. One can draw the aluminum upward into the bridge tube by pulling a partial vacuum in it, for example from a vacuum port at the top of the bridge.

This second embodiment has the disadvantage of potential aluminum product contamination by the second liquid metal, but the advantage of eliminating melt interface instability under current control in a single electrode. In the configurations of Figure 27, under constant current an increase in voltage results in higher heating, melting back the solid aluminum, which increases resistance and voltage in the electrode, rendering it unstable. The second embodiment in the configurations of Figure 28 does not have a melt interface and is more stable.

Note that under voltage control, the configurations of Figure 27 are stable: an increase in current results in higher heating, melting back the solid aluminum, and the resulting resistance increase lowers the current, reducing heating and letting the aluminum freeze back to its equilibrium interface position.

That said, consider a pot under current control which has multiple current collectors linked in parallel to a single shared bus bar. In this situation, if one current collector heats up and melts back, when its resistance increases, the current through it shifts somewhat to the other current collectors, reducing heating such that it freezes back. Thus each current collector can be considered to operate under something close to voltage control, and is stable, even though the aggregate is attached to a large bus bar under current control.

Optimal Current Collector Geometry

The Wiedemann–Franz law indicates that a good electrical conductor is a good thermal conductor. There is an optimum balance between electrical and thermal resistance of the current collector to minimize the sum of electrical resistance creating heat, and thermal resistance preventing excess heat loss from the cell. The resistivity of aluminum varies widely over the temperature range from the environment, say 20°C where it is $\rho = 2.65\times10^{-8}$ Ω-m to 927°C where it is 2.89×10^{-7} Ω-m[37]. Since for a material with constant properties, $\Delta V = \rho JL$ (J is current density, L is lead length), this bounds the product JL to be between about 6×10^{5} and 7×10^{6} A/m. Typical maximum industrial aluminum bus bar current density is around 10^{6} A/m² (*i.e.* 100 A/cm²); using this current density would result in optimal aluminum lead length on the order of 1 m.

Optimal geometry with a second metal, whether a high-melting connector such as copper or nickel or a low-melting metal such as tin, bismuth or lead, depends on its composition and temperature distribution.

A 1-D Finite Difference model of electrical and thermal conduction in the liquid and solid current collector helps to quantify the trade-off between electrical and thermal losses. This analysis assumes constant cross-section area of the liquid and solid aluminum ledge or conduit, uniform current density J throughout it, and uniform temperature in any cross-section. It neglects effects of convection and radiation, assumes a single flat liquid-solid interface, and assumes that the outside of the conduit is perfectly insulated. The latter assumption is a design goal for the conduit, as it is most effective at retaining energy when thermal loss from its sides is as low as possible.

This model takes into account temperature-dependent electrical and thermal conductivity of solid and liquid aluminum (per reference 37), and implicitly calculates the location of the melt interface. It sets constant-temperature Dirichlet boundary conditions: 1000°C at the liquid pad end (x=0) and 200°C at the environment end (x=L). It has 20 intervals along the length of the current collector.

Figure 29 shows the calculated temperature-distance relationship along the aluminum for J = 100 A/cm² and L = 0.8 m. Its curvature is high above the 660°C melting point and low below it because the liquid exhibits both higher resistivity and generation of heat, and lower thermal conductivity, than the solid. The slope at the environment end $\partial T/\partial x$ times thermal conductivity k gives the total energy loss per unit cross-section area of the conductor q. Total thermal energy loss per unit metal product is proportional to q/J, which has units of volts.

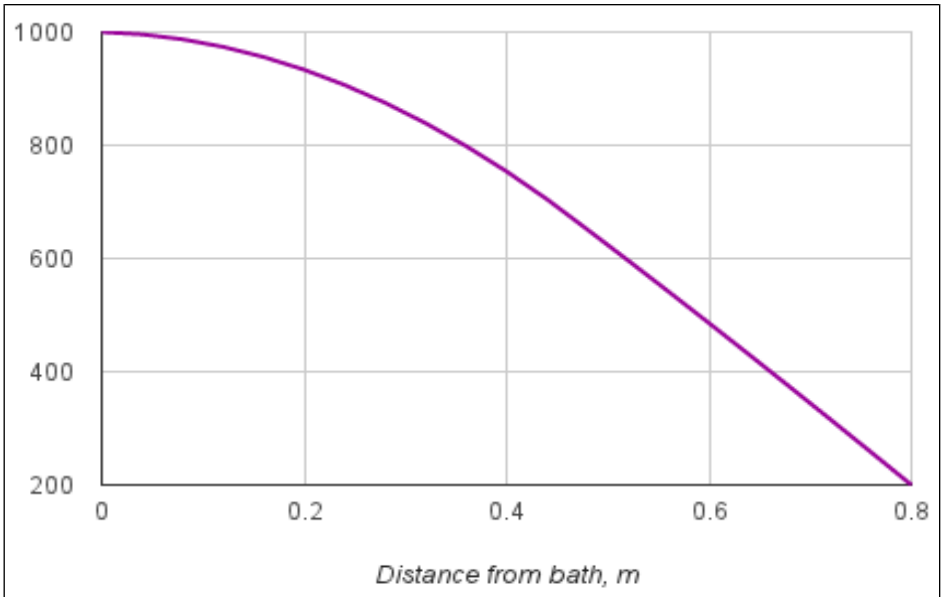


Figure 29: Calculated temperature distribution for an 0.8 m long aluminum lead with 100 A/cm² current density.

Figure 30 shows the calculated total energy loss q/J as a function of total current collector length L for several values of current density J. It indicates that minimum total energy loss is approximately 0.34 V (1.0 kWh/kg) for values of J from 40 to 100 A/cm². Note that minimum theoretical energy loss given by equation 18 on page 49 is $(P+Q)\div I$

= 2 $\sqrt{L\ T\ \Delta T}$, which for T=1273K and ΔT=800°C is 0.316 V, which is very close to this calculated value. In addition, the product of current density times optimal length at that current density is roughly constant at approximately 90 m·A/cm², *i.e.* 9×10⁵ A/m.

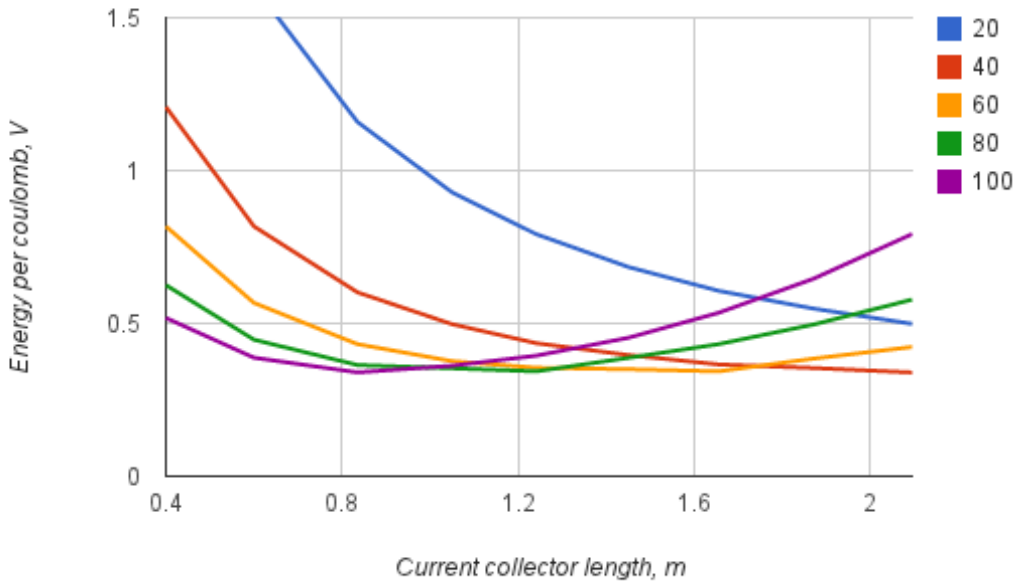


Figure 30: Calculated energy loss through the current collector vs. length for current density 20-100 A/cm².

For this reason, an aluminum solid/liquid current collector operating at 100 A/cm² should be approximately 0.85 m long. It loses approximately 16% more energy if it is 30% shorter, and 7% more energy if it is 30% longer, than this optimal length.

Issues and Challenges

Compared with Configuration 2, Configuration 3 has many additional challenges. It will require sheaths around the anodes, which must be inert to aluminum in the floating pad but insulating to avoid creating metal at cathodic potential. It will require a new cathode current collector, as traditional graphite-steel current collectors around the small edge interface area will have high resistance, and connecting TiB_2 cathodes at the top plate is complex and also leads to high resistance. Hoopes cell graphite current collectors should work, but have very high resistance. INFINIUM has a novel aluminum cathode current collector design with low resistance, described above starting on p. 39 which we have begun testing (p. 128).

That said, the advantages of Configuration 3 motivate some work into the above areas. Furthermore, if successful, this configuration could be used for recycling aluminum dross, which is a fine-grained metal-oxide mixture created as a waste product of aluminum melting and casting operations. The advantage would be that dross could sit on the bottom of the cell until dissolving. In Configuration 2, a dense oxide-metal dross mixture could sink through the bath and into the liquid aluminum pad, forming a sludge layer between the pad and cathode connection.

Sodium Mass Balance

Smelter grade alumina typically contains 0.5-0.6 wt% Na_2O. In a Hall-Héroult cell, one feeds AlF_3, and Na_2O becomes NaF in situ. In our cell with AlF_3-KF bath, high NaF content affects liquidus temperature, zirconia lifetime, etc.. The cost model's mass balance estimates that feeding bath to keep NaF dilute will require 222-268 kg bath/tonne Al product; this raises the modeled cost by 46-57¢/kg (p. 56).

There are seven options which can reduce the cost of closing the sodium mass balance. Using a baseline of 0.05 wt% Na_2O in alumina, options and their costs are:

- KOH precipitation can replace NaOH in alumina production. Unfortunately, the total world potash market is 33-38 million tonnes/year, which may not be sufficient for this purpose. As of December 2014, caustic soda prices on AliBaba center around $400/T, whereas potash on InfoMine is around $300/T, though on an equal molar basis, about 40% more potash mass would be required. An alumina plant could recycle its potassium, but this could be expensive.

  Assumed alumina impurities: 0.4 wt% K_2O, 0.1% Na_2O → 1¢/kg lower cost than baseline, though alumina will cost more. In the linear system, if alumina K_2O content exceeds 0.5 wt%, "KF input" flips negative, indicating excess potassium.

- Boric acid addition with calcining at 1200°C drives Na_2O content to 0.05 wt%. This increases energy use and reduces surface area and dissolution rate in bath.

  This is the baseline case, though alumina will cost more due to high-temperature calcining and boric acid. If boron enters the cell, it will volatilize as BF_3.

- Straight calcining at 1300° drives Na_2O to 0.1 wt% but with the same drawbacks. Increasing from 0.05 to 0.1 wt% Na_2O increases production cost by 5¢/kg.

- Alumina hot precipitation near 100°C reduces Na to 0.05 wt%, but with broader distribution and more fines – which is not a problem in a Configuration 3 cell.

- Improved Bayer process control using split seed streams can drive Na_2O to 0.1-0.3 wt% at little additional capital cost.

- A new acid process pioneered by Orbite can produce very low-soda alumina. Cost is reportedly low, and pilot trials are reportedly beginning in China.

- Develop an inorganic chemistry process to separate the bath out stream into sodium-rich and sodium-lean streams, and recycle the sodium-lean stream as some of the bath input.

In short, sodium balance presents a serious issue requiring some combination of upstream alumina innovation and recycling/separation, but there are several potential solutions already available. We will need to do some investigation into upstream alumina production, at least at the level of techno-economic analysis, to characterize these options, pick one or two winners, and determine the impact on our technology.

Techno-Economic Energy, Emissions and Technical Cost Modeling

Energy Use and Emissions Modeling

Overview

INFINIUM has completed an estimate of energy use and emissions associated with aluminum oxide reduction using electrolysis with Pure Oxygen Anodes™. This includes a detailed temperature-dependent estimate of electrolysis cell applied voltage, and the effect on this estimate of using natural gas fuel in the anodes. This includes sensitivity of anode production energy use and emissions assuming lifetime to cell lifetime from 4000 hours down to just 250 hours. At 250 hours, impact of production of the inert anode themselves is low but not negligible: 1.0 kWh/kg Al *vs.* 5.7 kWh/kg (chemical energy) for carbon anode production, and 0.22 kg CO_2e/kg Al *vs.* 3.3 for carbon anodes. Above 1000 hours lifetime, this becomes negligible.

Methods

The INFINIUM energy use model begins by estimating the overpotential of each component in the series from anode lead, to anode, zirconia, oxygen mass transfer, molten salt, cathode and cathode lead, plus the alumina dissociation potential (free energy of the reaction $AlO_{1.5} \rightarrow Al + ¾\ O_2$ divided by *nF,* number of electrons passed per atom times Faraday's constant). It does not include charge transfer overpotential due to its very low fraction of the total at high temperature. The sum of these overpotentials represents the total applied voltage required to achieve the desired current density. Multiplying this result by *nF/M* (electrons per mole, Faraday constant, molar mass) gives the total energy consumption passed as ionic current through the system per kilogram of aluminum product. Dividing this result by the current efficiency adds energy losses due to electronic current, which essentially acts as a resistance heater, to give the total energy consumed in the cell per unit of aluminum product.

The model's spreadsheet estimates each resistance in the series as accurately as possible. Some resistances are unknown and assumptions are based on educated guesses. Those assumptions include: cathode IR drop is negligible, and O^{2-} ion mass transfer adds 0.2 V. INFINIUM bath conductivity is 1.6 S/cm. FEA shows effective anode-cathode distance is 2 cm; further FEA results will refine this estimate. The model also assumed 95% current efficiency.

Next, the spreadsheet calculates necessary energy usage. This includes: enthalpy of the alumina reduction reaction, minimum electrode lead losses in a bipolar cell (*cf.* lower heat loss in a multipolar cell, which is not possible here), and energy required to heat incoming feedstock consisting of alumina raw material and, in the fueled anode case, natural gas. Fuel gas flow rate is taken to be the stoichiometric requirement plus 10% excess to maintain the required CO/CO_2 and H_2/H_2O ratios to stabilize metals in the anode and/or current collector/fuel inlet tube. In a plant, the excess CO and H_2 would be burned. The difference between heat generation by total voltage, and necessary

energy usage, is the waste energy required to keep the cell in balance.

The model uses the cell temperature as the reference temperature, such that it calculates the energy required to heat cell inputs to the cell temperature, and doesn't double-count by also calculating energy lost by exhausting hot products. This is simpler than using ambient temperature as the reference for two reasons: first, it is easier to estimate heating energy for one or two material inflows than for many outflows, particularly when heat exchangers could be involved; second, the heat contained in the liquid aluminum product is not necessarily "lost" as it can be cast in that state to save the energy of re-heating and re-melting.

The model does not yet include any heat exchange between inputs and outputs. Fuel heating is inherent to the design, as the incoming fuel and outgoing gases are separated only by a tube which is either metal or a high-thermal conductivity oxide such as alumina. This mechanism also provides the heat required to reform fuel using a portion of the combustion product gas by the reaction: $CH_4 + \frac{1}{3} CO_2 + \frac{2}{3} H_2O \rightarrow 4/3\ CO + 8/3\ H_2$. Using anode gas to heat alumina raw material is possible with the fueled or oxygen-generating configurations.

Temperature-dependent inputs

As mentioned above, several model inputs are temperature-dependent, as follows:

- Alumina dissociation: $AlO_{1.5} \rightarrow Al + \frac{3}{4} O_2$, ΔH = 1127.76 - 0.0077×(T – 900°C) kJ/mol O_2 product, ΔG = 832.3 - 0.27.36×(T – 900°C) kJ/mol O_2 product.[38]
- Alumina dissociation with natural gas: $AlO_{1.5} + \frac{3}{4} CH_4 \rightarrow Al + \frac{3}{4} CO_2 + 3/2\ H_2O$, ΔH = 727.05-0.0109×(T – 900°C) kJ/mol O_2 product, ΔG = 453.2 + 0.064×(T – 900°C) kJ/mol O_2 product.[38]
- Alumina enthalpy *vs.* environment temperature: ΔH = 99.87 + 0.1287×(T – 900°C) kJ/mol Al.[38]
- Methane enthalpy *vs.* environment temperature: ΔH = 51.2 + 0.0794×(T – 900°C) kJ/mol CH_4.[38]
- 6 mol% yttria-stabilized zirconia (6YSZ) conductivity: $\sigma_i = 29.91 \times \exp\left(-\frac{7692}{T}\right)$ S/cm[39] for off-the-shelf tubes. 8YSZ with standard sintering is: $\sigma_i = \frac{2.22 \times 10^7}{T} \exp\left(-\frac{14171}{T}\right)$ K·S/cm, about twice this with enhanced sintering.[40]
- 9 mol% scandia-stabilized zirconia (9ScSZ) electrical conductivity: $\sigma_i = \frac{363100}{T} \exp\left(-\frac{8731}{T}\right)$ S/cm.[41]
- Samaria-doped ceria (SDC15) conductivity: $\sigma_i = \frac{112202}{T} \exp\left(-\frac{7691}{T}\right)$ K·S/cm.[42]

- Silver electronic resistivity: $\rho_e = 16.5 + 0.015\times(T - 1250\text{K})$ Ω-cm.[43]

Minimum energy loss through leads in a bipolar cell

A high-temperature electrolysis cell loses heat through its leads if they have high thermal conductivity. If electrical conductivity is low, then lead resistance will waste energy. Unfortunately, thermal conductivity and electrical conductivity are coupled through the Wiedmann-Franz law, so a material which makes a good lead for thermal loss purposes is a bad one for electrical losses.

The goal is to minimize total thermal and electrical energy loss in the leads for a cell with current *I*, voltage *V*, lead with length *L*, cross section *A* and electrical and thermal conductivities σ and *k*, cell temperature *T* which is ΔT above that of the environment. Start with the equations we know.

Electrical resistance energy loss in the lead:

$$P = IV = I^2 R = \frac{I^2 L}{\sigma A} \tag{10}$$

Thermal energy loss through the lead:

$$Q = \frac{k A \Delta T}{L} \tag{11}$$

Wiedmann-Franz law for the electronic component of thermal conductivity:

$$k_{el} = L_{el} \sigma_{el} T \tag{12}$$

Solve (10) for σ:

$$\sigma = \frac{I^2 L}{P A} \tag{13}$$

Substitute (12) and (13) into (11):

$$Q \geq \frac{A L_{el} \sigma T \Delta T}{L} = \frac{A L_{el} T \Delta T}{L} \frac{I^2 L}{P A} = \frac{I^2 L_{el} T \Delta T}{P} \tag{14}$$

Resulting constraint:

$$PQ \geq I^2 L_{el} T \Delta T \tag{15}$$

The lowest energy cost[a] $P+Q$ is at $P=Q$ with both given by:

$$P = Q = I \sqrt{L_{el} T \Delta T} = I \sqrt{\frac{k \Delta T}{\sigma}} \tag{16}$$

This gives the material figure of merit as σ/k, which Wiedmann-Franz dictates is at most $1/(L_{el}T)$. This is best in a material whose thermal conductivity is mainly electronic, *i.e.* a high-conductivity metal such as copper, silver or aluminum. The total loss is minimized where:

a Assuming that *P* and *Q* carry equal "prices" for the cell design, otherwise adjust accordingly.

$$P = Q \rightarrow \frac{I^2 L}{\sigma A} = \frac{A k \Delta T}{L} \rightarrow \frac{L}{A} = \frac{\sqrt{k \sigma \Delta T}}{I} \geq \frac{\sigma}{I} \sqrt{L_{el} T \Delta T} \quad (17)$$

Equation (17) is therefore a design guideline for choosing the electrode dimensions for a given material and current. The minimum total energy loss *P*+*Q* from equation (7) gives the corresponding energy per coulomb as:

$$\Delta V = \frac{P+Q}{I} = 2\sqrt{L_{el} T \Delta T} \quad (18)$$

For a process running at T = 1373 K in a 298 K environment so ΔT = 1075 K, the value $L_{el} = 2.45 \times 10^{-8} \frac{V^2}{K^2}$ gives the energy as: $\Delta V = 2\sqrt{0.0362\, V^2} = 0.38 V$ which is quite high. (Note the *IR* voltage drop across the lead is just the *P* component of this energy which is 0.190 V; the additional *Q* component is heat loss from the cell through the lead to the environment.) This says that for a single cell with two leads to the environment, one must budget at least 0.76 V worth of energy for these lead losses—even if the cell itself is *perfectly* thermally insulated.

This should be an overestimate, because the lead temperature is not the cell temperature T throughout its length. Instead, for a constant-cross section lead, the temperature profile is a curve with $\partial T/\partial x = Q/Ak$ where the lead joins the cell and $(P+Q)/Ak$ where it reaches the environment, and variable Joule heating according to the local temperature. The average temperature is over half of the mean of cell and environment, so the loss is roughly 0.3 V in each lead. There may be an optimal variation in cross section area along the length of the lead, but such analysis is beyond the scope of this calculation – and such a lead would likely increase cost more than it would save in reduced energy use.

INFINIUM has devised a 1-D heat conduction model for pure solid and liquid aluminum in a cathode lead, using temperature-dependent electrical and thermal conductivities. Results on page 42 indicate that actual optimal energy ΔV is within 5-10% of this theoretical result. That model also shows that for T=1000°C and ΔT=800°C, optimal current density-length product $J \cdot L = 9 \times 10^5$ A/m.

It may be possible to further reduce heat losses by keeping the bus bars between cells hot, increasing the environment temperature. There are two limitations on this: first, metal electrical conductivity falls quite a bit at high temperature, increasing the resistance of this bus; second, it would be very difficult to seal the cell or even anodes (and fuel inputs) if the entire cell lid were above about 200°C.

A multipolar cell accomplishes this by connecting multiple cells in series within a single bath. Unfortunately a multipolar cell is not possible in Hall-Héroult because the anode is consumable. For solid oxide membrane technology it would require a high-temperature seal between the conductor and tube and is therefore not likely practical.

Silver anode layer thickness or cross-section

The silver anode overpotential calculation merits particular attention. Figure 31 shows

the oxygen-generating anode with silver between the outer zirconia tube and an inner oxygen transport membrane, such as a porous ceramic or mixed ionic-electronic conductor. Here we quantify the silver resistance, evaporation, and leasing/capital cost, and estimate the optimal thickness.

Expressed in terms of dollars per kilo-amp-hour (kAh, roughly ⅓ kg Al), silver leasing cost C_1 is given by:

$$C_1 = \frac{S \rho R t}{JU}, \tag{19}$$

where S is the silver cost per kg, ρ is its density, R is the silver lease rate, t its layer thickness, J anode current density, and U anode utilization rate. Cost of silver lost to evaporation C_2 is:

$$C_2 = \frac{p_{v,Ag}}{p_{O_2}} \frac{M_{Ag}}{n_{O_2} F} S, \tag{20}$$

Where $p_{v,Ag}$ is silver vapor pressure, p_{O_2} is oxygen pressure generally 1 atm, M_{Ag} is silver molar mass, n_{O_2} is number of electrons transferred per oxygen molecule *i.e.* 4, and F is Faraday's constant. Cost of electrical resistance C_3 is:

$$C_3 = \frac{HL^2 J}{2 \sigma t}, \tag{21}$$

where H is electricity rate in $/kWh, L is the anode immersion depth in the salt, and σ is silver electrical conductivity. Figure 32 shows these three costs as a function of thickness in graphical form.

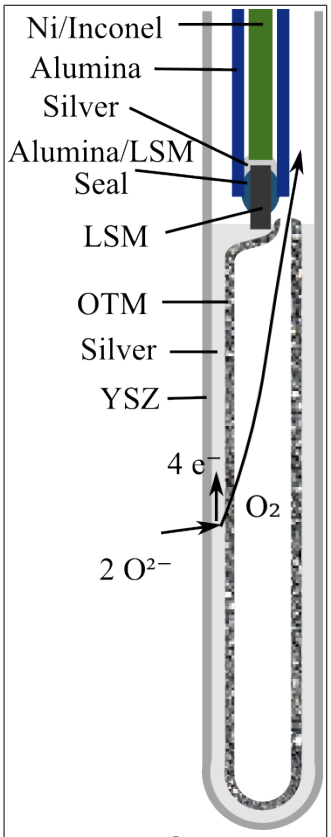


Figure 31: Oxygen-generating anode assembly.

Setting the sum of the cost derivatives to zero gives the optimal silver layer thickness:

$$t = \frac{LJ}{2} \sqrt{\frac{2HU}{S \rho R \sigma}}. \tag{22}$$

Note that this relationship holds whether the silver anode is liquid or solid. The model thus has a temperature-dependent conductivity which is discontinuous at the melting point, and the spreadsheet calculates a narrower optimal conductor cross-section when operating below the silver-oxygen eutectic temperature of 939°C. That said, below 939°C, one can likely use silver-coated nickel and decrease resistance at very low cost,

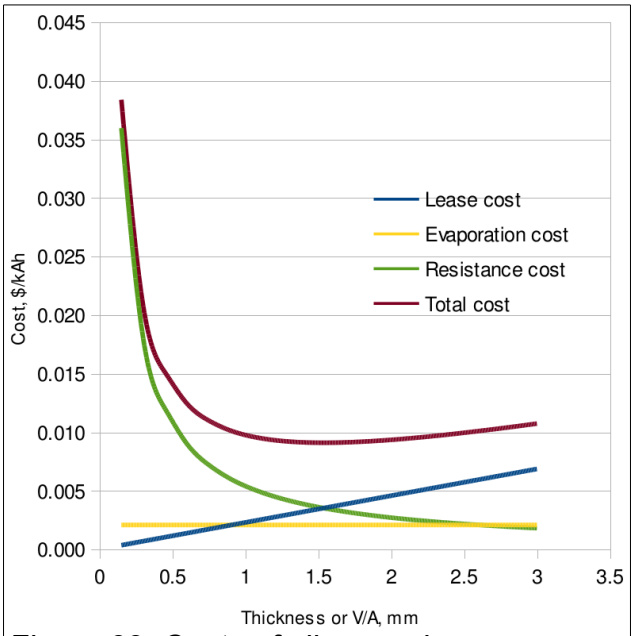


Figure 32: Costs of silver vs. layer thickness.

reducing the “Silver lease cost” and anode resistance.

A further note on silver leasing is warranted here. Precious metals lease at far lower rates than convectional bank loans due to the collateral of a valuable asset. Furthermore, the difference between current and futures prices often leads to strange results. For example, on June 25 2014, the one-year lease rate for gold was 0.40%, platinum 0.49%, palladium 0.77%, and silver -0.06%![44] We are not anticipating that anyone will pay us to hold onto silver for them, so our model sets a minimum lease rate of 3%.

Results and Discussion

In spite of the increased current density, the substitution of 8YSZ with conventional sintering[40] for 6YSZ and increased temperature lowers the zirconia resistance overpotential from 0.90 V to 0.40 V.

For silver layer thickness, using equation 22 with L=40 cm, J=1.06 A/cm², H=4¢/kWh, U=90%, S=$757/kg, ρ=9.21 g/cm³, R=3% and σ=5.4×10⁴ S/cm gives an optimal silver thickness of 1.57 mm, which visually corresponds with the minimum cost in Figure 32.

Effects of Temperature and Zirconia Thickness

Figure 33 shows the effects of temperature and zirconia thickness on cost of resistance losses. That figure shows a strong effect of zirconia type, with advanced firing producing dramatic increases in conductivity. It also shows monotonic decrease in cost with increasing temperature and decreasing zirconia thickness.

This illustrates a limitation of this model, which is that it is missing a lot of information. For example, increasing temperature requires use of more costly cell materials, and decreasing zirconia thickness reduces tube lifetime and increases longitudinal stress in the tube due to anode liquid weight. Building these into the model to generate a truly optimal result will take considerable time, and the noisy nature of such relationships as materials costs with temperature may make this better settled manually rather than by an analytical global optimization exercise.

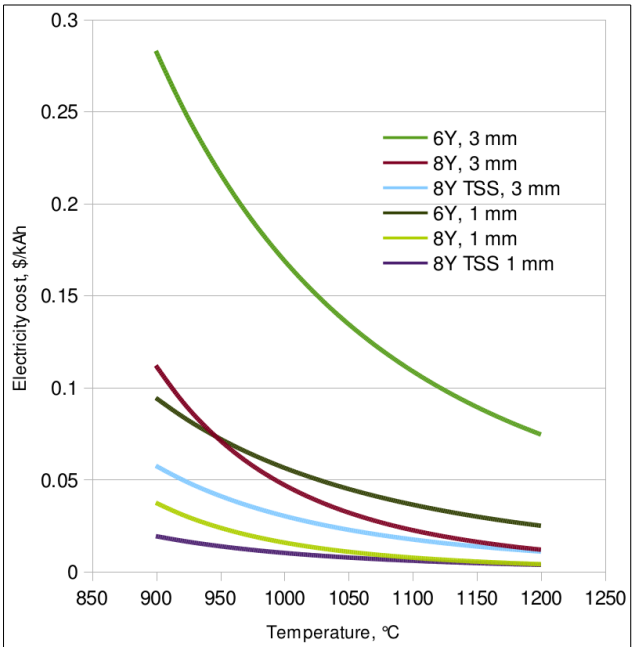


Figure 33: Effect of temperature, zirconia type and thickness on resistance cost.

Energy Budgets

Commodity Aluminum Plant Figure 34 shows the voltage and heat budgets for oxygen-generating and fueled anodes from 700-1100°C using the same conditions as above, but with a 0.9 mm 8YSZ

membrane based on thin membrane experimental results. At 1100°C, the resulting voltage is 3.98 V, down from 4.17, leading to total energy use of 12.48 kWh/kg, down from 13.08. At 1000°C, voltage is 4.20 V, with energy use of 13.17 kWh/kg, which is similar to the prior result at 1100°C. Reducing temperature by 100°C is a big advantage – if the membrane can survive to the target lifetime, and if an anode can be made to work at this temperature, since it's well above the 939°C Ag-O eutectic.

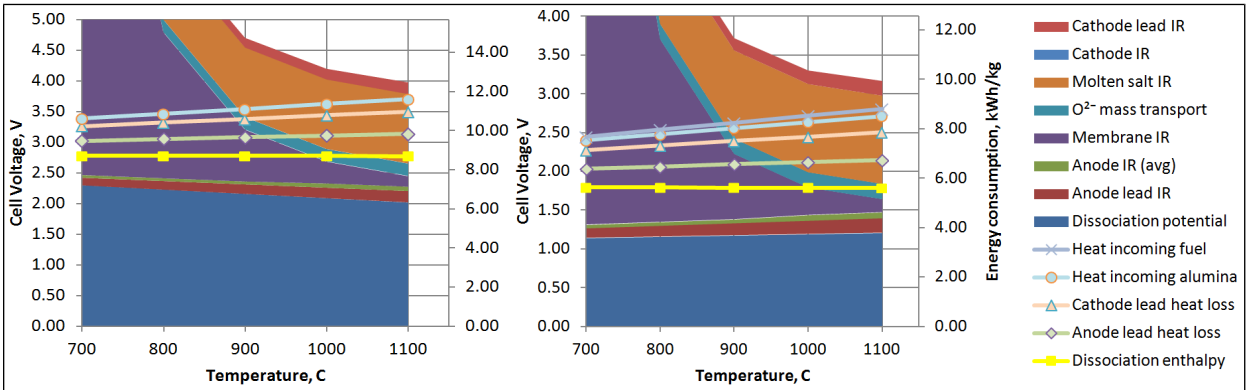


Figure 34: Stack graphs of voltage (solids) and heat (lines) budgets in 0.9 mm 8YSZ membrane for oxygen-generating (left) and fueled (right) anodes at 700-1100°C.

500 TPY Al-2Sc Full Electrolysis Plant Figure 35 shows the voltage and heat budgets for oxygen-generating and fueled anodes from 700-1100°C using 0.9 mm 9ScSZ membranes at 0.5 A/cm². In this case, the oxygen-generating cell at 700°C operates at 4.32 V, corresponding to 12.88 kWh/kg, and at 95% current efficiency this leads to 13.56 kWh/kg. Likewise, fueled anodes operate at 3.27 V, i.e. 9.75 kWh/kg, at 95% current efficiency this leads to 10.26 kWh/kg electrical energy (14.02 kWh/kg including the fuel). Both of these are close to this project's targets of 13 kWh/kg for oxygen-generating and 10.5 kWh/kg for fueled anodes.

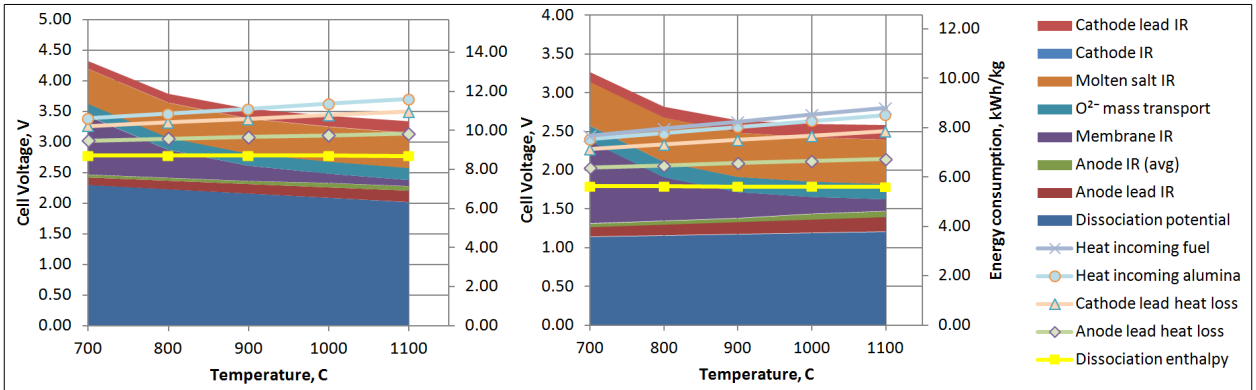


Figure 35: Stack graphs of voltage (solids) and heat (lines) budgets in 0.9 mm 9ScYSZ membrane for oxygen-generating (left) and fueled (right) anodes at 700-1100°C.

Overpotentials in these graphs are much lower than those in Figure 34 for three reasons: more conductive membrane, and lower current density. For a 2 mm 8YSZ membrane at 1 A/cm², 1000°C is a useful minimum operating temperature; this cell is

not self-heating above 900°C, and has a practical maximum temperature around 800°. This lower useful temperature range should work well for our relatively volatile KF-AlF_3 based bath.

Figure 36 shows the oxygen-generating plot using 0.9 mm 8YSZ and 9ScSZ. Total voltage is uniformly higher, and a good operating temperature is about 800-850°C. It also shows the effect of increasing current density to 1.0 A/cm² using 9ScSZ: again overpotential is higher, with a good operating temperature around 900°C.

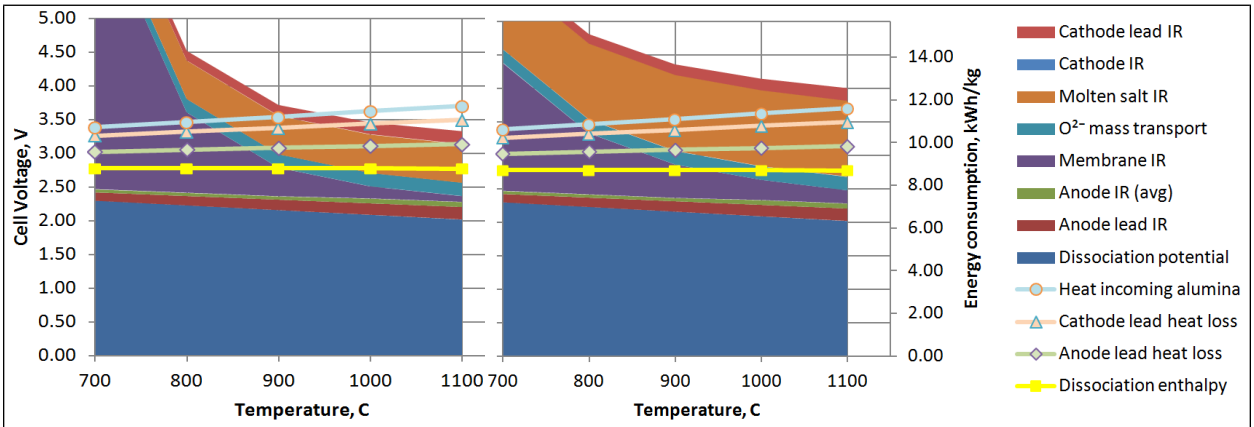


Figure 36: Stack graph showing energy budgets for oxygen-generating anodes using 0.9 mm 8YSZ membrane at 0.5 A/cm² (left), and 9ScSZ membrane at 1 A/cm² (right).

Dross Recycling All of these figures are per kilogram of electrolyzed metal, and are valid for a primary production cell. In a dross recycling cell with feedstock consisting of 50 wt% aluminum oxide and 50 wt% aluminum metal, the heat stack includes a significant use of energy to heat and melt the aluminum metal. At 1.0 A/cm² with a 2 mm 8YSZ membrane, side wall losses need to be just 0.37 kWh in the oxygen-generating case and 0.64 kWh/kg in the fueled anode case in order to maintain heat balance in a self-heated cell. 50 wt% metal thus represents a useful ceiling for the dross charge in this type of cell. Note however that if the dross is 50 wt% metal and 50 wt% oxide, then only one third of the aluminum is produced by electrolysis. Energy use per kilogram of total metal product is one third of these values, or around 4.4 kWh/kg for oxygen-generating anodes and 3.5 kWh/kg for fueled anodes.

Energy and Emissions from YSZ-Silver Anode Production

Anodes are the most novel aspect of this technology, and the shortest-lived cell components. As we eliminate other sources of energy use and loss, such as carbon anode production and side wall heat loss, we partially offset these reductions with new impacts of anode production. This section estimates these new impacts.

Zirconia production impact is surprisingly low given the extreme temperatures used to vaporize off silica in its production: about 150 MJ/kg ZrO_2 (40 kWh/kg) and 5 kg CO_2e/kg ZrO_2 according to Pehnt.[45]

For yttria, Koltun assumes that costs and impacts of ore mining, upgrading,

beneficiation, separation, and calcining are fixed for the whole mixed bag of rare earths. Thus one must allocate these impacts among the elements, such as by mass fraction, or by dollar value fraction. That said, the price of yttria is close to the mass-weighted mean of rare earths, they are relatively close. Mass basis: 179.49 MJ/kg Y_2O_3 (50 kWh), 34.49 kg CO_2e/kg; value basis: 213.03 MJ/kg (59 kWh), 37.87 kg CO_2e/kg.[46]

For 8 mol% YSZ, *i.e.* 13.8 wt% yttria, using the higher yttria value, the weighted average energy use is 42.6 kWh/kg YSZ, with emission of 9.54 kg CO_2e/kg.

A 2 mm thick tube with 250 hour lifetime running at 0.5 A/cm² passes 125 A·h/cm², *i.e.* about 40 g Al/cm² anode area, say 50 to account for head space. This is 250 g Al/cm³ YSZ, or 42 g Al/g YSZ. Anode production impact is thus about 1.0 kWh/kg Al, and 0.22 kg CO_2e/kg Al, which is small but not negligible. This scales with thickness, and inversely with current density and lifetime, so at 0.9 mm, 1.0 A/cm² and 1000-hour lifetime, it's 0.06 kWh/kg Al and 12 g CO_2e/kg Al, compared with 3.3 kg CO_2e/kg Al direct emissions in today's cells.

Our prospective scandia supplier uses a very different flow sheet without the complex separations of mixed rare earths, and with lower emissions. Estimation of energy and emissions will require more information, whether for anodes or alloy raw material.

Silver production emissions estimates are much harder to find. Most silver impact papers focus on recycling in substance flow analysis due to its scarcity, and emissions of silver itself due to toxicity. Co-production together with copper makes separate silver analysis very difficult. Copper dominates both the mass and value of products from that supply chain, and the marginal impact of recovering silver from electrorefining anode slime is very low. For now, we neglect this small contribution as relatively small compared to that of YSZ tube production.

Carbon comparison The carbon anode reactions are:

$$½\,Al_2O_3 + ¾\,C \Rightarrow Al + ¾\,CO_2 \quad (23)$$

$$½\,Al_2O_3 + 3/2\,C \Rightarrow Al + 3/2\,CO \quad (24)$$

With about about 80% of carbon following equation 23 and 20% equation 24. The CO will at some point oxidize to CO_2, so that's 0.8×¾ + 0.2×3/2 = 0.9 moles CO_2/mol Al, and 1.47 kg CO_2/kg Al (direct). Anode production emits another 1.83 kg CO_2e/kg Al,[47] for a total of 3.30 kg CO_2e/kg Al. Energy consumption is 5.71 kWh energy/kg Al.[48] Thus for a 2 mm YSZ tube at 0.5 A/cm² lasting just 250 hours, its energy use of 1.0 kWh/kg is about one-sixth that of carbon anodes, and emissions of 0.22 kg CO_2e/kg also is about one-fifteenth. As mentioned above, these impacts are much lower than carbon, but not negligible. Above 1000 hour anode lifetime, these impacts fall to negligibly low values.

Issues, Risks, and Challenges

This analysis gives us a framework for understanding energy flows in our process. It does not actually calculate all of the thermal energy losses from the process, but shows what those losses need to be in order to achieve energy balance in the cell. And as

noted above, several areas such as cathode resistance require further refinement.

Technical Cost Modeling

Overview

INFINIUM has taking two approaches to techno-economic cost modeling. The first "top-down" approach began with the parts manifest from the 200 kA industrial cell design (p. 27), and compared INFINIUM cell capital and operating costs with those of a conventional Hall-Héroult cell. We built this model in collaboration with an industrial partner (thus it is proprietary), and it has served to illustrate the difference in operating cost, with associated calculations such as return on investment (ROI).

The second "bottom-up" approach begins with the ARPA-E METALS program techno-economic analysis (TEA) spreadsheet, Version 2, by ARPA-E contractor Daniel Matuszak. We have modified the spreadsheet somewhat, adding an energy balance model sheet (described above), a linear system mass balance solver, other cell calculations unique to our technology, and a switchable anodes sheet.

We did considerable work to make this spreadsheet flexible enough to model both mature industries with very low labor costs such as Hall-Héroult aluminum plants, and new technologies rolling out at small scale. The latter are well suited to the spreadsheet's current high overhead structure; the former required some changes. We made the changes in a way which makes the large-scale plant cost model agree well with results from a similar cost model by Alton Tabereaux, a consultant on this project.

Cost Model Components

Mass Balance Methodology

The mass balance is implemented within the ARPA-E techno-economic analysis spreadsheet mentioned above. It uses the inputs in the list to the right to build a matrix of elements entering and leaving the reactor through five streams or sub-streams: alumina, bath in AlF_3, bath in KF, bath in Y-compounds, and bath out. The product of this matrix and the unknown vector of those stream flow rates equals the vector of element quantities in aluminum product. That is, the sum of (aluminum fraction of alumina times alumina input rate) plus (aluminum fraction of bath in times bath input rate) minus (aluminum fraction of bath times bath output rate) equals aluminum production. Applying the same balance to fluorine, sodium, sodium, potassium and yttrium leads to the following matrix equation:

Mass balance model inputs:

- Bath cryolite ratio
- Alumina soda (Na_2O) fraction
- Alumina K_2O fraction
- Alumina SiO_2 and Fe_2O_3
- Aluminum metal Na fraction
- Aluminum metal K fraction
- Aluminum metal Y fraction
- Bath target alumina fraction
- Bath maximum NaF fraction
- Bath target YF_3 fraction

$$\begin{pmatrix} Alumina\,\%Al & AlF_3\,\%Al & KF\,\%Al & Y\,comp\,\%Al & Bath\,\%Al \\ Alumina\,\%F & AlF_3\,\%F & KF\,\%F & Y\,comp\,\%F & Bath\,\%F \\ Alumina\,\%Na & AlF_3\,\%Na & KF\,\%Na & Y\,comp\,\%Na & Bath\,\%Na \\ Alumina\,\%K & AlF_3\,\%K & KF\,\%K & Y\,comp\,\%K & Bath\,\%K \\ Alumina\,\%Y & AlF_3\,\%Y & KF\,\%Y & Y\,comp\,\%Y & Bath\,\%Y \end{pmatrix} \times \begin{pmatrix} Alumina\ input\ rate \\ AlF_3\ input\ rate \\ NaF\ input\ rate \\ Y\,comp\ input\ rate \\ Bath\ flow\ rate \end{pmatrix} = \begin{pmatrix} Product\,\%Al \\ Product\,\%F \\ Product\,\%Na \\ Product\,\%K \\ Product\,\%Y \end{pmatrix} \quad (25)$$

All fractions are mass fractions (except cryolite ratio is the KF:AlF_3 mole ratio), and the stream flow rates are given in ratios to product flow rate, *e.g.* tonnes of alumina per tonne of metal product.

Solution of the unknown stream flow rates requires solving the linear system. The spreadsheet does this by inverting the matrix, and multiplying that inverse by the aluminum product composition. "Y comp(ounds)" can mean YF_3 or Y_2O_3, the system can use any fraction of each of these, and Y_2O_3 can cost half as much on a Y basis. Oxygen is omitted, and not needed: the spreadsheet balances it separately. The model uses the following prices: alumina 32¢/kg, AlF_3 $1.05/kg, KF $2.30/kg, Y_2O_3 $14/kg. Weighted-average bath price for the baseline case is $2.21/kg.

This mass balance shows that if alumina is 0.5-0.6 wt% Na_2O, which is a typical smelter-grade alumina range, then one must feed 222-268 kg bath per tonne Al product, which is very high. It raises the modeled cost by 46-57¢/kg, an unacceptable increase. This motivates a study of mitigation strategies described on p. 45.

Energy Balance and Utilities

The energy balance estimates the amount of energy consumption per unit of metal product. The Utilities sheet in turn uses this and other inputs to estimate total plant energy use. Electrolysis cells consume the overwhelming majority (>90%) of the plant energy, as described above in the energy use model (p. 46).

Consumables

The consumables model estimates the cost of producing membrane tubes, and will include other items as well such as cathodes and anode-cathode cluster modules. The membrane tube model consists of a matrix of unit operations and cost components of each unit operation. For ceramic tube production, those unit operations are:

- Powder and binder costs;
- Slurry mixing/milling to disperse deflocculant and binder and eliminate clumps;
- Extrusion including end closing;
- Firing green parts;
- Testing such as die penetration and pressure burst tests;
- Supervisory labor.

Each of the five unit operations has the following cost components: capital, labor, consumables, energy, and waste disposal including yield losses at each stage. Model inputs include production volume (tubes/year); tube dimensions; raw material costs; equipment capacity, cycle time, uptime and depreciation lifetime; operators per

machine; operator salary; and disposal cost.

A small plant using few tubes will show high per-unit cost. For this reason, the overall model caps tube cost at $500, which is half the cost of off-the-shelf tubes, and approximately the volume price which a couple of small producers have discussed with us.

Table 1 shows modeled per-tube cost by unit operation and cost category for production of 42,048 tubes/year with 3.7 cm OD and 0.9 mm thickness. The total cost of about $63/tube is 45% labor cost, with materials and capital at 31% and 18% respectively. This plant would employ five line workers and one supervisor.

This model will later build up to models of cathode, anode, cemented anode tube, and anode-cathode cluster production.

Tube cost by unit operation		Tube cost by category	
Powder	$19.04	Raw materials	$19.83
Binder	0.79	Capital	11.14
Milling/Mixing	6.43	Labor	28.84
Extrusion	16.97	Consumables	3.31
Firing	9.60	Energy	0.07
Testing	8.01	Disposal	0.03
Supervisors	2.38		
Total	$63.22	Total	$63.22

Table 1: Membrane tube production cost by unit operation and by cost category.

In a 100k TPY pure Al plant, the cost falls to just $31.83/tube for thin 8YSZ tubes. Production volume is over 235,000 tubes/year, which reduces labor cost from nearly $29/tube to just over $2/tube vs. the 42,000 tubes/year volume for a 500 TPY plant with low current density and lifetime (0.5 A/cm² and 250 hrs *cf.* the larger plant model at 4000-hour lifetime and 1 A/cm² current density).

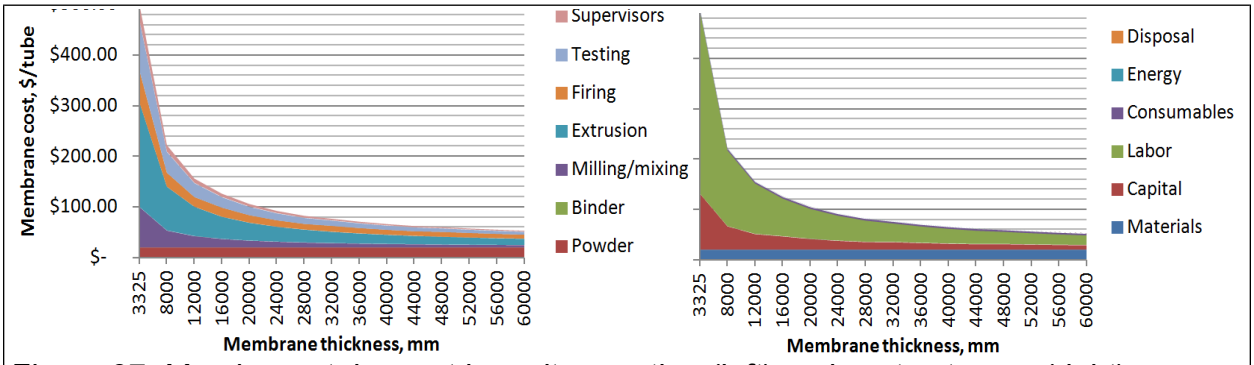


Figure 37: Membrane tube cost by unit operation (left) and cost category (right).

Labor

The labor cost model begins with the elements in the Opex sheet. Based on the cost model of consultant Alton Tabereaux, we set the cost based on 37 workers/shift in a 100,000 TPY plant, then scale the result for this plant's output, with a minimum of two workers/shift. This is not really the most rigorous way to do things, but based on Tabereaux' three decades of experience, we believe this is the most accurate way we know how to model a large smelter. Modern aluminum smelting plants are a marvel of logistical efficiency, and this roughly captures that. And a small plant such as this, with about ten 20 kA cells, should only need two workers around the clock. Between the two extremes of very small and very large plants, this does not work well. This is in addition to labor to produce consumables, such as the tube cost model above.

Capital

The capital cost model straightforwardly uses the Equipment and Capex sheets of the ARPA-E TEA model to build up total capital cost. Equipment includes electrolysis cells, bus bars, rectifiers, the salt decanter, and if using fueled anodes, the residual fuel burner.

Large Aluminum Plant Results

For the 100,000 tonnes/year plant using 200 kA cells modeled here, total estimated operating cost is just under $1638/tonne, or just under $1.64/kg. At the time of this study, world price was $1477/T (January 15, 2016), so this is very close to break-even without capital costs. That said, our process has no carbon emissions, and a Hall-Héroult plant emits at least 3 T CO_2/T Al product. At $50/T CO_2, this amounts to a 10% advantage. The anode plant also emits volatile organic compounds and other hazardous emissions, so this process eliminates all such problems.

The capital cost estimate is $216M, or just over $2000 per tonne/year of capacity. This is very low, partly reflecting the elimination of the anode plant. This is likely incomplete, and will require adjustment, likely mostly upward, in the coming months and years.

Small Aluminum (Alloy) Plant Results

For the second goal of this project, aluminum-scandium alloy production, one option is to co-electrolyze both 98% aluminum and 2% scandium in a single cell/plant. A cost model for a plant using this method would have a similar framework as described above, but just 500 tonnes/year output. In that case, total baseline estimated operating cost is $8153/tonne, or just over $8.00/kg, plus scandium oxide. Membrane tube production opex is $2510/T Al, or nearly 1/3 of opex, leaving considerable room for improvement. Today's master alloy cost is $80-100/kg, which is $20-40/kg above the $60/kg in scandia metals basis cost (~$2000/kg → $3070/kg metals basis). Our technology should thus have a cost advantage based on going from oxide to alloy in a single step.

Baseline conditions consist of: 526 TPY capacity, 95% utilization for ~500 TPY output,

120 tubes per cell with 333 cm² anode area, 250-hour tube lifetime, 0.5 A/cm² anode current density, and 8¢/kWh power cost. Thus there are ten cells at 20 kA each, with output and cell size held constant such that increasing current density results in fewer cells with higher current. Note this model does not include labor required to change the tubes with this much shorter lifetime than was assumed for pure aluminum production.

Figure 38 left shows how cost varies with production volume from 100 to 2000 TPY, and indicates that the biggest returns to scale happen up to 500 TPY.

Figure 38 right shows sensitivity to current density from 0.2 to 1.3 A/cm², reflecting fewer 120-tube cells, thus declining labor and tube costs, and rising energy cost with current density. Under current assumptions, total cost declines to 1.0 A/cm², then rises due to energy cost. However, when operating at high current density, one would likely run at higher temperature to reduce resistance, which is not captured in the model. Note that below about 0.4 A/cm², the cell will likely not be self-heated, and will either require supplemental heating (also not captured in this model), or higher internal or lead resistance to generate the required heat.

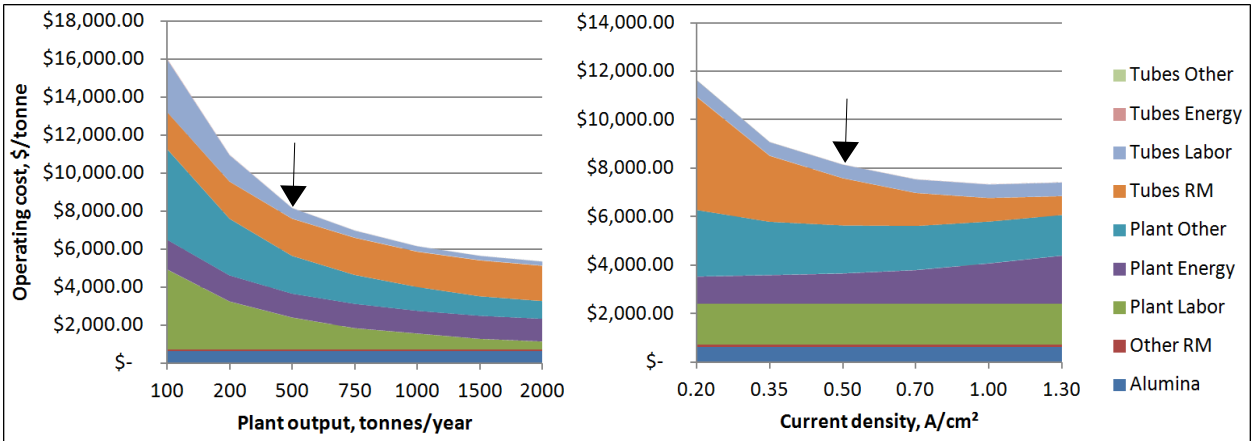


Figure 38: Operating cost stack vs. plant output (left) and current density (right). Black arrows indicate model baseline conditions.

Figure 39 left shows sensitivity to tube lifetime up to 2500 hours. The main effect is on raw material for tubes, tube casting labor does not change appreciably.

Finally, Figure 39 right shows sensitivity to electricity cost at 2-10¢/kWh. This simply scales the energy cost without appreciably affecting other areas.

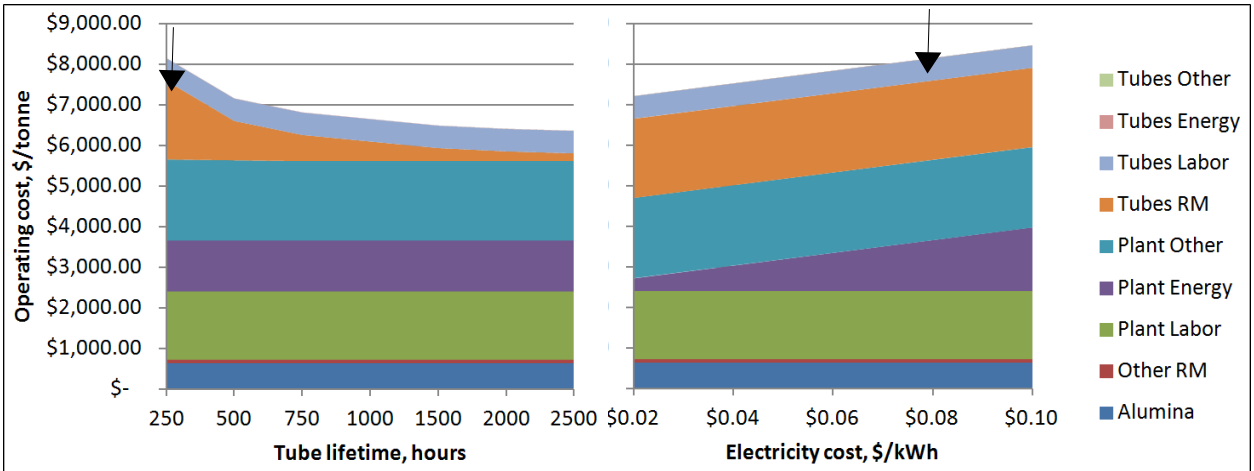


Figure 39: Operating cost stack vs. tube lifetime (left) and electricity cost (right). Black arrows indicate model baseline conditions.

To summarize, scale has the largest effect on total operating cost of production. That said, alumina, energy, and membrane tube costs do not decrease appreciably with scale. Alumina is a bulk commodity which cannot change much (and may cost a bit more at just 500 TPY metal, *i.e.* ~950 TPY alumina). Changing energy cost probably requires changing location. But more work needs to be done on tube cost – which represents nearly 40% of total opex at 2000 TPY – by understanding and reducing corrosion rate, and by reducing membrane resistance in order to improve current density.

Fixed capital investment (FCI) estimate is just over $12.5M, or about $27,900 per tonne/year of capacity. This is relatively high, reflecting the small plant scale, and is incomplete, though other estimates in the model such as buildings, start-up costs and plant overhead likely make up for much of the shortfall.

At an Al-Sc master alloy price $20/kg above the scandia metals basis cost, our margin is just about $12/kg, which for this plant is just under $6M/yr. With 50/50 debt/equity financing, and 4.5% interest on debt, this leads to a weighted average cost of capital (WACC) of 7.35%, first year levelized cost of production (LCOP) of $11.21, and internal rate of return on equity (IRROE) of 11.92%.

Note this model does not consider the high-purity oxygen by-product generated by the plant at 54 kg/hour (450 T/year at 95% plant utilization), which could potentially generate an additional $5-25 per 100 SCF depending on purity and market availability. That is equivalent to $0.91-4.55/kg O_2, *i.e.* $0.82-4.10/kg Al-2Sc, or $400k-2M/year in plant revenue.

If that same plant uses carbon anodes, costs increase by about $2-5/kg. Sealed cells with carbon anodes require inert gas dilution to prevent CO_2 reaction with cell components such as the lid or crucible. While this eliminates most emissions, particularly for the plant itself resulting in much better workplace safety, it is a significant expense – above that of carbon anodes if argon is required. The alternative to a sealed

cell is high-volume air withdrawal with dry scrubbing to remove HF and cyclone or similar separations to remove fluoride bath dust. Either way, CO combustion is required, and is not part of the cost model at this time.

That said, carbon anodes are considerably more reliable at this time. If a zirconia tube breaks and releases silver into the product, it must likely be tapped dry and restarted. This cost is not captured by the TEA, and will require significant advances in quality and reduction in lifetime variance before we can consider deployment. This can be considered an execution risk of this approach.

For a new plant reducing Sc_2O_3 into a liquid aluminum cathode with carbon anodes, the economics are quite different, requiring a new TEA spreadsheet, as described on page 216.

That said, capital cost for this facility is much lower than for production of all 500 TPY from Al_2O_3 and Sc_2O_3. With total primary equipment cost between $180,000 and $280,000, IRR on this plant is in the range of 150-200%. There is also much lower risk in the use of carbon anodes, such that ramp-up time is likely to be much faster than the three-year assumption of this spreadsheet, accelerating returns even further.

Conclusions

Operating cost for production of pure commodity aluminum using Pure Oxygen Anodes at $1.63/kg is comparable to median 2014 aluminum smelter production cost around $1.65-1.70. That said, the modeled cost remains below the greenfield opex estimate of a global aluminum partner shared with us two years ago. If carbon emissions are penalized in the future, adding 5-10¢/kg to the cost assuming $30/T CO_2 price, then INFINIUM technology could be compelling due to its lower emissions – provided we can meet the performance targets of 4000 hours lifetime at 1 A/cm² current density.

Experimental Apparatus: INFINIUM Quad Cell

Overview

INFINIUM completed construction of four cells and control system for high-throughput electrolysis testing, known as the “Quad Cell”. These cells have independent heaters and electrolysis power supplies, but shared gas, electrical, and water cooling infrastructure. Shared infrastructure provides cost savings and simplifies maintenance and operation. Independence of cells enables operation of simultaneous experiments at different temperature set points and start and end times. The system is designed for robust long-term operation and rapid turn-around between electrolysis runs.

Instrumentation and Controls Design

Figure 40 shows a preliminary piping and instrumentation diagram (P&ID) of a single cell. Included are the furnace and electrolysis power supply along with regulators, valves, rotameters, mass flow meters, pressure gauges, thermocouples, current transducers, voltage leads, and cooling water circuit.

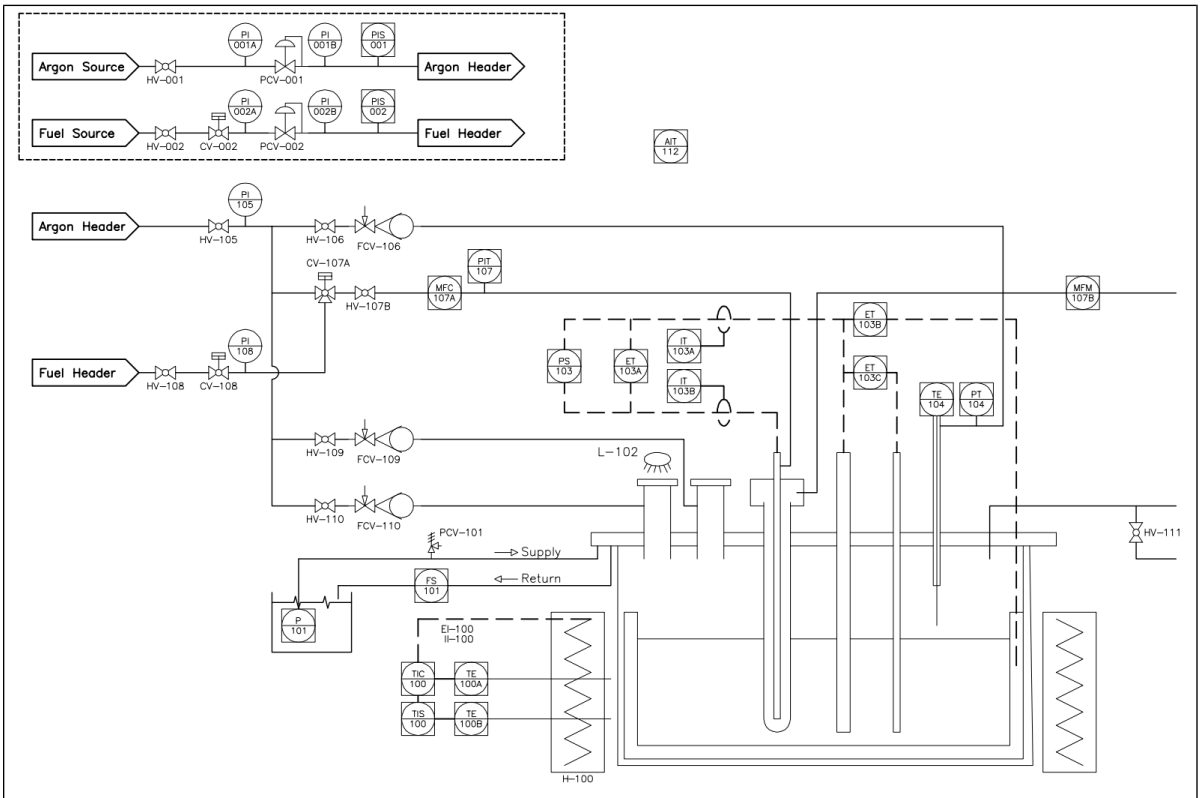


Figure 40: Preliminary P&ID for single cell

The entire system runs on a single programmable logic controller (PLC), allowing for the

control and monitoring of various temperatures, flows, voltages/currents, and cooling loops. The PLC also has numerous safety interlocks. For example, if cooling water temperature rises, it will first alarm, and then shut off furnace power to prevent an uncontrolled water spill or other failure mode. Any indication of fuel leakage will result in fuel shut-off and safe shut down of one or all of the experiments.

Container Design

The Quad Cell container design shown in Figure 41 is centered around a 6" schedule 10 pipe made from 310 stainless steel. The design incorporates custom flanges which allow slightly larger crucibles to be used and is appropriate for the scale of experiments in most of this project.

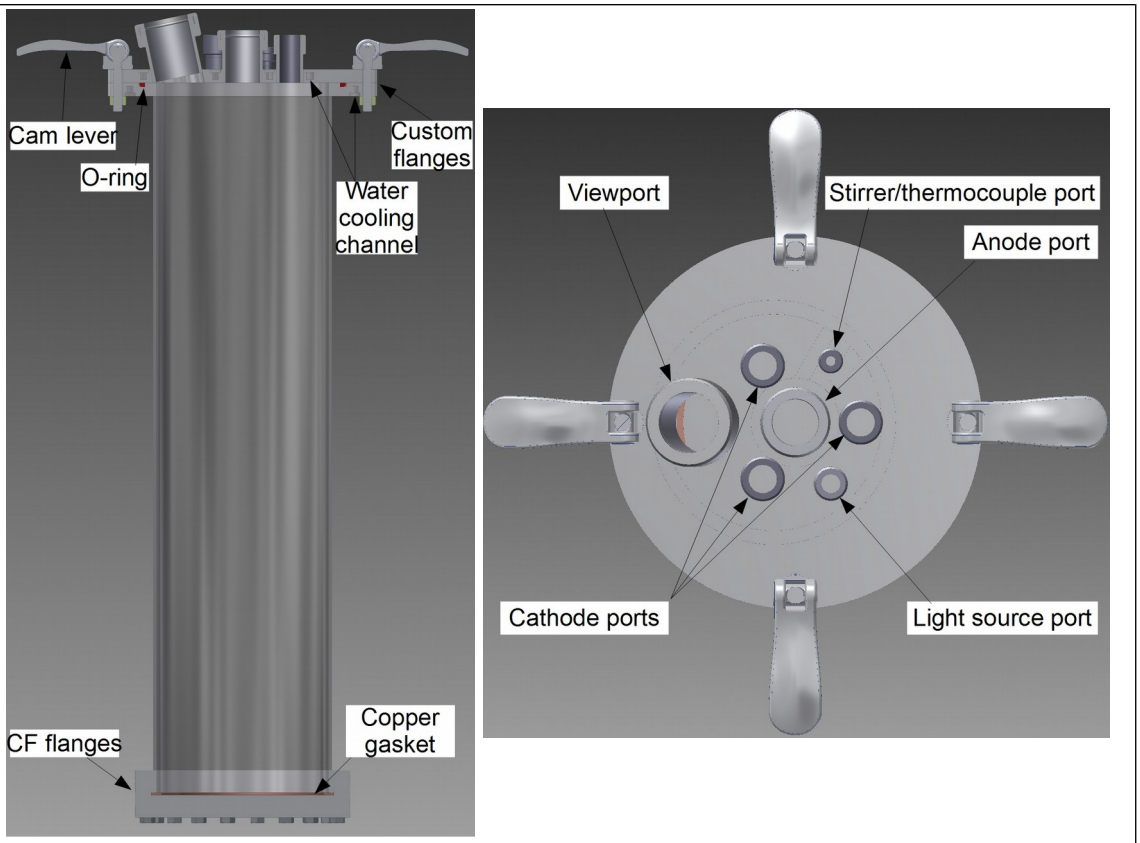


Figure 41: Container design: left: side view showing tube, bottom and top plates, and water cooling channel; (right: top plate showing clamps and port layout.

A CF flange made from 304 stainless steel seals the bottom of each pipe with a copper gasket while allowing it to be opened for cleaning if needed. This differs from previous designs which used a welded bottom to seal the pipe, and significantly reduces the time required to clean and reassemble the container. A set of custom flanges made from 304SS seals the top with a silicone o-ring and includes water cooling for both the o-ring and the fittings used to pass electrodes and other items into the crucible. The flanges also allow for the use of cam levers instead of bolts to compress the o-ring, which

significantly reduces the amount of time required to open and close the cell.Due to the increased bore of the flanges over the available stock tube sizes, the full inside diameter of the pipe can be used and a 6” OD crucible fits inside. The first design for the top flange has a port in the center for an anode up to 1.25” in diameter, with 3 evenly spaced ports around it so multiple cathodes can be tested. There is also a large viewport for observation, a port for a light source, and a port for a stirrer/thermocouple.

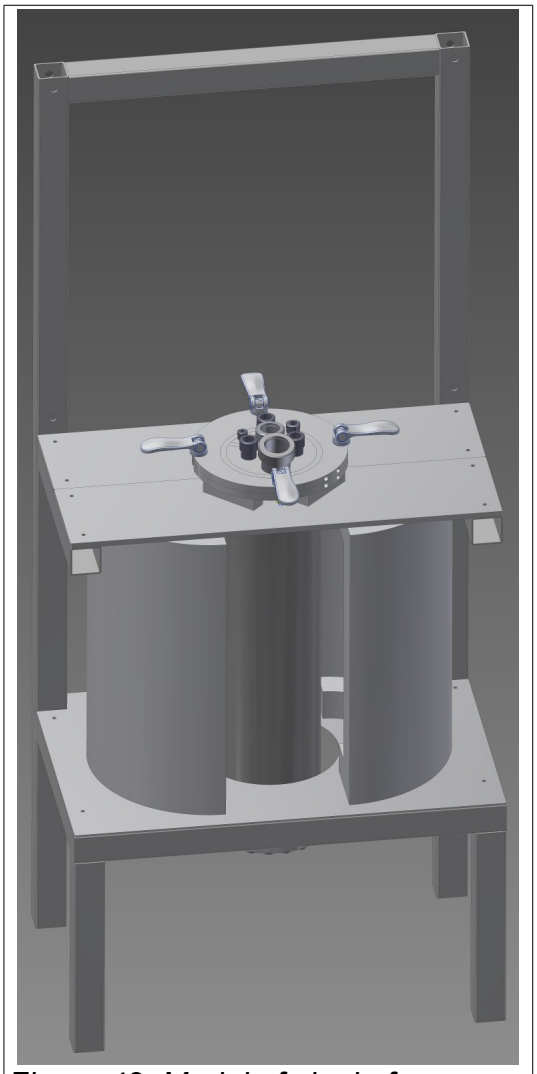


Figure 42: Model of single frame with furnace and container installed

Frame Design

The frames are made of 2” square steel tubing painted to prevent rust, and hold stainless steel plates, which are bolted in place, on the top and bottom. Each bottom plate supports the furnace insulation and heating elements while the top plate supports and provides height adjustment for the container and it's contents. Interlocking blocks insert between the container and the top plate, allowing for multiple possible heights of the furnace within the container. The frame also provides mounting holes for the gas panel and convenient locations to attach wires and hoses.

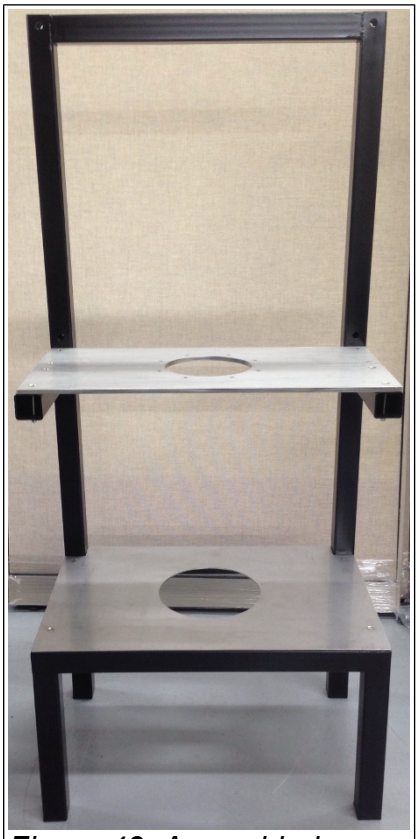


Figure 43: Assembled frame.

Two of these frames attach to the side of each gas panel in a side-by-side arrangement, with another two frames and gas panel arranged back-to-back. Each setup is thus accessible from three sides, allowing two people to work on an experiment at a time.

Each gas panel also has a touch screen attached to the PLC control system. Each display has data indicators and controls for the two experimental setups next to it.

Completed Apparatus

Figure 44 shows the completed Quad Cell apparatus. It

shows two of the four cells, with the other two hidden behind them. The control cabinet (gray) and main control PC (black cart) are on the right, along with electrolysis power supplies. The control computer provides a human interface to the PLC in the gray cabinet. At the time this photo was taken, a mass spectrometer was present in front of the gas/control panel.

Figure 44: INFINIUM Quad Cell, including the control cabinet (right), two complete cells, one gas panel with touch screen controls, and water cooling infrastructure (left).

Molten Salt Selection

The molten salt bath must satisfy multiple criteria, which are:

- High alumina solubility: one novelty of the Hall-Héroult cell was the use of a sodium cryolite molten salt composition with sufficient alumina solubility to operate at high current density.
- Electrochemical stability: bath anions and cations must be more stable than those of alumina, such that it consumes dissolved alumina alone, *i.e.* the anode consumes only oxide ions and the cathode only aluminum cations. Based on this, halide are the only anions available, and Li, Mg, Ca, Sr, Ba and rare earths are available.
- Low volatility: evaporation rate must be low.
- High mobility: to support high current density, electromigration mobility, *i.e.* ionic flux in response to an electric field, must be high. Similarly, high anion diffusivity allows oxide ions to diffuse into the depleted boundary layer adjacent to the anode.
- Low aluminum metal solubility: dissolved metal leads to electronic conductivity and low current efficiency. This problem is exacerbated by the reducing environment created by Pure Oxygen Anodes™, though this project indicated that aluminum is less problematic in this regard than other reactive metals such as magnesium and rare earths.
- Low melting point: though not necessary, ability to operate at lower temperature than 960°C would be helpful.
- Zirconia compatibility: described in detail above in the introduction on page 8.

For a traditional Hall-Héroult cell, low density is also required. Here it is helpful to have both low-density and high-density options for cell configurations 2 and 3 as defined in the industrial cell narrative on page 27. The primary low-density bath used here is potassium cryolite $KF{\cdot}AlF_3$, and the primary high-density bath is “BU Flux” based on MgF_2-CaF_2 eutectic.

Bath Volatility

The melting point and volatilization rate of high-alumina BU flux (composition 45wt% MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt% CaO and 4wt% YF_3) was studied with DSC/TGA analysis.

Methods

Pre-mixed flux powder was placed in a platinum crucible with the surface area of 1.217cm^2. The crucible was placed in the TA Instrument SDT Q600 TGA/DSC instrument. The sample was heated to 400°C and held for 2 hours under Ar atmosphere to remove water vapor. Then the sample was heated to 1100°C at a rate of 20°C/min and held for 2 hours, then cooled to room temperature at a rate of 20°C/min. Another sample was held at 1200°C for 2 hours in order to evaluate the volatilization rate of the flux at 1200°C.

Results

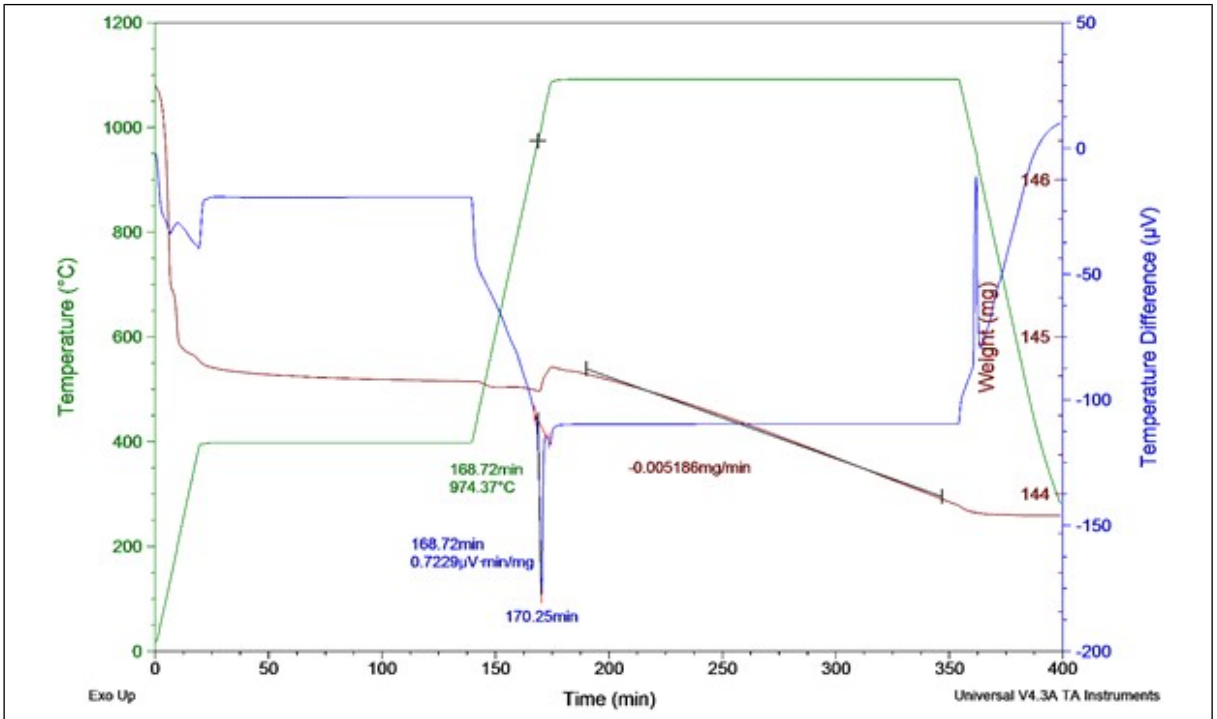


Figure 45: DSC/TGA profile of 45wt% MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt% CaO and 4wt% YF_3 flux at 1100°C.

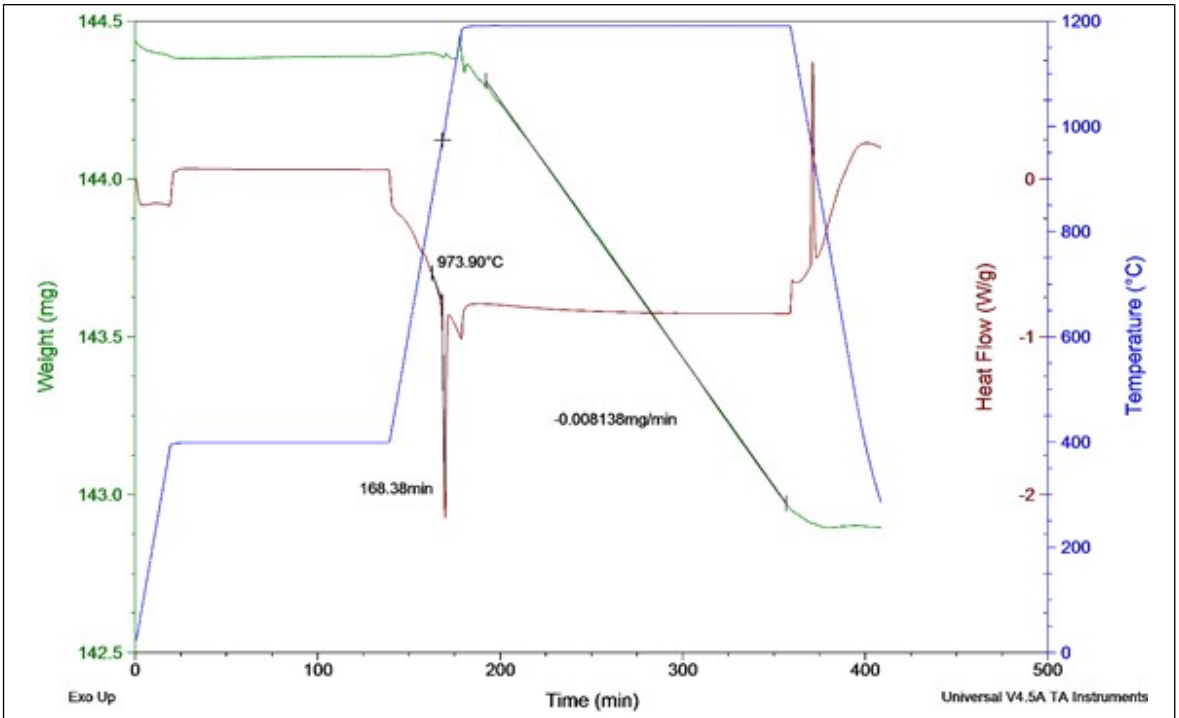


Figure 46: DSC/TGA profile of 45wt% MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt % CaO and 4wt% YF_3 flux at 1200°C.

Based on the DSC/TGA profile, the melting point of the flux is 974°C. The volatilization rates of the flux are 0.071µg/cm^2-s at 1100°C and 0.111µg/cm^2-s at 1200°C, as calculated from Figures 45 and 46, respectively. This data show that the volatilization rate of the flux increases with increasing temperature. Compared with the previous flux compositions with lower oxide content, about half as much alumina and calcium oxide, the flux volatilization rate decreased. These values can be found in Table 2.

Flux Oxide Composition	Temperature [C]	Volatilization [µg/cm^2-s]
5wt% Al_2O_3, 11wt% CaO	1200	0.156
9wt% Al_2O_3, 12wt% CaO	1100	0.071
9wt% Al_2O_3, 12wt% CaO	1200	0.111

Table 2: Volatilization at Different Temperatures and Compositions

Since the most volatile compounds in the flux are the fluorides, the volatilization rate is expected to decrease with decreasing fluoride content. In addition, the volatilization rate for all these samples is below 1 µg/cm^2-s, the target rate specified by ARPA-E. Thus, all flux compositions tested meet the specified volatilization requirement.

Conclusions

We are able to now easily analyze the zirconia solubility in various molten salt bath compositions. Specifically, the zirconia solubility of 45 wt% MgF_2 – 55wt% CaF_2 with 9wt % Al_2O_3, 12wt% CaO and 4wt% YF_3 flux was measured in this quarter using EDS spectra analysis. It was shown that the zirconia solubility in the salt was negligible. So, the salt is compatible with the zirconia membrane.

Alumina Solubility

Alumina capillary tubes containing multiple molten salt compositions were obtained from drawing the molten salt into the capillary tube using a pipette bulb. The alumina capillary tubes were with 0.25'' OD and 0.125''ID. They were cut into 1 - 1.5" length sections after the sampling.

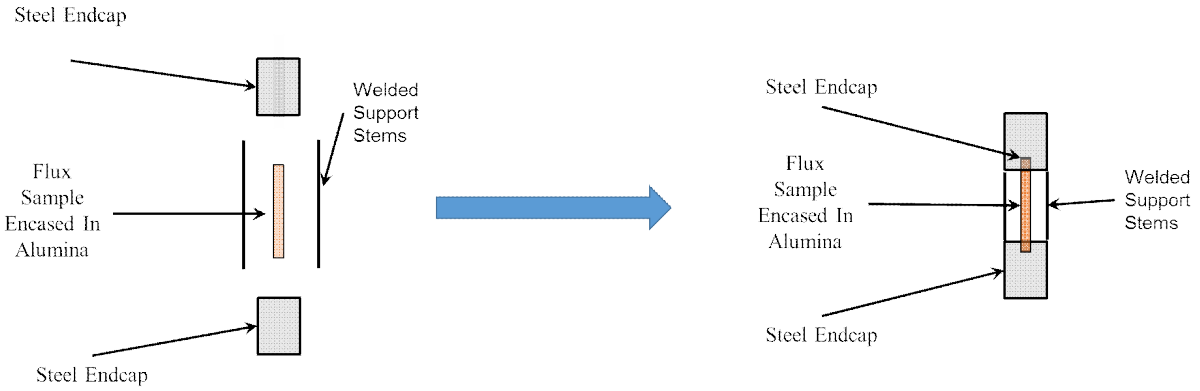


Figure 47: Two end caps design for aluminum solubility test

Two steel end caps were placed on both ends of the alumina capillary tube. Each end cap was made of steel with a hole drilled to accommodate the alumina capillary tube. Once the end caps were placed on the alumina tube, steel support rods were welded between the top and bottom end caps to ensure both caps stay in place. This design is shown in Figure 47. The assembly was placed in a vertical tube furnace and heated to the desired temperature. Forming gas was constantly purged in the tube furnace to ensure a reducing atmosphere. At a designated time interval, the sample was removed and quenched to room temperature. This sample was then cross-sectioned using a diamond saw.

The samples were later prepared for analyses by SEM, EDS and WDS analysis.

Alumina Sampling and Solubility Measurement: Results

In the last quarterly report, it was proposed that the alumina solubility in the molten flux would increase with higher CaO content. The flux composition with eutectic MgF_2-CaF_2, 12wt% CaO and 4wt% YF_3 was therefore studied. The flux was drawn into the alumina capillary tube and cut into 1.5" sections. The capillary tube was held at 1200°C for 29 hours with the two end-caps as outlined before. The capillary tube was then sectioned, and the flux cross-section was characterized. The SEM and EDS analysis was reported last quarter. The sample was further characterized with WDS and the alumina solubility was determined.

Figure 48 shows the back-scattering SEM image of the BU-12CaO salt held for 29

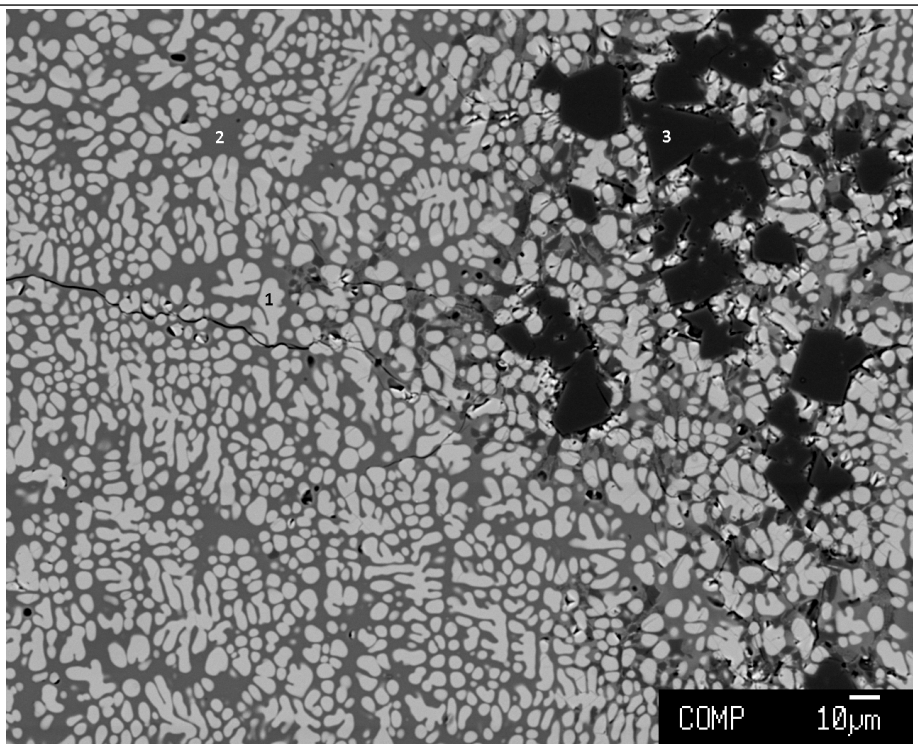


Figure 48: SEM image of 1200°C BU-12CaO salt 29 hours sample

hours at 1200°C. WDS analysis identified three phases as shown below. Phase 1 (Bright) is mainly CaF_2; phase 2 (gray) is eutectic MgF_2-CaF_2; phase 3 (dark) is $MgAlO_2$.

Bright Phase (1)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	0.316	67.978	1.073	0.029	0.772	29.82	0

Gray Phase (2)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	47.056	16.088	9.679	9.936	0.628	6.179	10.236

Dark Phase (3)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	57.858	0	12.421	29.651	0	0.046	0.003

Six radial line scans starting from the center of the flux to the interface between the flux and the alumina, each containing ten 30µm x 30µm spots were scanned with WDS to measure the average composition of the flux. The atomic percentage of each element is shown below. Attributing all aluminum to Al_2O_3, the Al_2O_3 solubility is calculated to be 11.68wt%.

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	30.10	37.74	5.502	5.273	0.552	16.109	4.605

The silicon and sodium were the impurities from the alumina tube and the alumina paste. The alumina capillary tube was 99.8% pure alumina, however there is typically 0.15wt% SiO_2 and 0.02wt% Na_2O in the alumina tube. Since the amount of flux is very small compared with the large surface area of alumina, it is possible for SiO_2 and Na_2O to dissolve in the flux along with the alumina. Silica is also a common additive in the alumina paste . The addition of CaO may have increased the solubility of SiO_2 and Na_2O along with that of alumina.

Comparing the alumina solubility of the BU-12wt% CaO salt with the earlier lower 6wt% CaO BU salt, it is shown that the alumina solubility increased from 3.16wt% to 11.68wt %

The alumina solubility of the BU-12wt% CaO salt was also measured at 1100°C. An alumina capillary tube with BU-12wt% CaO salt inside was held at 1100°C for 36 hours with two end-caps. The flux cross-section was characterized with WDS to calculate the alumina solubility.

Figure 49 shows the back-scattering SEM image of the BU-12wt% CaO salt held for 36 hours at 1100°C. WDS analysis identified three phases. The WDS analysis of the three phases is shown below. Phase 1 (Bright) is mainly CaF_2; phase 2 (gray) is eutectic MgF_2-CaF_2; phase 3 (dark) is $MgAlO_2$.

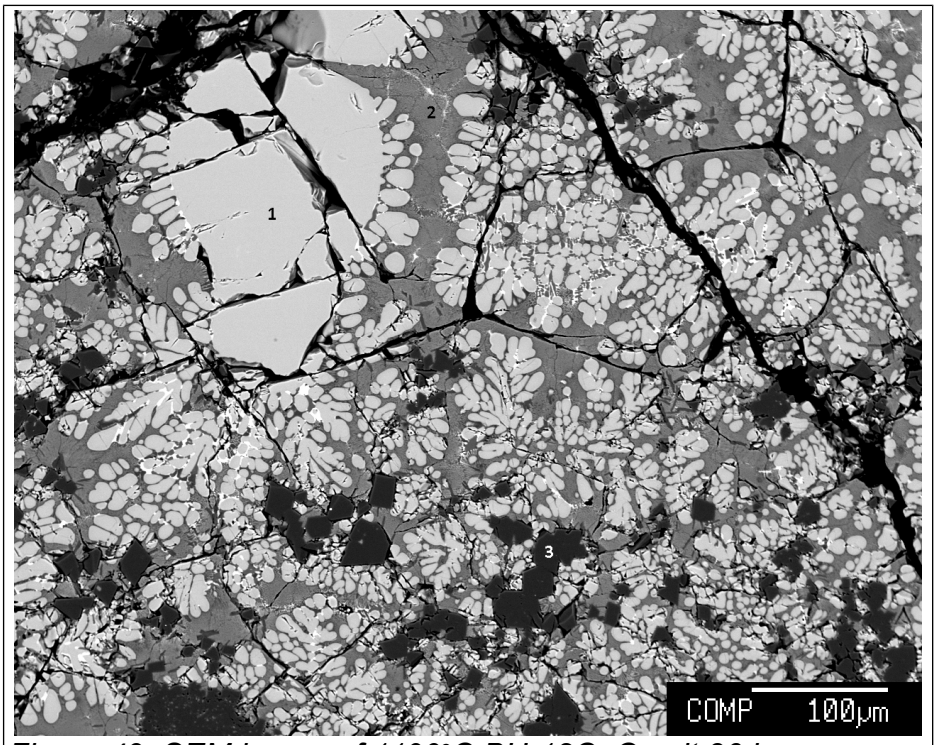


Figure 49: SEM image of 1100°C BU-12CaO salt 36 hours sample

Bright Phase (1)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	0.8228	66.717	1.27	0.001	0.845	30.344	0

Gray Phase (2)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	12.274	51.914	18.277	0.03	0.646	13.796	2.939

Dark Phase (3)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	54.136	0	14.69	30.991	0.008	0.145	0.029

Six radial line scans starting from the center of the flux to the interface between the flux and the alumina, each containing ten 30μm x 30μm spots were scanned with WDS to measure the average composition of the flux. The atomic percentage of each element is shown below. Attributing all aluminum to Al_2O_3, the Al_2O_3 solubility is calculated to be 9.24wt%.

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Si*
Atomic %	20.21	45.69	11.86	4.194	0.604	15.046	1.980

Alumina solubility test of Infinium Al-SOM salt

The composition of the Infinium salt was K_3AlF_6, 5wt% Al_2O_3. 48 hours and 87 hours solubility tests were performed with the Infinium salt at 850°C with the enclosed steel capsule method. The EDS analysis of 48 hour sample was reported in the last quarterly report. In this quarter, WDS analysis was used to measure the solubility of alumina in the Infinium salt. Figure 50 shows the SEM image of Infinium salt 48 hours sample.

8 radial line scans starting from the center of the flux to the interface between the flux and the alumina, each containing ten 30μm x 30μm spots were scanned with WDS to measure the average composition of the flux. The atomic percentage of each element is shown below. Attributing all O to Al_2O_3, the Al_2O_3 solubility is calculated to be 8.29wt%.

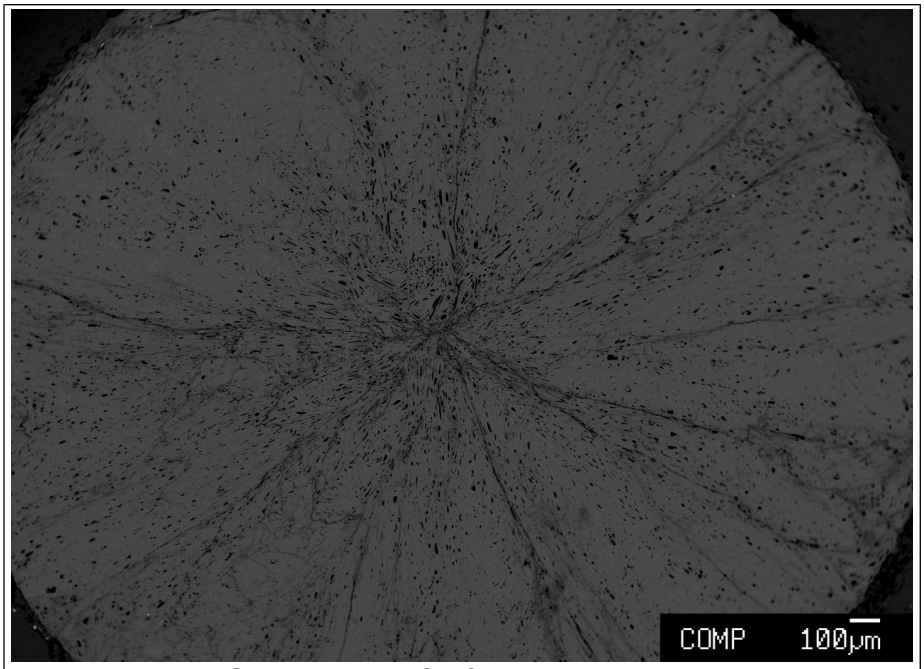


Figure 50: SEM Image of Infinium salt 48 hours sample

Elements	*O*	*F*	*Al*	*K*
Atomic %	5.68	62.11	16.65	15.55

Alumina solubility test of BU Al-SOM salt with 12wt% CaO addition

It is shown that the approach of adding AlF_3 to the flux did not increase the alumina solubility. In order to increase the alumina solubility to more than 5wt%, the flux composition must be tailored in another way. It is thought that in the BU flux system, the aluminum associated with oxygen from calcium oxide may form other complex ions such as AlO_3^{3-}, $Al_2O_5^{4-}$, $Al_3O_7^{5-}$, etc. The addition of CaO will thus increase the O^{2-} activity, and promote forming AlO_3^{3-} and breaking large Al-O network into smaller Al-O complex anions. Possible reactions are shown as follows,

$Al_2O_3 + 3O^{2-} = AlO_3^{3-}$; $Al_2O_5^{4-} + O^{2-} = AlO_3^{3-} + AlO_3^{3-}$.

The flux composition with eutectic MgF_2-CaF_2, 12wt% CaO and 4wt% YF_3 was therefore studied. The flux was drawn into the alumina capillary tube and sectioned into 1.5" sections. The capillary tube was held at 1200C for 29 hours with the two end-caps as outlined before. The capillary tube was then sectioned, and the flux cross-section was analyzed with SEM and EDS.

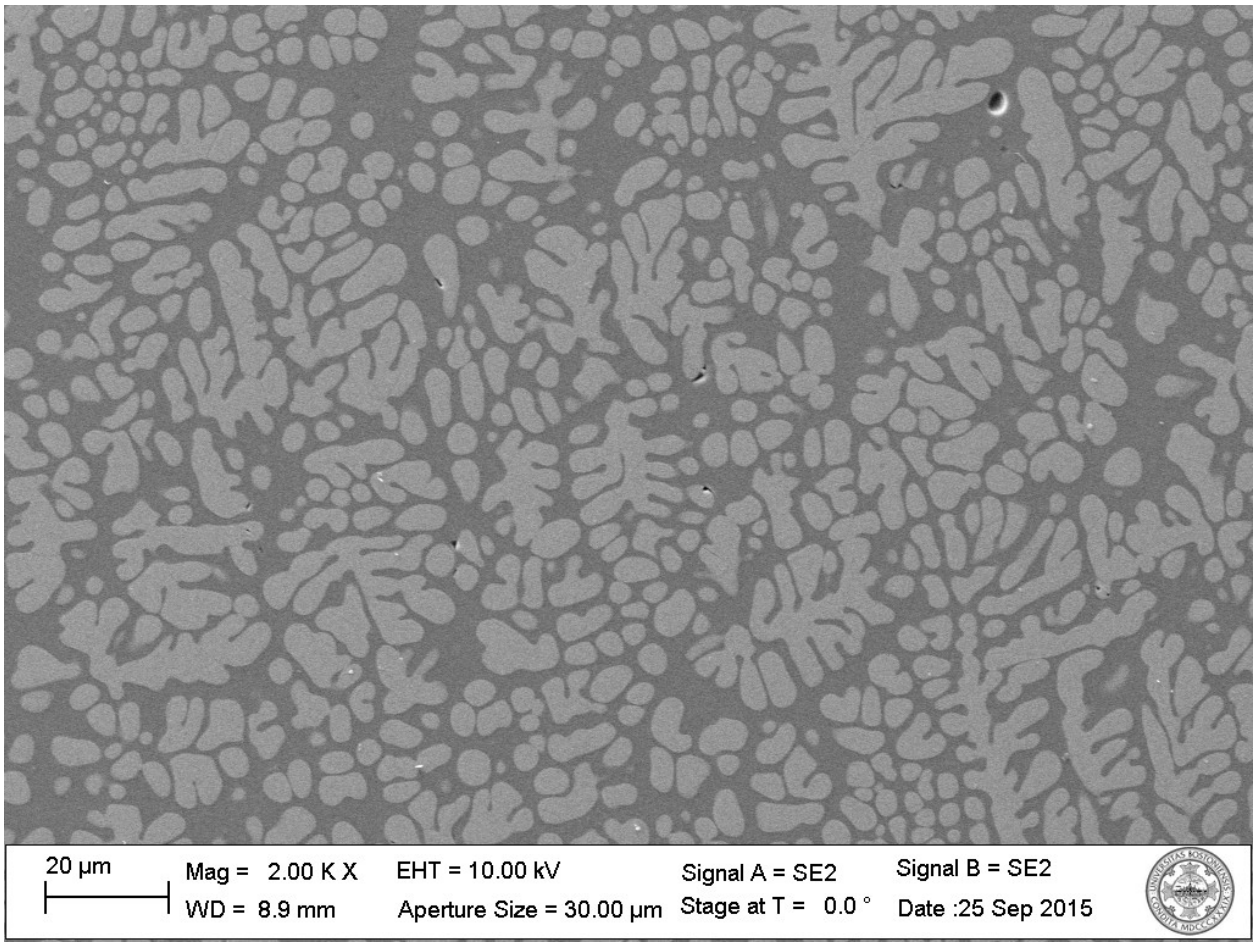


Figure 51: SEM image of BU-12CaO salt 29 hours sample

Figure 51 shows the SEM image of the BU-12CaO salt held for 29 hours. Quantitative EDS analysis of multiple flux areas were analyzed to acquire the Al content in the flux. The result is shown below.

Elements	O	F	Na	Mg	Al	Si	Y	Ca
Atomic %	22.9	39.21	0.27	6.23	5.29	5.69	0.67	19.72

The silicon and sodium were the impurities from the alumina tube. The alumina capillary tube was 99.8% pure alumina, however there is typically 0.15wt% SiO_2 and 0.02wt% Na_2O in the alumina tube. Since the amount of flux is very small compared with the large surface area of alumina, it is possible for SiO_2 and Na_2O to dissolve in the flux along with the alumina. The addition of CaO may have increased the solubility of SiO_2 and Na_2O along with that of alumina.

Comparing the aluminum atomic percentage of the BU-12CaO salt with the earlier lower 6CaO BU salt, it is shown that the Al content increased from 2.0at% to 5.29at% (both measured by EDS). Although the EDS analysis is not accurate enough to calculate the alumina solubility directly, it indicated that the Al content more than doubled. The alumina solubility in this flux will be further analyzed using WDS.

Zirconia Solubility in "BU Flux"

The zirconia capillary tube experiment was performed with BU-6wt% CaO-3wt% Al_2O_3 salt (eutectic MgF_2-CaF_2 with 3wt% Al_2O_3, 6wt% CaO and 4wt% YF_3). The tube was held at 1200°C for 24 hours with an improved method described as follows.

A graphite crucible was made such that the capillary tube could be inserted in the crucible and fit tightly. The flux powder was solvent mixed and subsequently dried. Then the flux was pre-melted and crushed several times to ensure homogeneous mixing. After the capillary tube was inserted in the graphite crucible, flux powder was filled in the capillary tube. Figure 52 shows the schematic of the experiment setup.

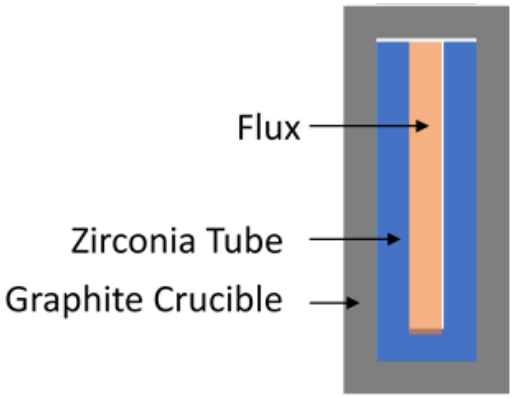


Figure 52: Schematic of improved capillary tube testing method

The improved experimental method is beneficial in several ways. Foremost, it provides a way to avoid using alumina paste containing SiO_2 since SiO_2 has the tendency to corrode YSZ membrane. It also allows use of a short section of zirconia tube in the experiment. The end-cap method for the alumina solubility measurements involved drawing molten flux into a long tube. Using a long zirconia tube significantly increases the cost of the experiment, and would increase the risk of the tube cracking under thermal shock or otherwise failing under mechanical stress.

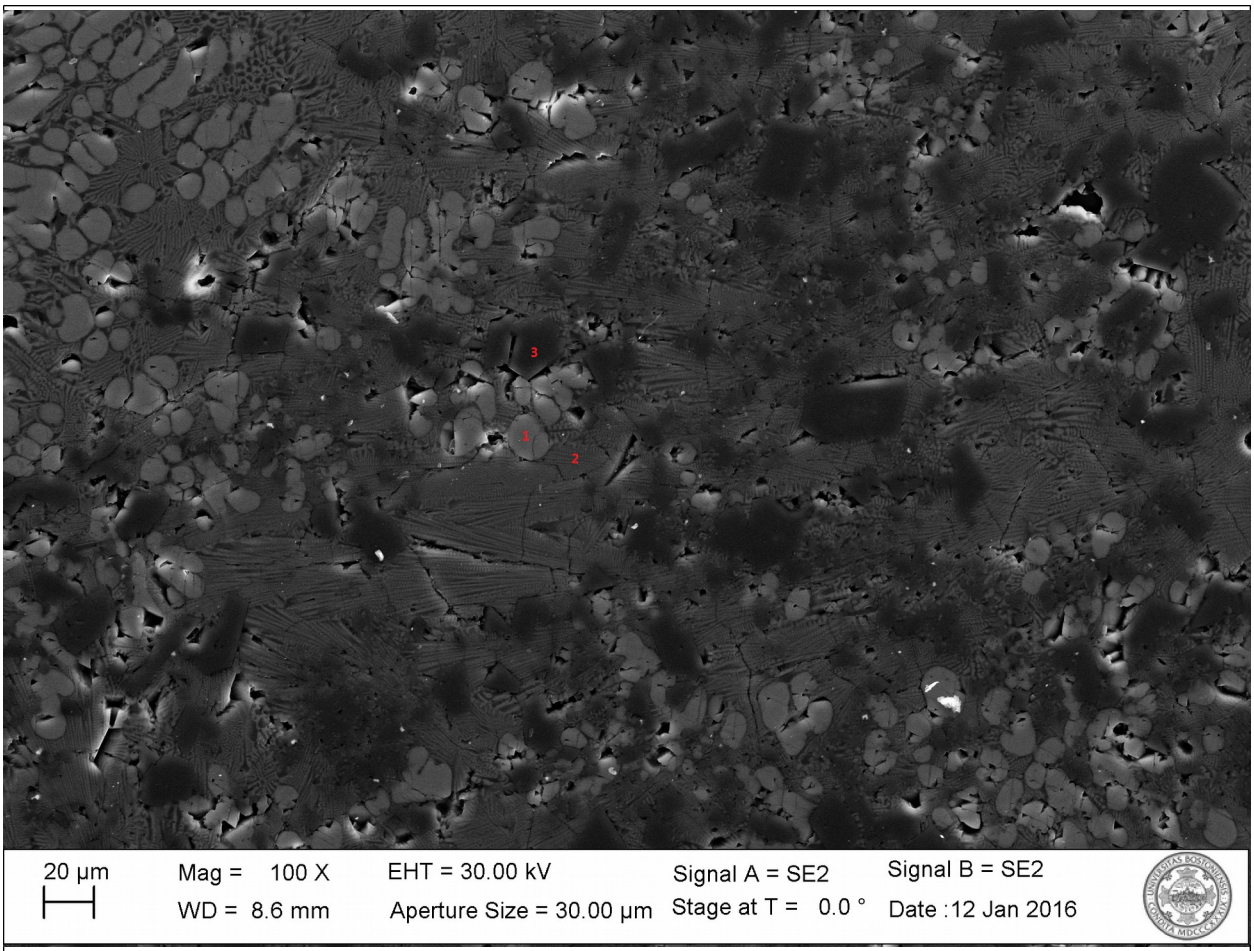


Figure 53: SEM image of BU-6wt% CaO-3wt% Al_2O_3 salt 24 hour ZrO_2 solubility test at 1200°C

The flux cross-section was analyzed with SEM and EDS to calculate the solubility of zirconia in the molten flux.Figure 53 shows the SEM image of the BU-6CaO-3Al_2O_3 salt held for 24 hours at 1200°C. Quantitative EDS analysis of multiple flux areas were analyzed to measure the Zr content in the flux. Three phases were identified from the SEM image.

Bright Phase (1)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Zr*
Atomic %	2.40	59.45	3.50	0.38	1.05	33.12	0.37

Gray Phase (2)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Zr*
Atomic %	3.47	57.81	20.50	0.22	0.53	17.42	0.06

Dark Phase (3)

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Zr*
Atomic %	46.95	0.25	17.78	34.36	0.13	0.46	0.08

Phase 1 (Bright) is mainly CaF_2; phase 2 (gray) is eutectic MgF_2-CaF_2; phase 3 (dark) is $MgAlO_2$.

A large area of flux cross-section (2.25mm x 3.01mm) was scanned with EDS to measure the average composition of the flux. The atomic percentage of each element is shown below. The Zr content is below the detection limit. It is confirmed that the ZrO_2 solubility is less than 0.4wt%. There is negligible ZrO_2 dissolved in the molten flux.

Elements	*O*	*F*	*Mg*	*Al*	*Y*	*Ca*	*Zr*
Atomic %	9.21	55.34	15.40	2.70	0.55	16.72	0.08

The cross-section of the zirconia tube was also analyzed with EDS line scan. Figure 54 shows that there is no yttrium or zirconium depletion between the boundary of the flux and the zirconia tube. Also the flux did not attack the zirconia.

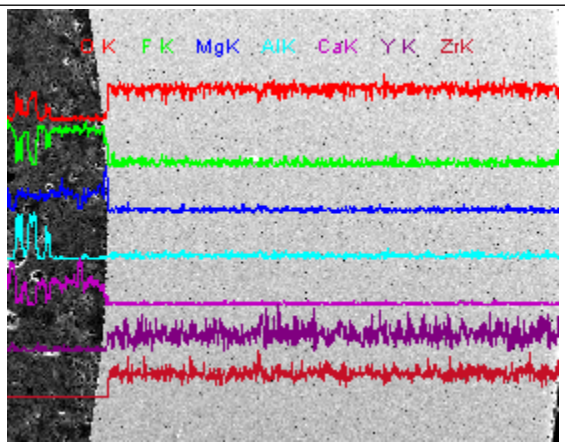


Figure 54: EDS line scan of ZrO_2 after 24 hour solubility test

Conclusions

We are able to now easily analyze alumina solubility and the zirconia solubility in various molten salt bath compositions. Specifically, the BU-12wt% CaO salt has an alumina solubility of 9.24wt% at 1100°C and 11.68wt% at 1200°C. It is now possible to operate the SOM cell with the BU-12wt% CaO salt at 1100°C instead of 1200°C and maintain high alumina solubility. The zirconia solubility of BU-6wt% CaO-3wt% Al_2O_3 salt was measured with EDS. It was shown that the zirconia solubility of BU-6CaO-$3Al_2O_3$ salt was negligible at 1200°C. The salt was compatible with the zirconia membrane.

Additional Tests

The zirconia capillary tube experiment was performed with the flux composition of 45 wt % MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt% CaO and 4wt% YF_3. The tube was held at 1100°C for 24 hours. The flux cross-section was analyzed with SEM and EDS to calculate the solubility of zirconia in the molten flux.

Figure 55 shows the SEM image of the 45 wt% MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt% CaO and 4wt% YF_3 salt held for 24 hours at 1100°C. Quantitative EDS analysis of multiple flux areas were analyzed to measure the Zr content in the flux.

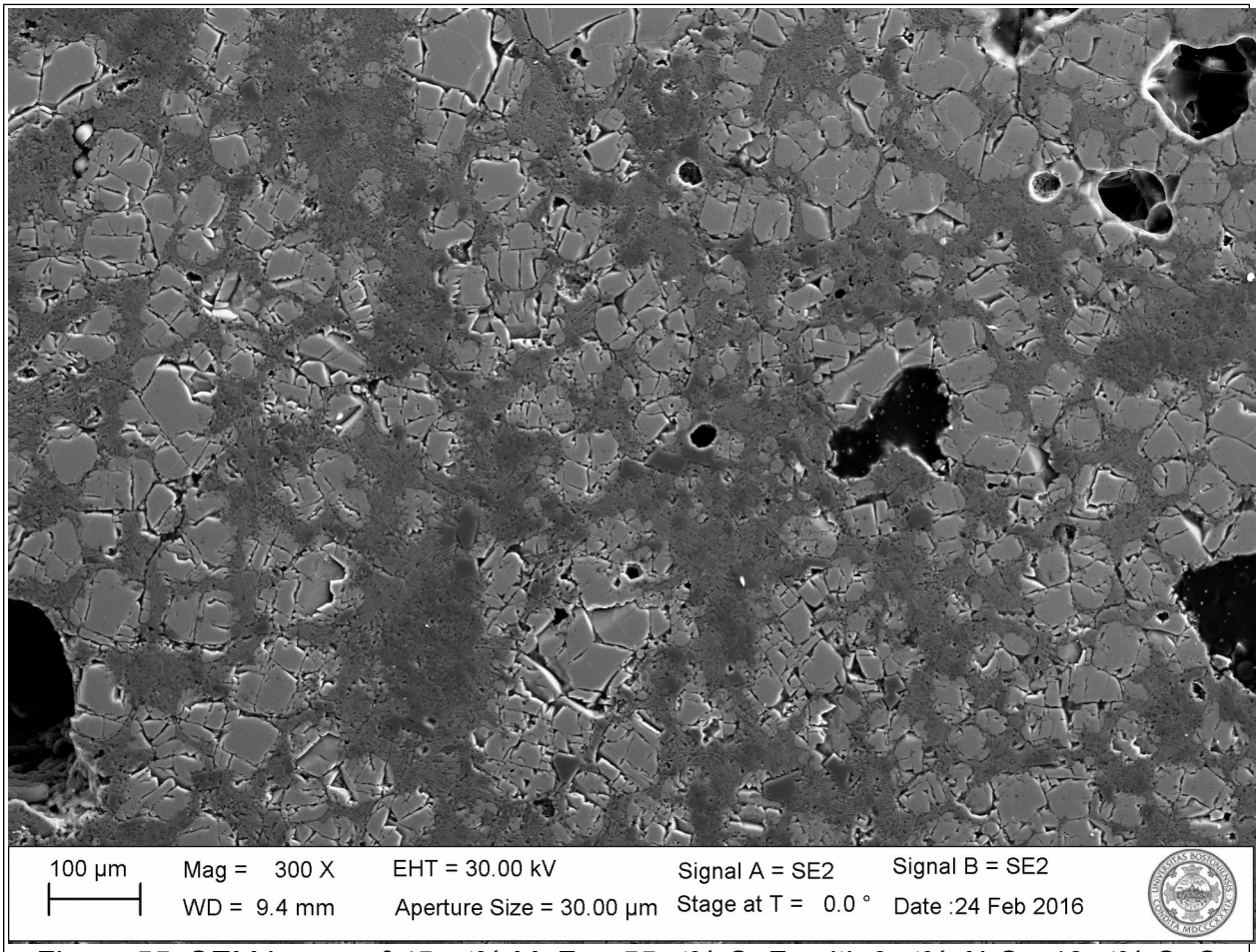


Figure 55: SEM image of 45 wt% MgF_2 – 55wt% CaF_2 with 9wt% Al_2O_3, 12wt% CaO and 4wt% YF_3 salt 24 hour ZrO_2 solubility test at 1100°C

Four large areas of the flux's cross-section (1.36mm x 1.02mm) were scanned with EDS to measure the average composition of the flux. The atomic percentage of each element is shown below. Since the Zr content is below the detection limit, the ZrO_2 solubility is less than 0.3wt%. This indicates that zirconia negligibly dissolves in the flux.

Elements (at%)	O	F	Mg	Al	Y	Ca	Zr
Area 1	17.66	42.57	17	4.23	0.45	18.03	0.06
Area 2	17.42	43.48	14.9	4.62	0.53	19	0.04
Area 3	18.64	41.85	15.64	5.61	0.51	17.68	0.07

Area 4	15.83	44.74	17.16	5.17	0.49	16.51	0.09
Average	17.39	43.16	16.18	4.91	0.50	17.81	0.07

The cross-section of the zirconia tube was also analyzed with an EDS line scan. Figure 56 shows that there is no yttrium or zirconium depletion between the boundary of the flux and the zirconia tube. Also, the flux did not attack the zirconia tube itself. This is indicative of the tube being stable in the presence of the flux.

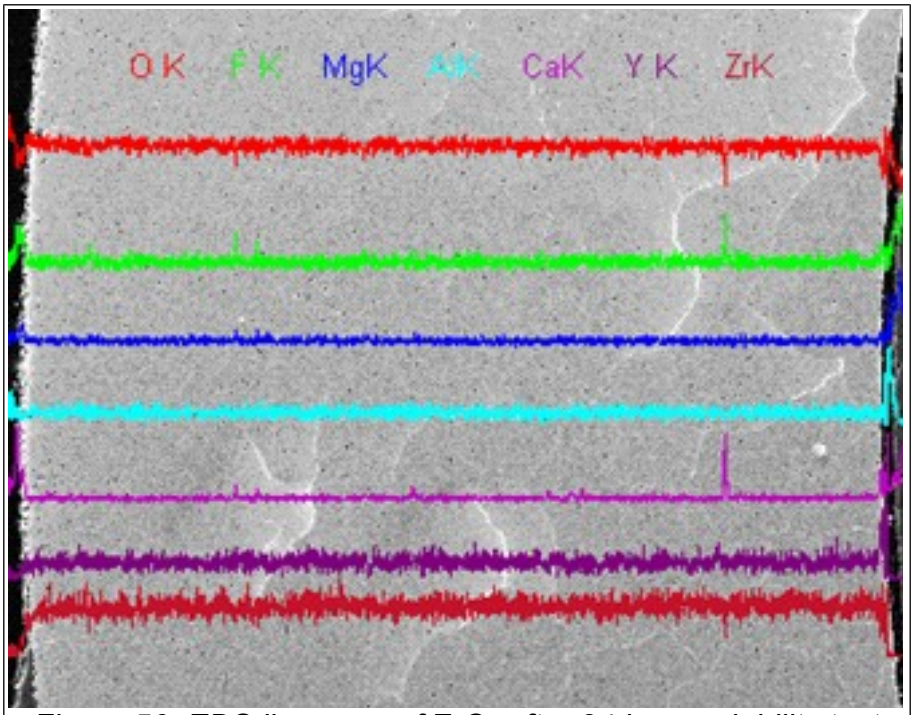


Figure 56: EDS line scan of ZrO_2 after 24 hour solubility test

Zirconia Stability in Potassium Cryolite

Overview

Three experiments were run in order to understand the stability of 8mol% YSZ tubes in the potassium cryolite flux. Salt corrosion was measured in the 1.3 cryolite ratio KF-AlF_3 flux with and without yttrium oxide additions quantitatively by flux penetration depth. An open and closed-ended tube were used to compare the degradation on the outer and inner edge. Banding on the inner edge of the tube has been seen during past experimentation, so testing both open and closed-ended tubes would verify if this only occurred when the tube is broken and exposed to flux.

The first static soak without Y_2O_3 used two YSZ tubes (Anode ID: 10B and 10C). Anodes 10B and 10C had initial wall thicknesses of 3.75mm and 3.8mm respectively. The second soak used the same parameters, but with the addition of 5wt% Y_2O_3. Finally, the last experiment ran with the same initial parameters, but with the addition of 10wt% Y_2O_3. Each experiment ran for 66 hours at a furnace set point of 760C. A PLC was used to control the power supply and furnace temperature. This was a static experiment so no electrochemical data was recorded.

Experimental Parameters

Anode: 2x 3/4" OD 8YSZ tubes with no silver (10C was closed-ended and 10B was open-ended)

Salt composition: KF-AlF_3 – 1.3 cryolite ratio with 14wt% saturated alumina

Salt Height: 4.5"

Crucible: Isomolded graphite – 5" OD x 4.5" ID x 12" High (11.5" Deep)

Furnace Set-point: 760°C

Crucible temperature: 720°C

Procedure (same for each experiment)

1. Break up pre-melted flux block into approximately 2" pieces and load into graphite crucible.
2. Insert graphite crucible into 6" OD steel containment and load into quad cell #3.
3. Attach top plate and insert anodes down through the top plate and rest on the top of the broken up flux pieces.
4. Insert an alumina bubbling tube and thermocouple. Plug any remaining ports on the top plate.
5. Add container argon purge to the view port and auxiliary argon purge to a reducing fitting on the alumina bubbling tube.
6. Turn the argon supply on at the manifold on the quad cell control panel.

7. Attach water cooling lines.
8. Heat furnace up to 760°C furnace set point.
9. Once crucible temperature has stabilized, lower the alumina bubbling tube 1/2" from the bottom of the crucible and allow to bubble for 10 minutes.
10. Remove the alumina bubbling tube from the flux and leave as an argon purge 1" above the surface of the molten flux.
11. Slowly lower each anode to the bottom of the crucible at a rate of 1/8" per minute.
12. Start the timer on the electrolysis screen.
13. After 66 hours remove the anodes quickly from the flux and support with a test stand. NOTE: wear respirator and full hot suit when removing from the flux.
14. Take flux sample by dipping a 1/8" stainless steel welding wire and quickly removing it from the furnace to quench the salt on the wire.
15. Power off the furnace and allow to quench overnight.

Results: Static Study with no Y_2O_3:

Anode 10B broke during extraction after 66 hours of soaking. The anode was quenched with the molten salt and was recovered by breaking the salt block carefully around the anode (Figure 57).

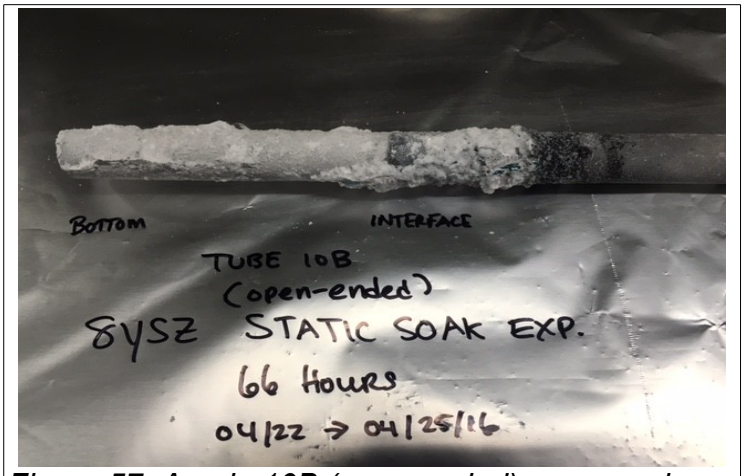


Figure 57: Anode 10B (open-ended) recovered from salt bath after 66 hour soak test

Anode 10C was removed successfully and visible white discoloration was seen (Figure 58) on the immersed part of the tube.

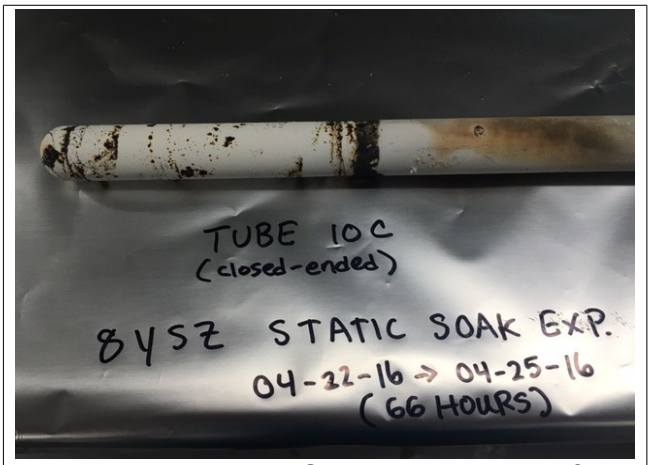


Figure 58: Anode 10C (closed-ended) after 66 hour soak test

The tubes were then mounted in epoxy and cut for SEM analysis. Both tubes were cross-sectioned and samples were cut from the top, middle and bottom of the tube. These samples were compared to a control sample taken from the top of the tube that was outside of the active area.

Table 3 highlights the change in bulk concentration, in weight percentages, from the middle of the tube for both anodes. The closed-ended tube increased Zr concentration by 18wt%, increased Y concentration by 22% and decreased oxygen concentration by 44%. The open-ended tube increased Zr concentration by 7wt%, increased Y concentration by 4% and decreased oxygen concentration by 26%.

	Tube	Width of Tube (mm)	Zr	Y	O
Closed-end	*Control*	3.8	57.79	12.58	29.63
	10C	3.9	68.23	15.32	16.45
Open-end	*Control*	3.8	65.74	14.61	19.65
	10B	4.0	70.32	15.14	14.53

Table 3: Bulk concentrations for control sample and active sample on both closed and open-ended tubes. Values reported in weight percentage unless otherwise noted.

Visible banding was observed on the outside of the closed-end tube (Figure 59), but no band on the inside (Figure 60). The outside band penetrated over 200 microns.

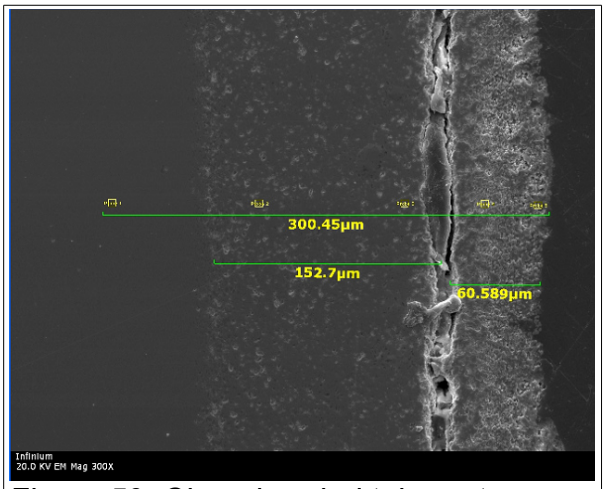


Figure 59: Closed-ended tube outer edge. Visible banding seen over 200 microns into the tube.

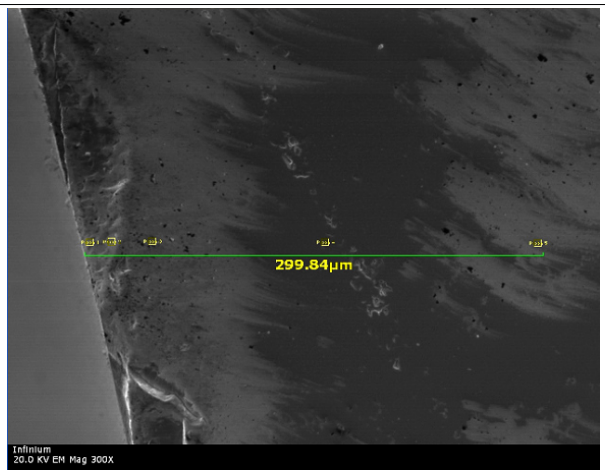


Figure 60: Closed-ended tube inner edge. No visible banding seen and no concentration change over 300 microns.

The open-ended tube showed visible banding both on the inside and outside edge, penetrating 187 and 237 microns respectively.

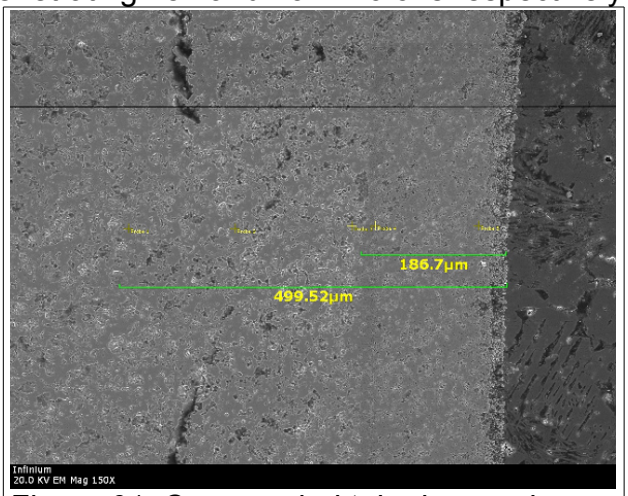


Figure 61: Open-ended tube inner edge. Banding present 186.7 microns into the tube.

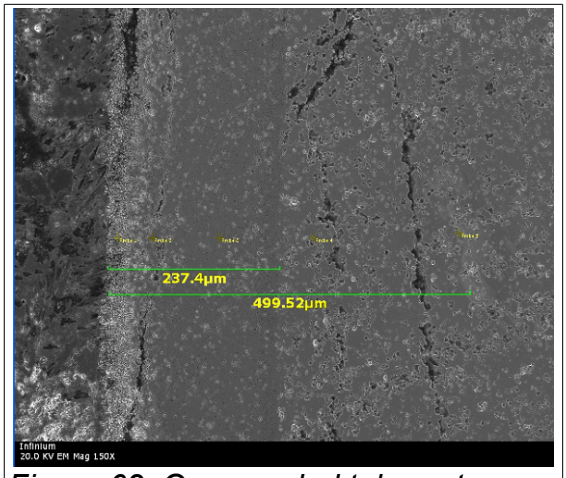


Figure 62: Open-ended tube outer edge. Banding present 237.4 microns into the tube.

Results: Static Study with 5wt% Y_2O_3:

The same procedure outlined in the first static soak was followed for this experiment. The only change was to add 5wt% yttrium oxide to the bath to try and stabilize the membrane. The soak ran for the same time (66 hours) and temperature (760C furnace setpoint) to compare results. Figure 63 shows the visible discoloration of the outside of the anode post experiment and Figure 64 is the SEM image showing the flux penetration. The flux penetrated the membrane 60 microns, which is an 80% decrease in penetration depth compared to the initial test.

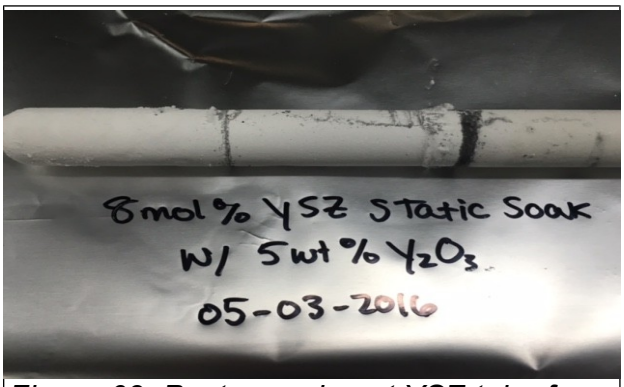


Figure 63: Post-experiment YSZ tube from 5wt% yttrium oxide static soak (66hrs).

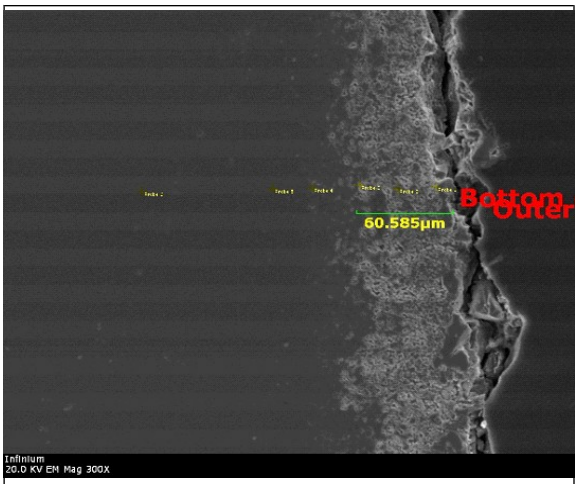


Figure 64: Outer edge of Anode 11B (3.77mm thick). Flux penetration seen 60 microns into the membrane.

Table 4 highlights the composition change of the anode over the course of the static soak. The yttrium concentration decreased 68%, while the oxygen increased 43%. This is likely due to the yttrium dissolving into the flux from the outer membrane interface.

Tube	Zr	Y	O	F	Al	K
Control	64.82	15.54	19.64	0	0	0
Bulk	63.95	4.91	28.01	0	0	0

Table 4: Bulk concentrations for control sample and active sample on Anode 11B (wt% unless otherwise noted).

Results: Static Study with 10wt% Y_2O_3:

The final static test was to add double the yttrium content in the bath to see if it further reduced the flux penetration in the membrane. The experiment was run with the same operating parameters as the first two static studies. Figure 66 shows the tube post experiment, while Figure 65 shows the penetration of the flux being roughly 57.5 microns into the membrane. It appears that the yttrium oxide concentration in the bath has reached an equilibrium with the degradation to the membrane. Adding any more yttrium oxide to the bath would increase the viscosity and risk getting yttrium as an impurity in the final product.

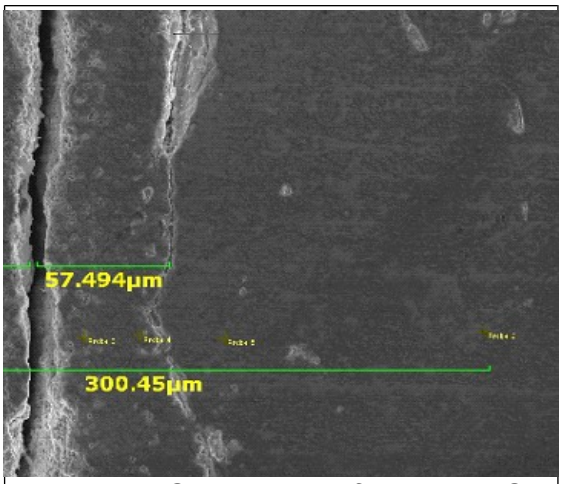


Figure 65: Outer edge of Anode 11C (3.71mm thick). Flux penetration seen 57.5 microns into the membrane.

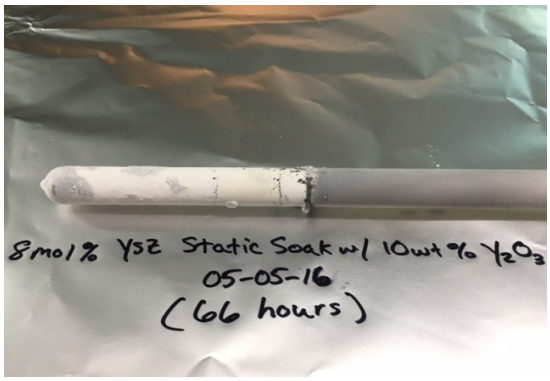


Figure 66: Post-experiment YSZ tube from 10wt% yttrium oxide static soak (66hrs).

Table 3 highlights the composition change of the anode over the course of the static soak. The yttrium and zirconium concentrations decreased 33% and 6% respectively, while the oxygen increased 36%. This shows that the higher yttrium concentration in the bath to begin with suppressed the leaching into the bath. The decrease in zirconium can be due to a phase change within the membrane as flux penetrates.

Tube	Zr	Y	O	F	Al	K
Control	64.82	15.54	19.64	0	0	0
Bulk	60.95	10.39	26.74	0	0	0

Table 5: Bulk concentrations for control sample and active sample on Anode 11B (wt% unless otherwise noted).

Summary

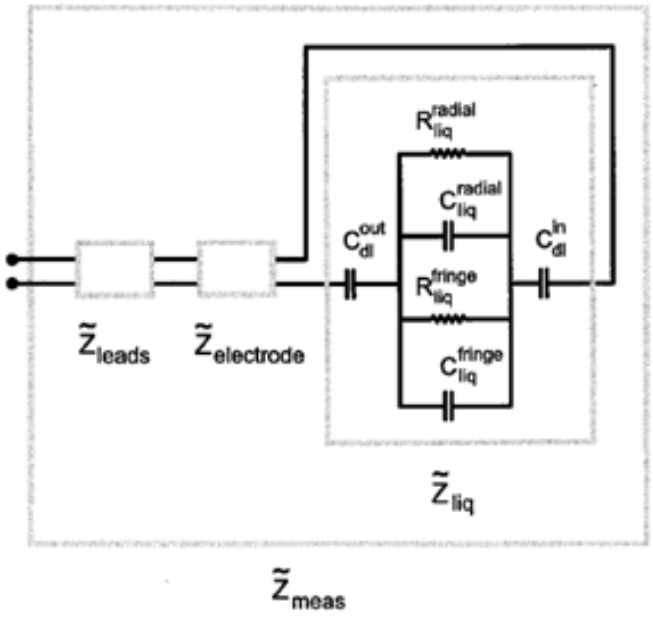
This static soak showed the stability of the membrane over 66 hours. Comparing a closed and open-ended tube in the first experiment proved successful in targeting the cause for banding. Banding was only seen on surfaces exposed to flux. Therefore, future banding on the inner membrane will be attributed to the tube breaking and allowing flux contact to the inner edge.

The concentrations of the tube changed drastically and flux penetration was seen 300, 60 and 57.5 microns into the membrane in the 0, 5 and 10wt% Y_2O_3 experiments respectively. This trend is most likely due to yttrium equilibrium with the flux, that is, the 8YSZ tube is in equilibrium with 5-10 wt% Y_2O_3 in the bath.

With 60 μm corrosion in 66 hours, that corresponds to a 1 μm/hour corrosion rate. However, for thin 0.6-0.9 mm tubes described in later experiments, slower corrosion rate is necessary. One method for accomplishing this is increasing cryolite ratio, resulting in a more basic bath. Electrolysis experiment 035 on page 199 does this to great effect, but we did not perform static soak tests in bath with this cryolite ratio. Another potential method described above is to increase grain size, in order to reduce grain boundary attack; we have not yet attempted this.

Conductivity Measurement

Electrical conductivity of a selected flux was determined by using two concentric coaxial electrodes in a test cell and measuring the impedance spectra as a function of immersion depth in the molten electrolyte, following the method of Schiefelbein et al.[49] Results showed that the accuracy of the conductivity measurements are very sensitive to the electrode resistance. From the equivalent circuit of the coaxial cell shown in Figure 67, it is obvious that in order to acquire the impedance of the molten salt (Z_{liq}), the lead-wire resistance and electrode resistance need to be subtracted from the impedance measurement (Z_{meas}). Lowering the lead-wire resistance and the electrode resistance will increase the accuracy of Z_{liq}.



$\tilde{Z}_{meas}$: total measured impedance
$\tilde{Z}_{leads}$: leadwire impedance
$\tilde{Z}_{electrode}$: electrode impedance
$\tilde{Z}_{liq}$: liquid impedance
R_{liq}^{radial} : radial component of liquid resistance
C_{liq}^{radial} : radial component of liquid capacitance
R_{liq}^{fringe} : fringe component of liquid resistance
C_{liq}^{fringe} : fringe component of liquid capacitance
C_{dl}^{in} : double-layer capacitance (inner electrode)
C_{dl}^{out} : double-layer capacitance (outer electrode)

Figure 67: Coaxial cylinder cell equivalent circuit.

Previous experiments have confirmed that using a setup made with concentric steel electrodes and molybdenum connecting rods has the lowest electrode resistance using conventional materials. Cryolite (sodium hexafluoroaluminate) was used to benchmark the accuracy of the method. Here, the geometry of the electrodes was changed to lower the weight of the instrument and potentially increase the sensitivity of the instrument. In particular, the outer diameter of the inner electrode was increased to 0.375 inches, which increased the salt resistance and the inner diameter of the outer electrode was increased to 1.750 inches which decreased the electrode resistance.

The modified coaxial concentric electrodes setup is shown in Figure 68. The results of the modified experimental setup are better in terms of accuracy. Using this newer approach, the conductivity of the new high alumina solubility BU salt (12% wt CaO, 9% wt Al_2O_3, 4% wt YF_3, remainder MgF_2-CaF_2 eutectic) was measured at three different temperatures. The results at each tempearture are on the same order of magnitude as the cryolite conductivity.

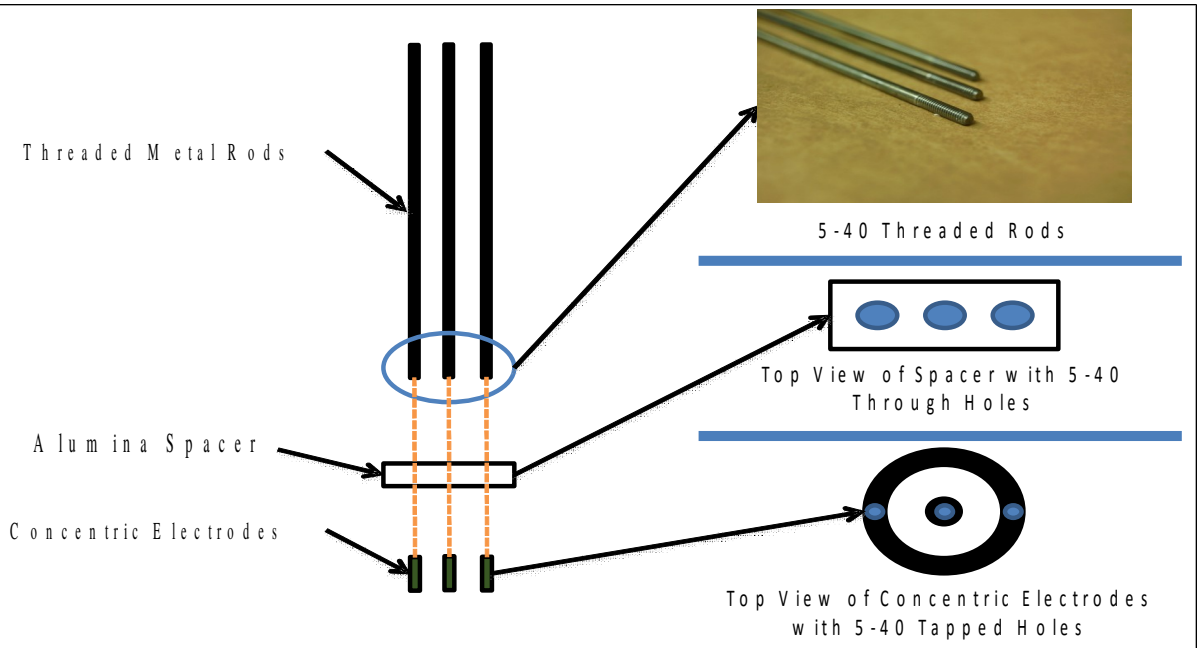


Figure 68: Detailed concentric coaxial electrode setup

Results and Discussion

Immersing the electrodes 0.125 inches at a time and taking three measurements, the electrodes were able to generate the following data sets at 1085°C, 1141°C, and 1185°C. This data is presented in a manner where the x-axis is given in centimeters [cm] and the y-axis is given in admittance [Siemens]. Admittance in this instance is defined as the inverse of the impedance of the molten salt. These values can be seen in Figures 69, 70, and 71.

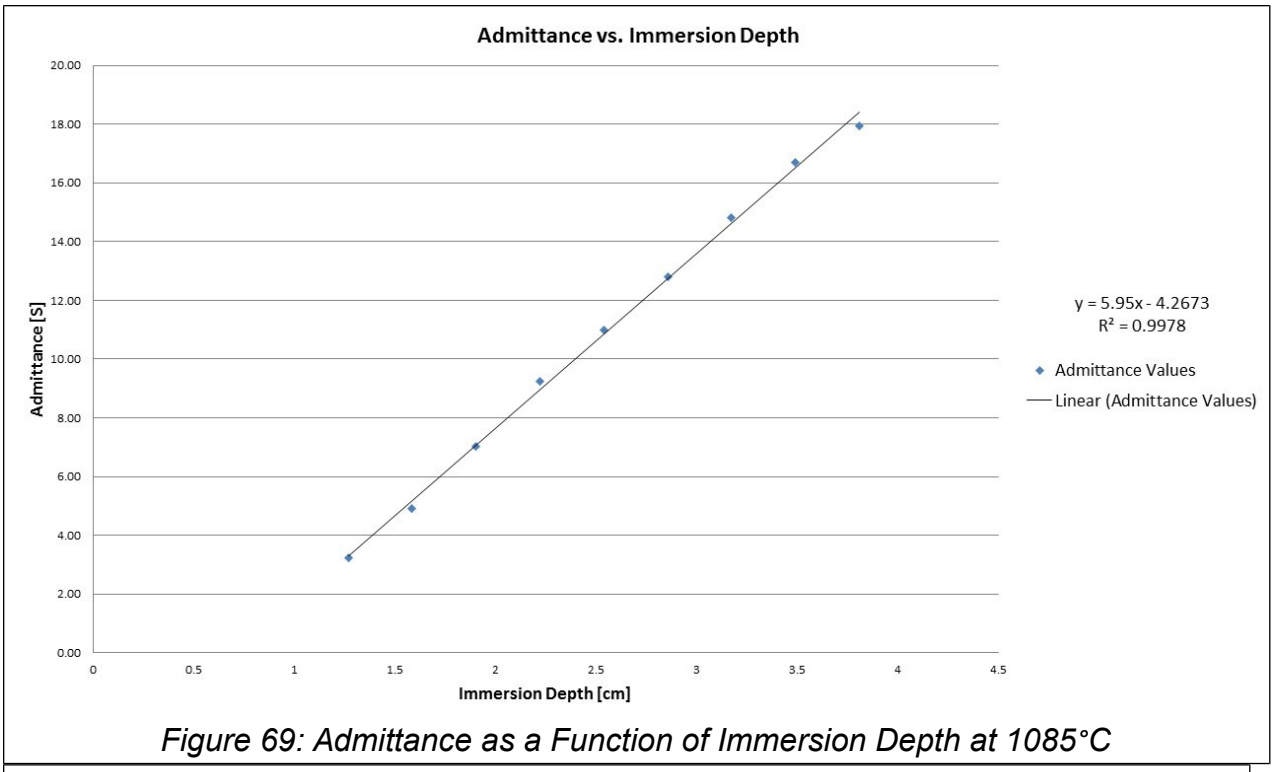


Figure 69: Admittance as a Function of Immersion Depth at 1085°C

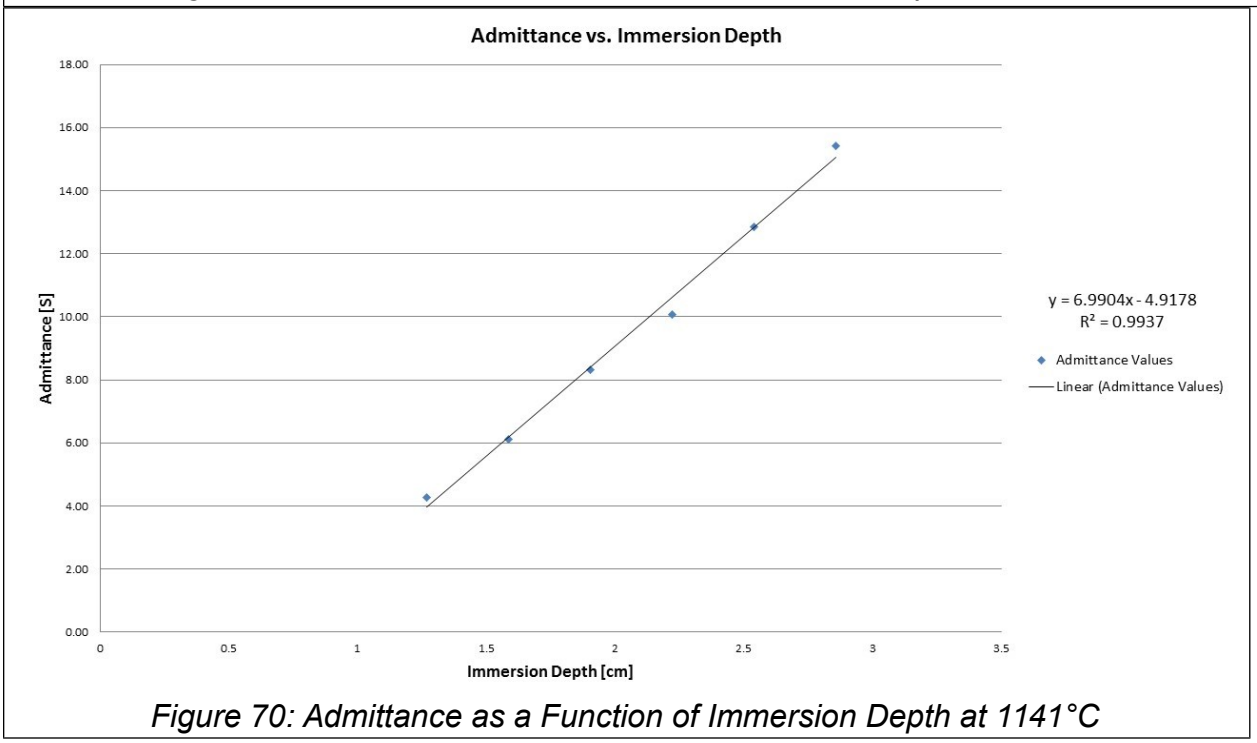


Figure 70: Admittance as a Function of Immersion Depth at 1141°C

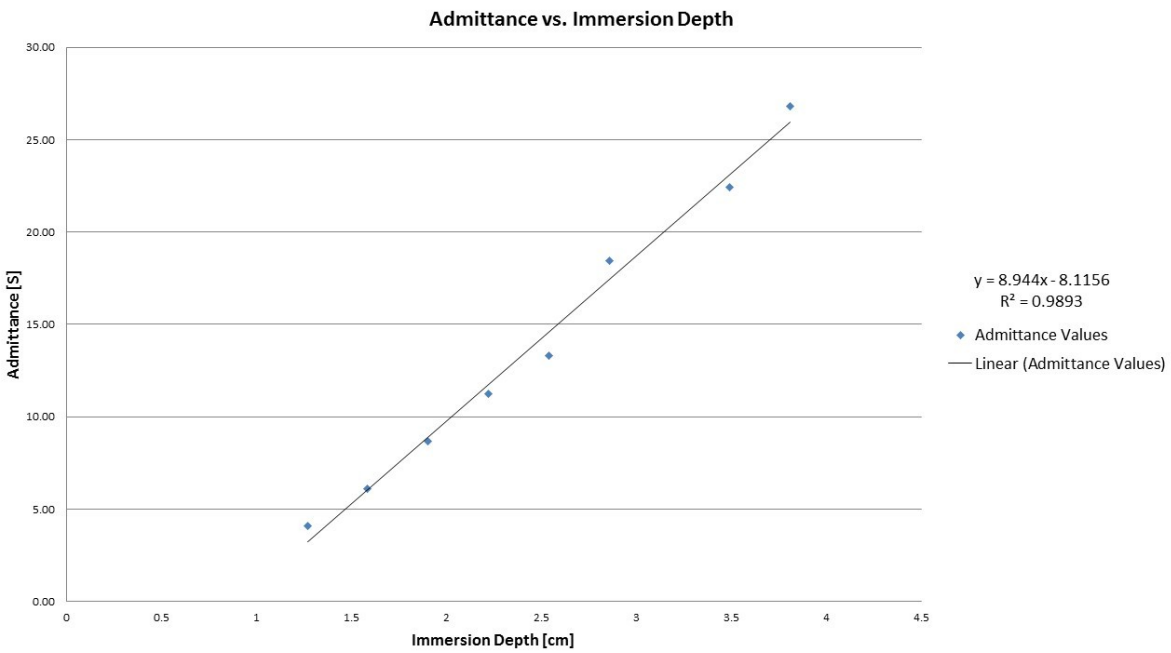


Figure 71: Admittance as a Function of Immersion Depth at 1185°C

Given the slopes of the plot and the dimensions of the electrodes, the values of conductivity are presented in Table 6.

Temperature [C]	Conductivity [S/cm]
1085	1.46
1141	1.71
1185	2.19

Table 6: Conductivity Values of Molten Salt at Various Temperatures

Using the conductivity values and inverse temperature values, an Arrhenius relationship was used to determine the activation energy of this flux system. The Arrhenius equation is given in Equation 26 and the plot is shown in Figure 72. From the slope of the plot, the activation energy is determined to be 79.0 kilojoules per mole.

$$\ln \sigma - \ln A = -\frac{E_a}{RT}. \tag{26}$$

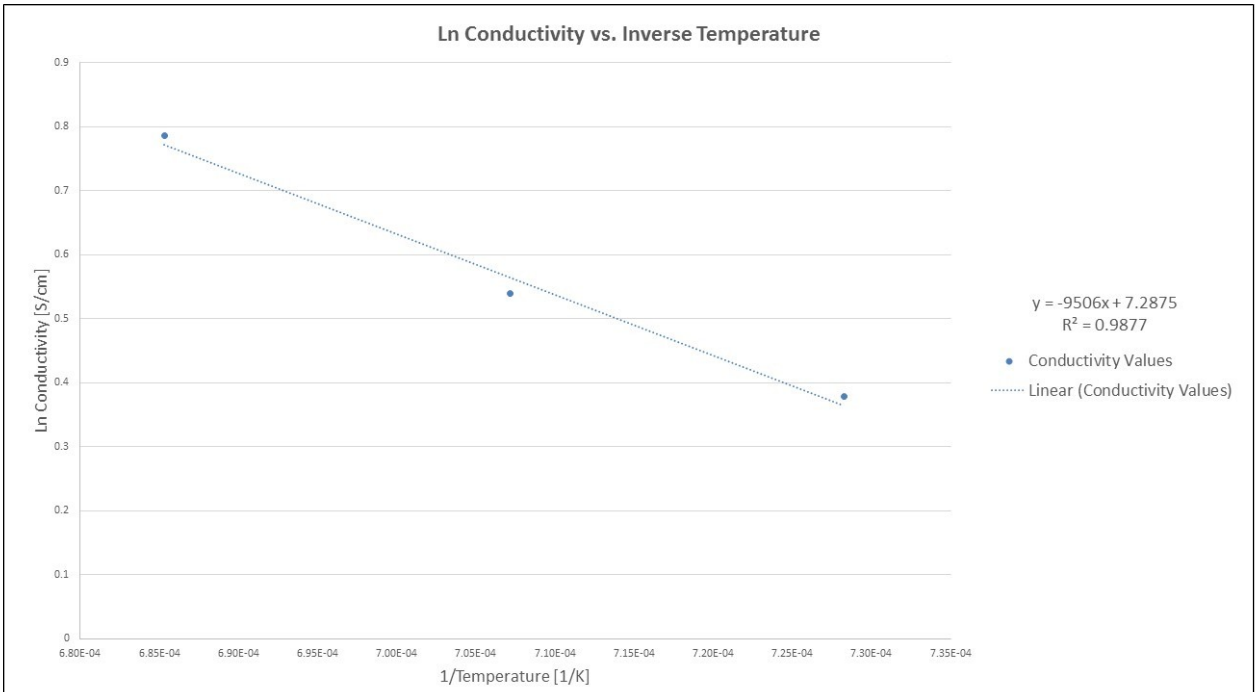


Figure 72: Ln Conductivity as a Function of Inverse Temperature

Compared to the previously studied salt (with an oxide content of 6 wt% CaO and 3% wt Al_2O_3), the newer, higher oxide content salt shows a decrease of conductivity of roughly 50% at comparable temperatures. Since the bulk of the salt is still fluoride based, the salt itself is still quite conductive and exceeds the target value of 1 S/cm. The salt's conductivity is dominated by the mobility of the fluoride anions but it established a negative correlation between the oxide content in the salt and the conductivity.

Conclusions

Using a previously established and calibrated technique, the conductivity of a new high alumina solubility BU salt was found at several temperatures. At 1085°C, the conductivity was found to be 1.46 S/cm. At 1141°C and 1185°C, the conductivity values are 1.71 S/cm and 2.19 S/cm, respectively. Using these values, the activation energy of this salt was found to be 79.03 kilojoules per mole. Qualitatively, a relationship between the oxide content of the salt and its conductivity has been established. It is seen that an increase in the oxide content of the flux results in lowering its conductivity, which makes sense as the oxygen can form something closer to a covalent linked polymer system with lower conductivity and (likely) higher viscosity.

Zirconia Tube Development

Large Thin Tubes

In order to cast anodes less than 1 mm in thickness, a new 8YSZ aqueous slurry was developed. The previous recipe has worked well when casting anodes greater than 1.5 mm in thickness. It does not work for thin anodes (< 1 mm) because the green bodies crack/break during drying and handling.

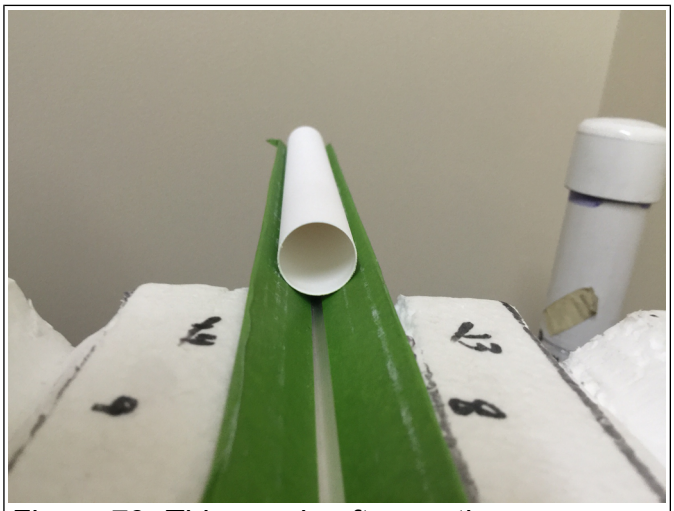


Figure 73: Thin anode after casting

The new recipe uses a modified binder with a new plasticizer. The process of making the slurry was modified to exclude any binder/plasticizer during the initial milling of 24 hours. After milling, the slurry is filtered into a separate container, and the correct amount of binder and plasticizer is added. The slurry is then sheared-rolled (without media) for 12 hours before use.

The casting process is similar to the previous which includes adding a de-foaming agent before pouring into the mold. Because the de-foamer phase-separates from the slurry and floats, it is poured out just before casting to keep voids from forming.

We have also developed a new method of casting tubes with a thickness gradient that increases from bottom to top of the tube. This was possible with a jig that allows for the mold to be inverted during casting which contains the slurry and keeps the pressure

Figure 74: Green body after casting. The dark ring inside the tube is the visible thickness gradient.

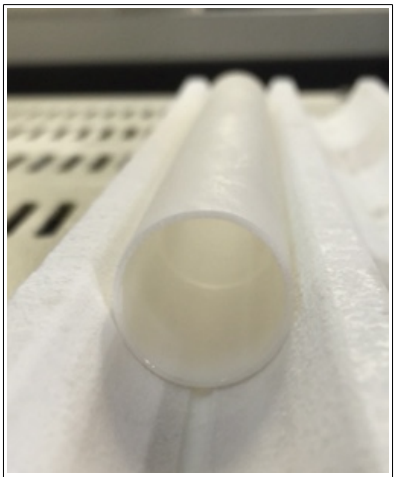


Figure 75: Sintered anode with the thickness difference clearly visible.

Figure 76: Difference in thickness clearly visible under light.

regulated at 1 atm inside the mold. Because green tubes often crack at the top, this prevents the most common pre-firing failure mechanism. It also allows the tube to handle greater thermo-mechanical stresses above the molten salt where radiation baffles exist. This method and apparatus constitute a potentially patentable invention.

We used this method to cast multiple tubes with outer diameter from 0.75” to 1.25” (1.9-3.2 cm) and 0.85-0.95 mm thickness, and 20 cm maximum length. The Pilot Cell will require at least 60 cm long tubes, which will undergo larger bending forces and stresses in the green state, and may require additional innovation to cast successfully.

Liner Material

In contrast to the mildly oxidizing environment of the Hall-Héroult cell with its CO_2 production, the reducing environment created by Pure Oxygen Anodes™ represents an opportunity to create a new type of cell with much lower energy loss. In the Hall-Héroult cell, the only cell liner material which can withstand both the bath-metal and bath-atmosphere interfaces is frozen bath; maintaining this frozen bath layer inside the cell requires extracting considerable thermal energy. The key to making this work is therefore to find a liner material which can withstand these interfaces.

At the bath-atmosphere interface, bath compositions which dissolve alumina are excellent catalysts for metal oxidation by rapidly dissolving any protective oxide. They tend to also be highly corrosive to traditional refractory ceramics such as magnesia, and zirconia reacts with liquid aluminum to form intermetallics. Boron nitride is expensive and fragile. At the bath-metal interface, the bath also dissolves aluminum carbide, ruling out carbon as a liner material.

Ceria is relatively stable in contact with liquid aluminum, and due to its low cost, ceria brick could act as a suitable liner. A follow-on study would do well to investigate this option.

The reducing environment here opens up the option of a metallic liner, which can act as both cathode connection and “refractory” surface. The best such material for liquid aluminum compatibility is titanium diboride, TiB_2. It is too costly to fabricate a complete liner out of TiB_2, therefore this project focused on creating a high-quality (adhesive and impermeable) coating of TiB_2 on another metal sung as steel using electrodeposition.

Bath Surface Tension and Titanium Diboride Wetting

Using two developed analytical techniques, the interfacial characteristics between titanium diboride and selected electrolysis fluxes were determined. The maximum bubble pressure technique uses a steel tube, which is immersed in a selected molten flux, and blows forming gas through the tube to acquire a surface tension value between the gas and the liquid flux by examining the peak pressure exerted on the gas by the liquid. The second technique, a sessile drop method, uses a camera and optical setup to take an image of a molten flux piece resting on a cathode material. This allows a contact angle at a three phase interface to be measured. Using this contact angle, a solid-liquid surface tension can be calculated, assuming gas-liquid and gas-solid surface tension values are already known.

Maximum Bubble Pressure Setup: Methods

The maximum bubble pressure experiment was performed in a vertical tube furnace in which a mild steel crucible with a selected flux (molten salt) was placed on top of an alumina spacer and stainless steel post. This crucible was placed in a way that allows it to be in line with the furnace's hot zone. The furnace environment was then purged with forming gas (Ar-5% H_2) while a thermocouple was placed next to the crucible and a

stainless steel tube was placed above the flux. The tube had a 0.25” OD and a 0.18” ID. Both of these implements were run through the upper cooling plate and sealed off from the atmosphere by using appropriately sized Swagelok fittings. All other openings on the upper cooling plate were sealed off with metal rods. The steel tube was connected by a series of compression fittings to an Omega pressure gauge that was capable of taking measurements with an accuracy of 0.001 PSIG (Gauge PSI - PSI above atmospheric pressure). This line was connected to a source of forming gas (Ar-5% H_2), which was opened during the experiment. Figure 77 shows a representation of the setup.

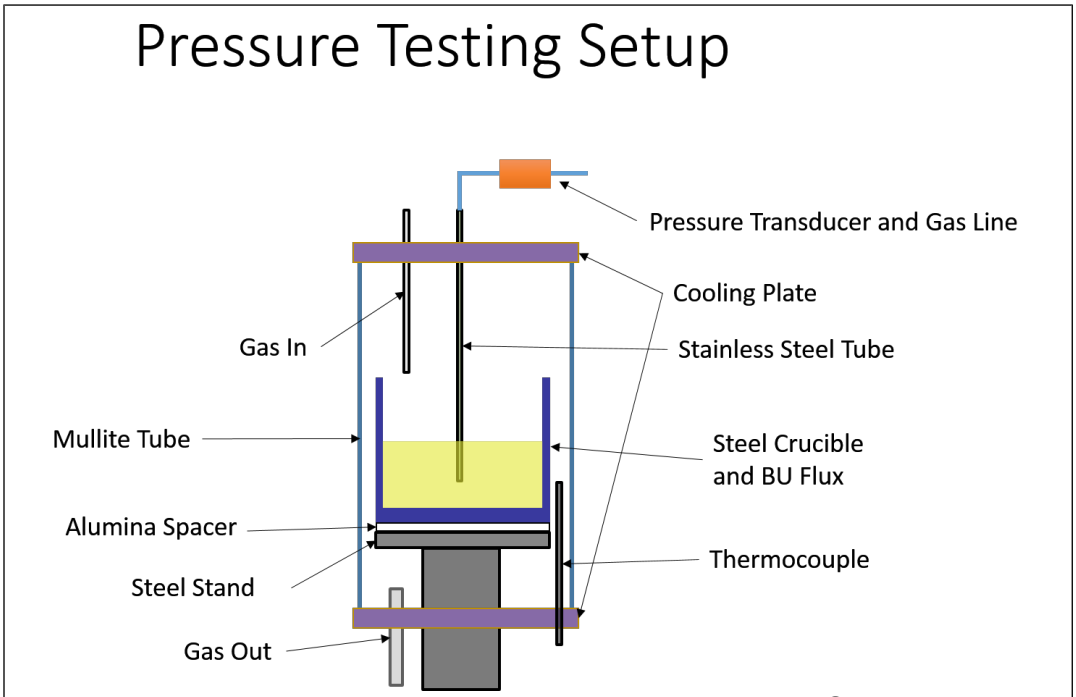


Figure 77: Maximum Bubble Pressure Testing Setup

High Temperature Sessile Drop Setup: Methods

The high temperature sessile drop experiment was performed in a horizontal tube furnace in which a 1” OD, 0.75” ID alumina tube was placed. On one end of the alumina tube, a 1” Swagelok tee was placed so that a thermocouple could be run into the hot zone of the furnace and forming gas (Ar-5% H_2) feed lines could be run into the system using a 1/4” Swagelok tee. On the other end of the alumina tube, a 1” Swagelok tee assembly holding a glass or quartz viewing port was installed with O-rings, creating an airtight seal. In addition, a Nikon D610 camera was placed on a rising scissor-lift stand and elevated to be collinear with the flux sample on the side with the viewing port. Figure 79 shows a schematic of the setup. Inside of the alumina tube, a yttria-stabilized zironia flat plate sample with flux on top was inserted. To verify that the sample was properly upright, levels were used to ensure that the flat plate was parallel to the ground. Additionally, by leveling the imaging camera, the camera was also used to ensure that the sample was level.

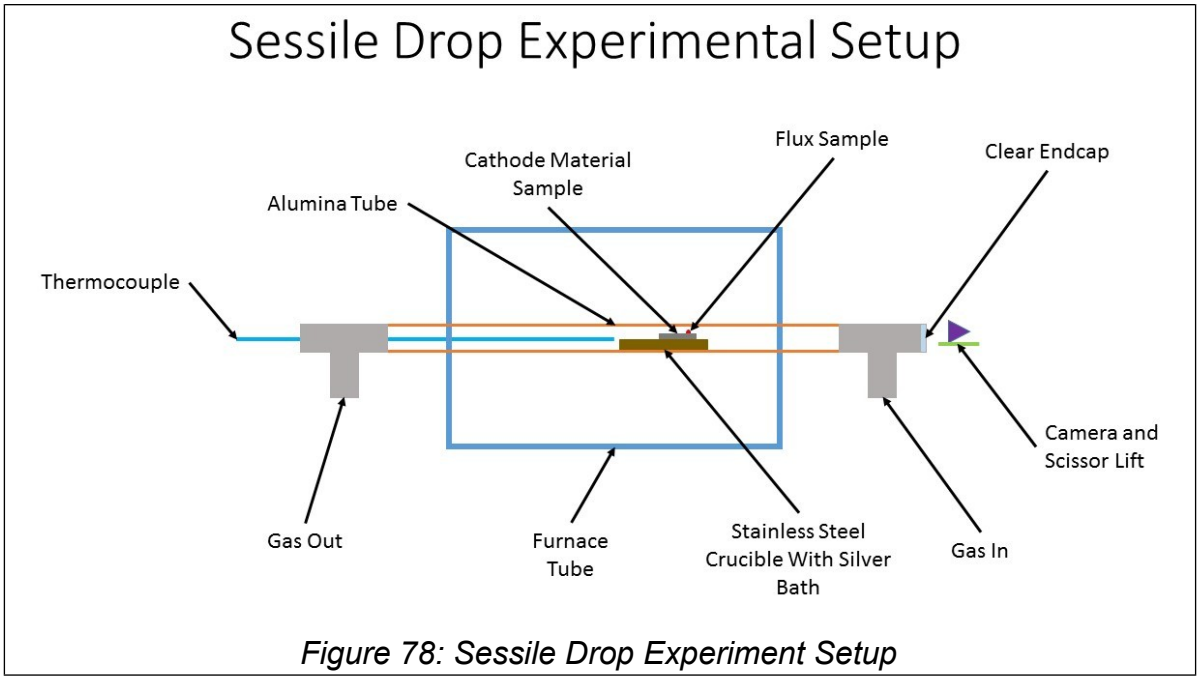


Figure 78: Sessile Drop Experiment Setup

Maximum Bubble Pressure Experiment

The experimental setup for the maximum bubble pressure experiment was heated to 1200ºC to melt a sample of the BU flux and reach an optimal electrolysis temperature. Once that temperature was reached and the system was allowed to equilibrate for an hour, the steel tube attached to the pressure gauge was purged with a feed of forming gas. Then, one port on top was opened and the instrumentation was tared to the atmosphere. Once completed, the tube was lowered until a back pressure was detected by the pressure gauge. The tube was subsequently lifted until that pressure reached zero PSIG. This established the location of the top of the flux. The open port was sealed and the steel tube was dipped 0.125 inches into the flux. The rotameter feeding the forming gas into the tube was opened so that a minimal amount of gas was allowed to flow through, creating a pseudo-static equilibrium. As this occurred, the pressure gauge is constantly collecting data. Once a set of data points were collected, the tube was lowered another 0.125 inches. The data collection process was repeated again and the tube is lowered again by 0.125 inches after each data set was collected.

High Temperature Sessile Drop Experiment

The high temperature sessile drop experiment started by placing a sample of BU flux (11wt% CaO, 4wt% YF_3, 5wt% Al_2O_3, remainder eutectic CaF_2-MgF_2) on a flatten piece of yttria-stabilized zirconia (also known as YSZ). The sample and zirconia substrate were subsequently put in the hot zone of the furnace and examined to make sure that they are level. Then, forming gas was used to purge the alumina tube for 15 minutes. Meanwhile, the camera was brought into collinear alignment with the flux sample sitting on the cathode material. The furnace was then turned on and heated to 1200ºC. As the temperature of the furnace increased, images were taken of the sample at

approximately 3 minute intervals to observe how the flux behaves once the luminosity became high enough in the tube. When the luminosity of the sample became too high to properly observe, the shutter speed of the camera was progressively increased to the minimum shutter speed of 1/4000th of a second. At 1200ºC, the sample was imaged over the course of 15 minutes after allowing the sample to equilibrate, taking pictures every minute to ensure that equilibrium at that temperature was reached. Afterwards, the temperature was decreased by 50ºC, and once the sample reached equilibrium, images were taken at 3 minute intervals. The temperature decrease and imaging was performed once more. After the experiment, the images were processed in Adobe Lightroom to develop the contrast between the flux sample and the background of the image. These images were later processed through the ImageJ plug-in DropSnake to determine the contact angle of the flux samples on the YSZ substrate.

Maximum Bubble Pressure Experiment Results

At 1200ºC, each sample depth was held for approximately 15 to 25 minutes with many bubble events occurring every minute. To determine which set of data points were valid, the first set of points above the zero PSI threshold were used. A selection of raw data for a tube depth of 0.75 inches can be seen in Figure 80.

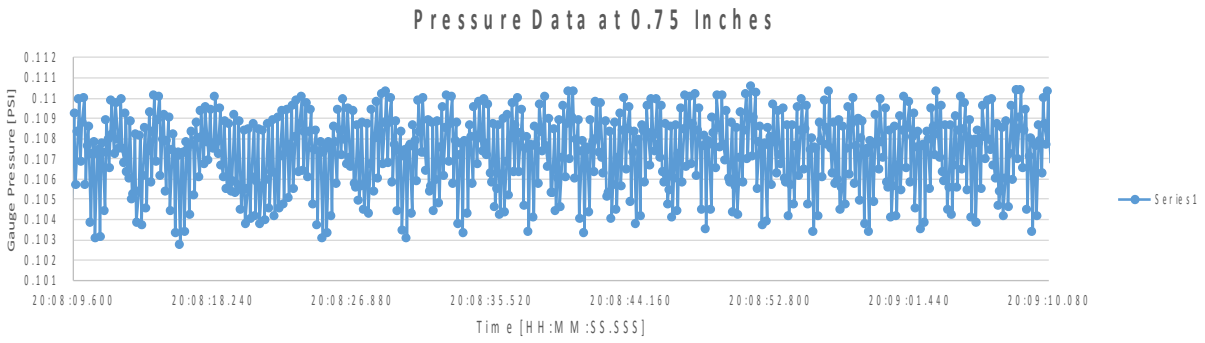


Figure 79: Pressure Data at 0.75 Inches

Looking at the selected data, each peak value was found per bubble event and graphed. The peak value was determined by finding the highest value before a rapid drop in pressure occurred. The average of the peak values at a specified immersion depth was then found. A graph was created that shows the average peak pressure as a function of depth within the flux. Figure 81 shows the data graph, slope, and y-intercept of the system, while Table 7 has the data values used to create the graph in both imperial and metric units.

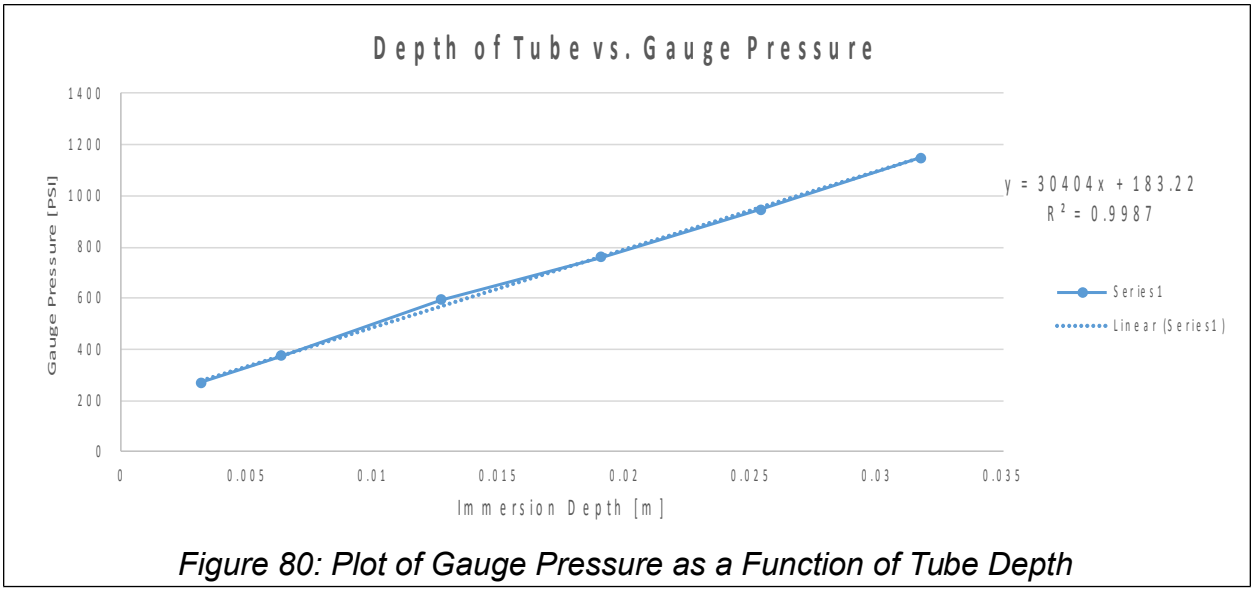


Figure 80: Plot of Gauge Pressure as a Function of Tube Depth

Tube Depth [in]	Tube Depth [m]	Peak Gauge Pressure [PSI]	Peak Gauge Pressure [Pa]
0	0	0.03	183.22
0.125	0.003175	0.0391	269.91
0.25	0.00625	0.0541	373.31
0.50	0.0127	0.0860	592.89
0.75	0.01905	0.1101	759.23
1.00	0.0254	0.1374	947.40
1.25	0.03175	0.1667	1149.09

Table 7: Tube Depth vs. Average Peak Pressure Data Values

Applying a linear fit to the data, it can be seen in Equation 27 that each component of the linear fit corresponds to a portion of the total pressure. The slope of the line equals the density of the flux times the gravitational constant. Whereas, the y-intercept represents the surface tension pressure.

$$P_{max} = P_{st} + P_{hydro} = \frac{2\gamma}{r} + \rho g h \tag{27}$$

Extrapolating to the y-intercept gives a value of 183.22 pascals. With no hydrostatic pressure present, Equation 27 simplifies to Equation 29 where the P_{max} is 183.22±10.55 pascals. In addition to this, the high temperature of the system causes the tube radius to change due to thermal expansion. The thermal expansion equation is presented in Equation 28. Using the thermal expansion coefficient of 304 stainless steel, the effective radius from Equation 28 is then applied to Equation 29 to calculate the surface tension between liquid flux and forming gas interface at 1200ºC.

$$r_{eff} = \sqrt{r^2(1 + 2\alpha_{stainless}\Delta T)} \tag{28}$$

$$\gamma_{(l-g)} = \frac{P_{max} r_{eff}}{2} \tag{29}$$

Using these equations and the associated error, the surface tension of the gas-liquid interface is 212.36±15.62 mN/m. By comparison, the gas-liquid surface tension of the old BU flux (6wt% CaO, 4wt% YF_3, 3wt% Al_2O_3, remainder eutectic CaF_2-MgF_2) was 92.68±7.06 mN/m, indicating that a higher quantity of oxides increases the surface tension of the fluoride based flux.

Sessile Drop Experiment Results

Images were taken of the BU flux on a sample of yttria-stabilized zirconia (8wt% yttria) when the furnace was at 1085ºC, 1135ºC, and 1185ºC. These image sets show a solid piece of flux that has melted and wetted the YSZ in Figures 81, 82, and 84. Each image set has five pictures that show the sample at three minute time intervals. Figure 81 shows the sample at 1085ºC, Figure 82 shows the sample at 1135ºC, and Figure 84 shows the sample at 1185ºC. The interface angles measured from the experiment are shown in Tables 8, 9, and 10 for temperatures of 1085°C, 1135°C, and 1185°C, respectively. These images illustrate the formation of a contact angle which relates three different surface tension values as shown in Equation 30.

$$\cos\theta = \frac{\gamma_{(s-g)} - \gamma_{(s-l)}}{\gamma_{(l-g)}} \tag{30}$$

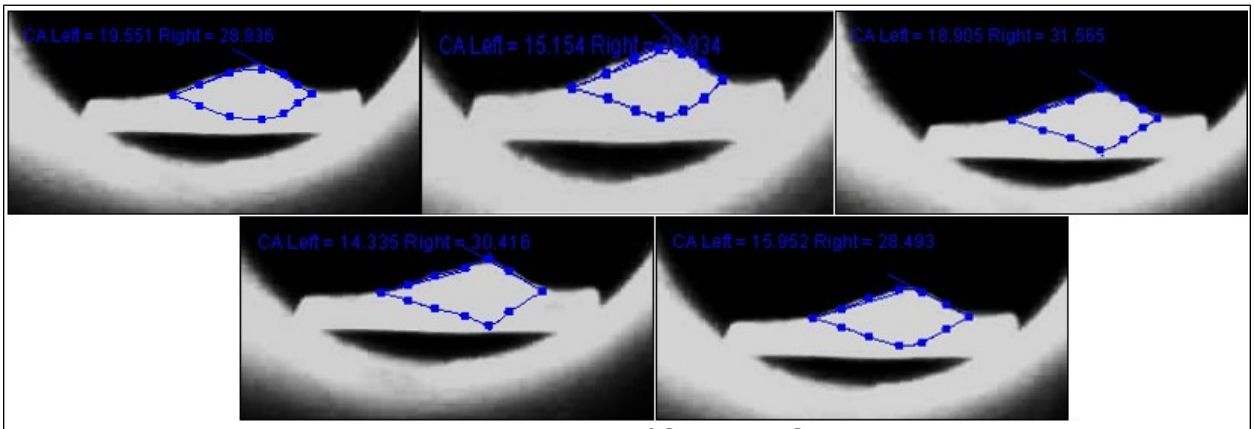


Figure 81: 1085ºC Image Set

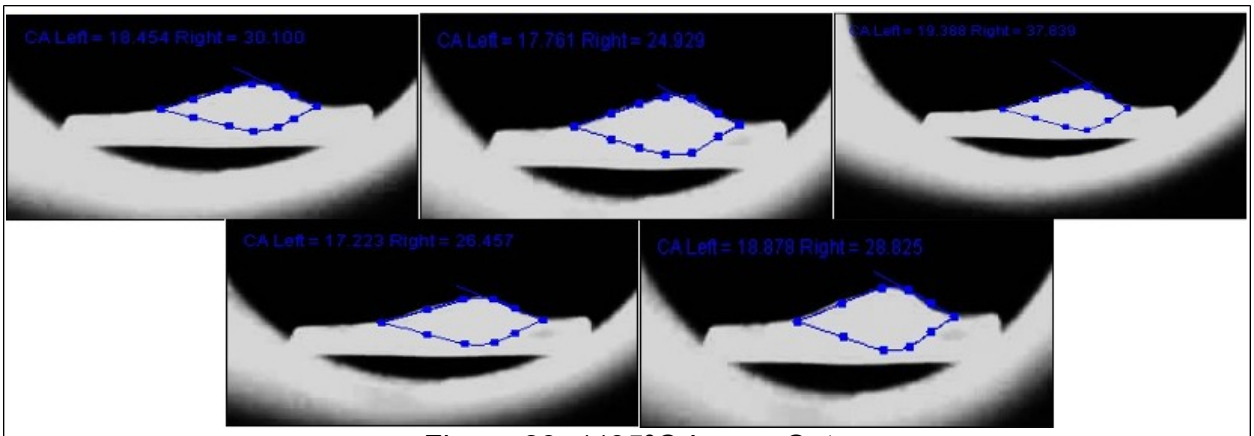


Figure 82: 1135ºC Image Set

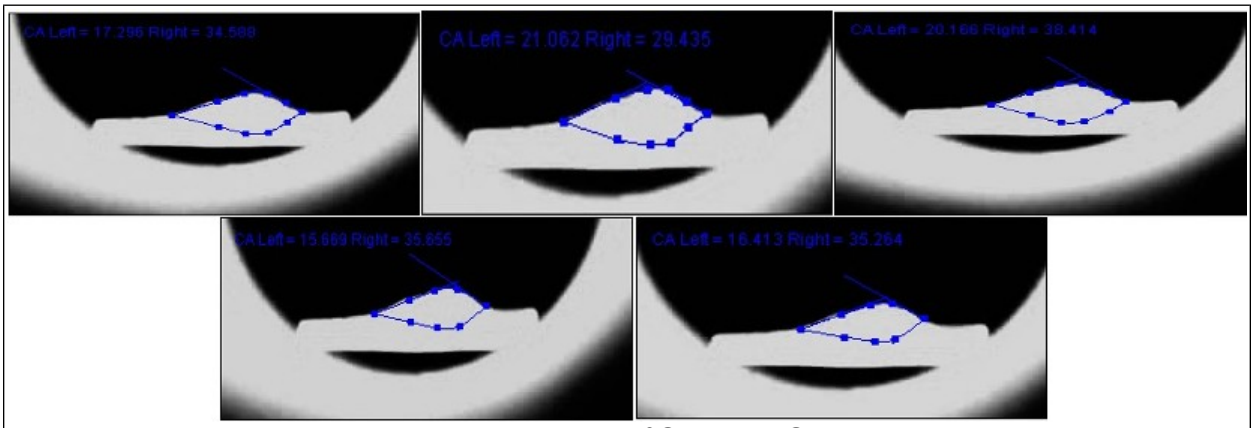


Figure 83: 1185ºC Image Set

1085ºC Sample	Receding Angle [Degrees]	Advancing Angle [Degrees]
1	19.551	28.936
2	15.154	28.934
3	18.905	31.565
4	14.335	30.416
5	15.952	28.493
Average	16.779	29.669
Standard Deviation	2.318	1.286

Table 8: 1085ºC Sessile Drop Data

1135ºC Sample	Receding Angle [Degrees]	Advancing Angle [Degrees]
1	18.454	30.100
2	17.761	24.929
3	19.388	37.839
4	17.223	26.457
5	18.878	28.825
Average	18.341	29.63
Standard Deviation	0.864	5.011

Table 9: 1135ºC Sessile Drop Data

1185ºC Sample	Receding Angle [Degrees]	Advancing Angle [Degrees]
1	17.296	34.588
2	21.062	29.435
3	20.166	38.414
4	15.669	35.655
5	16.413	35.264
Average	18.121	34.671
Standard Deviation	2.367	3.269

Table 10: 1185ºC Sessile Drop Data

When looking at the contact angles, it is important to note that the contact here is asymmetric, creating phenomena known as a receding angle and an advancing angle. The receding angle here is where the droplet tapers off into the substrate, and the advancing contact angle is where the drop quickly rises from the substrate. These features are labeled in a contact angle exemplar as seen in Figure 84.

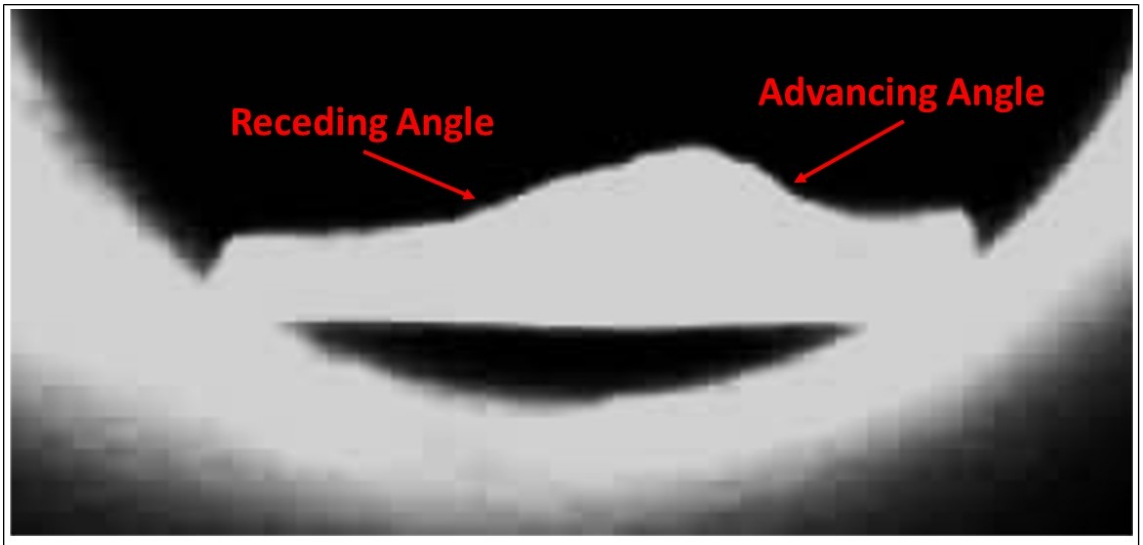


Figure 84: Contact Angle Exemplar

The difference in these two angles can be attributed to contact angle hysteresis. The origin of the hysteresis is generally attributed to a combination of liquid sorption, retention, and penetration effects between the liquid and the solid substrate.[50] Since most ceramics have imperfect surfaces and the BU flux may not permeate the substrate perfectly, this would indicate why a receding and advancing contact angle are present.

Given that the contact angle and previously acquired gas-liquid surface tension, there are still two unknown surface tension values, the gas-solid surface tension and solid-liquid surface tension. Unfortunately, a high temperature gas-solid surface tension value is not readily available for yttria-stabilized zirconia and forming gas, meaning that the solid-liquid surface tension cannot be calculated currently. Qualitatively, though, the equation shows that the gas-solid surface tension value is higher than solid-liquid surface tension, taking both contact angles into account. In addition, the results of the experiment give a value for the receding contact angle between the BU flux and the yttria-stabilized zirconia of 16.779±2.318°, 18.341±0.864°, and 18.121±2.367° for temperatures of 1085°C, 1135°C, and 1185°C, respectively. For the advancing contact angles, values of 29.669±1.286°, 29.63±5.011°, and 34.671±3.269° were found for temperatures of 1085°C, 1135°C, and 1185°C, respectively. Based on this data, it can be concluded that at higher temperatures, the contact angle is lower, but a quantitative relationship cannot be determined between temperature and resulting contact angle.

However, qualitative evidence indicates indicates that 11wt% CaO and 5wt% Al_2O_3 BU flux and YSZ sample wet very well, clinging to the YSZ surface in a previous SOM run. Considering that the BU flux tends to stay below 35° even at the advancing contact angle in this experiment, it can be concluded that good wetting occurs between the BU flux and the YSZ anode.

Summary

Using the maximum bubble pressure experiment, the gas-liquid surface tension for the

new BU Flux (11wt% CaO, 4wt% YF_3, 5wt% Al_2O_3, remainder eutectic CaF_2-MgF_2) was determined to be 212.36±15.62 mN/m. Comparing this to the old BU flux (6wt% CaO, 4wt% YF_3, 3wt% Al_2O_3, remainder eutectic CaF_2-MgF_2), which had a surface tension of 92.68±7.06 mN/m, it is indicative that a higher oxide content in the flux increases the surface tension. Additionally, a sessile drop experiment found a receding contact angle value of 16.779±2.318°, 18.341±0.864°, and 18.121±2.367° at temperatures of 1085°C, 1135°C, and 1185°C, respectively, between the new BU flux and the YSZ substrate. The same experiment also found that the advancing contact angle for this system was 29.669±1.286°, 29.63±5.011°, and 34.671±3.269° for temperatures of 1085°C, 1135°C, and 1185°C, respectively. Using the empirical observations from the previous SOM experiments, it would appear that the new BU flux and YSZ wet very well. Furthermore, the overall contact angles, particularly the advancing angle, were lower than 35°, indicating that the new BU flux wets the YSZ. Additionally, it was noted that as the temperature of the system increased the contact angle of the new BU flux generally became smaller; however, no quantitative relationship was determined for this system.

Next Steps

To ensure compatibility with new alloys, INFINIUM will run static bath-metal interface corrosion tests using liquid Al-2 wt% Sc alloy with both KF-AlF_3-Sc_2O_3 and ScF_3-LiF-Sc_2O_3 bath compositions and both graphite and (HIP) TiB_2. If these confirm that TiB_2 is a useful lining and cathode material, then we will pursue further electrodeposition experiments.

Medium-Term Crucible Liner Material Static Test

Unlike ceria and TiB_2, alumina and isomolded graphite crucibles are available off-the-shelf for <$500 in sizes needed for laboratory testing, and <$1000 for the 200 A cell. In case we have difficulty making ceria or TiB_2 for the pilot cell, we ran a 250-hour static test of small alumina and isomolded graphite crucibles in contact with aluminum metal and alumina-saturated bath, to determine whether these will meet the Pilot Cell goals.

Figure 85: (Left) static test components: (clockwise from top left) graphite crucible with 50 g flux, empty graphite crucible, alumina crucible liner, 10 g aluminum shot, 50 g bath; (right) both crucibles in the argon-purged container.

Experimental Procedure

1. Prepare two 7.5 cm tall 7.5 cm OD 5.1 cm ID isomolded graphite crucibles.
2. Insert an alumina liner into one of them.
3. Place about 50g pre-melted alumina-saturated $KF\text{-}AlF_3$ bath and 10g aluminum metal into each.
4. Place the two crucibles in a container to be purged with argon, and place the container into a kiln.
5. Start the argon purge into the container aiming for about 1 psi excess pressure.
6. Turn on the kiln power supply, and ramp its temperature to 850°C at 5°C/minute.
7. Using appropriate PPE, briefly (<30 seconds) open the kiln and argon-purged container about 65, 120, 165 and 230 hours after reaching 850° to determine whether there has been a breach which could lead to container and kiln damage.
8. 250 hours after reaching 850°, ramp temperature back to room temperature at 5°/minute, shut off argon, open the kiln and container, and examine the materials. In particular, focus on characterizing the crucible-bath-metal triple line.

Experimental Results

An attempt to check the experiment status 70 hours after reaching 850°C did not work: the container lid was stuck to the container, likely because of $KAlF_4$ volatilization.

The furnace controller did not start cooling the apparatus after 250 hours, so we

manually intervened and started it cooling down after 256 hours.

After the apparatus cooled, the containment vessel was removed from the kiln (Figure 85). Oxidation and condensed flux was seen on the outside of the crucible which explained why the vessel would not open.

The graphite crucibles did not experience notable air burn, but there was a yellow discoloration on the inside of the crucibles (Figure 86). SEM EDS analysis showed that this was a sulfur impurity which likely leached out of the graphite at temperature.

Figure 86: (Left) Crucible with alumina liner still in tact. (Right) Crucible without alumina liner.

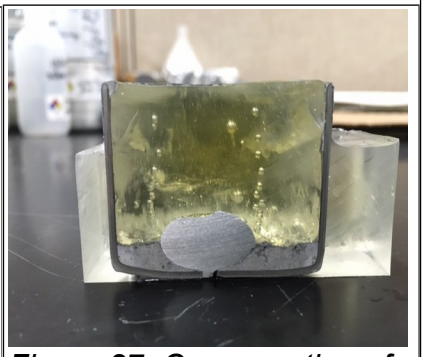


Figure 87: Cross section of alumina crucible. Alumina dissolved completely at the aluminum metal interface.

Figure 88: Crucible without alumina liner. The flux appears much darker in color.

The alumina liner was removed from the graphite and mounted with epoxy. The sample was then cut to analyze the cross-section of the liner (Figure 87). The figure distinctly shows the interface between the aluminum metal and alumina dissolved during the course of the experiment. There was also an alumina-rich layer of flux on the bottom of the liner that did not evaporate.

The aluminum metal in the crucible without the alumina liner (Figure 86, Right) was not touching the bottom of the graphite crucible, because the molten salt wet completely under it. Upon removal of the aluminum metal, there was a dark flux layer underneath (Figure 87). The aluminum metal recovered weighed 5% less than the original amount which could indicate metal dissolution. Aluminum from the crucible with the alumina liner was not weighed to avoid disturbing the metal-liner interface for analysis.

Summary

Isomolded graphite and alumina both seem to be viable crucible material candidates for a 250-hour pilot cell run. Isomolded graphite performance was very good as expected. The small hole in the alumina crucible at the metal is somewhat worrying, particularly because the mechanism is not clear, but aside from that the alumina crucible survived intact.

Titanium Diboride Electrodeposition

Overview

INFINIUM conducted one experiment to plate TiB_2 on a steel surface from KBF_4 and K_2TiF_6 precursors in a LiF-NaF-KF (FLiNaK) bath. This experiment did not succeed due to a strong increase in resistance, likely caused by use of a perfluorocarbon passivation of the carbon anode used here in an oxide-free bath. A follow-on experiment using a titanium anode is proposed.

Background

At this point, INFINIUM has identified two possible liner materials. The Q5 report discussed undoped ceria as a liner material, which in a partially-reduced state appeared to hold up well to bath-metal interface.

This project has also identified titanium diboride (TiB_2) as an excellent wetting cathode material from the beginning, due to its stability in both aluminum and fluoride salts and good electrical conductivity. Since recognizing bath-metal interface corrosion as a major issue in Q2-Q3, we have also considered TiB_2 as a potential liner material. However, its cost has seemed high in general and prohibitive for the liner application in particular, as traditional processing routes involve expensive hot isostatic pressing of expensive powder.

Recently a new processing route came to our attention involving direct electrodeposition of TiB_2 from molten salt. A 2011 doctoral thesis by Nataliia Rybakova[51] describes the history of this method beginning in 1929, and adds considerable new experimental data and innovative modifications to this coating production method. Based on this work, it is now possible to reproducibly make high-quality strong adherent fine-grained TiB_2 coatings on several substrates.

TiB_2 synthesis involves complex anions of titanium and boron reduced by electrons from the cathode. There are two reduction steps with the overall reactions as:

$$TiF_6^{2-} + e^- \rightarrow TiF_6^{3-}, \tag{31}$$

$$TiCl_{6-x}F_x^{3-} + 2\,BF4^- + 9\,e^- \rightarrow TiB_2 + (6-x)Cl^- + (8+x)F^-. \tag{32}$$

New to this work is the addition of a small account of tantalum in the form of $TaCl_5$. This leads to deposition of a TiB_2-TaB_2 mixture, which is apparently a solid solution (XRD lines are too close to distinguish) with striated composition, fine grained structure, and high strength.

This study produced its best coatings under the following conditions:

Supporting electrolyte: FLiNaK, which is a eutectic mixture of 46.5 mol% LiF, 11.5% NaF and 42.0% KF, (29.3 wt% LiF 11.7 wt% NaF 59.0 wt% KF) melts at 454°C. It has a very wide electrochemical stability window with excellent ionic and low electronic conductivity, and low viscosity and volatility. And it is an excellent solvent for TiB_2

precursors KBF_4 and K_2TiF_6. Some experiments also used Group I chlorides in the supporting electrolyte.

Electrolyte and precursor pre-treatment: researchers mixed the FLiNaK component salts, dried them at 180°C for 48 hours, further dried them at 400°C, melted them at 650°C, and cooled and crushed the mixture. They then added 3-10 mol% of vacuum-dried K_2TiF_6, KBF_4 and $TaCl_5$ precursors (mole ratio 1.0:2.0:0.06, mass ratio 47:49:4), and eliminated electronegative impurities by pre-electrolysis at 0.2 A/cm² on a separate cathode.

Cathode material: the study used molybdenum, tungsten carbide and steel cathode substrates. Degreasing and either pickling or sand blasting to remove oxides improve coating uniformity and adhesion.

Crucible: this study used nickel and glassy carbon, and noted corrosion problems with using Inconel, but did not elaborate on those problems.

Temperature: TiB_2 deposition from a FLiNaK melt seemed best at 650±10°C. Below 600 there was no deposit, below 630 coatings exhibited cracks and poor adhesion.

Environment: due to reactivity and HF formation with water and rapid metal corrosion in air, it is necessary to dry the electrolyte and precursors very thoroughly before use, and to operate the electrodeposition process in an inert environment.

Current pulsing: periodic current modification on a very short time scale can dramatically change the deposit structure, and increase maximum deposition rate with a smooth deposit. This can be either periodic pulsing with intermittent shut-off to modify boundary layer transport behavior, periodic reversal of current to dissolve roughness, or other periodic change in current density.

In pulsing mode, current should be in the range 0.2-0.7 A/cm², with pause time longer than pulse time, in order to produce a homogeneous coating. This results in four times faster deposition than with the maximum steady DC current which would produce a uniform coating. Reversal of current after every fifth cycle dramatically reduces roughness and improves homogeneity.

This practice results in excellent coatings on steel and molybdenum substrates, even on edges, when using the following parameters: pulse current density 0.3-0.5 A/cm², 50-100 Hz, pause/pulse duration ratio 2-4, 5 cathode cycles and one anode cycle. With lower or higher frequency, roughness and grain size both increase; with anode pulses, roughness decreases. The 50-100 Hz frequency range also corresponds with lower hardness. Current efficiency is better at low frequency, but at 100 Hz is it 94%, which is quite good. More frequent anode pulses decrease homogeneity and deposition rate, less frequent anode pulses increase surface roughness.

Presence of tantalum as $TaCl_5$ seems to catalyze formation of Ti^{3+}, as shown by violet color of the electrolyte, and results in double the maximum current density in cyclic voltammetry. It also leads to strong (001) TiB_2 texture. X-ray diffraction peaks for TiB_2 and TaB_2 are very close, making it hard to distinguish between them. For a deposit

made in pulsed mode with no anodic cycles, SEM analysis in backscatter mode shows layers of high tantalum concentration with layer thickness on the order of 0.1 micron. EDS measurement (measurement spot size much larger than this layer size) shows average tantalum content goes from ~2.5 mol% near the substrate to ~4.2 mol% more than 5 microns away from it. Finally, tantalum reduces surface roughness scale and grain size by up to a factor of 10, increases current efficiency from 60-80% to 75-99%, reduces crashing and porosity, and improves reproducibility of deposit quality, with no observed loss of hardness.

Overall deposition rate is about 0.5-1 μm/minute. Films grew to thicknesses of 10-30 μm over 15-30 minutes, and thermal stress increased with film thickness to a maximum well over 1 GPa at 30 μm thickness. Using anode pulses reduced residual stress, but it is not clear whether it is possible to deposit very thick films using this method.

Liquid aluminum soak tests of a 10 μm coating on Mo at 720°C show excellent coating stability. The test was conducted in air however, and the coating oxidizes in air.

Experimental Procedure

Based on the above data and findings, we planned to conduct two experiments to try to duplicate the above findings, and ran the first one. The first experiment made a coating of TiB_2 without tantalum in the bath, and with pulse current though no current reversal. The second (to appear in the next report) will include tantalum in the form of $TaCl_5$, and will utilize reverse current pulses (Procedure step 7 below).

Based roughly on the procedures outlined in the Rybakova thesis (p. I.42), we prepared the salts using the following procedure:

1. Vacuum dry FLiNaK salts and coating precursors (KBF_4, K_2TiF_6) at 180°C and 3 mbar (argon bleed) for 48 hours;
2. Mix FLiNaK electrolyte powders in the ratio above and place the mixture into a stainless steel crucible;
3. Add coating precursor powders K_2TiF_6, KBF_4, $TaCl_5$ with mole ratio 1.0:2.0 (49 wt % K_2TiF_6, 51 wt% KBF_4) or 1.0:2.0:0.06 (see above), and with 19:1 mole ratio (83:17 mass ratio) of FLiNaK to precursors, and blend well.

Electrolysis runs attempt to coat the inside bottom 1.5 cm of a 304 stainless crucible, using the apparatus shown in Figure 89. This consists of a stainless steel crucible with 6.0 cm ID and 8.0 cm inner height, with a centered graphite anode (producing CCl_4 or CF_4 gas) with 1.27 cm diameter. A secondary cathode made of iron 6 mm diameter is centered 2 cm off the crucible center. Flux depth was just 1.50 cm, resulting in inner crucible area of 66 cm^2. Procedure is as follows:

1. Fill the crucible to within 1 cm of the top with the electrolyte-precursor mixture;
2. Purge/dehydrate the apparatus and electrolyte/precursor powder mixture at 180°C with argon flow at 50 SCCM for at least 10 hours (overnight);
3. Heat the crucible with mixed electrolyte-precursor powders to 400°C at

10°/minute, and hold that temperature for 2 hours to complete dehydration;

4. Heat the crucible to 650°C at 10°/minute to melt the mixture;
5. Run pre-electrolysis by shorting the anode to a separate cathode for 2 hours to remove metallic impurities;
6. Remove the pre-electrolysis cathode from contact with the liquid electrolyte-precursor mixture;
7. Run electrolysis for 90 minutes using the crucible as cathode, and the following waveform:
    - Run at 20 V for 5 ms, then 0 V for 5 ms; repeat;
    - (Some runs) After every fifth on-off cycle, run reversed at -0.35 A/cm² for 5 ms, then 0 A for 5 ms.
8. Cool the apparatus, maintaining argon purge.

Figure 89: TiB_2 electrolysis setup including crucible.

This should produce a uniform homogeneous TiB_2(-TaB_2) coating on the inside of the crucible about 15-35 µm thick.

Results and Discussion

The cleaning process generated a stable current of 81mA and did not appear to decay after 1 hour of electrolysis. After the hour was completed the cell was cooled down and the cathode was removed from the cell. There was a black deposit notice on the end of the cathode that was immersed in the salt.

As a result of the geometric constraints in INFINIUM's inert atmosphere apparatuses, the surface area of the crucible exposed to the salt was 100cm^2. Likewise, the only power supply with sufficient bandwidth had a 20A and 20V max outputs. Therefore the max current density that could be reached was 0.20A/cm^2. Because of these limitations, the waveform was altered from 0.35A/cm^2 for 3ms and 0a/cm^2 for 7ms to 0.20A/cm^2 for 5ms followed by 0A/cm^2 for 5ms to preserve the same average current density over the length of each cycle.

The current observed in cell was ~3A on average, significantly lower than the 10A average over each cycle. This was likely a result of several resistive losses in the cell. One cause of the high over potential may have been a result of the formation of CF_4 at the anode. CF_4 may have formed a passivation layer of gas at the anode. This caused the power supply to apply 20V but still only generate a small current. This theory was further supported by the periodic spikes in current which may have been a result of the

gas layer becoming large enough to dislodge from the anode. In the aluminum industry when cells run low on oxide an anode effect can occur which causes the formation of perfluorocarbon polymers. This phenomenon drives the operating voltage in the cells to over 25V to maintain the constant current applied to the cells. This phenomenon observed was not reported in the thesis, likely because pure fluorine gas was produced on the platinum anode which may not have the same passivation effects as perfluorocarbons on graphite.

Unfortunately, no TiB_2 could be detected on the cathode, likely due to this anode effect.

Future Work

Here we propose a new experiment to that limitation by using a titanium metal anode, which will form TiF_6^{3-} complex ions by the half-reaction:

$$Ti + 6F^- \Rightarrow TiF_6^{3-} + 3e^-. \quad (33)$$

This does not maintain a balanced system, as the anode reaction is:

$$TiF_6^{3-} + 2BF_4^- + 9e^- \rightarrow TiB_2 + 14F^-, \quad (34)$$

so with charge neutrality, for each mole of TiB_2 deposited, three moles of TiF_6^{3-} will join the bath. This will increase the Ti/B ratio in the bath, and at some point it could be helpful to address this via either: (1) a TiB_2 anode with the reaction 34 in reverse; (2) feeding additional KBF_4 into the cell; (3) a second boron anode with the reaction:

$$B + 4F^- \rightarrow BF_4^- + 3e^-. \quad (35)$$

The boron anode would run at 4/3 timse the titanium anode current in order to replace titanium and boron stoichiometrically. Boron conductivity is about 10 S/cm at 1000 K (727°C), which is usable, but falls off quickly at lower temperature,[52] so this anode would need to connect to a current collector in the high-temperature portion of the cell.

Experimental Procedure

This experiment will attempt to reproduce Rybakova's experiment with the best results, except for the steel cathode instead of molybdenum or tungsten carbide, titanium anode instead of platinum, and isomolded graphite crucible instead of nickel or glassy carbon.

1. Prepare eutectic LiF-NaF-KF salt mixture in 29.3:11.7:59.0 mass ratio.
2. Prepare K_2TiF_6-KBF_4-$TaCl_5$ precursor salt mixture in 1.0:2.0:0.06 mole ratio (47:49:4 mass ratio) with 10% of the total metal ion moles of FLiNaK salt (precursor:FLiNaK mass ratio 30:70).
3. Prepare two baking soda-blasted 1.27 cm diameter stainless rod cathodes, 1.27 cm diameter titanium rod anode, and an isomolded graphite crucible.
4. Assemble with bath in the crucible and anode and both cathodes above the bath.
5. Heat to 650°C at 10°C/minute.

6. Insert the titanium anode to the bottom and back it out 1-2 cm.
7. Measure the bath height by inserting the first stainless cathode until resistance to the anode rises, indicating the surface, and then measuring distance to hitting the bottom.
8. Remove electronegative impurities from the bath by lowering the first cathode until it hits the crucible bottom then withdrawing it to 1-2 cm from the bottom and running electrolysis at 0.2 A/cm² cathode area for two hours.
9. Remove the first cathode above the bath, and lower the second cathode to the bottom of the crucible, then back it out 2 cm.
10. Deposit TiB_2 onto the second cathode by pulsing at 0.35 A/cm² cathode area for 2-3 ms, then opening the circuit for 8-7 ms, for total 10 ms per cycle (100 Hz). Every fifth pulse should reverse the current at the same current density. Do this for one hour, which should deposit a 40-60 μm TiB_2 film.
11. Withdraw the second cathode and anode from the bath.
12. Lower the furnace set point to room temperature at 10°C/minute to cool down.
13. Section and analyze both steel cathodes.

LSM Anode Current Collector

BU and INFINIUM developed this current collector for use in zirconia solid electrolyte tubes.[53] Here we ran an experiment to characterize stability and longevity under constant current operation. Results showed very significant instability, relative to long-term stability under constant voltage operation.

In a cell or cluster with ten or more anodes connected in parallel, each anode would see roughly constant voltage conditions, even if the entire cell or cluster operates at constant current. Therefore, although this test demonstrated significant instability for a single anode current collector under constant current, this is not an indicator of performance in a cluster of anodes.

Long-Term Anode Current Collector Test

The goal of this experiment was to test the lifetime and performance of LSM current collectors by simulating their conditions inside of an anode for use in an Aluminum production cell. A graphite cylinder was machined to have three wells arranged in a circular pattern to be filled with molten silver. This is similar to how zirconia tubes would be filled with molten silver to make electrical contact to the current collector. In two of the wells, solid LSM current collector prototypes will be used to make electrical contact, and a graphite reference electrode in the third well of the same dimensions. This whole setup was lowered into a closed stainless steel shell with viewport, and purged under argon. Three separate power supplied were connected to each of the three current collectors (with the circuit completed through a common connection to the graphite) and lowered into the silver pools. Figure 90 shows a schematic of this setup.

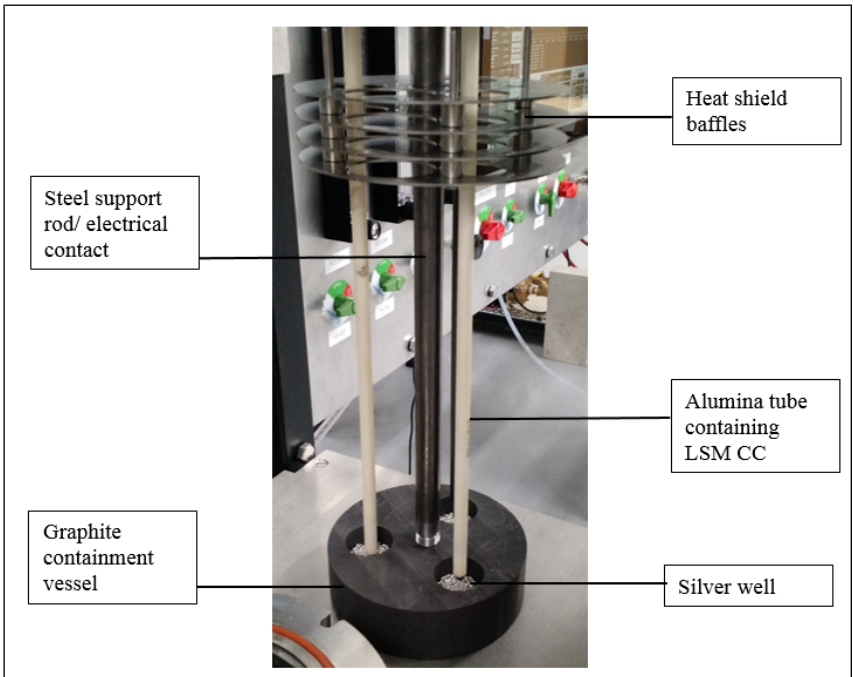


Figure 90: LSM Current collector test setup

Experimental Parameters

- 500 hour goal for passing a constant current through each of the three current collectors
- Graphite test fixture was isomolded graphite from graphitestore.com, free of

sulfur contamination. A threaded 1018 steel rod was used to hold it in place as well as provide the electrical contact for the current through the current collectors to return

- Lanthanum Strontium Manganate (LSM) current collector manufacture specifications (produced in-house)
  - 0.51cm diameter, 3cm long LSM cylinders cemented into 3/8in outer diamter alumina tubes
  - 0.5g LSM powder loaded into alumina tubes to make contact with solid LSM
  - 5g silver loaded on top of the LSM powder
  - 3/16” Inconel 601 rods were used to make contact with the silver
- Graphite current collector machined to match the diameter of the LSM cylinder for equal contact surface area to the silver pool
- Current density fixed for each current collector
  - 10 A/cm_2 current density on LSM current collector 1
  - 30 A/cm_2 current density on LSM current collector 2
  - 20 A/cm_2 current density on graphite reference current collector 3
- Furnace temperature set to 1000°C, corresponding to an internal temperature of 988°C
- Argon atmosphere in the test cell

Results and Discussion

Initial resistance measurements were taken for each of the three current collectors (immersed in liquid silver at temperature) to the steel rod connected to the graphite fixture. These measurements are given in Table 11.

Current collector	**CC 1 (10 A/cm_2)**	**CC 1 (30 A/cm_2)**	**CC 1 (20 A/cm_2)**
Resistance	11Ω	16Ω	5Ω

Table 11: Initial resistance values

Once resistance values were determined, the necessary current was driven through each of the current collectors to get the desired current density values. These initial values are given in Table 12.

Current collector	**CC 1 (10 A/cm_2)**	**CC 1 (30 A/cm_2)**	**CC 1 (20 A/cm_2)**
Current	4.8A	14.62A	9.6A
Voltage	2.03V	3.50V	1.30V

Table 12: Initial current and voltage values for each current collector

Applying Ohm’s law to these initial values we find that the resistance of CC 1 is 0.42 Ω, CC2 is 0.24 Ω, and CC 3 is 0.14 Ω.

The experiment ended up running for 142 hours and was stopped due to unstable performance and demands on the test cell for electrolysis experiments. Table 13 gives these average data points over the test period.

Current collector	CC 1 (10 A/cm$_2$)	CC 1 (30 A/cm$_2$)	CC 1 (20 A/cm$_2$)
Current	4.83A, σ=0.01	14.79A, σ=0.01	9.6A, σ=0.01
Voltage	1.61V, σ=0.49	1.95V, σ=0.26	1.21V, σ=0.51
Resistance	0.343 Ω	0.136 Ω	0.26 Ω

Table 13: Averaged data for full experimental duration

As can be seen from the small standard deviation in the data, the voltage remained relatively constant throughout the experiment, and actually ended up lower at the end of the experiment compared to the initial values. This points to there being some mechanism for "breaking in" the current collectors and allowing the resistance to go down over time.

We can also observe a lower resistance in CC 2 as opposed to CC1. This could be explained by nonlinearity of the resistance with voltage. Another possible cause could be variations in the construction of the current collectors as well as in the integrity of the LSM. More repetitions of the experiment would have to be carried out with same conditions on each CC to investigate further.

Trend Data

Figure 91 Is a sample section of the CC voltage plot near the middle of the experimental run over a period of about 8 hours. Units are not shown as this figure is meant to give an example of the general behavior of the voltage over the course of the experiment.

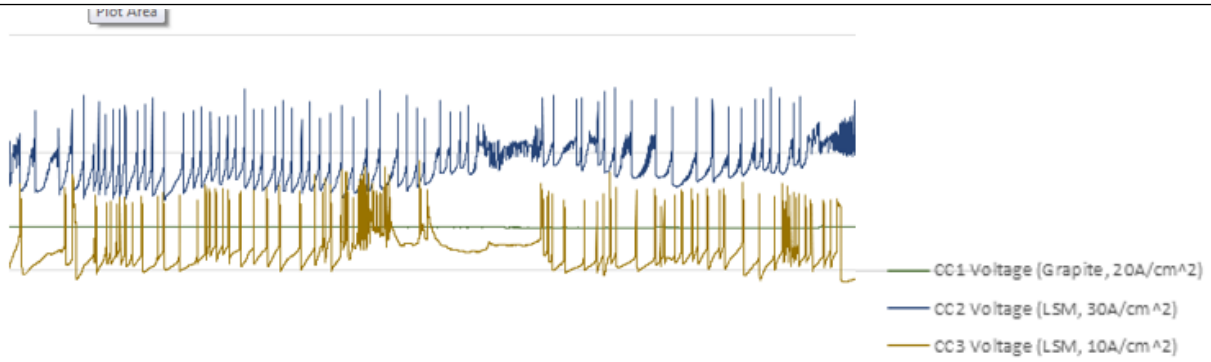


Figure 91: Current collector voltage plot sample data. X-Axis: Time, Y-Axis: Voltage

Several different patterns emerged in the voltage signal over the course of the observation time, the most comprehensible of which was a periodic voltage rise and sudden drop off. This pattern can be seen in both CC2 and CC3 trend lines in the above sample data. The period of this voltage pattern varied wildly throughout the experiment,

typically lasting for about 10 minutes but lasting up to a period of 2 hours.

Several mechanisms could explain this periodic voltage shift. One possibility is the formation of a bubble in the silver contact inside of the alumina tube that slowly grows and reduces the contact area to the LSM and/or the Inconel rod. Bubble release would explain the sudden drop of voltage close to the original point it was at. Another possible mechanism could be the formation of some resistive barrier layer on the surface of the LSM in contact with the silver pool. This barrier layer could steadily form and be broken through at a certain higher voltage.

Note this periodic behavior does not occur under constant voltage conditions. This could represent an instability present under constant current, and warrants further study.

Issues, Risks, and Challenges

An issue with this test is that the materials require it to be carried out under a reducing atmosphere (Argon in this case) to prevent the graphite and steel from being oxidized. This is different from the conditions that the current collector would see inside of an oxygen producing anode. This oxidizing atmosphere would likely have a different effect on the LSM and alumina tube. For this reason analysis of the surface of the LSM from the experiment is unlikely to provide any useful data – just the electrical signals.

A difficulty in setting up this experiment was keeping the surface area of the current collectors in contact with the silver pools constant across all three. The LSM current collectors in this case are extremely small and their distance into the pool needed to be controlled to within a couple hundredths of an inch. The wetting properties of the current collectors are also not well defined and the silver could have had a higher contact area than expected. In future experiments it would be more convenient to have larger current collectors or an easier way to confirm their height in the silver pools.

Finally, although this was carried out under constant current conditions, an anode in a cell would more likely see constant voltage, as described at the start of this section.

Wettable Cathode Material Selection

Overview

A vendor for Titanium Diboride Cathode material has been selected and 10 of them have been obtained for initial testing. A great deal of effort has gone into designing and testing means for making the connection between the cathode and electrode lead. As of the writing of this report the first long term cathode integrity test has just been completed and the results look promising. This test was designed with the intention of characterizing the resistance inherent to the proposed cathode assembly over time.

Using a high temperature sessile drop setup and computer image analysis technique, the surface tension interactions between cathode materials and various molten salts and metals were characterized. The sessile drop technique uses a camera and horizontal tube furnace to take images of molten salt drops or liquid metal drops resting on a piece of cathode material. The images are then processed with software packages to determine the contact angle between the liquid and solid interface.

Low-Current (<20 A) TiB_2 Cathode Design and Testing

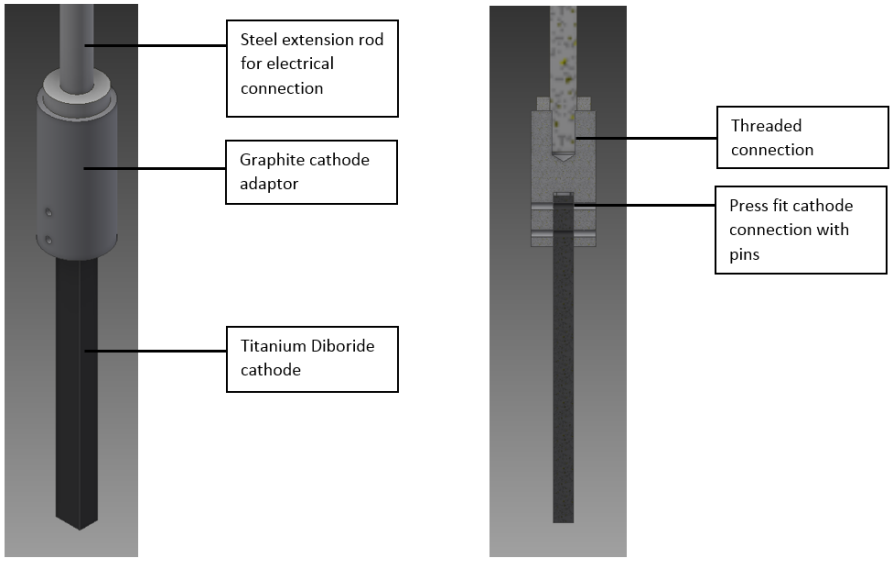


Figure 92: Titanium Diboride cathode schematic: Left - Isometric view, Right - Half section view

The current design for the cathode and associated electrical connections is shown in Figure 92. Titanium diboride (TiB_2) is machined into a rectangular bar with two holes to allow it to be pinned in place. It is press fit into an isomolded graphite rod machined to accept the cathode on one side. The graphite adaptor fits to a threaded steel extension rod on the other side. The TiB_2 makes contact with the graphite adaptor on all four sides for good electrical contact. When heated, the TiB_2 expands more than the graphite, creating compressive force which reinforces this connection. The steel extension rod has a threaded end with a washer welded above it, which when screwed down makes contact with the top of the graphite adaptor. This design replaces others based on steel screws through the TiB_2, and has proven to be most robust in terms of low contact resistance and robust operation.

Cathode long term integrity test objectives/ parameters

The goal of this experiment is to run the proposed cathode assembly design in a test cell for ≥500 hours in effort to determine how its resistance changes over time. This was accomplished by using a graphite crucible filled with aluminum metal placed in an inert (under argon), 750°C environment. Three electrodes were immersed in the molten aluminum metal: 2 graphite reference electrodes, 1 TiB_2 cathode assembly. A constant 10A current was passed using the TiB_2 as a "cathode" and the two graphite rods as "anodes" in parallel. See Figure 93 below for a picture of the experimental setup.

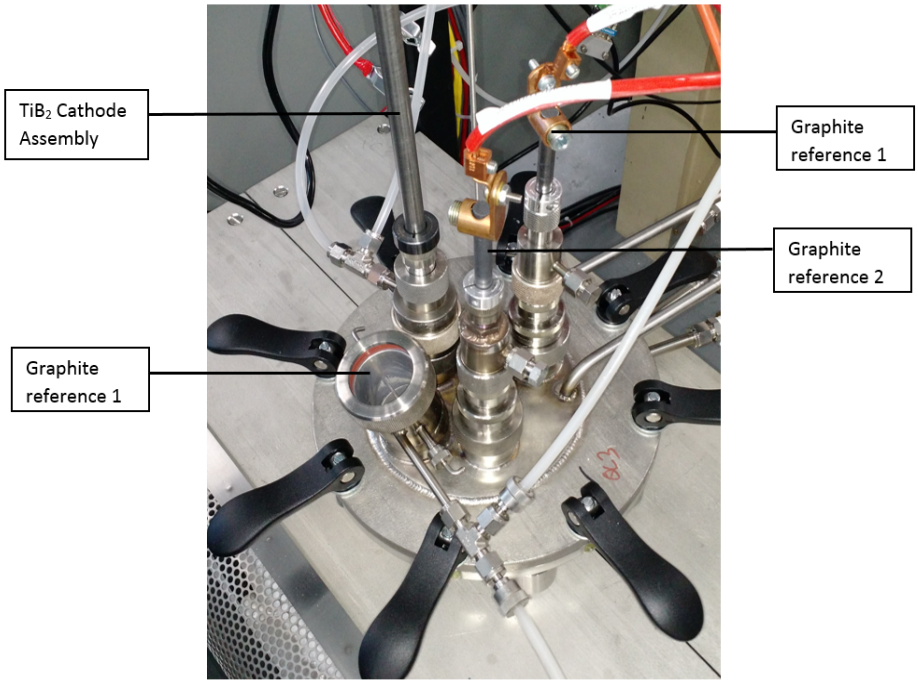


Figure 93: Long term cathode integrity test setup

This setup is meant to simulate similar conditions to electrolysis. The primary difference is that this setup utilizes aluminum as a conductor between the electrodes instead of an aluminum-molten salt bath mixture. Since the purpose of this experiment is to gauge the resistance in the TiB_2 cathode assembly only, the means through which current passes through the rest of the setup is less important. Figure 94 Details the electrical setup used for the long term integrity test setup.

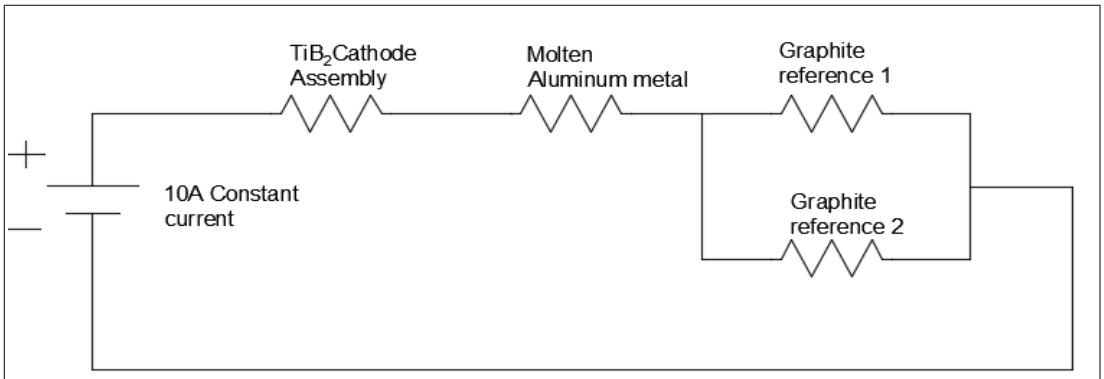


Figure 94: Test setup current path

This electrical setup was used through the course of the experiment. However, it was necessary to disconnect this constant current setup periodically to test the resistance of each electrode pair. This data is the focus of the subsequent analysis. The resistance of each electrode pair was taken by connecting a separate external power supply and running a current sweep over a two minute period from 1A to 20A. This data was logged and averaged to obtain periodic resistance measurements for each pair.

Results/Analysis

The data collected was analyzed to isolate the resistance of the cathode assembly and track it over time. This was accomplished by treating each electrode pair in the experiment as a series of resistors – which allowed a system of equations to be generated and solved for the resistance of the cathode assembly. Figure 95 Gives the results of these calculations whose inputs are the electrode pair resistance values taken periodically throughout the course of the experiment. The values on 1/6/16 immediately jump out as outliers on this plot – this was determined to be because the furnace malfunctioned and was operating 10°C cooler than normal. These results will ignore this value.

Table 14 Gives the overall values for each of the electrodes, the main variable of interest being the rate of change of resistance in the cathode assembly. This will allow us to predict it's change in an electrolysis scenario and account for that shift in further data analysis.

The data show a general positive correlation between time in use and resistance value for each electrode, as expected. The resistance of the TiB_2 cathode assembly changed by only 0.018 Ω over the course of the experiment, *i.e.* less than 10%. This leads us to believe that this design will hold up during aluminum electrolysis runs over time. The values determined here will also be taken into account for those experiments to filter out what small noise may be introduced by the changing resistance of the cathode assembly.

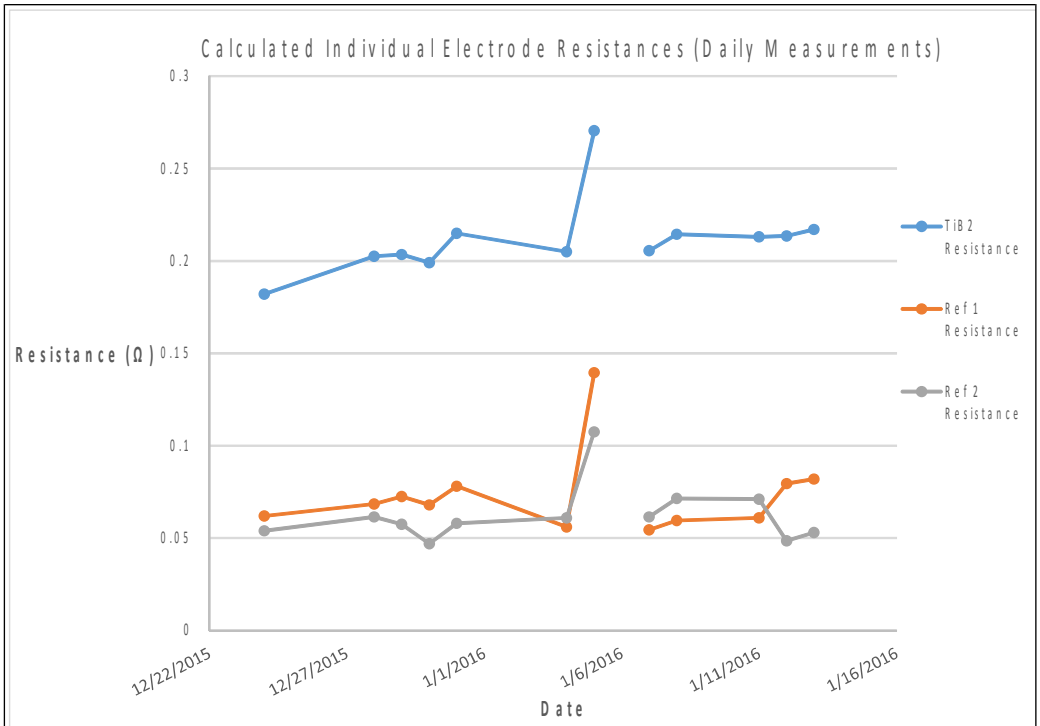


Figure 95: Cacluated electrode resistances taken from periodic measurements

Electrode	0.000	0.000	0.000	Rate of change (Ω/day)
TiB_2cathode assembly	0.199	0.217	0.206	0.0009
Graphite reference 1	0.062	0.082	0.067	0.0010
Graphite reference 2	0.054	0.053	0.059	-0.00005

Table 14: Electrode resistance overall statistics

An interesting result comes when you look at the data in more depth. Figure 96 shows the current-voltage plot for an individual day's resistance sweeps for each of the electrode pairs. They show behavior different from what was expected – that the voltage did not increase exactly linearly with current in all cases. The sweeps between the TiB_2 and graphite references consistently gave slightly nonlinear results. The resistance seems to be slightly higher at lower currents. This could be the result of some kind of semi-conductor behavior being exhibited by the TiB_2.

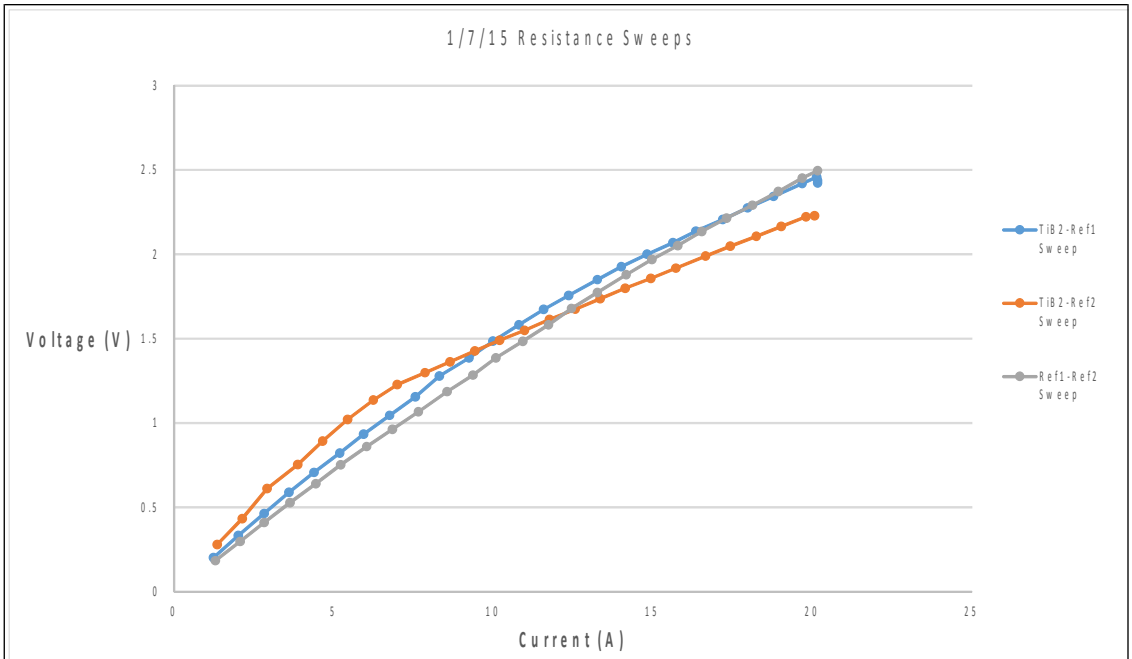


Figure 96: Daily electrode pair resistance sweeps for 1/7/15

The experiment was cooled down after the 500 hour test duration and a visual inspection was completed on the electrodes. Figure 19 Shows the electrodes taken out of this experiment. Some aluminum was still stuck to them but was easily peeled off after this picture was taken. The connection between the graphite and titanium diboride looks unchanged and fully intact. The titanium diboride itself seems to have darkened in color, as well as gaining a glossy looking finish.

Next Steps

More exploration needs to be done looking into the reason for the slightly nonlinear resistance of titanium diboride over a range of voltages. This behavior is not fully understood, but may end up having minimal impact on the practical operation of an aluminum cell.

It is clear from the outlier on the electrode resistance plot that there is likely a strong correlation between temperature and resistance. If temperature changes come into play in further experiments, it could be useful to investigate this relationship further.

The change in appearance of the titanium diboride over the course of the experiment is cause for further study as well. The change could be explained by interactions with the impurities in the aluminum metal or graphite used, or from some air getting into the cell. Tests better controlling these variables could shed more light on the cause of this change, or determine if the change has any effect on cathode performance at all. Means of cleaning off the surface of the titanium diboride post-use also need to be refined.

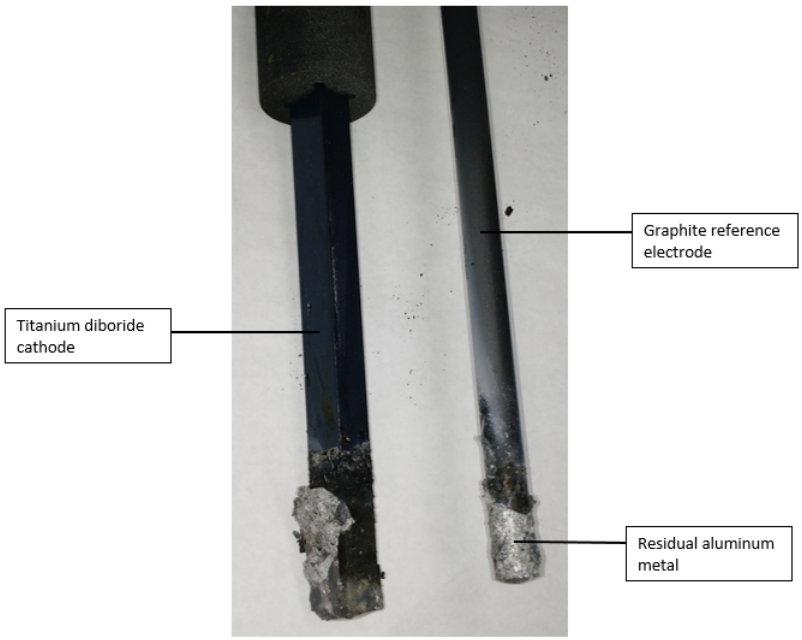


Figure 97: Post-experiment picture of electrodes from long term cathode test experiment

Surface and Interfacial Tension Tests

High Temperature Sessile Drop Setup: Methods

The high temperature sessile drop experiment was performed in a horizontal tube furnace in which a 1” OD, 0.75” ID alumina tube was placed. On one end of the alumina tube, a 1” Swagelok tee was placed so that a thermocouple could be run into the hot zone of the furnace and forming gas (Ar-5% H_2) feed lines could be run into the system using a 1/4” Swagelok tee. On the other end of the alumina tube, a assembly holding a glass or quartz viewing port was secured with O-rings and vacuum grease creating an airtight seal. In addition, a Nikon D610 camera was placed on a rising scissor-lift stand and elevated to be collinear with the flux sample on the side with the viewing port. Figure 98 shows a schematic of the setup. Inside of the alumina tube, a stainless steel crucible filled with silver is used to hold the titanium diboride, flux, and aluminum samples. This crucible is used primarily to naturally level the samples using gravity to prevent any misalignment with the camera.

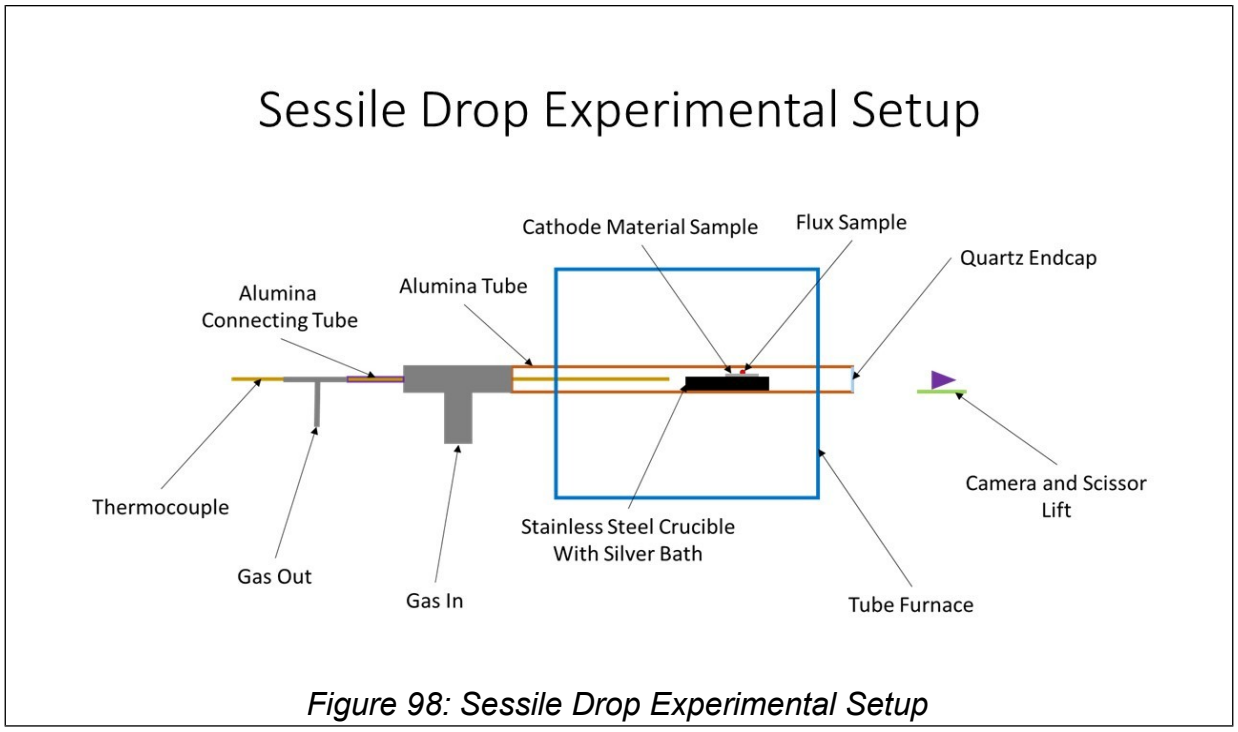


Figure 98: Sessile Drop Experimental Setup

High Temperature Sessile Drop Experiment

The high temperature sessile drop experiment started by premelting 40 grams of silver into the stainless steel crucible. Afterwards, a sample of titanium diboride with a piece of the BU flux (6wt% CaO, 4wt% YF_3, 3wt% Al_2O_3, remainder eutectic CaF_2-MgF_2) or aluminum was placed on top of the silver in the steel crucible. The crucible was subsequently put in the hot zone of the furnace. Then, forming gas was used to purge the alumina tube for 15 minutes. Meanwhile, the camera was brought into collinear alignment with the flux sample sitting on the cathode material. The furnace was then turned on and heated to 1200ºC. As the temperature of the furnace increased, images were taken of the sample at approximately 30 second intervals to observe how the flux behaves once the luminosity became high enough in the tube. As the luminosity of the sample became too high to properly observe, the shutter speed of the camera was progressively increased to the minimum shutter speed of 1/4000th of a second. At 1200ºC, the sample was imaged every minute over the course of 15 minutes after allowing the sample to equilibriate. After the experiment, the images were processed in Adobe Lightroom to develop the contrast between the flux sample and the background of the image. These images were later processed through the ImageJ plug-in DropSnake to determine the contact angle of the flux samples on the titanium diboride.

Sessile Drop Experiment Results

Images were taken of the BU flux on a sample of titanium diboride while the furnace was heating to 1200ºC. These images show a solid piece of flux melting and then wetting the titanium diboride as the flux sample spreads out as a thin film in Figures 99,

100, 101, 102, and 103. Each image represents a three minute time interval. The interface angles measured from the experiment are shown in Table 15. These images illustrate the formation of a contact angle which relates three different surface tension values as shown in Equation 36.

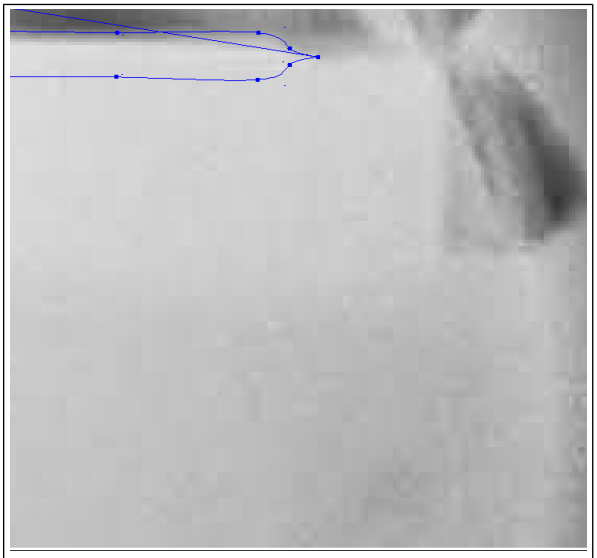


Figure 99: Sessile Drop Image 1

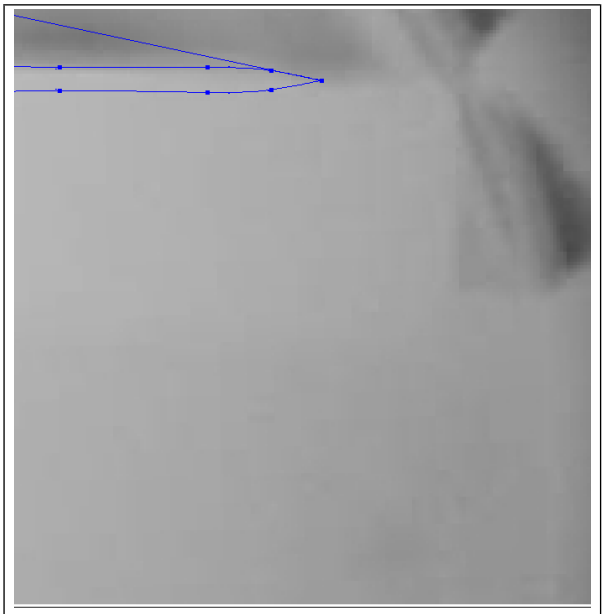


Figure 100: Sessile Drop Image 2

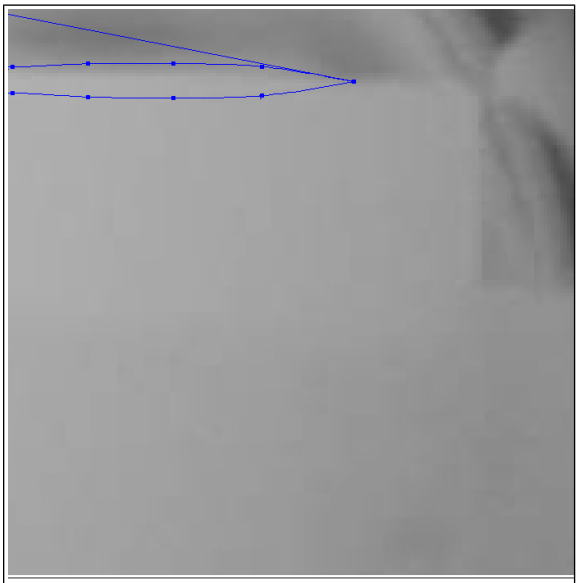


Figure 101: Sessile Drop Image 3

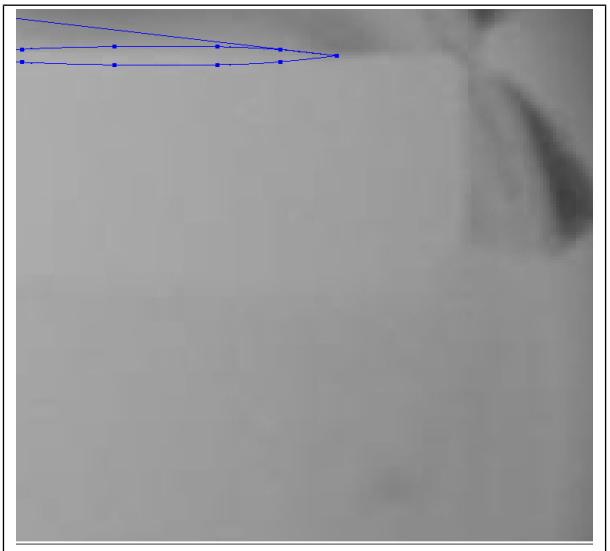


Figure 102: Sessile Drop Image 4

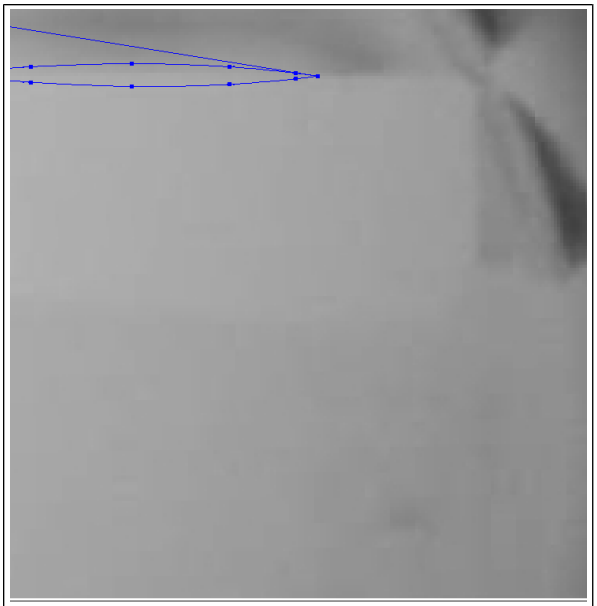


Figure 103: Sessile Drop Image 5

Sample	Angle [Degrees]
1	8.473
2	11.697
3	10.739
4	7.022
5	8.839
Average	9.354
Standard Deviation	1.357

Table 15: Sessile Drop Data

$$\cos\theta = \frac{\gamma_{(s-g)} - \gamma_{(s-l)}}{\gamma_{(l-g)}} \tag{36}$$

Given the contact angle and previously acquired gas-liquid surface tension, there are still two unknown surface tension values, the gas-solid surface tension and solid-liquid surface tension. Unfortunately, a high temperature gas-solid surface tension value is not readily available for titanium diboride and forming gas, meaning that the solid-liquid surface tension cannot be calculated currently. Qualitatively, though, the equation shows that the gas-solid surface tension value is higher than solid-liquid surface tension. In addition, the results of the experiment give an acceptable value for the contact angle between the BU flux and titanium diboride of 9.354±1.357°.

The same experiment was run for an aluminum shot sitting on titanium diboride. However, the oxide layers surrounding the aluminum could not be removed in-situ,

meaning that an accurate contact angle measurement could not be made. An example of this is seen in Figure 104. Attempts were made to reduce the oxide layer using magnesium shavings, but the trial proved unsuccessful.

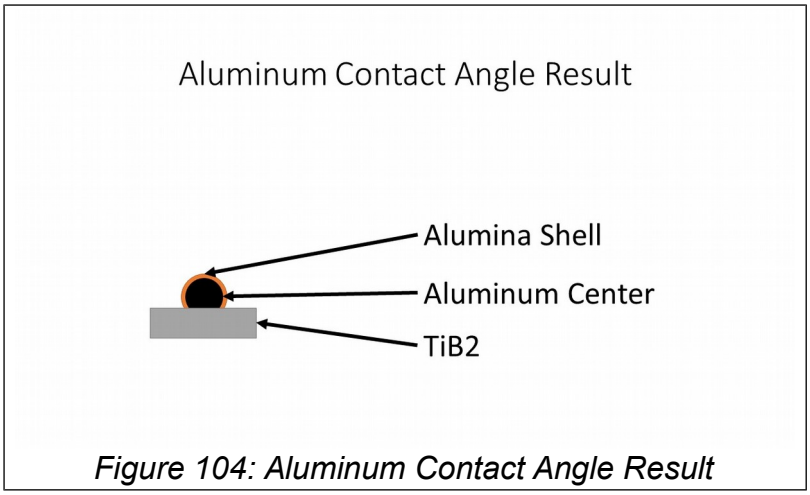


Figure 104: Aluminum Contact Angle Result

However, qualitative evidence and literature survey indicate that aluminum and titanium diboride wet very well. In looking at a previous SOM run, it was noted that the liquid aluminum would crawl up the titanium diboride cathode. In the sessile drop experiment, the aluminum droplet adhered very well at the aluminum-titanium diboride interface, practically similar to a weld. Furthermore, some literature indicates that aluminum can perfectly wet titanium diboride or have a low contact angle of under 10° based on how the titanium diboride is manufactured.[54]

Conclusions

Using the sessile drop setup, a contact angle value of 9.354±1.357° was measured between the BU flux and titanium diboride. Since the error of this high temperature measurement is approximately 10%, this value is adequate until more other experimental evidence is acquired. Using the empirical observations from the previous SOM experiments and fused aluminum sample as well as literature values, it would appear that titanium diboride and aluminum wet very well; however, a quantitative value for the lab's specific form of titanium diboride could not be ascertained. It can be assumed, though, that the contact angle is below 10°.

Next Steps

Future work with these experiments deals mostly with collecting more data and advancing the lab's capability of measuring the contact angle of different material samples under different gaseous environments. Secondary tests using the BU flux would be helpful in validating the contact angle between titanium diboride and the BU flux. In addition, the images produced in the experiments can be used in conjunction with numerical solving methods to determine the value of the gas-liquid interface as another way to better define the system.

Aluminum Cathode Current Collector

Overview

INFINIUM invented a new cathode current collector design using solid and liquid aluminum to connect to the aluminum pad. The section of this report on industrial electrolysis cell design) describes this current collector invention in detail on page 39. With 10-100 times higher electrical conductivity than graphite, this design enables a very high current density connection with lower overpotential than graphite, which is particularly important when using a floating liquid aluminum pad, whose advantages are described on page 29. (Incidentally, this high current collector design is also potentially a tremendous improvement over graphite for the Hoopes cell for aluminum purification.)

]INFINIUM ran two aluminum current collector experiments to test aspects of this current collector. Unfortunately, in the first oxide/sulfide contamination prevented complete melting of the aluminum, resulting in poor connection to graphite leads, and measured resistance was far higher than model predictions.

The second experiment used clean liquid aluminum, and also tested the device as a method for metal withdrawal. We also tested resistance of the graphite tubes, and of the tube-lead contacts, in order to better compare resistance of the actual graphite and liquid-solid aluminum components vs. theoretical predictions.

Experiment 1

Figure 105 shows a schematic of the current collector experiment, and Figure 106 shows a photograph of the apparatus before insertion into the furnace tube. The 1-D model predicted that optimal voltage would be about 0.3-0.33 V, and at just 61 cm aluminum electrode length, this should lead to about 150 A/cm² current density.

Experimental Procedure

1. Plug the top fittings, where copper rods are shown in Figure 105. Purge the apparatus with argon for at least an hour, until oxygen partial pressure is below 10^{-3} atm. The purge should reach up into the tubes, so one can act as inlet and the other exhaust, or both exhaust with a separate inlet.
2. Heat the crucible until the liquid aluminum temperature reaches 850°C. All components are very thermally robust, so the furnace can heat up quickly.
3. Lower the graphite tubes into the liquid aluminum in the graphite crucible, hitting the bottom and backing out approximately 2 cm.
4. Attach leads to the graphite tubes, and ramp voltage from 0 to 0.4 V at 0.001 V/sec (over ~7 minutes). Current should rise linearly from 0 to about 50-60 A.
5. Hold voltage at 0.4 V, 0.3 V, 0.2 V and 0.1 V for about 20 minutes each to allow current to stabilize, and measure current over time.

6. Use thermocouples in each of the graphite tubes to measure temperature vs. height every 10 cm, starting at the bottom with a quick dip in the aluminum pool while the thermocouple is cool. Temperature distribution should be about the same in both tubes.
7. On one graphite tube, connect the vacuum source to the fitting at the top, without opening the valve to allow vacuum connection.
8. Attach one lead to a copper rod, insert it into the top fitting, and quickly lower it to where its bottom end is about 15 cm below the top of the graphite tube.

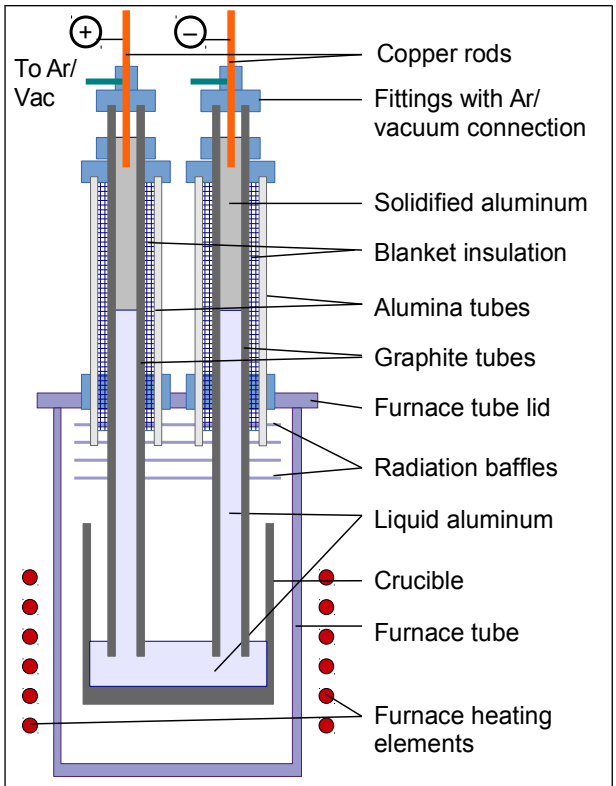


Figure 105: Schematic of experimental apparatus for demonstrating liquid/solid aluminum connection (not to scale).

Figure 106: Experiment apparatus.

9. Quickly open the vacuum source to the tube fitting, which should raise the liquid aluminum upward, connecting with the copper. The top of the copper rod should quickly heat up.
10. Repeat steps 4-5 with one lead on the copper rod of step 8. Current should rise to about 30% higher during the ramp than it did in step 5, and may decline slightly over time at 0.4 V due to aluminum melting in the tube.
11. Repeat steps 7-9 for the second graphite tube.
12. Repeat steps 4-5 with both leads on the copper rods. Current should rise to about 200 A (or the power supply maximum of 180 A) during the ramp, and may decline slightly over time at 0.4 V due to aluminum melting in the second tube.
13. Turn off the power supply and furnace, and allow the apparatus to cool to room temperature.

Experimental Results

Figure 107: Components post-experiment: alumina tube, graphite tubes, solidified aluminum rods attached to copper, melted aluminum with sulfur compound, crucible.

Due to surface oxidation of the aluminum granules used in the experiment, and contamination from an unknown source of sulfur (likely the graphite crucible), the aluminum did not melt completely, and likely made poor contact with the graphite leads. Figure 107 shows experimental components after the experiment, with detail showing this sulfur compound. X-ray diffraction of the yellow odorous powdery substance confirmed that it was Al_2S_3. As a result of this contamination, measured current was about a factor of ten lower than was predicted. That said, the aluminum core carried well over half of the current, that is, well more than the graphite tubes which had six times their area, demonstrating the much higher potential current density of this design.

An industrial cell would have a much higher aluminum cross-section fraction than <1/7 in this experiment. These dimensions allowed for a large graphite conductor to carry current in the event of a sudden failure of the liquid-solid aluminum cores. In an industrial cell, there would be multiple electrodes, and failure in one would merely increase the current in the others.

Figure 65 shows the voltage and current vs. time during steps 4-5 in the above procedure. Measured current is lower than predicted, due to poor electrical connection to the graphite tubes, and incomplete melting and consolidation of the aluminum in the bottom of the crucible. That said, resistance is roughly constant at 35 mΩ, except for a sharp drop early on likely due to an aluminum melting or oxide breakthrough event.

Figure 109 shows the voltage and current vs. time during a second set of runs. First voltage was increased to 4 V, resulting in about 150 A current across the graphite. Second, voltage was reduced to 0.4 V. While connected to the graphite holding at total system voltage of 0.4 V, at about 0.4 hours into the experiment, vacuum drew liquid aluminum into one tube, causing it to flow upward into it, attach to the copper, and partially solidify from the top. This increased the current from 14.5 A to a steady state value of 19.9 A, and reduced resistance from 27.6 mΩ to 20.1 mΩ. At about 0.7 hours, vacuum drew liquid aluminum into the second tube, causing it to flow up into it, attach to the copper, and partially solidify from the top. This increased the current to 25.6 A, and resistance to fall to 15.8 mΩ, which is about 43% lower than with no aluminum in either tube. Thus about as much current flowed through the aluminum as through the

graphite, despite having only about one-sixth the cross section area. This demonstrated the very high current density in this liquid-solid aluminum current collector with a graphite conduit.

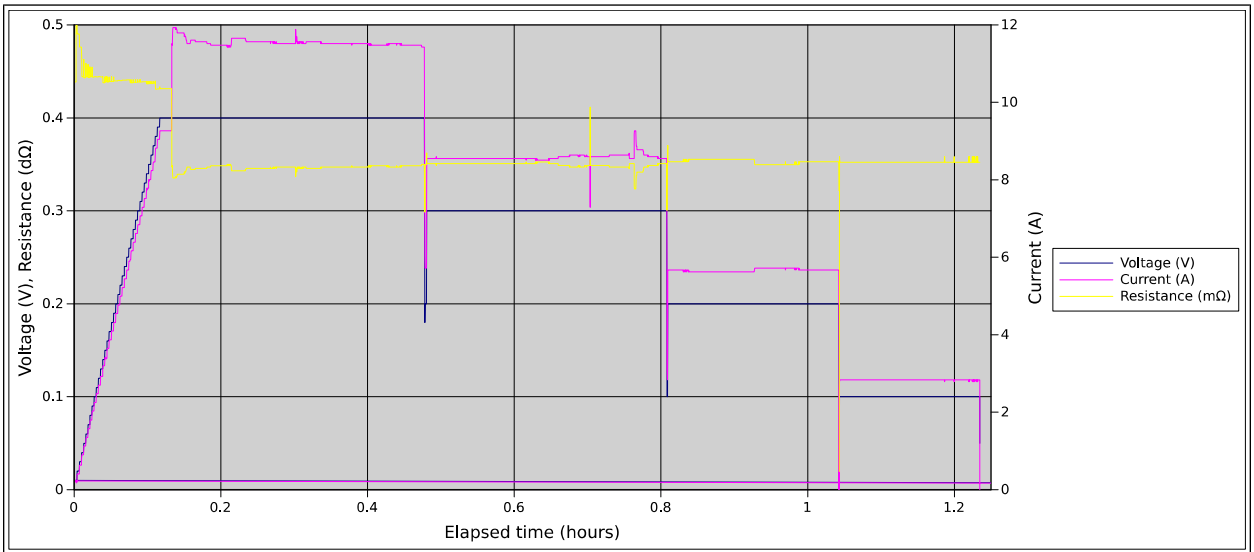


Figure 108: Applied voltage, current, and cell resistance with empty graphite tubes.

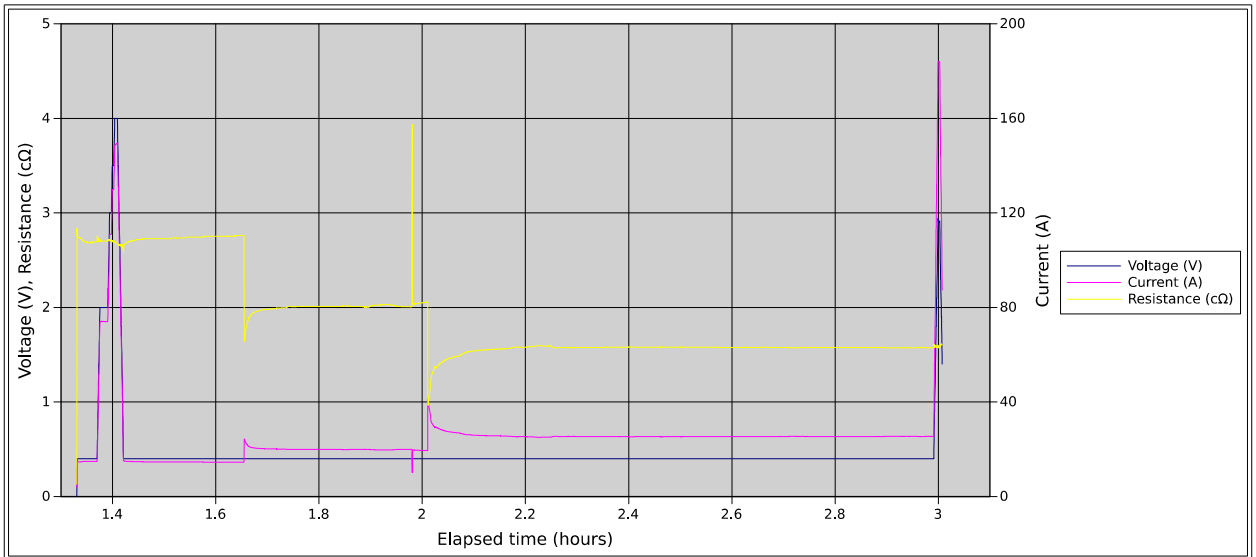


Figure 109: Applied voltage, current, and cell resistance adding aluminum to one tube and then the other.

A subsequent increase in voltage to 2.9 V caused current to reach 184 A, indicating about one half the prior resistance. This also demonstrated the ability of this device to carry current at about 100 A/cm^2 in the aluminum core.

Figure 110 shows the voltage and current in a ramp from 0 to 2.0 V, resulting in current increase to a maximum of 165 A, again demonstrating very low resistance and high current density. Cell resistance fell during this ramp from 12.8 mΩ to 11.3 mΩ, again likely due to melting and oxide breakage of the aluminum in the crucible caused by high

resistance heating.

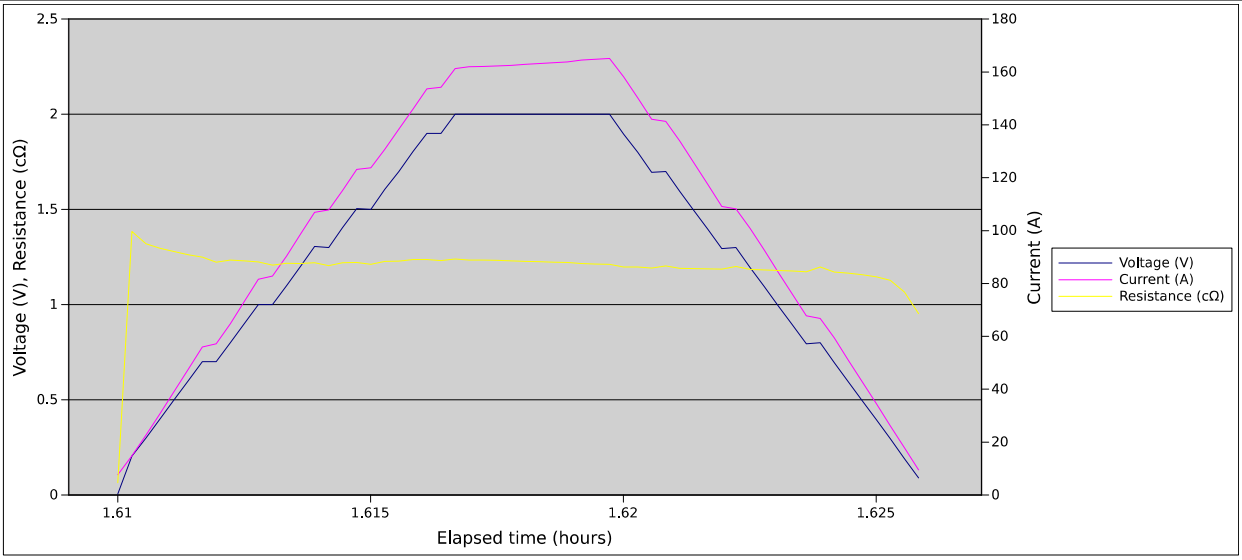


Figure 110: Applied voltage, current, and cell resistance during a voltage ramp with aluminum in both tubes.

Summary

The apparatus worked as expected in drawing liquid aluminum up into the tubes and attaching it to the copper leads. The ratio of resistances with and without aluminum in the tubes was as expected. Although cell resistance didn't fall to 2 mΩ as expected, it came within a factor of 6 of that value, and continued to decline. Higher resistance than expected was likely due to the observed sulfur contamination and oxide film on aluminum granules in the pool, and declined over time.

For the 200 A cell and beyond, higher current density in this device will allow us to use a much smaller cathode current collector than if we were to use graphite. That said, it will require at least two current collectors, and probably three, in order to avoid problems if one is disconnected. We should conduct long-term testing of these devices to determine their reliability before committing to a design.

Experiment 2

This used the same current collector apparatus shown in Figures 105 and 106.

Before running this experiment, we first measured the resistance of components, and connections, particularly the graphite tubes and copper rods. This can be done by measuring resistance along the entire length, and half the length, using the leads and connections which will be used in the experiment. The difference is the resistance of half the component; the whole minus twice half is the sum of the contact resistances.

Experimental Procedure

1. Plug the top fittings, where copper rods are in Figure 105. Purge the apparatus with argon for at least an hour, until oxygen partial pressure is below 10^{-3} atm. The purge should include the tubes, so one can act as inlet and the other outlet, or both outlets with a separate inlet.
2. Heat the crucible until the liquid aluminum temperature reaches 750-850°C. Components are very thermally robust, so the furnace can heat up quickly.
3. Lower the graphite tubes into the liquid aluminum in the graphite crucible, hitting the bottom and backing out approximately 1-2 cm. Shut off purge valve, and start recording voltage and current.
4. Attach leads to the graphite tubes, and ramp voltage between them from 0 to 1 V at 0.003 V/sec (over ~5 minutes). Current should rise linearly from 0 to about 50 A.
5. Adjust voltage by hand until current reaches about 60 A. Hold for one minute to allow current to stabilize, then gradually drop the voltage back to 0 V and shut off the power supply.
6. Disconnect the power supply from one of the graphite tubes, and insert a copper rod into it so the rod bottom is about 30 cm below the bottom of the top fitting, and switch the power supply connection from the graphite tube to this rod.
7. Ramp to the voltage used in step 5 over about 10 seconds, then slowly open the vacuum reservoir valve to draw aluminum upward into that tube. Current should rise to about 90 A, then fall back to zero linearly with voltage. After current rises, close the vacuum reservoir valve.
8. Reduce voltage back to zero, then shut off the power supply.
9. Repeat steps 7 and 8 for the second reservoir-tube valve. Current should rise to about 170-180 A, then fall back to zero linearly with voltage.

10. Disconnect the power supply. Loosen one of the top fittings from the copper rod. Wearing appropriate PPE, slowly (~3 mm/sec) draw the copper rod upward through the fitting, until aluminum is visible. If necessary, remove the fitting from the top of the graphite tube. Keep drawing it upward another 30 cm after aluminum is visible.
11. Reconnect the power supply, and repeat steps 7-8 (without opening the valve).
12. Repeat step 10 for the other copper rod.
13. Reconnect the power supply, and repeat steps 7-8 (without opening the valve).
14. Turn off the power supply and furnace, and allow the apparatus to cool to room temperature.
15. Measure the resistance of the aluminum rod and leads and contacts attached to it.

Experimental Results

This experiment was conducted using a 2 ft long copper rod with a ¼ inch diameter to be inserted into a 2 ft long graphite rod with 1 inch outer diameter and 3/8 inch inner diameter. The copper rods were measured to have about .00175ohms resistance prior to being loaded in the assembly pictured in figure 4. The graphite rod started with about .0425 ohms resistance but increased to about .0820 ohms after 1 minute under voltage as the rod produced heat. This was likely a source of increased resistance during the experiment.

Conducted at 850C, 3 V was necessary to pulse 60 A through the system. The liquid aluminum in the crucible was likely superheated and did not fill the graphite rod smoothly when pulled by the vacuum. It made less reliable connection to the copper rod and therefore did not conduct current as well as expected. After vacuum was pulled in the first graphite rod, the current peaked at 75A and slowly dropped back to 60A. When vacuum was pulled into the second graphite rod, the current peaked at about 80A and dropped back down to 67A. When the vacuum was removed, the current in the system dropped back to 60A, indicating an open circuit. Although the high resistance likely has to do with the state of the liquid aluminum and perhaps vacuum being pulled too quickly, the second copper rod was also omitted from this experiment run by mistake and compromised the results of the experiment.

The experiment was repeated at 750C with 1.5kg of pure aluminum and all other parameters unchanged. First, leads were connected to each graphite rod and voltage was ramped to 3 V at 0.003 V/sec, which resulted in 66 A across the system, as represented in figure 111. Voltage was reduced back to zero to add the first copper rod and pull vacuum in the graphite fitting. With the graphite tube filled with frozen aluminum, 3 V yielded 72 A which eventually dropped to about 67 A. The voltage was then reduced to zero again to draw aluminum into the second graphite rod. With both leads still attached to the graphite, 3 V yielded 78 A and then dropped to and held at 72 A. Transferring the leads to the copper rods resulted in dramatically less resistance. As

seen in figure 112, 1 V yielded 107.5 A and 1.5 V yielded over 120 A. The power supply was shut off and the copper rods extracted.

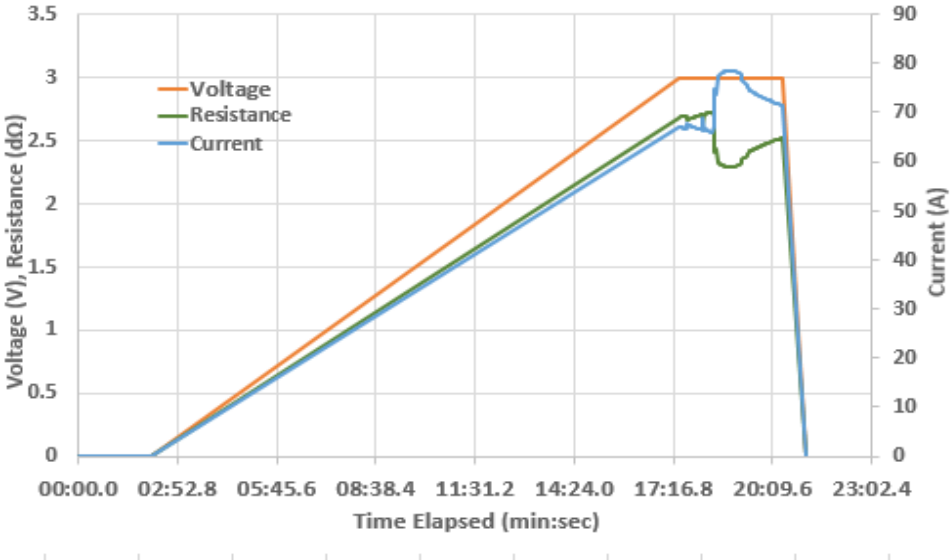


Figure 111: Applied voltage, current, and cell resistance with leads attached to graphite rods during a 3 V ramp with Aluminum.

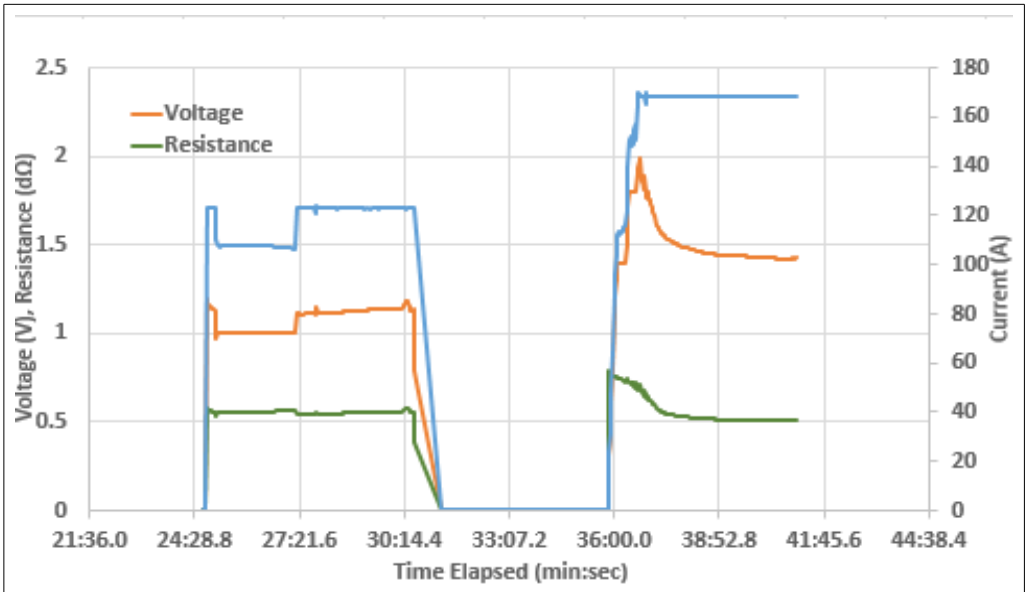


Figure 112: Applied voltage, current, and cell resistance with leads attached to copper rods.

Figure 113: Copper rods from both trials cell post-experiment.

First attempt at pulling a copper rod out was too fast and promptly broke any connection the frozen aluminum had to the liquid aluminum in the bottom of the cell. The copper rod was lowered back in place and max amperage was applied through the system again to reestablish the lost connection, this can be seen in the last peak in figure 112. After shutting off the power supply, the copper rod was again extracted at about 3 mm/sec. The rod resisted being pulled out because of the tight fit between the diameter of the frozen aluminum and the inner diameter of the graphite. The bottom rod pictured in Figure 113 was the results of the first run of this experiment at 850C, the top two rods from the experiment at 750C harvested more aluminum with a much smoother surface finish. Both rods from the second experiment broke off from the liquid aluminum at the same length indicating a possible mechanical barrier breaking contact within the cell.

Summary

This current collector worked as expected in extracting liquid aluminum from the cell. Resistance was higher than expected, which could be due to an oxide layer building up on the aluminum, poor surface contact between components in the cell, or excessive heat being created. It was determined that the current collector functioned better at a lower temperature and with a very slow vacuum pull to extract the liquid aluminum.

Though this is a robust and viable method of extraction, all qualifiers mentioned above stand; in terms of the 200A cell and beyond, it could be beneficial to have several smaller cathode current collectors in the cell to avoid disruption of the cell if one fails. This current collector system still requires long term testing to confirm it's viability.

Aluminum Electrolysis Experiments

Small-scale experiments have three goals:

1. Test and improve materials and subsystems in key performance metrics;
2. Validate an FEA model for design of a 200 A cell and the industrial cell (p. 27);
3. Measure tube corrosion rate as a function of temperature, current density and bath composition for use in a comprehensive techno-economic cost model in Subtask 4, toward eventual design optimization.

Performance Improvement Strategy

INFINIUM pursued two approaches to system performance improvement, which we refer to as: materials and engineering approaches. Experiments will test hypotheses related to these two approaches, and provide data for cost optimization in Task 4. Engineering approaches will seek to enhance tube lifetime, test larger tubes at higher current, and support modeling for scale-up design.

Tube Lifetime

The Introduction on page 8 describes the materials approach to improving tube lifetime in detail, with strategies moving forward enumerated on page 10. The engineering approach to tube lifetime seeks to improve the oxide ion concentration at the membrane-bath interface, using supersaturation and/or vigorous stirring. This is based on the observation that static tests with bath saturated in alumina perform much better than those with below-saturation alumina concentration, and that oxide ion concentration in bath during electrolysis falls both locally in a boundary layer around the membrane and globally due to removal through the membrane.

Based on these observations, our hypothesis is that accelerated membrane corrosion during electrolysis vs. static testing is at least partially due to low oxide concentration in the bath at the membrane interface. The other possible explanation is bath electronic conduction with low membrane electronic conductivity as described on page 10. But the high current efficiency observed in many experiments suggests that bath electronic conductivity is too low for this to be a major problem.

In our experiments, supersaturation could address depletion over time by providing excess alumina beyond saturation, which would stay in suspension and dissolve as electrolysis removes oxide from the bath. In an operating Hall-Héroult cell, this is not an option because suspended oxide with higher density than the bath could settle and get under the liquid aluminum pad, forming a sludge layer increasing resistance between the cathode and pad. This also may not be an option in our system in Configuration 2 (Figure 59 p. 85).

Stirring would thin the boundary layer in the bath adjacent to the membrane with low alumina concentration, resulting in faster oxide ion (or related complex ion) mass transfer, and higher oxide and alumina concentration at the membrane-bath interface.

Bath Characterization

To follow on the description of the potassium cryolite bath structure in the Introduction (page 11), we ran a series of experiments with an alternate anode, as described starting on page 149.

Anode Assembly Scale-Up Behavior

One goal of these experiments is to support design of the 200 A cell and beyond, which will require multiple high-current anode assemblies (including the tube, anode, and current collector). Based on this, maximum current per anode assembly is an important key performance metric.

The current cell design can accommodate membrane tubes up to 1.25" OD 1.0" ID, with up to 6-10" immersion depth. INFINIUM can slip cast tubes of this size using existing tooling. The maximum immersion depth depends on uniformity of the hot zone in the crucible. With these tubes at 6" immersion depth, anode-membrane interface area is about 121 cm², leading to 25 A at just over 0.2 A/cm², or 50 A at just over 0.4 A/cm². These cells can therefore test anode assemblies with performance close to the design goals of the 200 A cell electrodes.

Summary of Experiments and Results

Our first goal was to establish a sound baseline experimental technique, which we can use as the basis for factor space experiments moving forward. Toward that end, we ran 3 electrolysis experiments using CaF_2-MgF_2-based bath composition. After Experiment 1, we changed from steel to graphite containment in order to avoid iron contamination in the aluminum product, though we retained steel secondary containment. We also used an alumina sheath to contain the aluminum product.

We also ran 5 experiments using potassium cryolite flux in order to establish a baseline in that system.

Following this, ten electrolysis experiments were conducted, seven using membranes thicker than 1.9 mm and three using thin membranes (650-950 μm). The goal of these experiments was to contrast the performance of thin (<1mm) versus thick (2-4 mm) membranes. We decided to analyze in detail seven of the ten experiments; 21, 27-29 (thick anodes) and 30-32 (thin anodes).

Casting and handling thin fragile green ceramic membrane bodies was challenging. That said, thin membranes have performed significantly better, both in charge passed and in lowering the total potential required to make aluminum from 8V to 3.5V.

In experiment 21, two anodes roughly 2 mm thick were used. A constant current density of 0.1 A/cm^2 (6A/anode) was pulsed on average for 10 seconds followed by a 60 second healing period at 0.5A/anode. Healing is when the anode is made cathodic vs a graphite electrode, or the aluminum cathode. This pulse sequence resulted in an anode-cathode voltage that varied between 6.5-7V, with each anode producing 22-25 sccm of oxygen as detected by a mass flow controller. This is equivalent to 3.5-4 sccm

of oxygen per ampere for each anode. Post mortem analysis on the membranes revealed loss of yttrium on the anode-flux interface and on the silver-membrane interface.

Experiments 27 and 28 used 3-4 mm thick anodes and needed a higher overpotential to get current to flow. The pulse sequence shown in the results section highlights the anode-cathode voltage evolution as one Amp is pulsed with an upper voltage cutoff value of 9V. In experiment 27 the voltage kept increasing during each pulse reaching the cutoff value after 20 pulses and taking less than one second to reach the limit at the 100^{th} pulse. That said, when using reversed “healing” pulses, the system reached 120 pulses without exceeding cutoff voltage, qualitatively proving the benefit of reversing current. The same was observed in experiment 28 as can be seen in the results section. Experiment 29 did not have the cryolite flux fully dried and water was observed condensing on the cell exhaust. This severely degraded the 1.6 mm tube and very little current was passed. Both anodes in 27 and 28 exhausted ~12 sccm of oxygen per Amp which was excessive compared to experiment 21 (~4sccm/A) with thinner anodes.

The conclusion of experiments 21, 27-29 was that the current anodes are too thick to operate continuously and meet the target metrics with our new low-temperature system, therefore we need to fabricate anodes as thin as possible, preferably with thicknesses less than 1 mm.

We next successfully made sub-mm membrane tubes 3/4” OD and 8” in height. Experiments 30-32 clearly show the enhanced performance of these thin membranes. Compared to thick ones, current did not drop off nor did the voltage spike up to high values during the first hour of electrolysis. Experiment 30 operated at 3.2-3.5V holding an average current of 1.5A for over 8 hours. To verify that metal was made at this overpotential, the cathode was inspected after 4 Ah of charge passed, and it indeed had aluminum with silicon being the biggest impurity. Experiment 31 was voltage controlled, and was setup to see if the anode flow could be measured using its natural liquid seal. A flow of around 3 sccm/A was detected but after three hours of electrolysis the current started to decay. After this we switched to current control, we held it at 2A for five hours (anode exhaust also increased) before stopping the experiment.

Learning from the first two thin anode experiments, we set up experiment 32 with pulsing in mind. It was the most successful experiment which operated until hitting the control system's default 100 A·h limit, which effectively ended the experiment. (This limit was caused by an unfortunate programming error.) The anode-cathode pair was pulsed at constant current for 600 seconds and relaxed (not healed) for 100 seconds. The next day the cathode was inspected which was very metallic, and aluminum was harvested by dipping a clean TiB_2 cathode through the port. After testing the metal made, silicon was again the biggest impurity.

By the final electrolysis experiment with added yttria in the bath, we were able to achieve nearly a seven-fold increase in amp-hours of charge passed, up to 680 A·h, which is a significant step toward the final Subtask 3.1 goal of 8333 A·h (200 A 250 hrs divided by 6 anodes).

Experiment 0: First Quad Cell Electrolysis Trial

Goal: Run a long tube lifetime experiment using BU flux at relatively low temperature.

Parameters:

- Membrane: 0.75” OD YSZ closed-end tube in 1.25” OD steel sheath
- Anode: liquid silver, LSM-alumina-Inconel current collector
- Cathode: TiB_2, 1/8” steel rod connection
- Bath: 1.7 kg CaF_2-MgF_2 basis (49 wt% CaF_2-40 MgF_2-5 CaO-3 Al_2O_3-2 YF_3)
- Bath depth: 2.5” total, 0.5” steel sheath immersion, 1.5” YSZ tube contact, 0.5” clearance to crucible bottom.
- Crucible: steel tube 13” long 5” OD welded to ½” thick base, steel secondary container inside 5” long 6” OD also welded to ½” thick base, both ~1018 steel.

Results:

- Retort tube did not seal with just four clamps, required four additional C-clamps.
- 7 kW furnace can reach 1050°C temperature easily, with less than 80% power.
- Water-cooled lid remained cool throughout.
- Viewport is clear (low-volatility bath, adequate shutter coverage), provides excellent field of view.
- Bath is very transparent, silver from broken anode tube is clearly visible below it.
- Membrane broke before we could start electrolysis.

Conclusions:

- Apparatus works well in general, possibly as high as 1200°C max temperature.
- We should add additional clamps to other retorts – and later to this one as well.
- We need to be more careful about membrane tube handling.

Figure 114: Apparatus at temperature (top view) before lowering membrane tube.

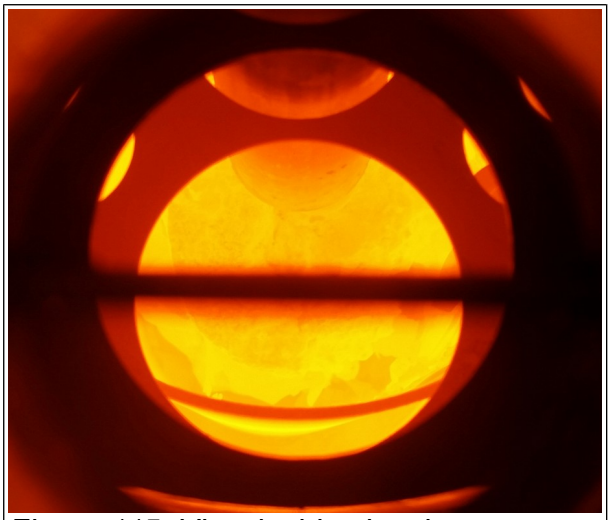


Figure 115: View inside showing upper crust of melting bath attached to sheath.

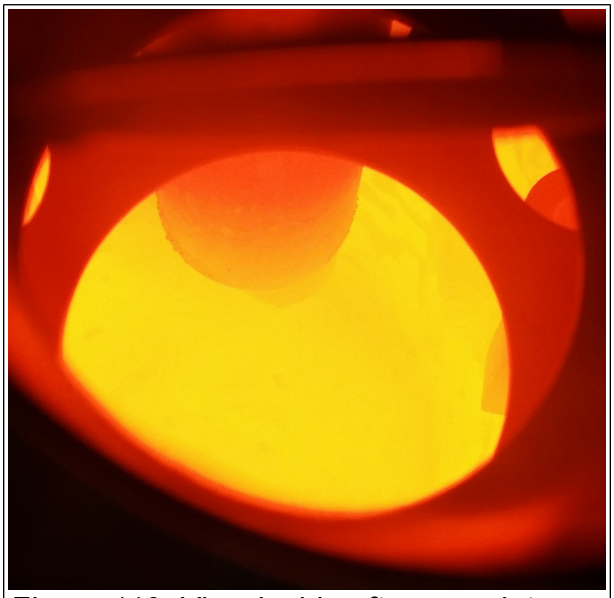


Figure 116: View inside after complete bath melting.

Figure 117: View inside after membrane tube breakage, showing silver inside tube below bath surface.

Figure 118: Upper tube fragments after cool-down (lower section remained in bath).

Experiment 1: First Quad Cell Electrolysis

Goal: Successful electrolysis using same parameters as Experiment 0.

Parameters:

- Membrane: 0.75” OD YSZ closed-end tube in 1.25” OD steel sheath
- Anode: liquid silver, LSM-alumina-Inconel current collector
- Cathode: TiB_2, 1/8” steel rod connection
- Bath: 1.7 kg CaF_2-MgF_2 basis (left over from Experiment 0)
- Bath depth: 2.5” total, 0.5” steel sheath immersion, 1.5” YSZ tube contact, 0.5” clearance to crucible bottom.
- Crucible: steel tube 13” long 5” OD welded to ½” thick base, steel secondary container inside 5” long 6” OD also welded to ½” thick base, both ~1018 steel.
- Electrolysis: sweep 0 to 3 V over 7 minutes, hold 3 V.

Figure 119: Anode cross section after Experiment 1.

Results:

- 100 hours of electrolysis at 0.1-0.5 A, Figure 120 shows voltage and current.
- YSZ tube broke while trying to remove silver anode (to prevent tube breakage). The cracks from the stress can be seen in figure 119.

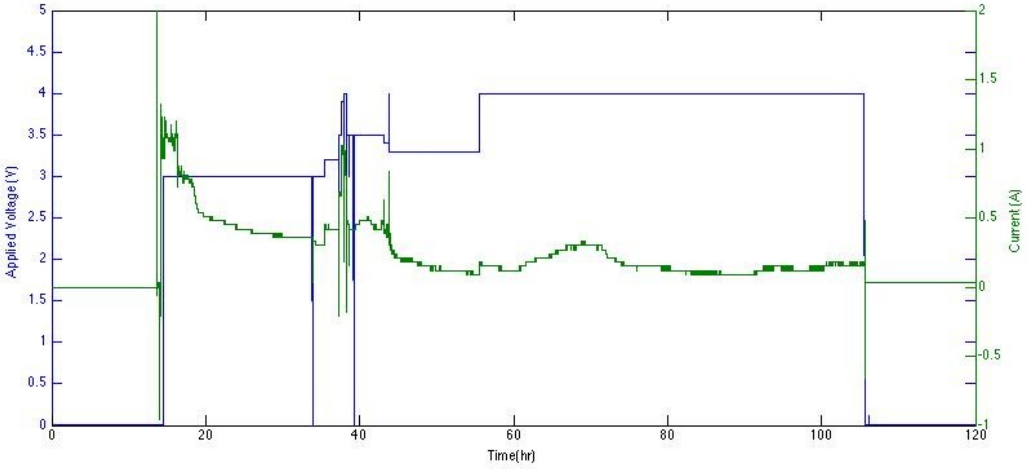


Figure 120: Current vs Voltage of first electrolysis experiment.

- EDX analysis of the membrane (figure 121) 100 microns from the silver interface revealed Ca, Mg, and F had intercalated into it. Silver was also present at 40 microns from the interface (figure 122).

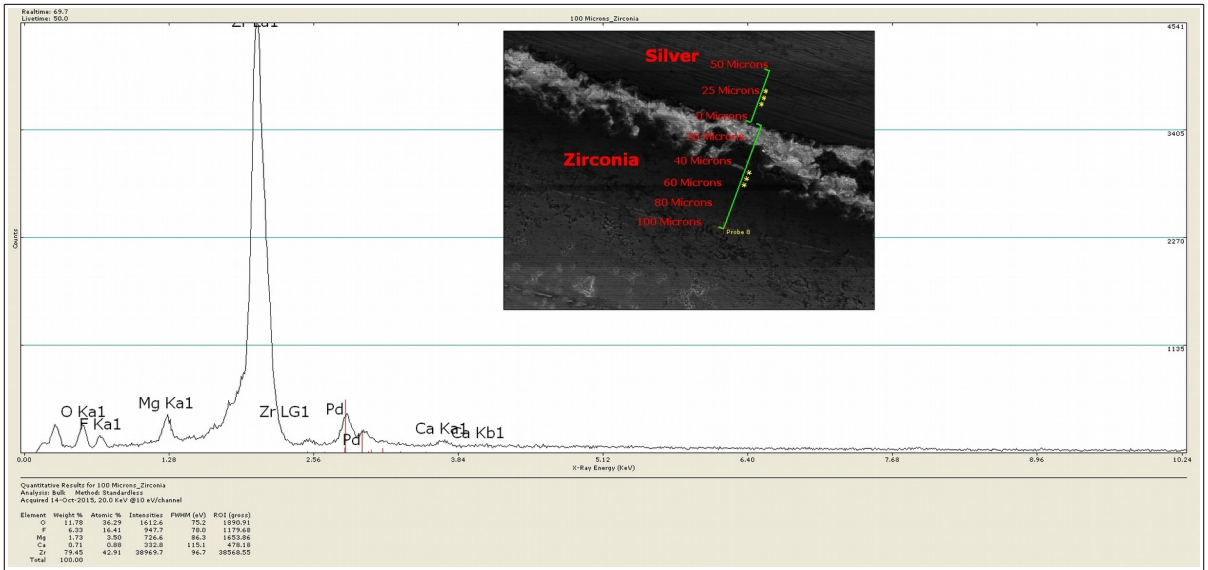


Figure 121: EDX of YSZ membrane 100 micrometers from the silver interface.

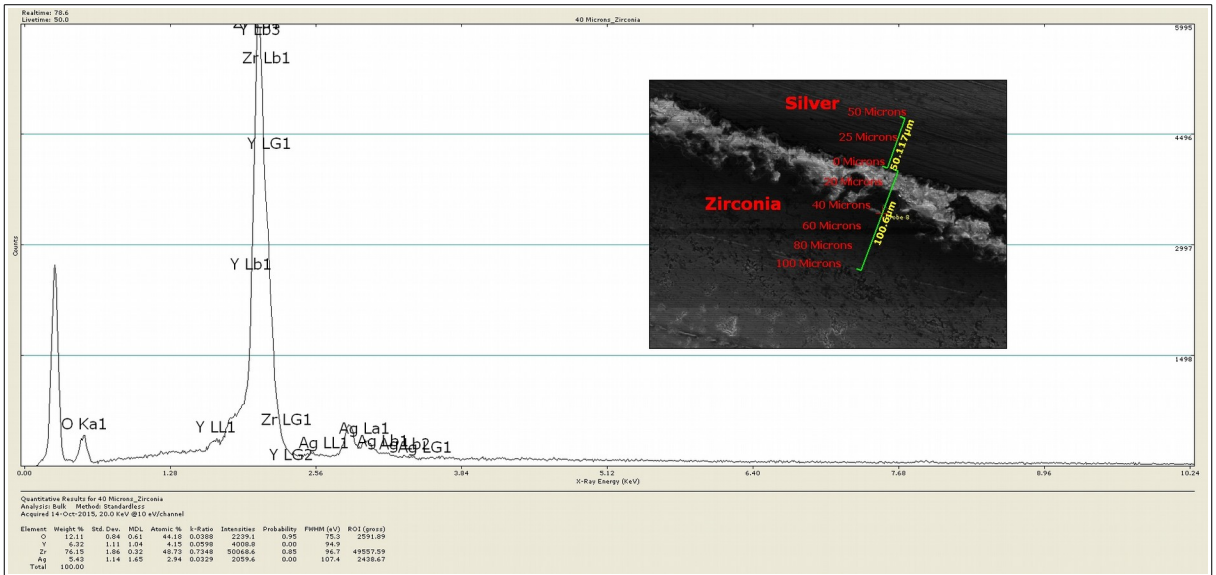


Figure 122: EDX of YSZ membrane 40 microns from silver surface. It shows silver intercalation into the membrane.

- Metal product contains significant magnesium, about 12% by SEM-EDS (figure 123). Cathode is shown on figure 124).
- CaF_2-MgF_2 flux is very non-volatile, the glass viewport remained clear over 120 hours at temperature.
- LSM current collector broke at some point (figure 125).

- Bath went from white before experiment, to clear with gray “cloud” during experiment, post experiment it's very gray indicating metal dissolution.

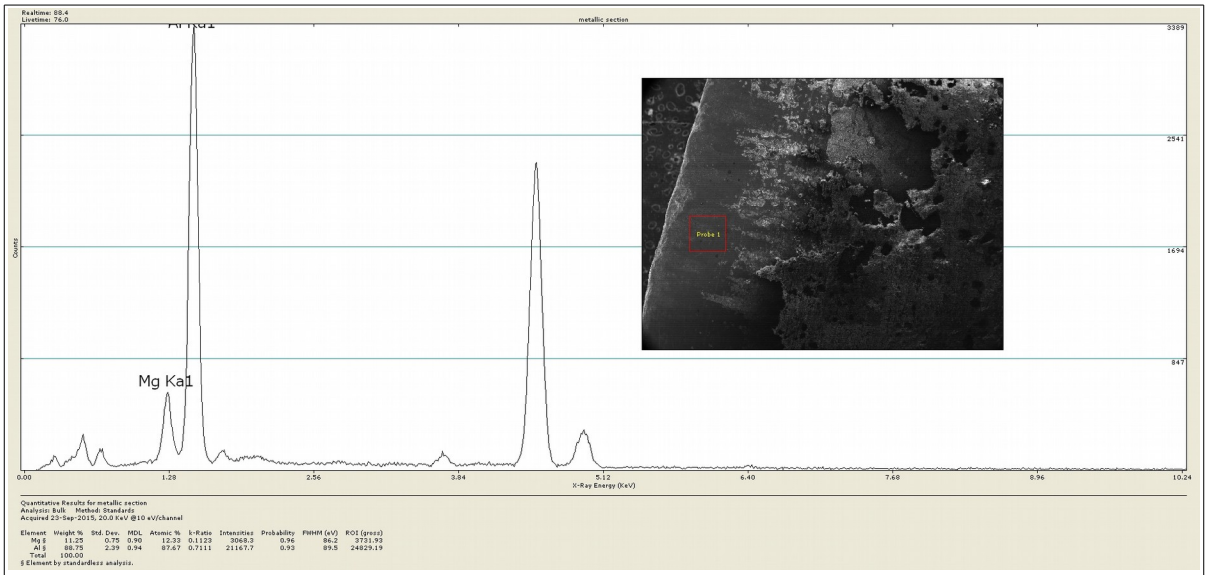


Figure 123: EDX of metallic surface on TiB2 cathode. 88 wt% Al - 12 wt% Mg alloy.

Figure 124: Cathode after Experiment 1.

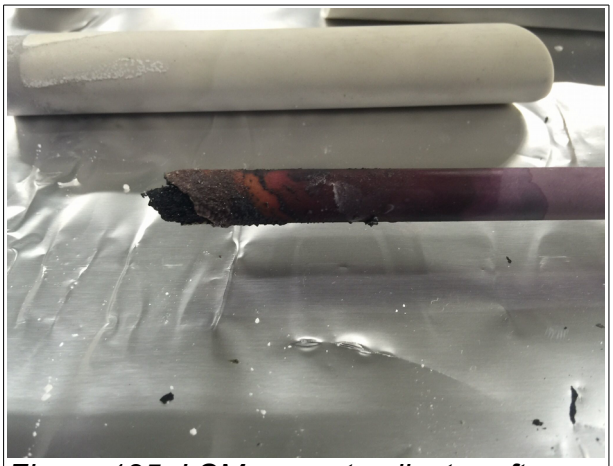


Figure 125: LSM current collector after Experiment 1.

Conclusions:

- The salts were not pre-dried before the experiment, so we established a standard operating procedure on salt preparation to ensure that they are dried and accurately weighed before each experiment.
- We need to be careful about the location of the LSM tube.
- We need to reconsider how we cool the tube, or withdraw the silver.
- Future flux compositions should not include MgF_2.

Experiment 2: Electrolysis Update

Goal: Electrolysis using better a crucible material, salt preparation procedure, and an alumina sheath around the cathode at constant current to have better metal collection. This will prevent any aluminum made from alloying or being contaminated by the crucible.

Parameters:

- Membrane: 0.75” OD YSZ closed-end tube NO sheath this time.
- Anode: liquid silver, LSM (1/4”x1”)-alumina-Inconel current collector (figure 28).
- Cathode: TiB_2, 1/8” steel rod connection in ¾” alumina sheath to collect metal
- Bath: CaF_2-MgF_2 pre-dried individually, then melted and blended before use.
- Bath depth: 4” total, 3” YSZ tube immersion, 1” clearance to crucible bottom.
- Crucible: graphite crucible 5” OD 12” height, ~1018 steel secondary container inside 5” inner height 6” OD welded to ½” thick base.
- Electrolysis: Set current limit to 6A (0.1 A/cm²) and voltage limit to 5V.

Results:

- Initial current was the limit of 6A with the voltage starting around 4V (figure 126). As time progressed, the voltage reached the limit of 5V and the current started to decay.

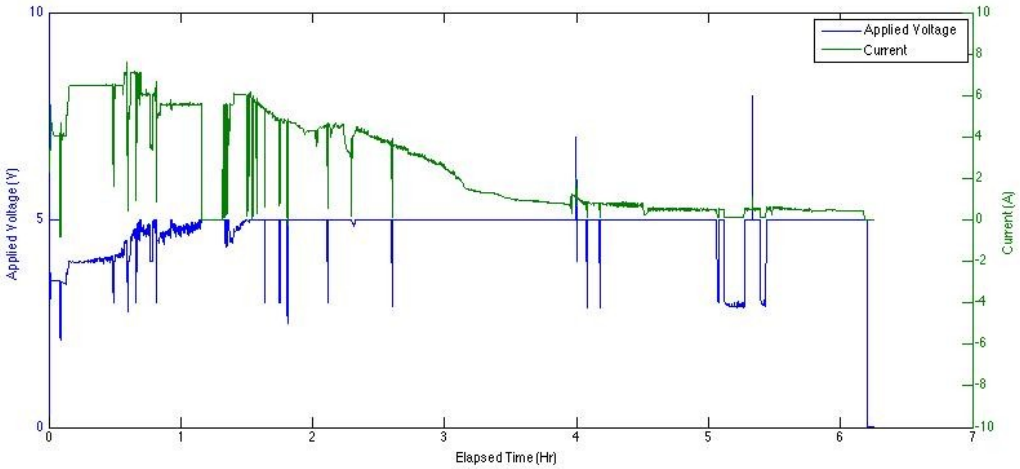


Figure 126: Electrolysis results using both current and voltage limits.

- Experiment was stopped after passing 15.6Ah due to the current being less than 1A.
- Setting the current to 0A resulted in an open circuit voltage of 2.8-3V indicating that we did not have metal solubility.

- At the end of the experiment the system was cooled to 940°C and the anode was lifted to prevent it from cementing to the flux.
- The TiB_2 cathode became detached from the current collector during electrolysis. We were not able to retrieve pure aluminum metal.
- The cathode was metallic (Figure 21), and EDX analysis showed (Figure 128) a Mg-Ni-Al alloy. The nickle contamination came from the cathode current collector since it was lowered into the melt to make contact with the dislodged TiB_2.
- Ca, Mg, F detected via EDX on membrane 100 microns from silver interface (figure 144).
- No Ca (only Mg, F) was detected 60 microns from silver surface (figure 144).

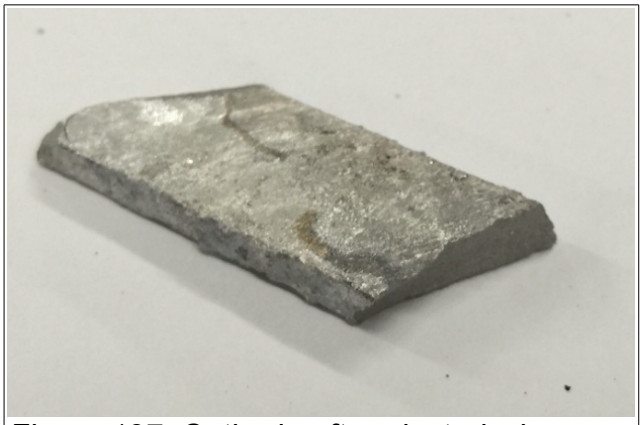


Figure 127: Cathode after electrolysis. Surface was metallic.

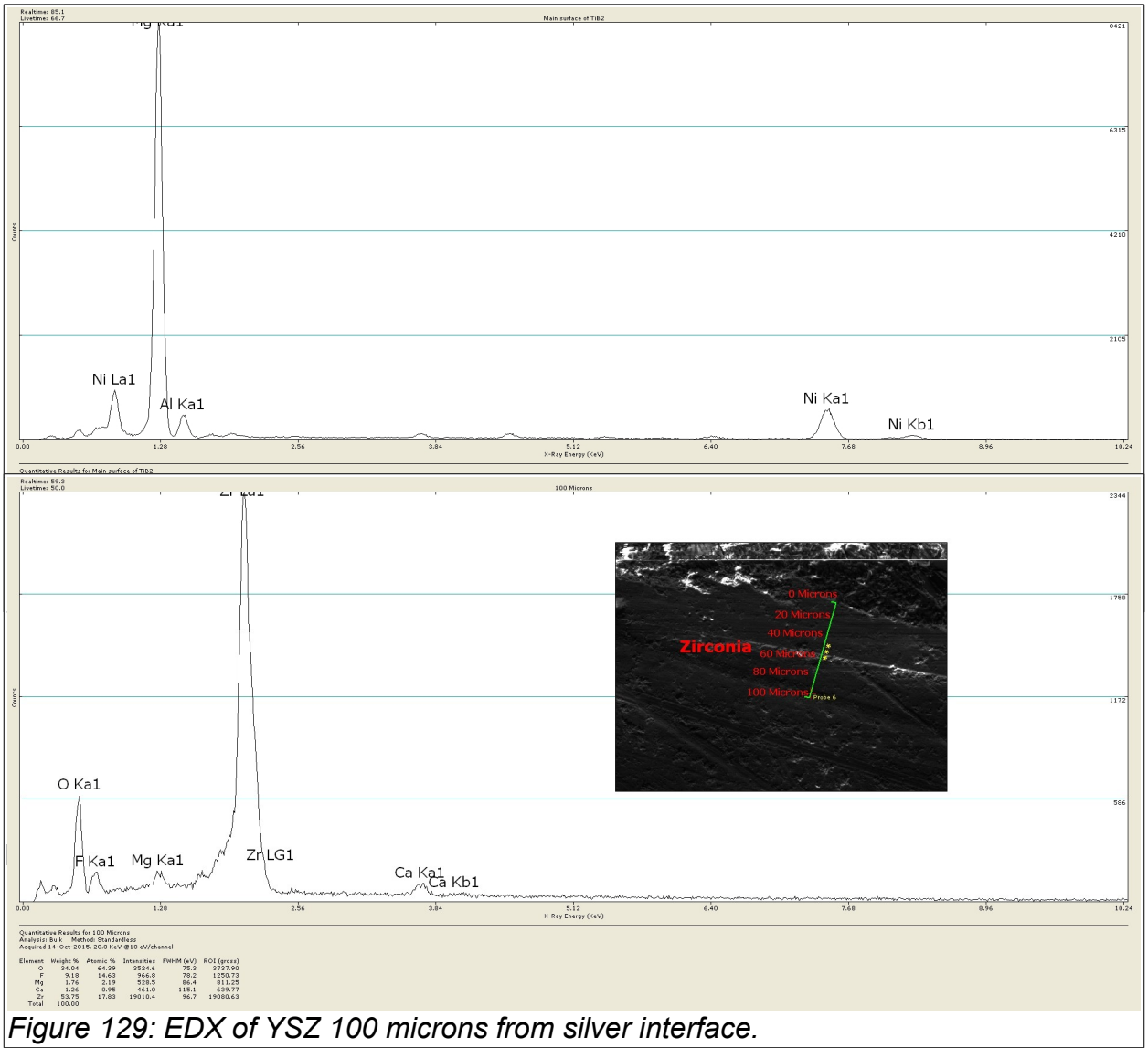


Figure 129: EDX of YSZ 100 microns from silver interface.

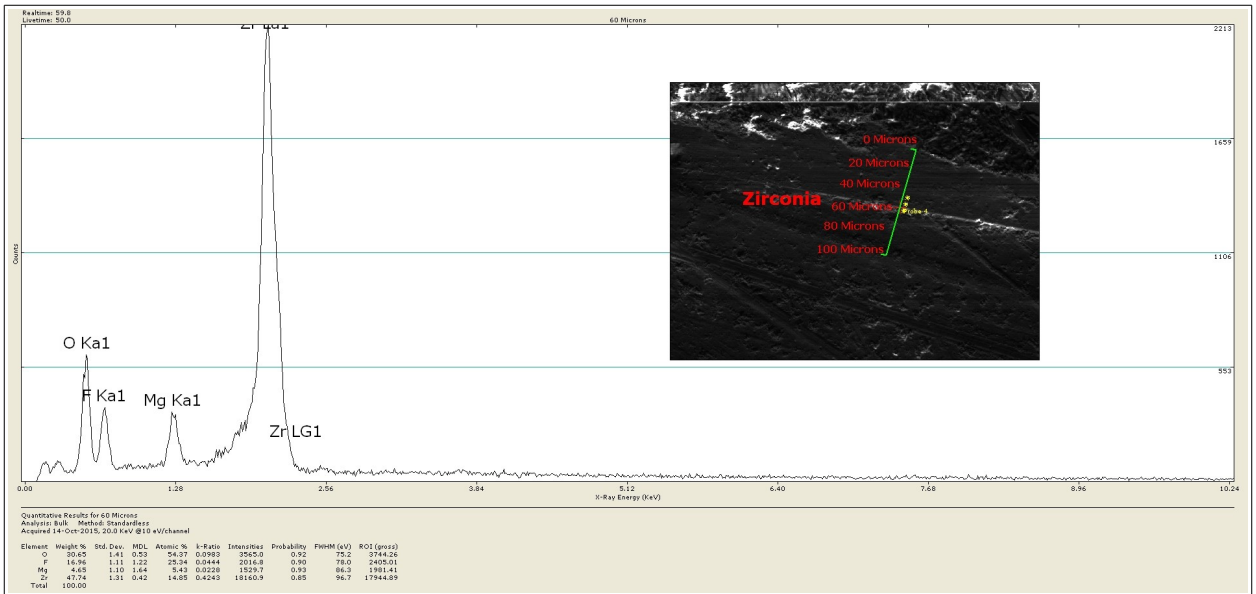


Figure 130: EDX of YSZ 60 microns from silver interface.

Conclusions:

- Unsure if the current decay overtime is due to the LSM current collector failing during electrolysis. Will be focusing some effort in making a better electronic connection between the LSM-Ag-Inconel interface.
- The YSZ membrane did not break during electrolysis, the experiment was stopped due to the drop in current and the cathode failing.
- The lack of plating aluminum is associated with our operating temperature. We will retry the experiment at 1200C.
- We believe the increase in current with respect to the first experiment is due to the LSM current collector being roughly three times more in surface area.
- Lifting the zirconia membrane after the silver solidified allowed us to keep it from solidifying with the flux.

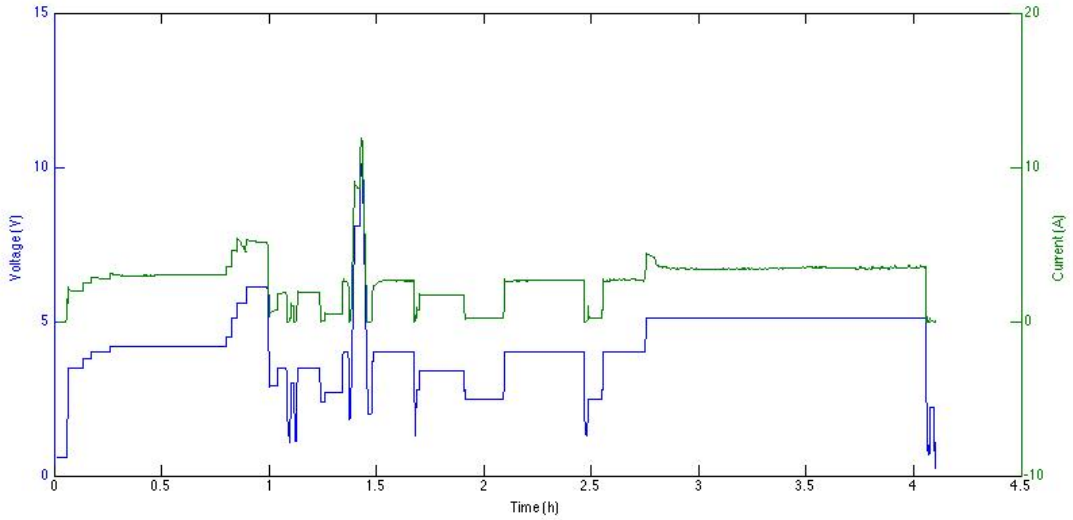
Experiment 3: Bath Characterization

Goal: Use an inert anode to generate oxygen using CR=1.8.

- Anode: inert electronic conducting rod
- Cathode: TiB_2, 1/4” steel rod connection
- Bath: 3kg KF-AlF3, 100g alumina.
- Bath depth: 7” total
- Crucible: steel tube 13” long 5” OD welded to ½” thick base, steel secondary container inside 5” long 6” OD also welded to ½” thick base, both ~1018 steel.
- Electrolysis: sweep 0 to 3 V, hold at 3V. Step up to 10V.

Results:

- No detectable oxygen from anode.



- Very low current at 3V, induced 10A at 10V. Probably not good, did this to develop understanding of the system.

Conclusions:

- System current was linear with voltage, no mass transport limitations observed at 10V.

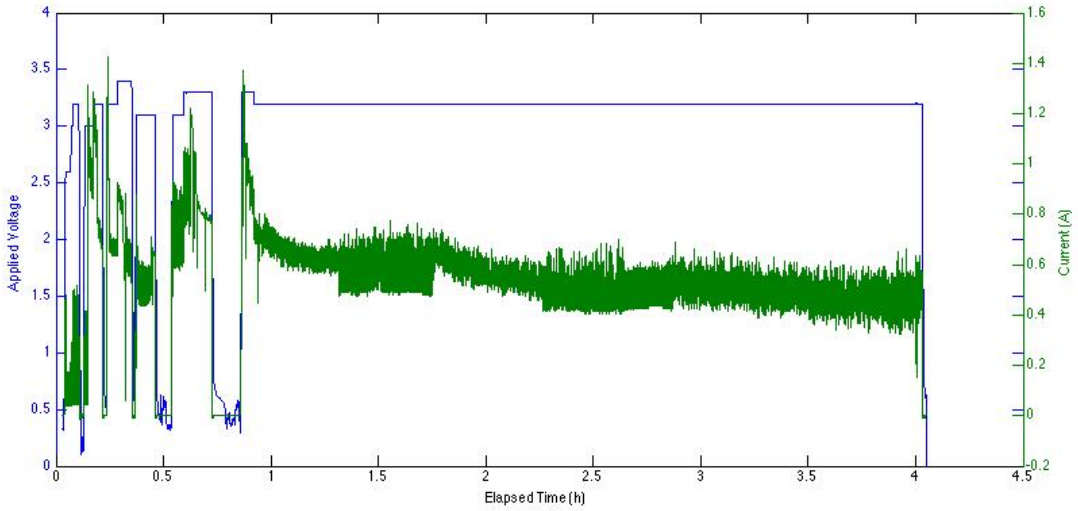
Experiment 4

Goal: Electrolysis using an alumina sheath around the cathode at constant current to have better metal collection. This will prevent any aluminum made from alloying or being contaminated by the crucible.

Parameters:

- Anode: Same as experiment #1.
- Cathode: TiB_2, 1/4" steel rod connection.
- Bath: Same as experiment #1.
- Bath depth: 7" total.
- Crucible: Steel crucible 5" OD 12" height, ~1018 steel secondary container inside 5" inner height 6" OD welded to ½" thick base.
- Electrolysis: Set to 3.2-3.6V

Results:



- Current was low, unsure if due to cathode or anode limitation.
- Cathode became coated with "black" crud that was tested to be K3AlF6.
- No oxygen detected during electrolysis.
- Anode was partially consumed.

Conclusions:

- Need to sheath anode to create separate volume to sample from using the mass spectrometer.
- Fe impurities found on anode, could be from steel crucible.

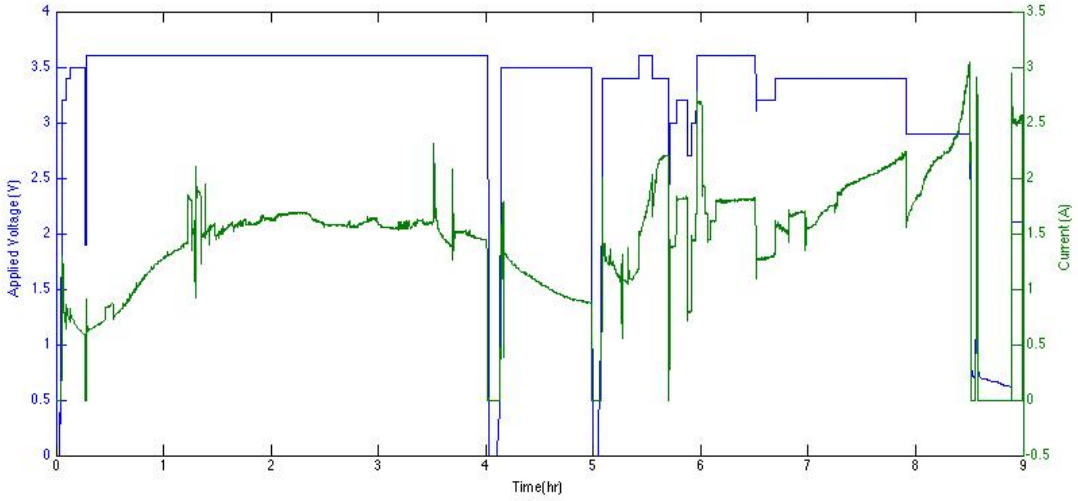
Experiment 5

Goal: Sheath anode using alumina tube and purge volume with argon at 5 sccms. Sample gas stream with mass spec during electrolysis. Super-saturate flux to minimize sheath dissolution.

Parameters:

- Anode: New 1/4” inert anode rod casted inside 3/4”OD alumina sheath.
- Cathode: TiB_2, 1/4” steel rod connection.
- Bath: New 3kg CR=1.8 KF-AlF3 with 400g alumina.
- Bath depth: 7” total.
- Crucible: Steel crucible 5” OD 12” height, ~1018 steel secondary container inside 5” inner height 6” OD welded to ½” thick base.
- Electrolysis: Set to 3.2-3.6V

Results:



- Detected oxygen at anode.
- Anode looked black after electrolysis, surface was dark and we detected Fe and Al on surface. Unsure if it is from the flux or alloying.

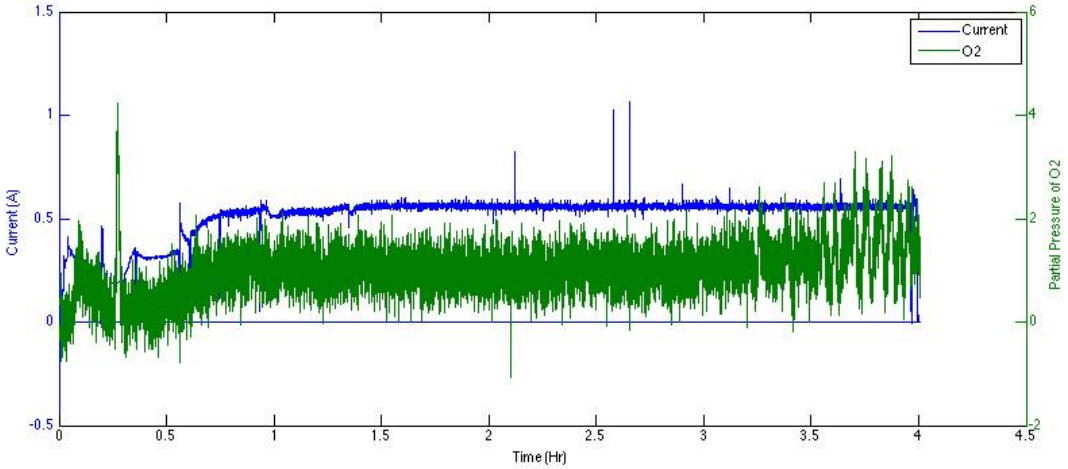
Experiment 6

Goal: Made oxygen, repeat same procedure on anode. Have not plated aluminum on the first three experiments, will add LiF to see the effect this has. We believe LiF will help keep the KF ratio low at the cathode to help lower the ratio locally.

Parameters:

- Anode: New 1/8” anode wire casted inside 3/4”OD alumina sheath.
- Cathode: TiB_2, 1/4” steel rod connection.
- Bath: Same from #3, 3kg CR=1.8 KF-AlF3 with 400g alumina, with 100g LiF.
- Bath depth: 7” total.
- Crucible: Steel crucible 5” OD 12” height, ~1018 steel secondary container inside 5” inner height 6” OD welded to ½” thick base.
- Electrolysis: Set to 3.0-3.2V

Results:



- Made oxygen at the anode. It was consistent with current, no CO/CO2 detected.
- Still very low currents, will need to double the anodes used to see if the current scales proportional to the anode surface area.

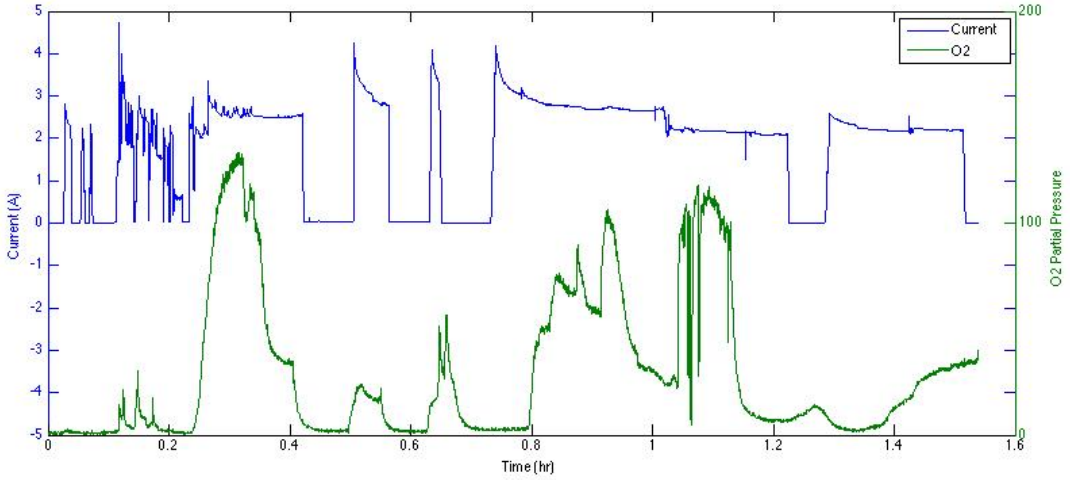
Experiment 7

Goal: Made oxygen, repeat same anode procedure using two anodes. Focus on cathode. Have not plated aluminum on the first four experiments, will use a sheath around cathode with a BN plug at the bottom to collect Al if made. Added 30g AlF3 inside cathode sheath to locally lower cryolite ratio.

Parameters:

- Anode: New 1/8” wire casted inside 3/4”OD alumina sheath.
- Cathode: TiB_2, 1/4” steel rod connection.
- Bath: Same from #3, 3kg CR=1.8 KF-AlF3 with 400g alumina, with 100g LiF and 30g AlF3.
- Bath depth: 7” total.
- Crucible: Steel crucible 5” OD 12” height, ~1018 steel secondary container inside 5” inner height 6” OD welded to ½” thick base.
- Electrolysis: Set to 3.0V

Results:



- Made oxygen at the anode. Pulsed the voltage to show the oxygen decrease when electrolysis was stopped. This is clear confirmation of the anode being inert since it is generating oxygen.
- Current was higher when using second anode. Sheath acts as a mass-transport overpotential.
- No aluminum collected. Could be due to not passing sufficient current during the test.

BU Flux Electrolysis Experiment

Methods

A laboratory scale SOM electrolysis employing liquid aluminum cathode and high CaO content flux were performed. Figure 131 shows the SOM cell schematic.

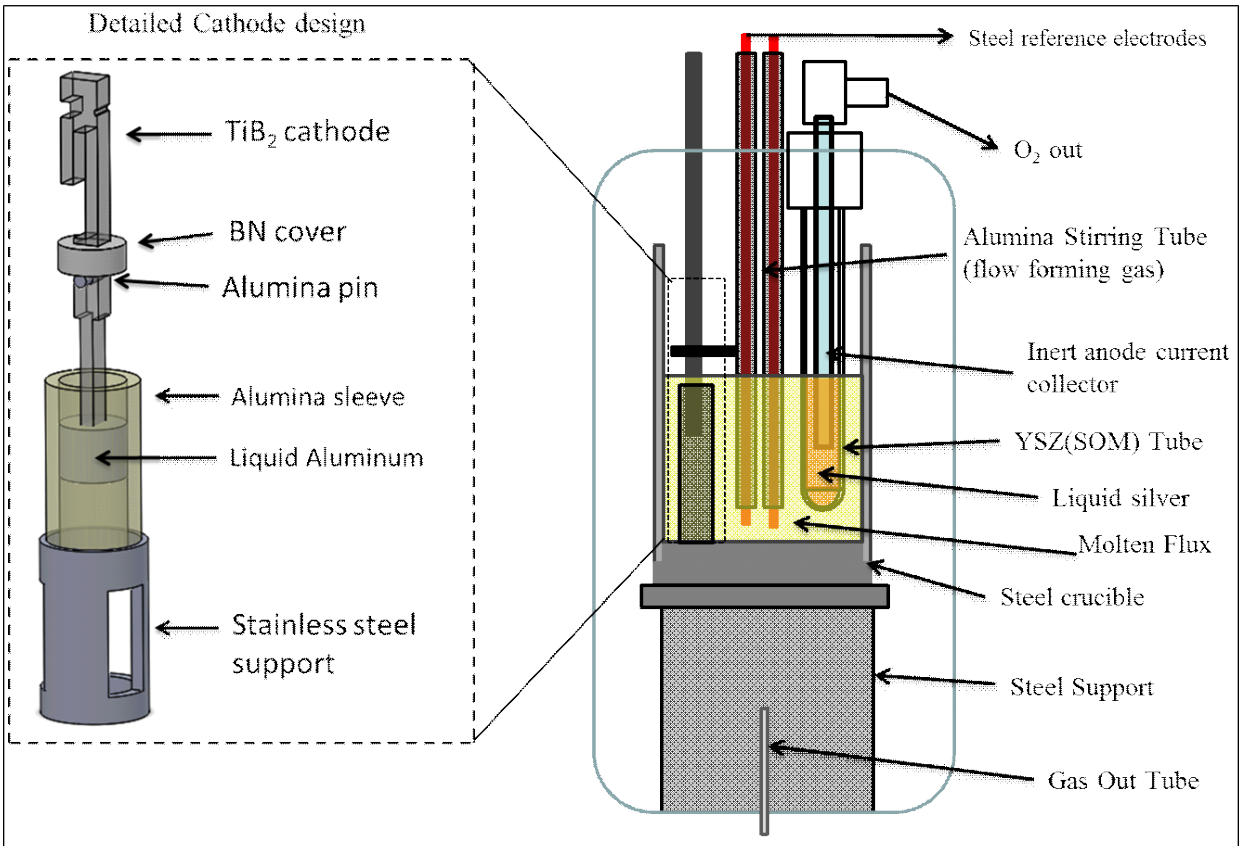


Figure 131: Schematic of electrolysis cell for SOM aluminum experiment.

The flux composition is eutectic MgF_2-CaF_2 flux with 9wt% Al_2O_3, 12 wt% CaO, and 4wt % YF_3. The electrolysis was performed at 1200°C in forming gas (95%Ar-5%H_2) atmosphere. An Al_2O_3 dielectric tube serves as a cathode separator as well as a source of Al_2O_3. Thus the activity of Al_2O_3 near the cathode maintained a high level during the electrolysis. The Al_2O_3 dielectric tube was inserted in a stainless steel support welded to the bottom of the stainless steel crucible. The stainless steel support has multiple opening at the side to allow flux go through the support and contact with the cathode. The flux was pre-melted in the steel crucible. 4.5g of Al shot was added prior to the electrolysis as the cathode. When the applied DC potential exceeded the dissociation potential of Al_2O_3, liquid aluminum was produced at the cathode. The cathode current collector has the same design with previous electrolysis using TiB_2. A stainless steel crucible welded to a stainless steel rod was connected with the TiB_2. The crucible part of the current collector contained a layer of liquid silver that formed a liquid connection with

the TiB_2 cathode. During the electrolysis, the cathode current collector was immersed in the liquid aluminum cathode.

The electrochemistry behavior of the cell was characterized by potentiodynamic scan (PDS), electro interstitial scans (EIS) and potentiostatic hold. Post-experimental analysis was performed using scanning electron microscopy (SEM) and Energy-dispersive X-ray spectroscopy (EDS).

Results and Discussion

Prior to electrolysis, a potentiodynamic scan was performed between the cathode and the anode as shown in Figure 132. The only dissociation potential identified from the scan at 2V was that of alumina which appears to close to the Nernst potential (2.08V) for the reaction Al_2O_3 = 2 Al + 3/2 O_2(g) at 1200ºC. A pre-electrolysis was perfomed at 2V for 3 hours in order to remove the impurities in the flux. The PDS after the pre-electrolysis (Figure 133) shows that the leakage current prior to the alumina dissociation is negligible, indicating that all impurities were removed by the pre-electrolysis.

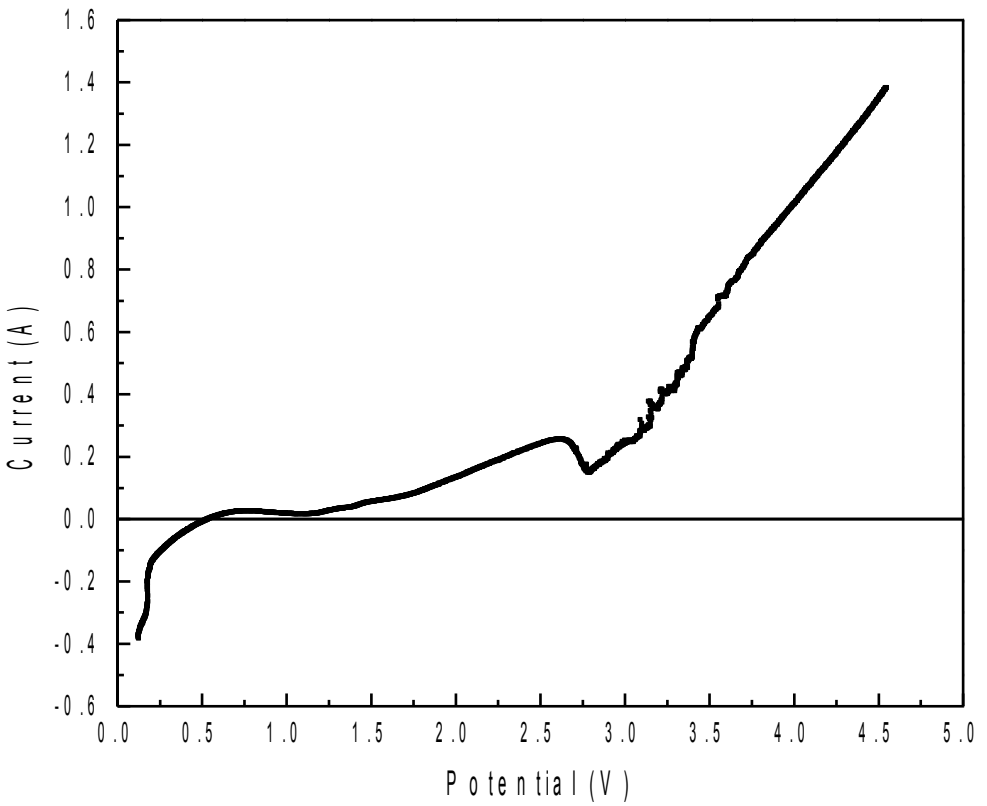


Figure 132: Potentiodynamic scan before electrolysis.

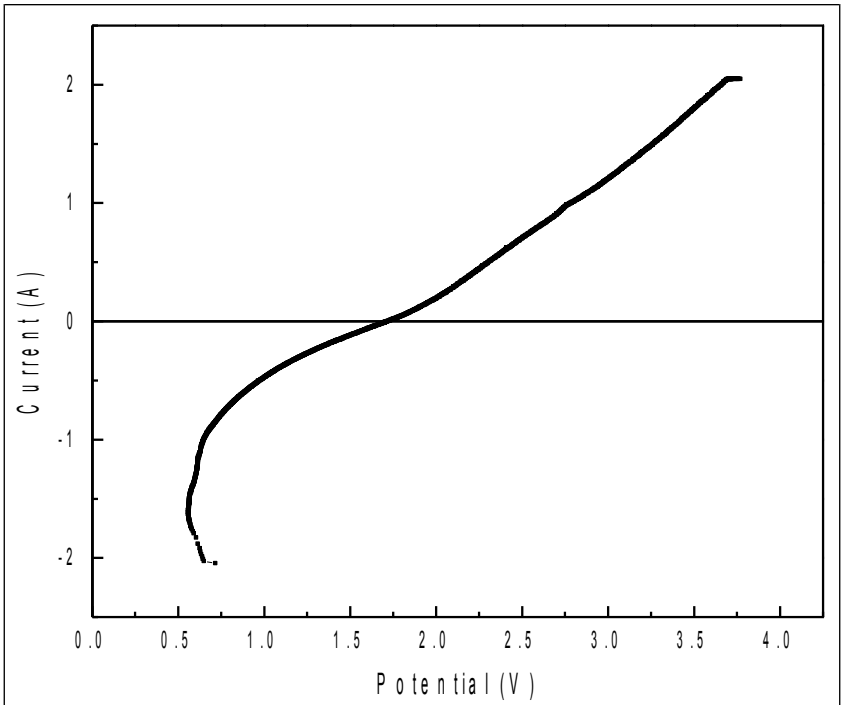


Figure 133: Potentiodynamic scan after pre-electrolysis.

Electrolysis at 3V was performed for 8 hours. Current efficiency was monitored. Figure 134 shows the current and current efficiency as a function of time.

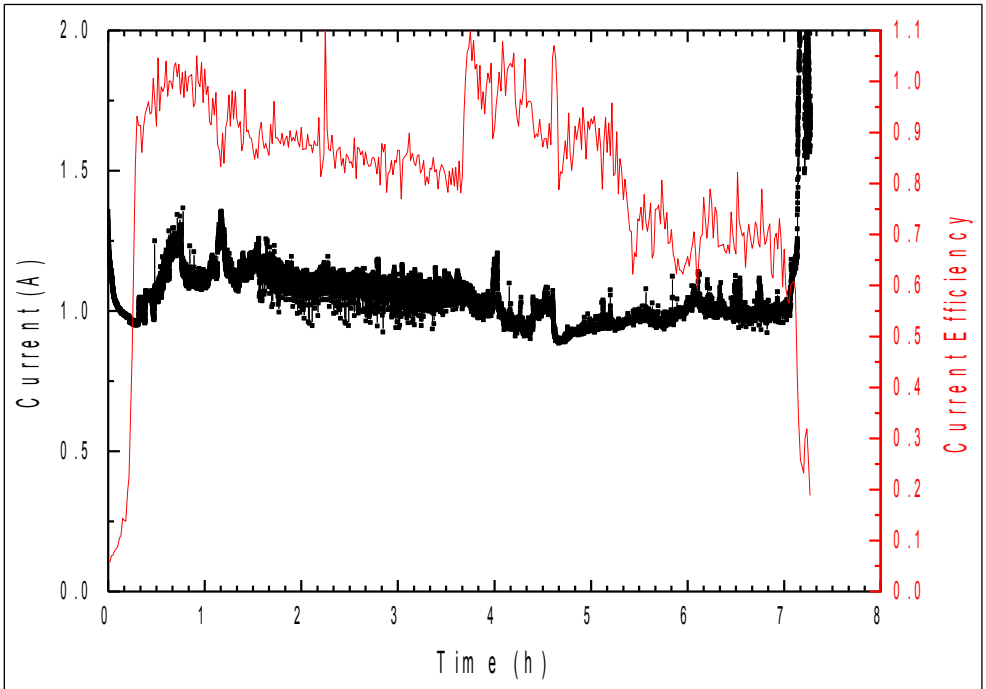


Figure 134: Current and current efficiency during the first electrolysis.

At the end of the electrolysis, the current increased significantly over a very short period of time. At the same time, the current efficiency dropped to ~20%. We believe that the aluminum at the cathode leaked out of the alumina dielectric tube and shorted the cell. The electrolysis was stopped when this shorting phenomena was observed. The system was held at the operating temperature for 3 hours to let cell to equilibrate. During that time, the majority of the aluminum leaked out of the cathode area was likely oxidized in the flux due to high oxygen partial pressure. Then the second electrolysis was performed at 3V for 8 hours. Figure 135 shows the current and current efficiency as a function of time during the 2nd electrolysis.

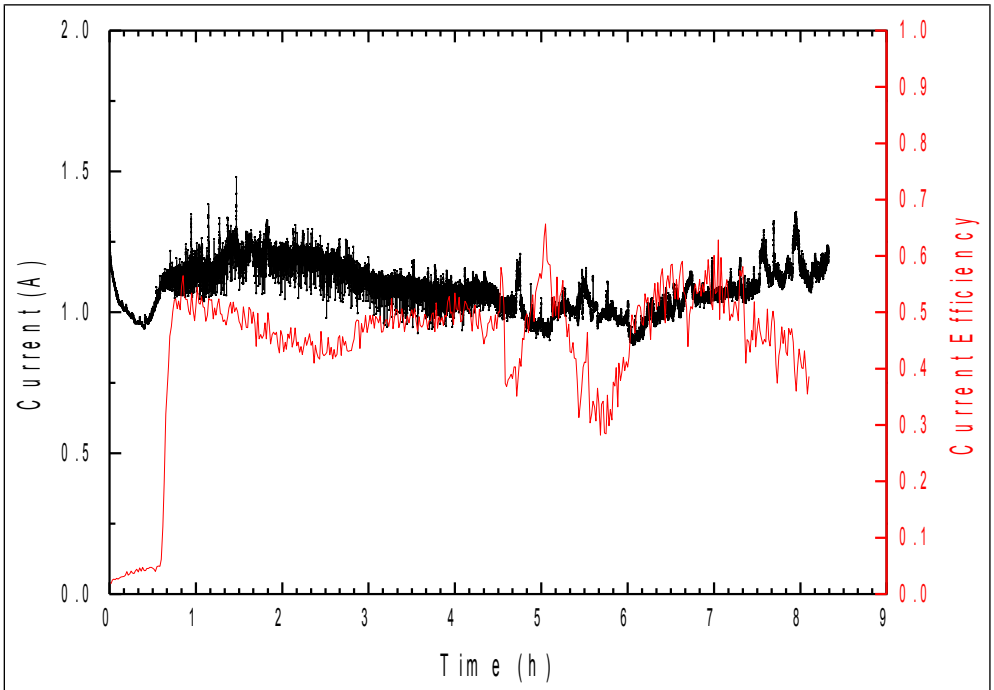


Figure 135: Current and current efficiency during second electrolysis.

EIS scans were performed between the cathode and the anode after the pre-electrolysis and after the 2nd electrolysis (Figure 136). The EIS scan showed that the total resistance of the cell was around 0.5~0.6Ω. This results agrees with the total cell resistance measured in previous SOM runs.

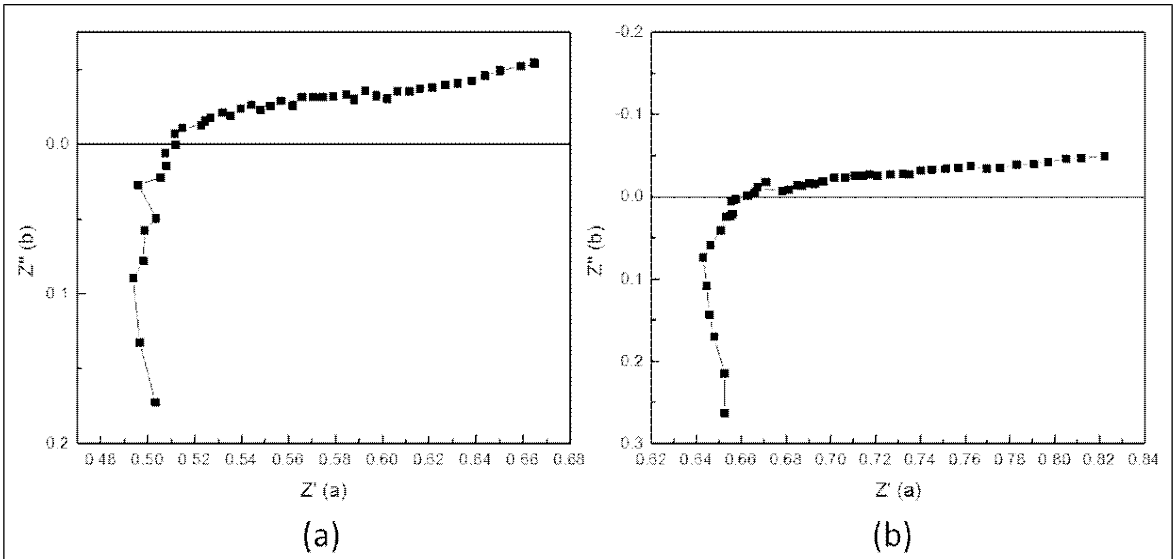


Figure 136: EIS scans: (a) after pre-electrolysis, (b) after the 2nd electrolysis.

After SOM electrolysis, aluminum was found near inside the cathode alumina dielectric tube. Quantitative EDS analysis showed that the aluminum product was 88wt% pure, with less than 12wt% magnesium impurity. The magnesium impurity increased compared with previous SOM runs. We believe that it was due to the change of the cathode design. In the previous SOM run, we noticed that the aluminum was evaporating from the liquid aluminum cathode. In order to prevent the aluminum evaporation and increase the yield of the product, the BN cover fully covered the top of the alumina dielectric tube. In fact, after the electrolysis, we found that the BN cover was fused with the alumina dielectric tube. While this design prevented possible aluminum evaporation, it also prevented the magnesium impurity to evaporate from the aluminum product. Since the magnesium did not evaporated from the aluminum product, it may dissolve in the flux. This also contributed to the low current efficiency in the 2nd electrolysis, after a significant amount of aluminum as well as magnesium was produced.

The YSZ membrane stability was analyzed after the experiment using SEM and EDS line scan shown in Figure 137. It is shown from the line scan that there is no yttria diffusion from the YSZ membrane. The bulk of the YSZ was stable after the electrolysis. There is minor no flux attack occurred at the interface between the flux and the membrane. This is likely due to the high current at the end of the 1st electrolysis.

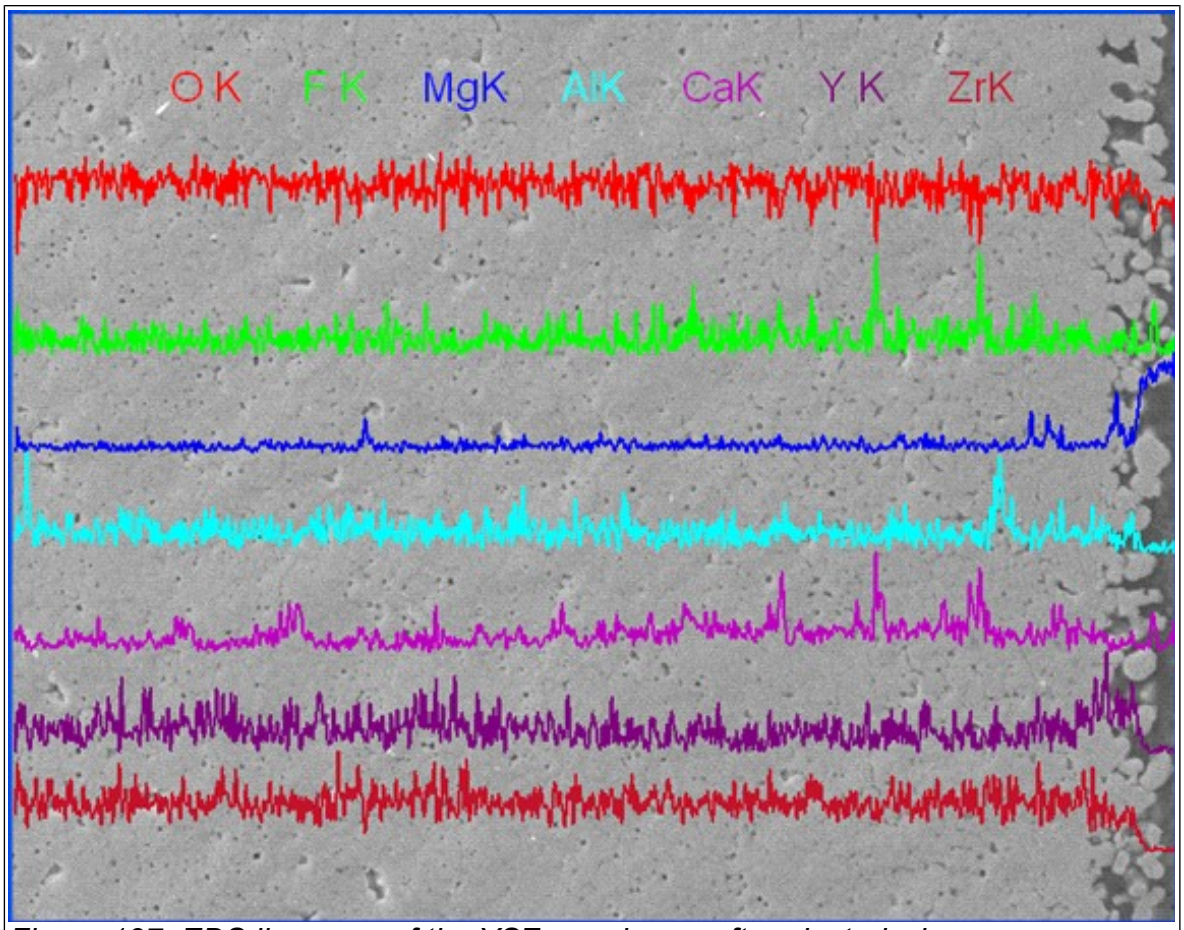


Figure 137: EDS line scan of the YSZ membrane after electrolysis.

Experiment 021: Switching experiment

Overview The goal of this experiment was to test the performance of two 3/4" OD 1.9 mm thick 8YSZ (8 mol% Y_2O_3) anodes while pulsing at constant current and healing in-situ. The experiment ran for roughly 5 hours and passed 12.8 Ah and 13.2 Ah through anode A and B respectively. A graphite electrode was used to measure the system voltages against and to heal the anodes periodically.

Experimental Parameters

Cathode: TiB_2 (0.5" L x 0.2" W x 3" H) with 1/2" dia. 1018 steel rod and screw connection

Cathode immersion: 1" below alumina sheath immersed 2" below the flux interface

Anode(s): Two 3/4" OD ~2mm thick 8YSZ tube with 150g mixed silver particles (0.05"-0.15" diameter)

Anode current collector: 30" L x 1/8" dia. silver wire (99.99%)

Anode immersion: 6"

Salt composition: KF-AlF_3 – 1.3 cryolite ratio with 14wt% saturated alumina

Salt Height: 4.5"

Crucible: Isomolded graphite – 5" OD x 4.5" ID x 12" High (11.5" Deep)

Crucible temperature: 720ºC

Results and Discussion

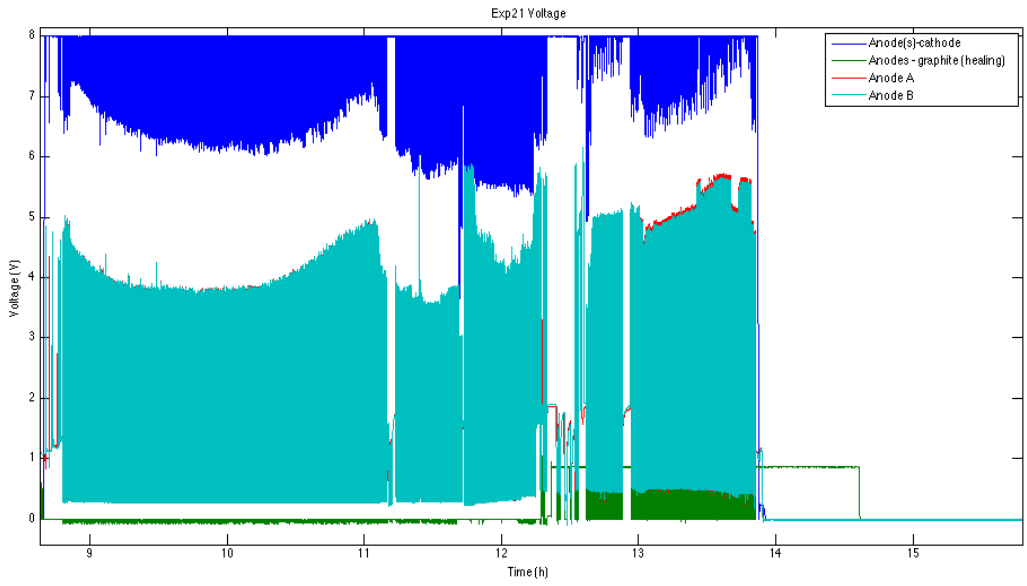


Figure 138: Voltages

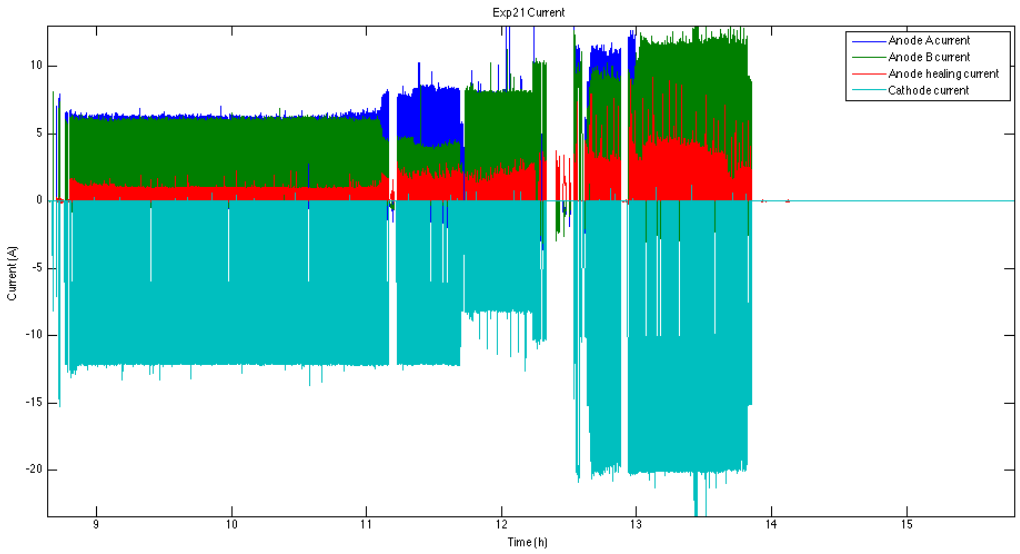


Figure 139: Currents

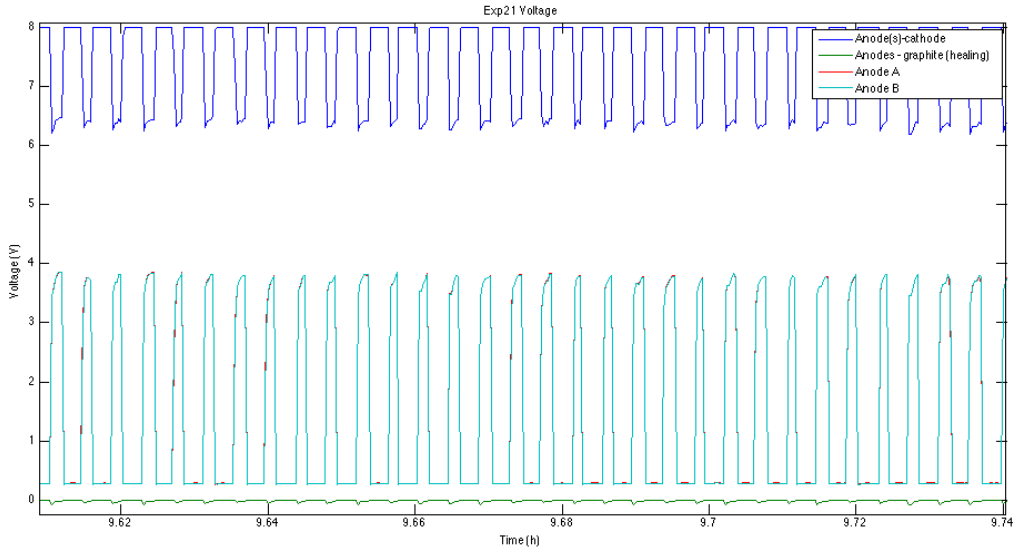


Figure 140: Voltage profiles during pulsing zoomed in.

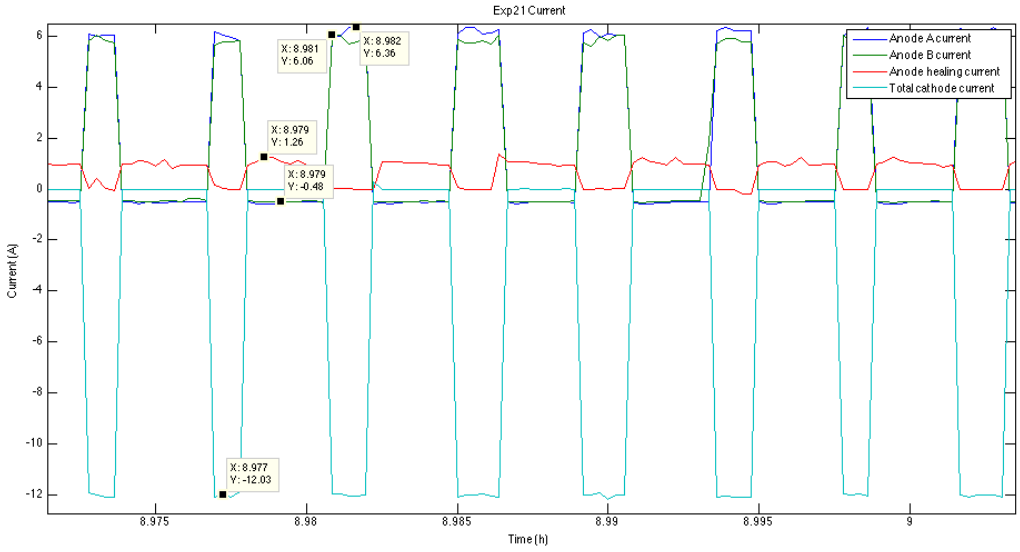


Figure 141: Current profile during pulsing.

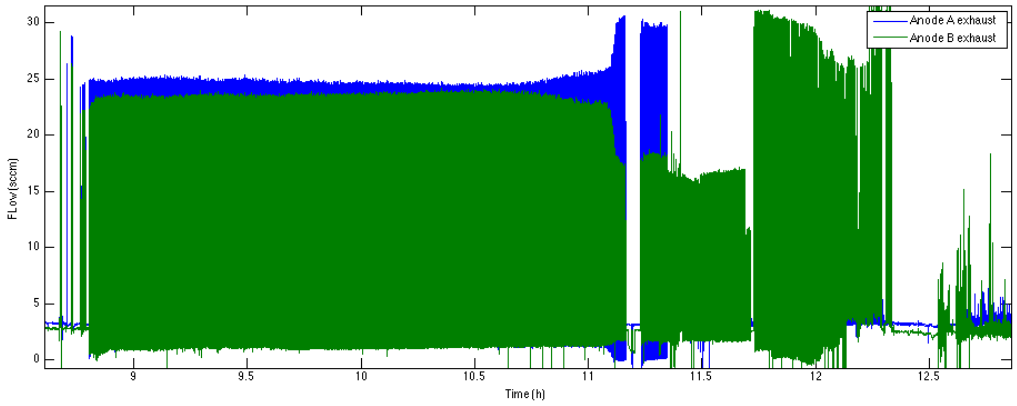


Figure 142: Anode A (blue) and anode B (green) exhaust flow.

Anode Both of the anodes had visible cracks post-electrolysis (Figures 143 and 144). A control sample was taken from the top of each tube for comparison with the active area.

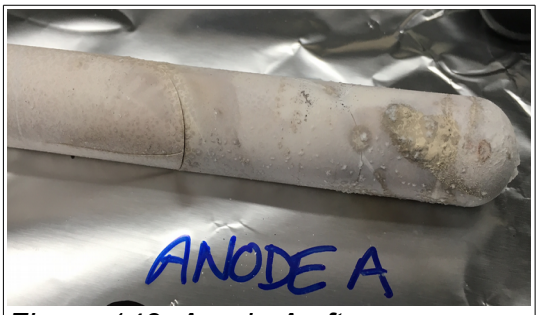


Figure 143: Anode A after electrolysis

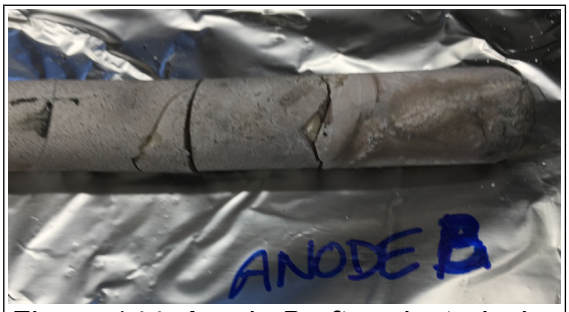


Figure 144: Anode B after electrolysis

After cross-sectioning anode A, there was no visible banding in the tube. Table 16 highlights the composition change across the membrane. There was a 10% decrease in zirconium, 5% decrease in yttrium and a 30% increase in oxygen. Presence of aluminum and potassium were also seen on the outer edge.

Anode B appeared to be much more degraded. EDS confirmed this (Table 17) by showing over a 50% decrease in zirconium and 90% decrease in yttrium concentration on the inner edge. This is due to the flux penetrating the membrane after it broke and dissolving the yttrium from the tube.

Table 16: Anode A cross section EDS analysis of the membrane post-electrolysis

	Zr	O	F	Al	K	Y	Ag
Control	66.02	21.05	0	0	0	12.93	0
Probe 1	58.94	27.94	0.89	0	0	12.23	0
Probe 2	59.93	27.67	0.45	0	0	11.95	0
Probe 3	60.1	27.36	0.33	0	0.11	12.1	0
Probe 4	60.02	27.33	0.38	0	0	12.27	0
Probe 5	60.6	26.58	0.51	0	0.09	12.22	0
Probe 6	60.27	26.57	0.74	0	0.03	12.4	0
Probe 7	60.58	26.36	0.85	0	0.03	12.18	0

	Zr	K	Al	F	O	Y	Ag
Control	66.79	0	0	0	20.7	12.51	0
Probe 1	29.57	9.49	8.55	30.81	19.93	1.65	0
Probe 2	59.73	0.15	0	6.48	24.82	8.82	0
Probe 3	62.27	0	0	0.49	26.71	10.53	0
Probe 4	60.96	0	0	0.49	28.12	10.43	0
Probe 5	59.98	0	0	0.15	29.16	10.72	0
Probe 6	58.65	0.16	0	0.12	30.98	10.09	0
Probe 7	58.54	0	0	0.12	31.24	10.11	0
Probe 8	58.54	0	0	0.13	31.2	10.09	0

Table 17: Anode B cross section EDS analysis of the membrane post-electrolysis

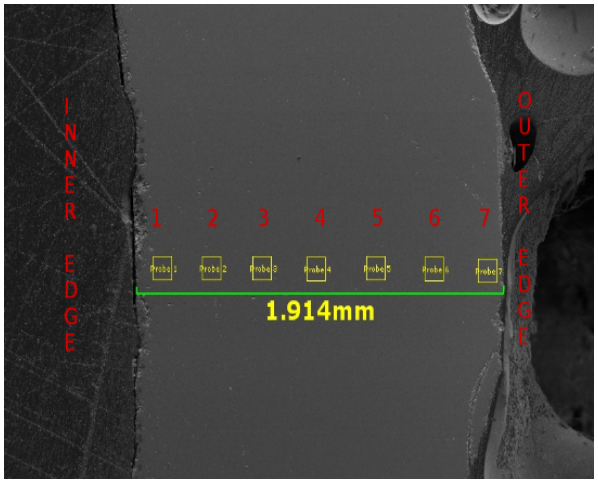


Figure 145: Anode A post-electrolysis.

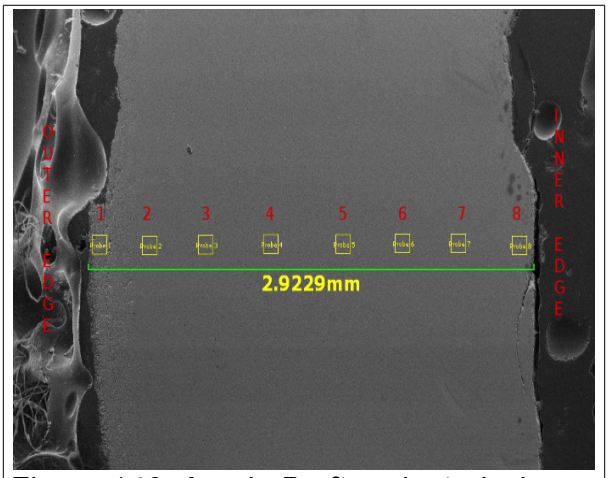


Figure 146: Anode B after electrolysis with no visible banding.

Cathode The cathode was removed from the cell during electrolysis to find a metallic coating (Figure 147). This metal was scraped off and analyzed in the ICP for impurities. Three replicate samples were run in the ICP and a report was generated that takes the average weight percentages of metals detected (Table 18).

The aluminum metal was 98.94% pure, with zirconium constituting the highest impurity at 2.43%. Trace amounts of iron and potassium were detected in the metal as well.

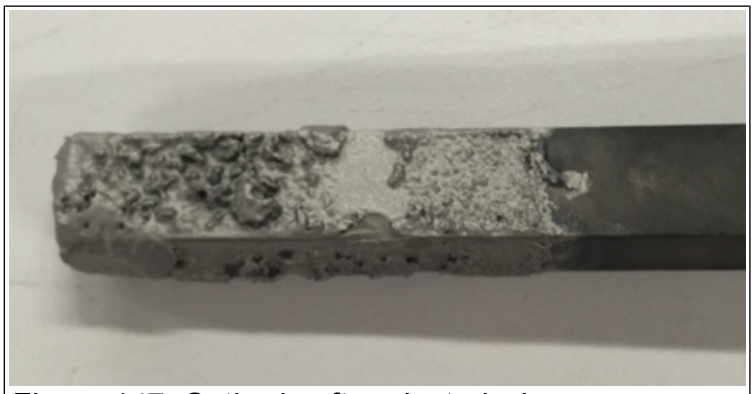


Figure 147: Cathode after electrolysis.

	Material:	Aluminum Metal			
	Experiment:	21			
Sample I.D.	**Weight (g)**	**Volume (mL)**	**Dilution**	**Element**	**Conc. (ppm)**
Replicate 1	0.0956	50.00	10	Al	9.772E+05
				Fe	8.421E+02
				Zr	2.382E+04
				Y	0.000E+00
				K	7.592E+02
Replicate 2		50.00	10	Al	9.962E+05
				Fe	8.213E+02
				Zr	2.256E+04
				Y	0.000E+00
				K	5.762E+02
Replicate 3		50.00	10	Al	9.948E+05
				Fe	8.426E+02
				Zr	2.659E+04
				Y	0.000E+00
				K	7.427E+02
	Al	*Fe*	*Zr*	*Y*	*K*
Average (%)	98.94	0.08	2.43	0.00	0.07

Table 18: ICP analysis of metal sample taken off of TiB2

Experiment 027-029: Thick (>2mm) anode electrolysis

Overview The goal of this experiment was to test the performance of 3/4" OD 1.6mm, 3mm, and 4mm thick 8YSZ (ZrO_2-8 mol% Y_2O_3) anodes. Experiments 027 and 028 used the same molten salt and hot-swapped both anodes in in quad-cell 1. Experiment 029 used the same molten salt composition but was run in quad-cell 3.

Experimental Parameters

Cathode: TiB_2(0.5" L x 0.2" W x 3" H) with 1/2" dia. 1018 steel rod and screw connection

Cathode immersion: 1" below alumina sheath immersed 2" below the flux interface

Anodes: 3/4" OD 3 um thick 8YSZ tube with silver particles (0.05"-0.15" diameter)

Anode current collector: 30" L x 1/8" dia. silver wire (99.99%)

Anode immersion: 4"

Salt composition: KF-AlF_3 – 1.3 cryolite ratio with 14wt% saturated alumina

Salt Height: 4.5"

Crucible: Alumina-lined isomolded graphite – 5" OD x 4.5" ID x 12" High (11.5" Deep)

Crucible temperature: 720ºC

Exp027 Results and Discussion

Figure 148 shows the anode-cathode voltage and anode flow during constant current pulsing. The pulse sequence used was a 5 second 1 A pulse followed by 100s of open circuit relaxation. The system was controlled by a voltage cutoff value of 9V for which after the 20th pulse it became active. In other words, the first 20 pulses reached the 5s limit and after it was controlled by the 9V cutoff for which it took only 0.8s by the 100th pulse (Figure 150).

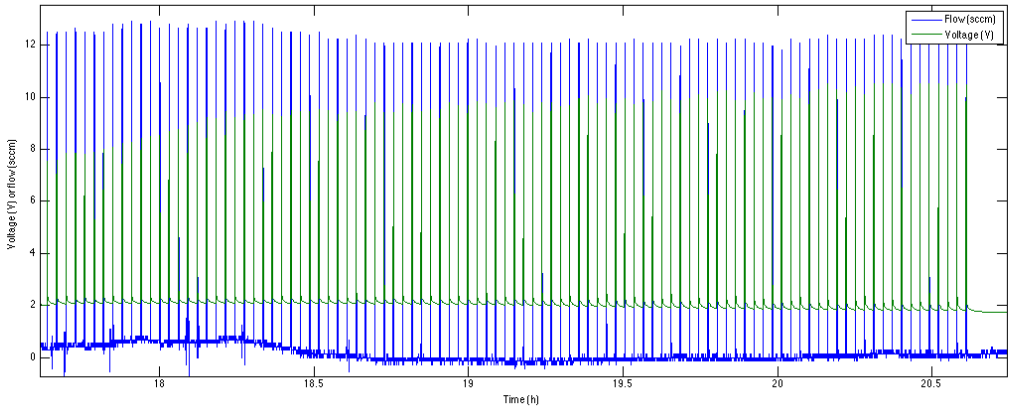


Figure 148: Anode-cathode voltage and anode exhaust flow during 1A pulsing.

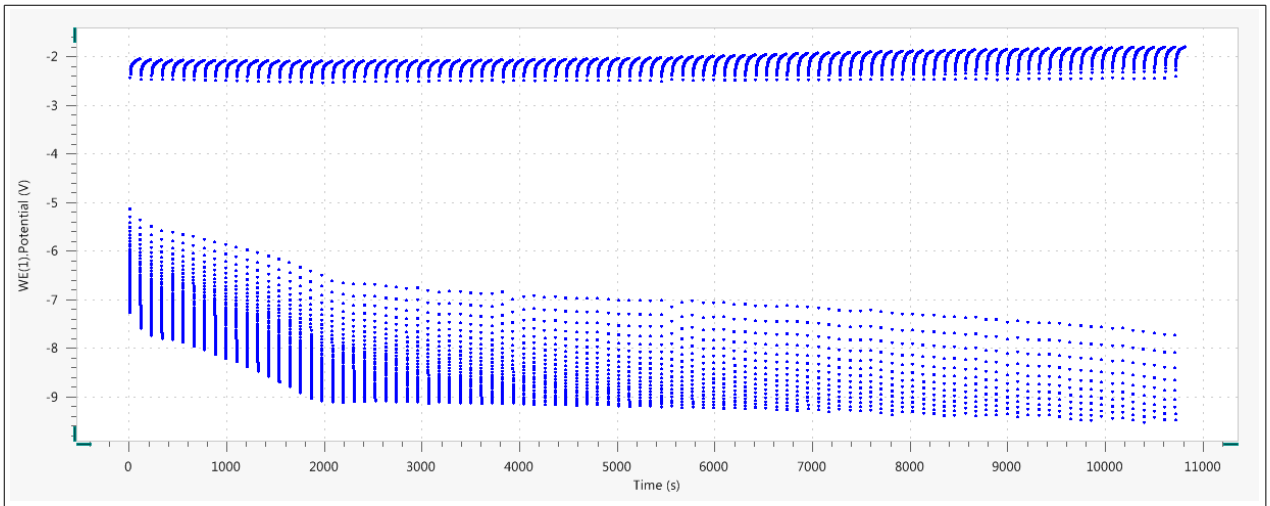


Figure 149: Voltage pulse sequence

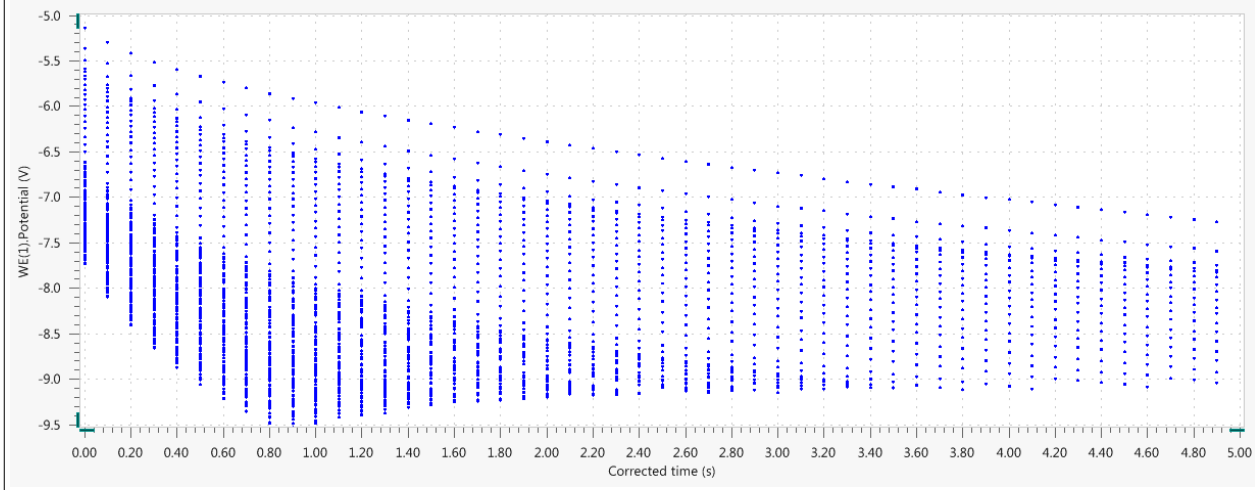


Figure 150: Anode-cathode evolution of each current pulse.

Membrane analysis Anode tube thickness = 4.00 mm

- Loss of Y (max = 50%) on inner (up to 15 µm inward) and outer (up to 50 µm inward) edges small band (~ 50 µm thick) on outer edge
- Rind of flux on outer edge
- Inner edge more porous visually than rest of tube

Figure 151: Exp027 anode after electrolysis.

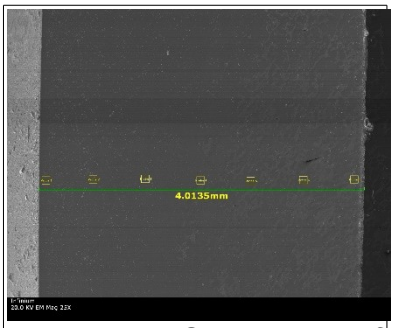


Figure 152: SEM image of bulk anode

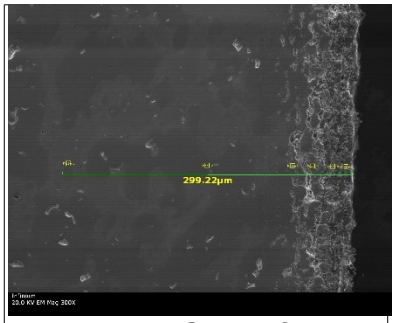


Figure 153: SEM of anode-flux interface.

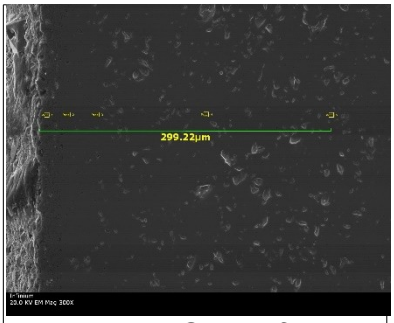


Figure 154: SEM of anode-silver interface

	Zr	O	F	Al	K	Y	Ag
Bulk Anode	71.5	13.33	0.51	0	0	14.66	0
Anode/Flux Interface	73.96	12.95	2.95	2.15	2.65	5.34	0
Anode/Silver Interface	71.52	14.4	3.24	1.5	1.08	7.87	0.4

Table 19: EDS analysis of membrane for Exp027

Exp028 Results and Discussion

1A 60 second pulses with 10V cutoff

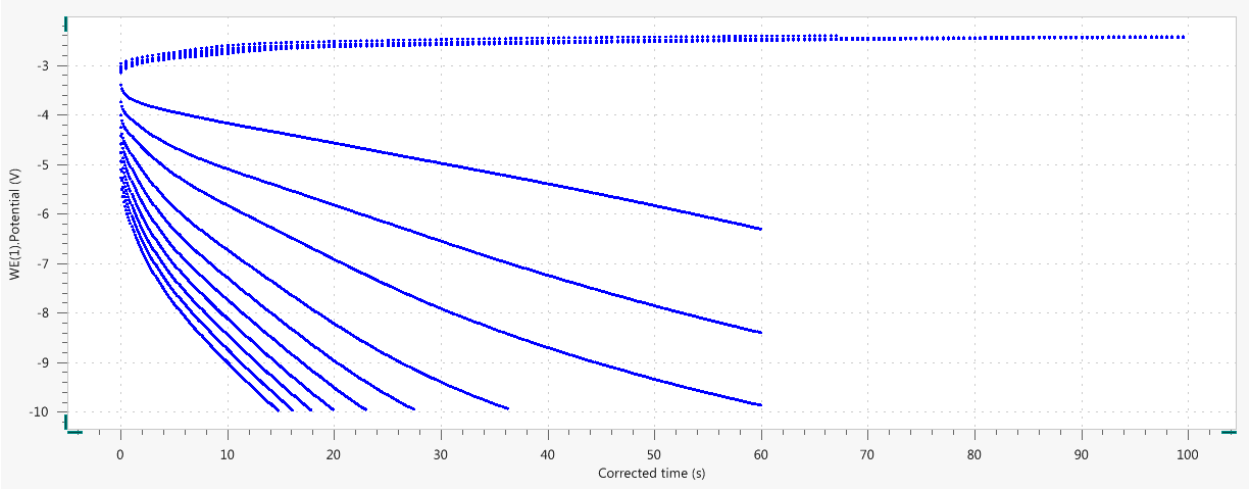


Figure 155: Voltage profiles during 1A pulsing.

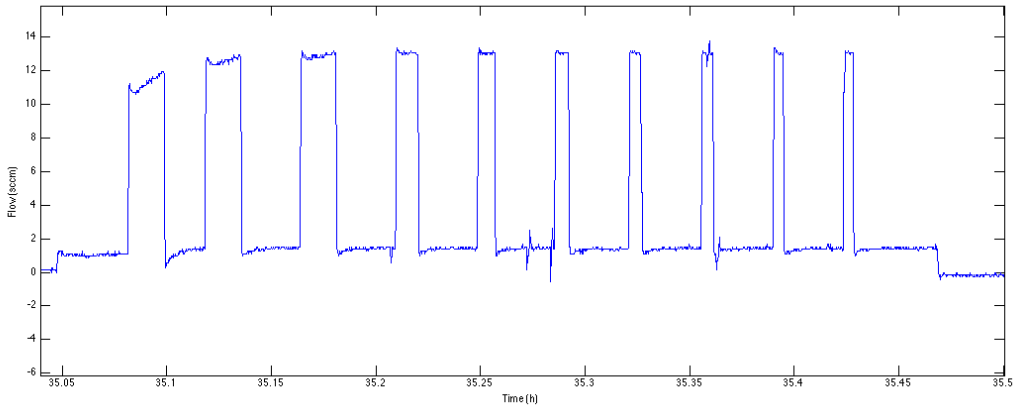


Figure 156: Anode flow corresponding to 1A pulses

0.5A 120 second pulses

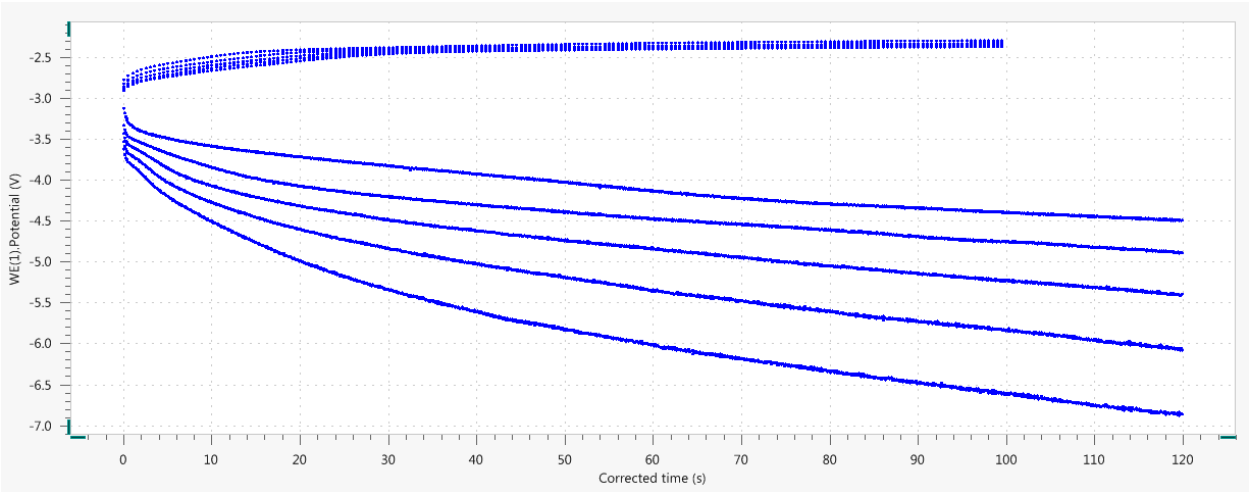


Figure 157: 0.5A 120s pulses after healing.

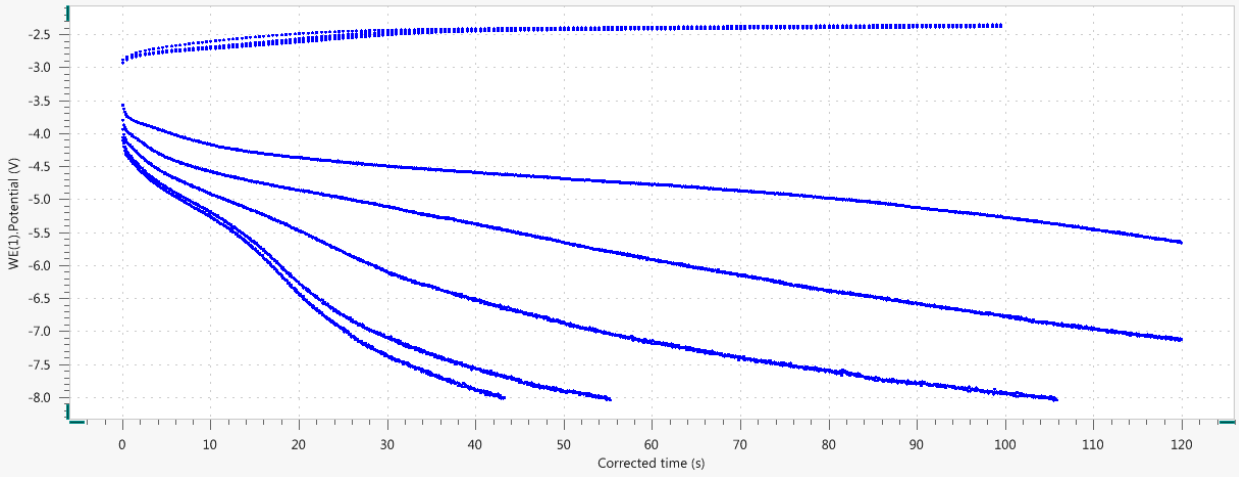


Figure 158: 0.5A 120s pulses with no healing.

0.5A 30 second pulse

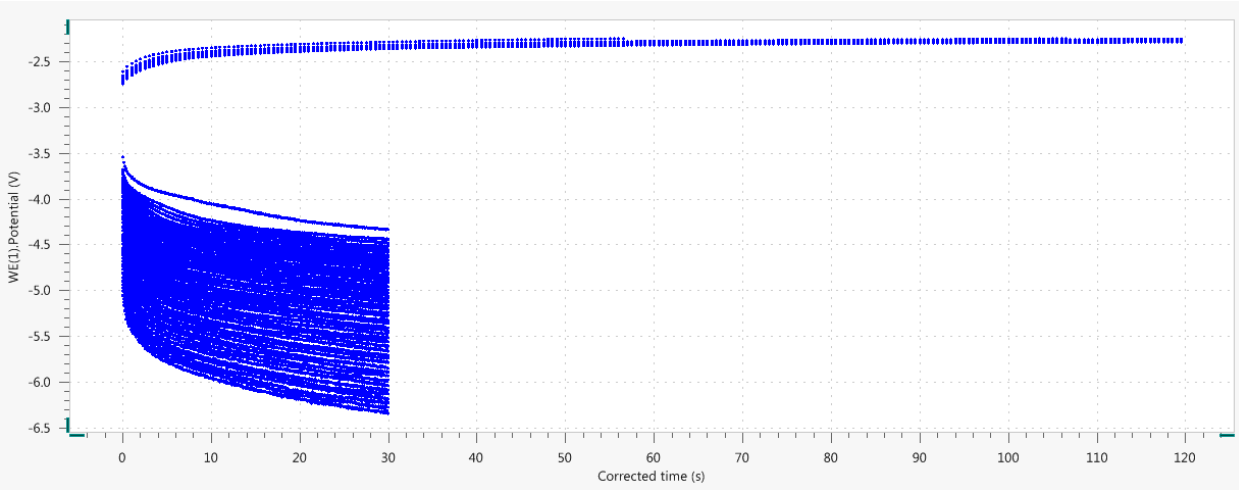


Figure 159: 0.5A 100 30s pulses

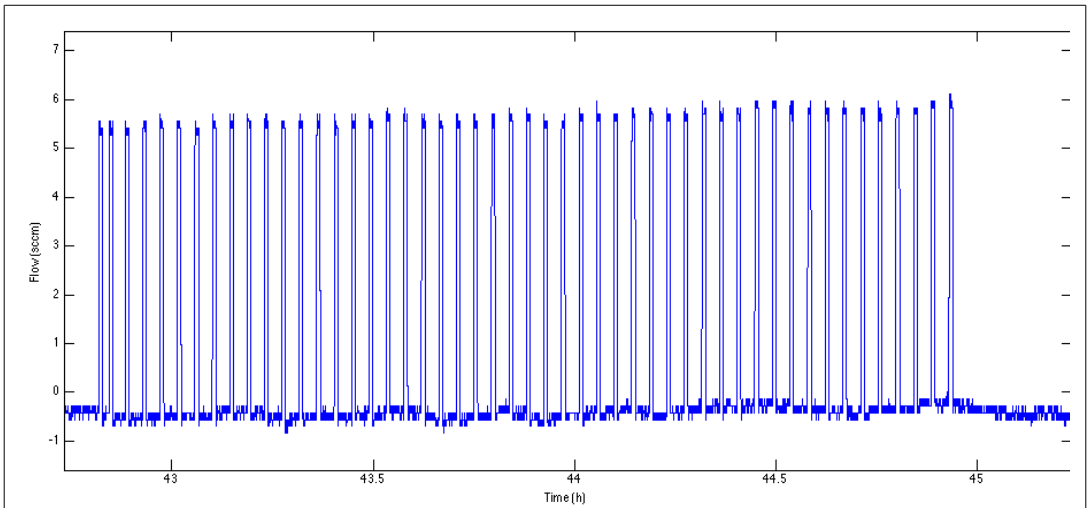


Figure 160: Anode flow during the 0.5A 100 30s pulse sequence.

Membrane analysis

Anode tube thickness = 3.00 mm

- A0: No change in membrane composition
- A1:
  - No loss of Y on inside edge
  - Rind of flux on outside of tube with visible cracking
  - Loss of Y (max = 50%) up to 150 μm inward from outer edge
- A2: Significantly wider at bottom → 4 mm
  - Visible crack ~500 μm inward from outside edge
  - Rind of flux on outside of tube

- Loss of Y (max = 50%) up to 50 µm inward from outer edge

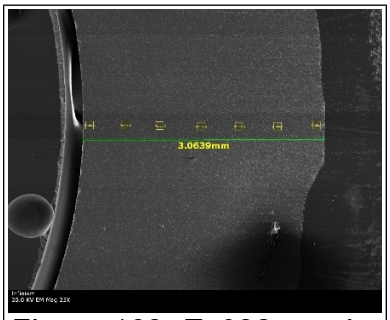


Figure 162: Ex028 anode cross section.

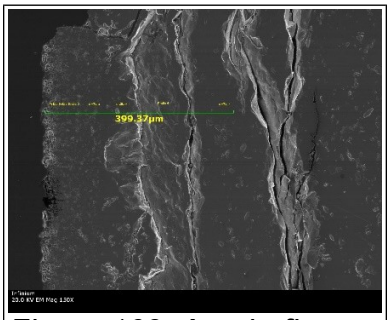


Figure 163: Anode-flux interface

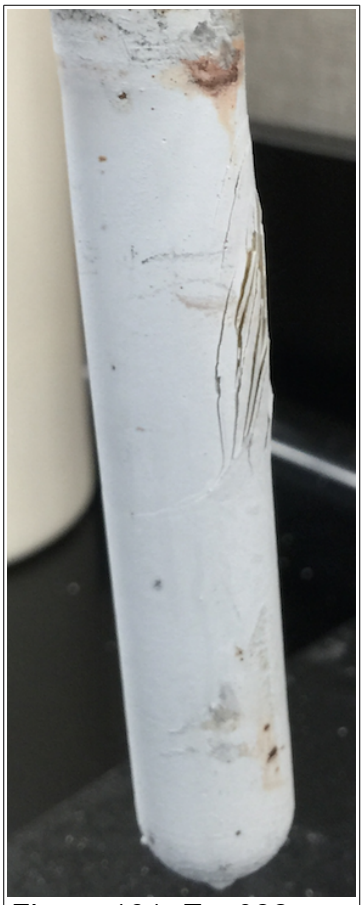


Figure 161: Exp028 anode after electrolysis. Outer surface delaminated while cooling down after hot-swap.

	Zr	O	F	Al	K	Y	Ag
Control	67.8	17.61	0	0	0	14.59	0
Anode/Flux Interface	67.96	15.94	5.51	2.05	2.9	5.64	0
Anode/Silver Interface	68.66	15.81	0.41	0	0	15.11	0

Table 20: EDS analysis of membrane for Exp028

Exp029-Results and Discussion

During the initial startup of experiment 029 water was observed condensing on the cell exhaust. This was a result from improperly drying the flux. The system was then cooled to 300C and allowed for the salts to dry before ramping back up to 720C. This experiment was not conclusive due to the water.

Figure 164: Exp029 anode after electrolysis with damage likely due to water attack.

Membrane analysis

1.6 mm tube thickness
A0: No change in membrane composition
A1: Some loss of Y with presence of F (~ 15 mol %) on outer edge (found up to 300 μm inward from edge)
Have loss of Y (~ 50%, found within small outer band) and presence of Al & K on inner edge
Noticeable visible band (smooth surface = presence of F in tube)
A2: Have visible band
Outside edge has loss of Y (~ 50%, found up to 25 μm inward from edge) with presence of Al & K
Found nodule of YSZ with high F (20 mol%) and presence of Al & K across membrane (locations match visible banding from outside of tube)
Bottom of tube is very degraded
Flux penetration into tube
~ 50% addition to tube thickness from Flux and Al_2O_3
Fluoride present throughout cross section of tube

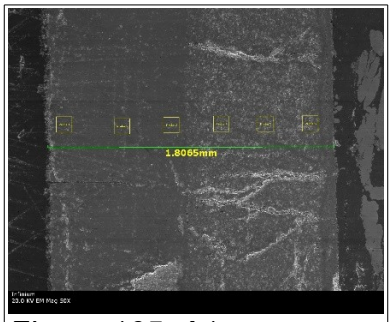


Figure 165: A1

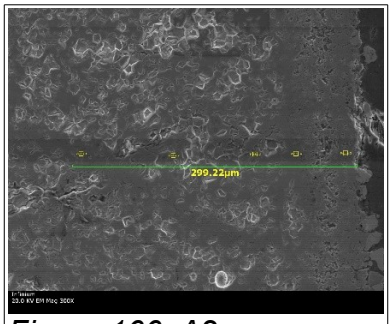


Figure 166: A2

	Zr	O	F	Al	K	Y	Ag
Control	66.77	19.52	0	0	0	13.7	0
Anode/Flux Interface	70.52	10.44	3.2	0	0.19	15.64	0
Anode/Silver Interface	69.58	15.11	1.44	0	0	13.57	0.31

Table 21: EDS analysis of membrane for Exp029

Experiment 030: 1st Thin (sub-mm) Anode. High Purity Al Produced Using an 8YSZ Anode

Overview

The goal of this experiment was to test the performance of a 3/4" OD 950 um thick 8YSZ (8 mol% Y_2O_3) anode. The experiment ran for roughly 20 hours and passed 21 Ahrs. Titanium diboride was used as the cathode and a steel rod with a screw connection as the current collector. Pre-dried and pre-molten KF- AlF_3 with a cryolite ratio (CR) of 1.3 and saturated with alumina was loaded into an alumina lined isomolded graphite crucible and lowered into a closed stainless steel shell purged under argon. A PLC was used to control the power supply and furnace temperature. The voltage was set to 3.2V initially, then to 3.5V to run overnight. This yielded a current ranging between 1-1.5 A. Aluminum metal was verified after passing 2 Ah (Figure 167) which was the first time we made aluminum at a potential lower than 8V. A plot of the electrolysis is outlined below in Figure 168.

Figure 167: Aluminum deposited on cathode after 2Ah.

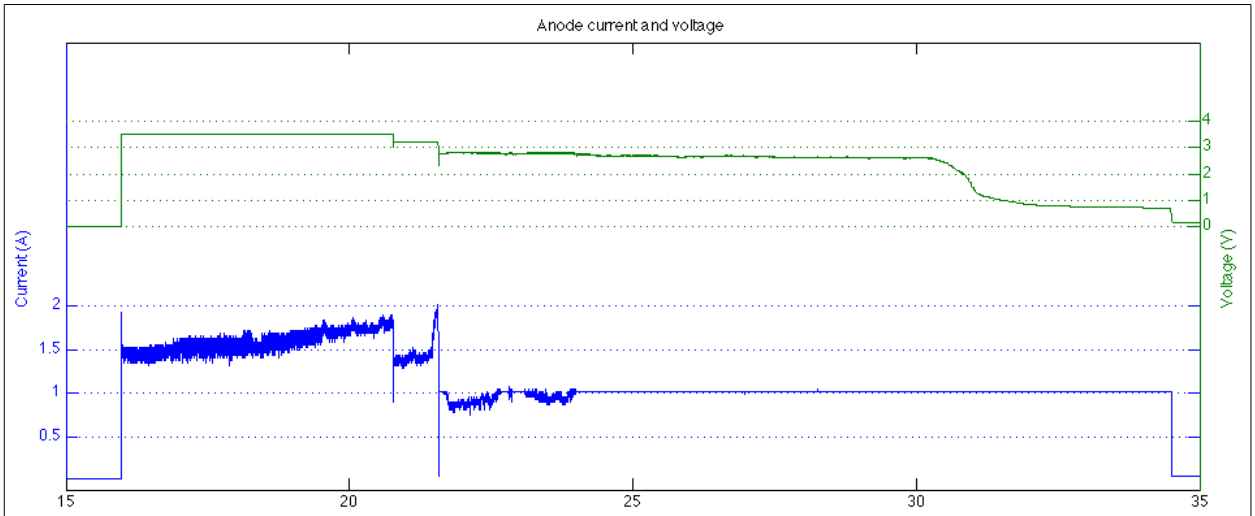


Figure 168: Voltage and current during experiment 30.

The anode maintained a constant current of 1.5A when set to 3.5V for five hours before it became unstable.

Experimental Parameters

Cathode: TiB_2 (0.5" L x 0.2" W x 3" H) with 1/2" dia. 1018 steel rod and screw connection

Cathode immersion: 1" below alumina sheath immersed 2" below the flux interface

Anode: 3/4” OD 950 um 8YSZ tube with 150g mixed Ag particles (0.05”-0.15” diameter) inside alumina support

Anode current collector: 30” L x 1/8” dia. silver wire (99.99%)

Anode immersion: 6”

Salt composition: KF-AlF_3 – 1.3 cryolite ratio with 14wt% alumina saturation

Salt Height: 4.5”

Crucible: Alumina-lined isomolded graphite – 5” OD x 4.5” ID x 12” High (11.5” Deep)

Crucible temperature: 700ºC

Results and Discussion

Anode After electrolysis the anode was prepped for analysis. Membrane samples were taken from inside the alumina sheath, at the bottom of the crucible in the flux, and a control from the inactive part of the tube. Table 22 highlights the EDS analysis taken from each sample.

	Y	O	ZR	AG	AL	K	F
CONTROL	13.91	18.98	67.1	-	-	-	-
TUBE IN AL2O3 SHEATH	5.73	18.43	73.97	-	-	0.17	1.65
TUBE IN FLUX	9.26	18.47	65.46	-	0.06	1.09	5.67

Table 22: EDS analysis of membrane samples taken post-electrolysis. All values are reported at 300x zoom and weight percentage unless otherwise noted.

The control sample showed no composition change across the membrane. The tube recovered from inside the alumina sheath lost 59 wt% yttrium and showed very slight KF penetration on the edges. An SEM image of the membrane (Figure 169) shows visible banding 20 microns thick on the inner edge and 100 microns thick on the outer edge.

The active membrane sample recovered from the flux decreased in yttrium concentration by 33%. Visible banding can be seen on the SEM image in Figure 170. Fluorine concentration increased 240% from the sample inside the alumina sheath. The band thickness on the outer and inner edges did not change from the values reported from the the tube sample in the alumina sheath.

Cathode Aluminum metal was collected by pulling the cathode out during electrolysis and scraping a sample off of the TiB2 (Figure 172). EDS analysis (Figure 171) showed that the deposit was mostly aluminum, with silicon and silver impurities. The silicon

impurity was traced back to minor scaling of the stainless steel vessel while the silver impurity came from the anode.

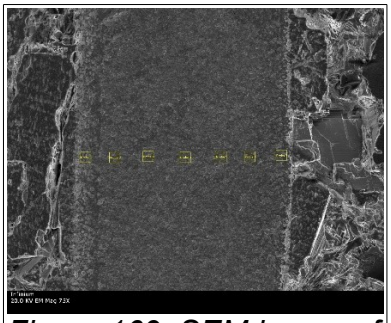


Figure 169: SEM image of anode piece that was inside the alumina sheath after electrolysis.

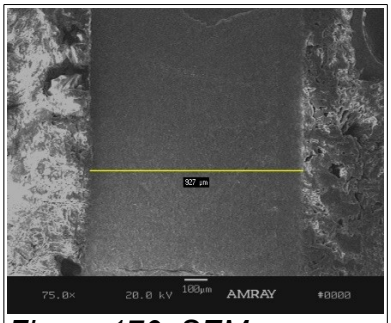


Figure 170: SEM cross section of anode section that was active during electrolysis.

Figure 171: Aluminum sample scraped from cathode after 4Ah passed at 3.2V

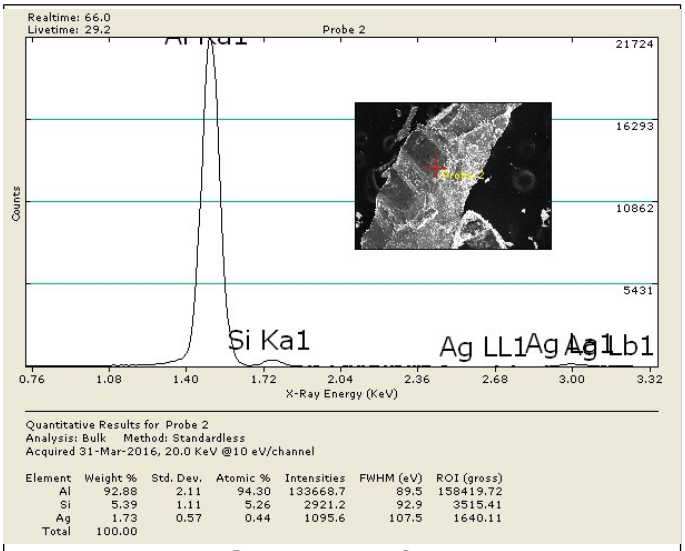


Figure 172: EDS analysis of metal sample scraped off of TiB2 cathode

Conclusions

The 950 um 8YSZ tube performed well during electrolysis. We made aluminum metal with the lowest membrane overpotential to date (3.2-3.5V total). Being a short run, more testing is needed to verify its stability over time. SEM verification of aluminum deposits showed the viability of using DKKK 8YSZ as an anode for metal production.

Experiment 031: 2nd Thin (sub-mm) Anode.

Overview

The goal of this experiment was to test the performance of a second sub-mm 8YSZ (8 mol% Y_2O_3) anode that was 3/4" OD, ~670 um thick while monitoring the exhaust flow with a mass flow controller. The experiment ran for roughly 10 hours and and passed 10.7 Ahrs. Titanium diboride was used as the cathode and a steel rod with a screw connection as the current collector. Pre-dried and pre-molten KF- AlF_3 with a cryolite ratio (CR) of 1.3 and saturated with alumina was loaded into an isomolded graphite crucible and lowered into a closed stainless steel shell purged under argon. A PLC was used to control the power supply and furnace temperature. The voltage was initially set to 3V, then to 3.2V when the current started to decay. A plot of the electrolysis is outlined below in Figure 173.

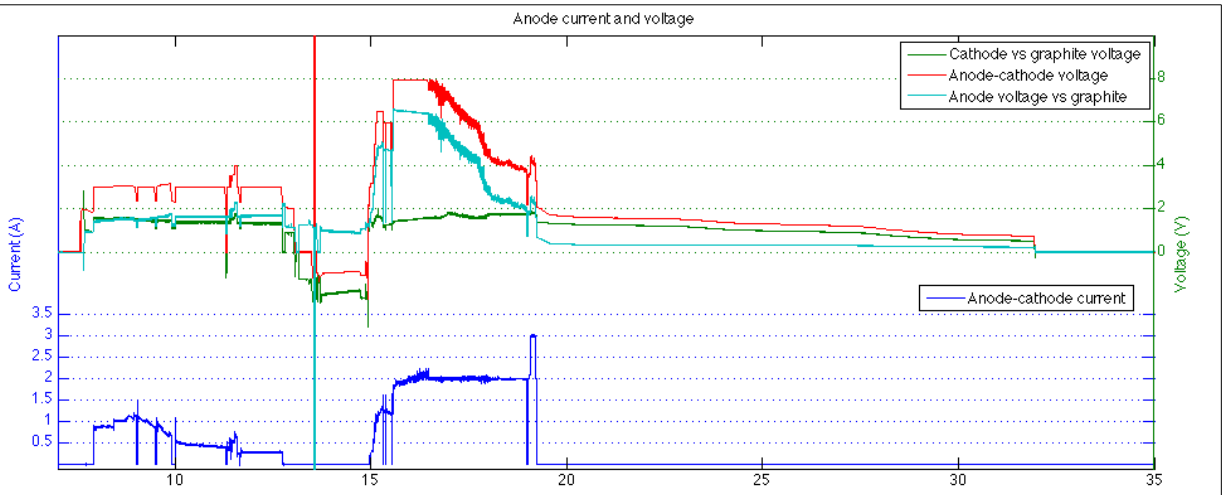


Figure 173: Voltage, current and anode oxygen flow rate during electrolysis.

Experimental Parameters

Cathode: TiB_2 (0.5" L x 0.2" W x 3" H) with 1/2" dia. 1018 steel rod and screw connection

Cathode immersion: 1" below alumina sheath immersed 2" below the flux interface

Anode: 3/4" OD 670 um 8YSZ tube with 150g mixed Ag particles (0.05"-0.15" diameter) inside alumina support

Anode current collector: 30" L x 1/8" dia. silver wire (99.99%)

Anode immersion: 5"

Salt composition: KF-AlF_3 – 1.3 cryolite ratio with 14 wt% alumina saturation.

Salt Height: 6"

Crucible: Alumina-lined isomolded graphite – 5" OD x 4.5" ID x 12" High (11.5" Deep)

Crucible temperature: 700ºC

Results and Discussion

Anode After electrolysis the anode was prepared for analysis.

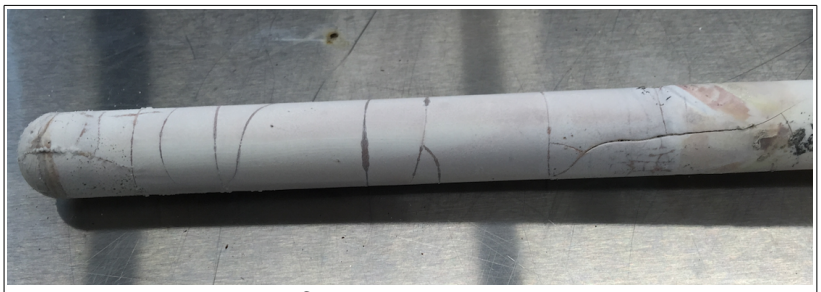


Figure 174: Anode after electrolysis. The cracks seen are filled with flux, which we believe are sealed in-situ.

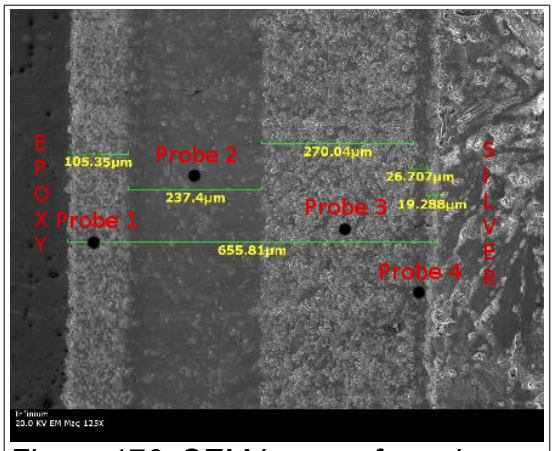


Figure 176: SEM image of anode showing distinct banding. The bands are lower in yttrium compared to the center.

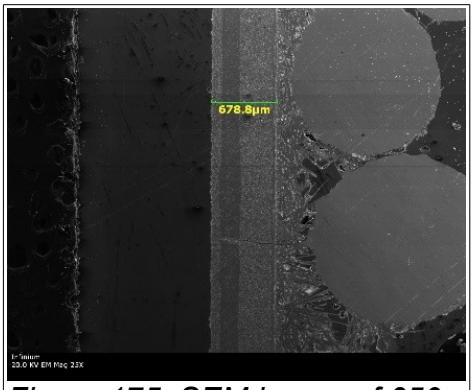


Figure 175: SEM image of 650 um anode after electrolysis. Circles are silver shot.

	Y	O	ZR	AG	AL	K	F
CONTROL	13.2	18.8	68	-	-	-	-
PROBE 1	7.22	11.45	78.6	-	0.06	0.61	2.06
PROBE 2	14.06	9.69	72.44	-	-	0.06	3.75
PROBE 3	13.76	13.05	72.08	-	-	-	1.1
PROBE 4	5.89	12.01	80.28	-	-	0.31	1.51

Table 23: EDS analysis of the banding seen in the tube post-electrolysis. Zoom is 100x and numbers are in weight percentage unless otherwise noted.

At the inner edge (Figure 176, Probe 4), the membrane showed a 20 micron thick band

that had a 55% reduction in yttrium from the control sample. In this band, there was also presence of potassium and fluorine indicating that there must have been some penetration of flux through the membrane during electrolysis. In Figure 5, a 240 micron thick black band can be seen. Probe 2 from Table 23 shows a 48% decrease in oxygen and a 3.75% presence of yttrium. This indicates that the dark band represents a conversion of yttrium oxide to yttrium fluoride. Finally, probe 1 represents the outside of the tube where it was in contact with the molten flux. Probe 1 shows penetration of potassium, aluminum and fluorine 50 microns into the membrane. Yttrium concentration decreased 45% from the control due to minor dissolution of yttrium into the flux.

Conclusions

The thin zirconia membrane proved to hold up well during electrolysis. The flux was able to seal any cracks that started to propagate during the run. SEM analysis of the membrane post-electrolysis showed no decrease in the thickness of the membrane, but slight banding due to destabilization of the yttrium concentration. Future experiments will be done to optimize the stability of yttrium in the flux.

Experiment 032: 3rd Thin (sub-mm) Anode.

Overview

The goal of this experiment was to test the performance of a thin 8YSZ (8 mol% Y_2O_3) anode that was 3/4" OD, ~670 um thick under pulsing conditions. For the first 15 hours the current was pulsed to 3.5 A for 600 seconds while relaxing at open circuit for 100s. The experiment ran for roughly 35 hours and passed 100 Ahrs. The system had an Ah limit of 100 and unfortunately when it reached it (hour 55) it zeroed the voltages for approximately three hours and the performance never recovered.

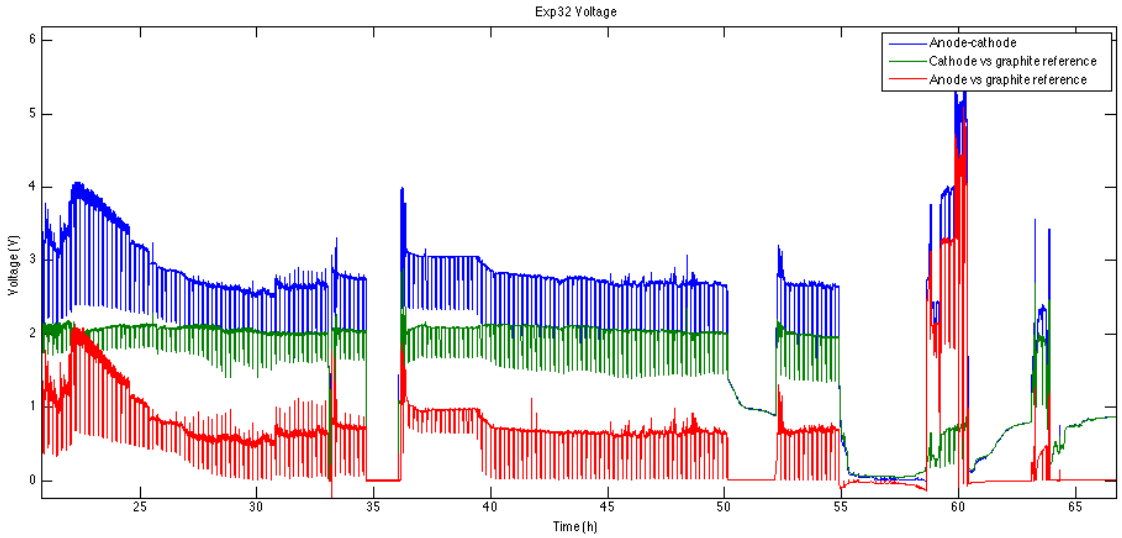


Figure 177: System voltages.

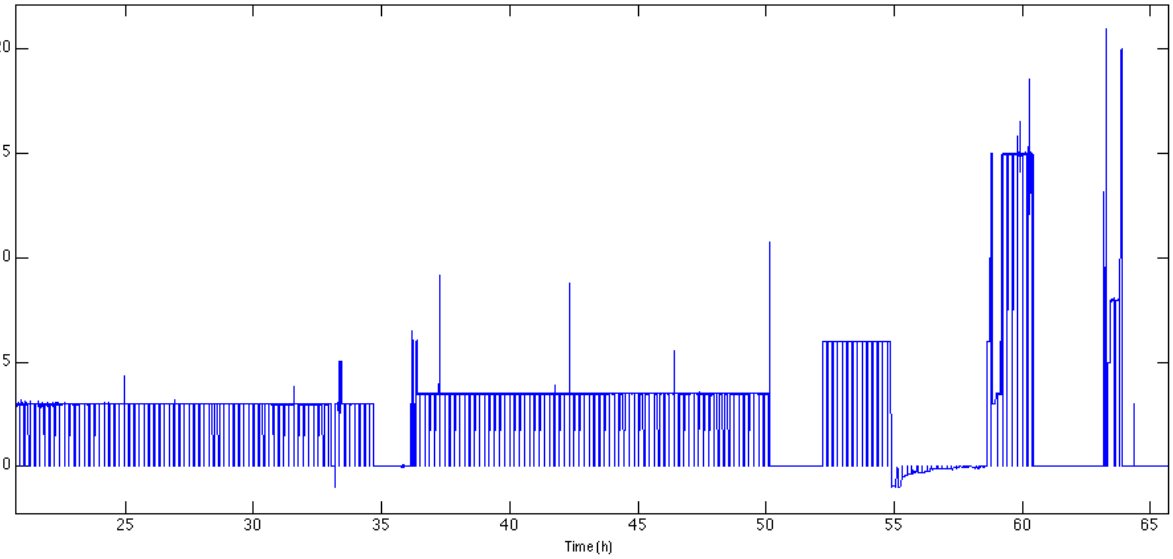


Figure 178: Current and anode exhaust during electrolysis.

Figure 178 shows the pulsed current.

Experimental Parameters

Cathode: TiB_2 (0.5” L x 0.2” W x 3” H) with 1/2” dia. 1018 steel rod and screw connection

Cathode immersion: 1” below alumina sheath immersed 2” below the flux interface

Anode: 3/4” OD 670 um 8YSZ tube with 150g small Ag shot (1-2mm diameter) inside alumina support

Anode current collector: 30” L x 1/8” dia. silver wire (99.99%)

Anode immersion: 5”

Salt composition: $KF\text{-}AlF_3$ – 1.3 cryolite ratio with 14wt % Alumina saturation

Salt Height: 6”

Crucible: Alumina-lined isomolded graphite – 5” OD x 4.5” ID x 12” High (11.5” Deep)

Crucible temperature: 700ºC

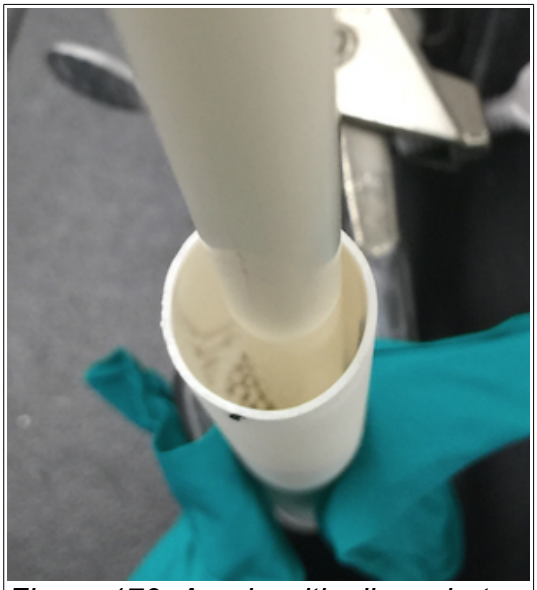


Figure 179: Anode with silver shot inside.

Results and Discussion

Anode The anode was not in one piece after electrolysis. This was likely attributed to the voltage forcing the electrolysis to run in reverse for three hours. Figure 180 shows the membrane recovered from the bottom of the flux block upon tear-down. Discoloration can be seen on the membrane which was analyzed via SEM and EDS. The sample was mounted in epoxy, polished down to 1 micron and sputter coated prior to analysis.

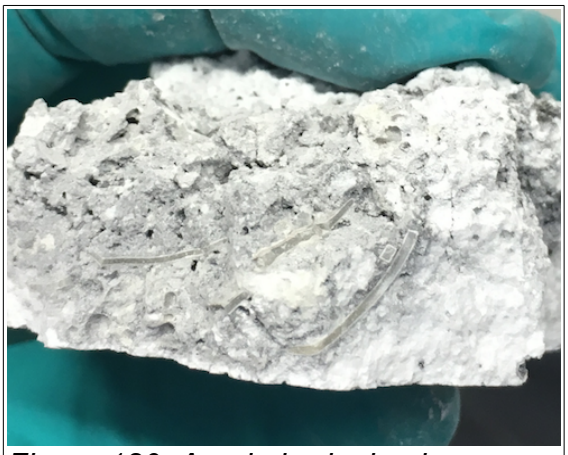


Figure 180: Anode broke in pieces, we believe this happened when the voltage was set to zero after reaching the Ah limit.

At the inner edge (Figure 181, Probe 5), the tube lost 65% yttrium from the control. Potassium, aluminum and fluorine were also present 65 microns into the membrane, confirming minor dissolution of the tube and flux penetration into the membrane. Figure 181 showed a 30 micron thick band starting 65 microns into the membrane. The composition of the band (Table 24, Probe 4) revealed a 45% reduction in oxygen coupled with the presence of 1% fluorine. This confirms that the band is experiencing a conversion from yttrium oxide to yttrium fluoride. A similar band (Probe 2) appeared 100 microns from the outer edge and measured 60 microns in width. The concentration of the outer band had a 150% increase in fluorine and 14% decrease in oxygen. This shows that the outer edge band experienced a higher conversion of yttrium oxide to yttrium fluoride. This makes sense because the outer edge is the first to experience oxygen migration through its vacancies which is more prone to yttrium oxide conversion.

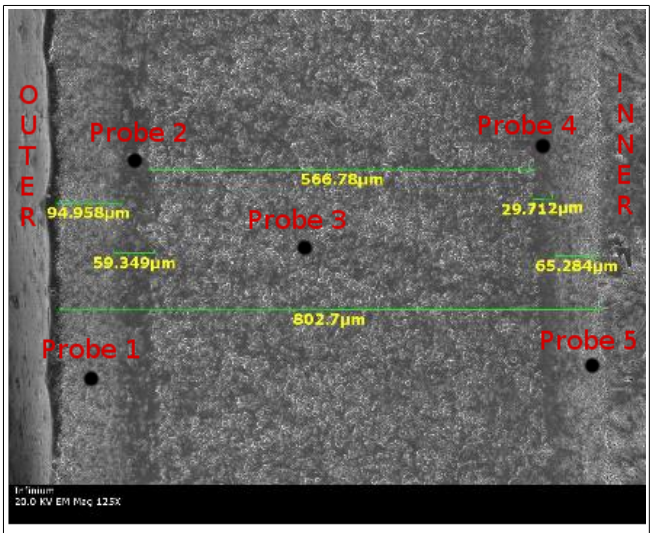


Figure 181: SEM image of anode that was active during electrolysis. Distinct banding can be seen.

	Y	O	ZR	AG	AL	K	F
CONTROL	13.56	18.16	68.28	-	-	-	-
PROBE 1	4.87	10.38	78.06	-	1.59	2.05	3.05
PROBE 2	14.07	8.63	74.81	-	-	-	2.48
PROBE 3	14.41	8.98	75.71	-	-	0.25	0.65
PROBE 4	14.16	10.01	74.75	-	-	0.09	0.99
PROBE 5	4.7	10.81	79.77	-	0.59	1.81	2.31

Table 24: EDS analysis of the banding seen in the tube post-electrolysis. Zoom is 125 and all values are reported as weight percentages unless otherwise noted.

Cathode EDS analysis showed that the deposit was mostly aluminum (Figure 182). Zirconium, silicon and silver alloyed with the aluminum at concentrations of 1.17%, 10.26% and 3.4% respectively. High concentration of silicon is most likely due to scaling of the stainless steel vessel or airborne insulation settling into the crucible.

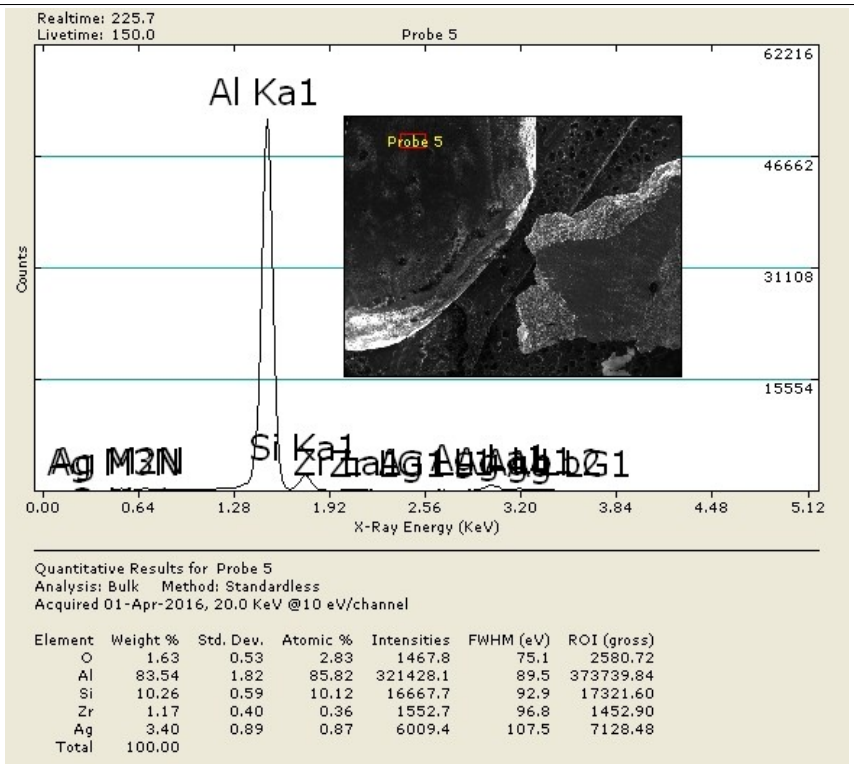


Figure 182: EDS analysis of the metal collected off of the TiB2 cathode

The cathode was inspected after running electrolysis in pulse mode for roughly 14 hours (Figure 184) which was the most metallic to date. We dipped a clean TiB_2 electrode (Figure 183) in the cathode port to see if we could recover any aluminum that had dripped inside the cell during electrolysis. 8 grams of aluminum metal was recovered from the cell prior to shutdown. .

Figure 183: Clean TiB2 electrode being dipped through the cathode port to recover aluminum made during overnight electrolysis.

Figure 184: Cathode after overnight electrolysis (hour 34).

Figure 185: Metal droplets recovered after with clean TiB2.

Conclusions The current thin anode configuration was successful in creating the most metal in any experiment to date. The purity of the metal needs to be optimized by engineering ways to isolate the impurities. The cathode wet the aluminum metal very well, which was showcased when dipping a clean piece of TiB2 into the cell and being able to remove the metal without it dripping off. Another experiment will need to be run to verify these results. An upgrade to the system parameters will also be done to prevent experiments from reversing when the amp-hr limit has been reached.

Exp 033: Thin-membrane Electrolysis with Powder and Shot Anodes

Overview

This experiment had two goals. One was to test the performance of yttria additives in the salt bath during electrolysis. The second was to test both silver powder and shot to see how the change in contact area effects both the resulting electrolysis parameters and degradation of the anode. The experiment ran for 95 hours and passed 682.9 Ahrs between two thin-membrane 8YSZ (8 mol% Y_2O_3) tubes. A thick 8YSZ tube was used as a reference electrode during electrolysis.

Experimental Parameters

Quad cell: 3

Anode(s): 2 3/4" OD 8YSZ with 4" Ag powder (Anode A) and Ag shot (Anode B) in between alumina sheath and anode

Cathode: TiB_2 (0.5" L x 0.2" W x 3" H) with 1/2" dia. 1018 rod and titanium screw connection

Flux: CR 1.3 + 5wt% yttria

Crucible: 5" OD isomolded graphite

Crucible Temperature: 720°C

Procedure

1. Make sure vessel is not grounded
2. No yellow alumina inside flux
3. Bubble for 10 minutes using steel bubbling tube through thermocouple port.
4. After 10 mins, carefully lower both anodes and reference electrode until 0.25" from the bottom inside open ended alumina exhausts
5. Back off alumina exhaust tubes to expose 4-5" of anodes
6. Connect each anode to a mass flow meter to correlate the flow to current
7. Pulse anode at 4V for 10 minutes, then relax for 10 minutes while pulsing anode B at 4V
8. Continue alternating pulsing and relaxation periods until target Ahrs reached

Results and Discussion

Figures 186 and 187 show current and voltage signals in anodes and the cathode.

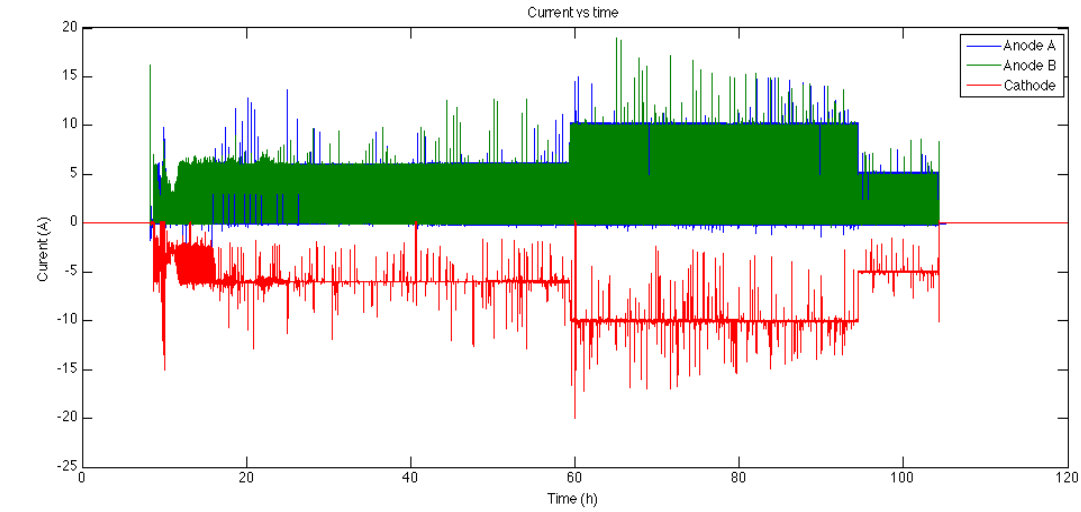


Figure 186: Anode and cathode currents during exp 033.

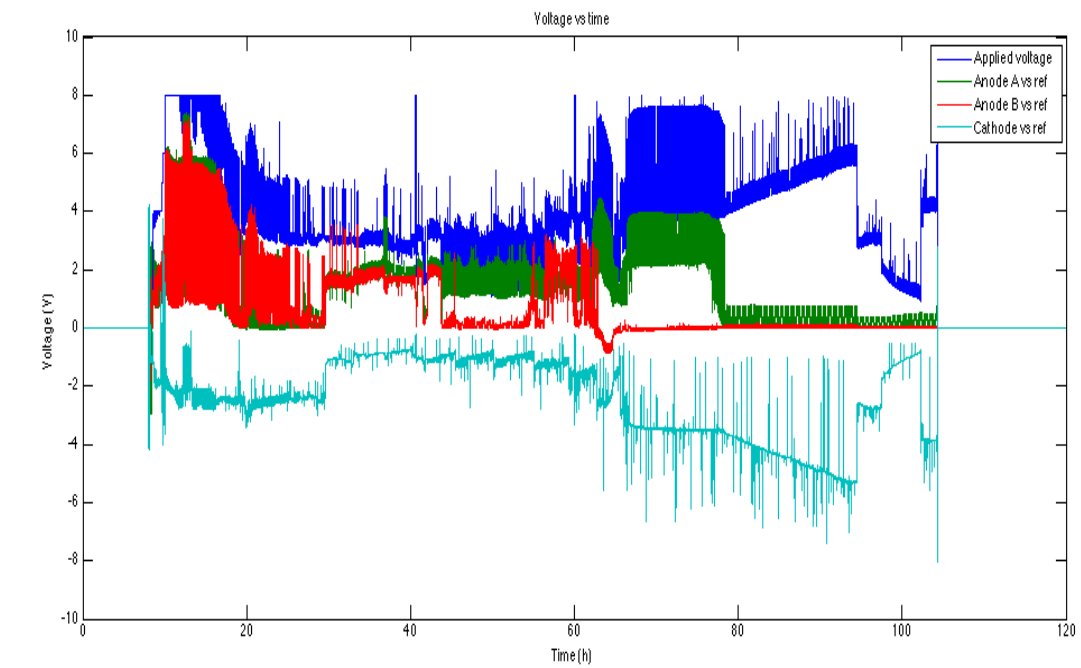


Figure 187: Anode and cathode voltage signals during exp 033.

Anode Characterization

The membranes for both the powder and shot anodes appeared severely degraded upon tear-down (see Figure 188 and Figure 189 below). Metal migration from the cathode to the anodes ultimately caused the cell to short and the experiment to fail.

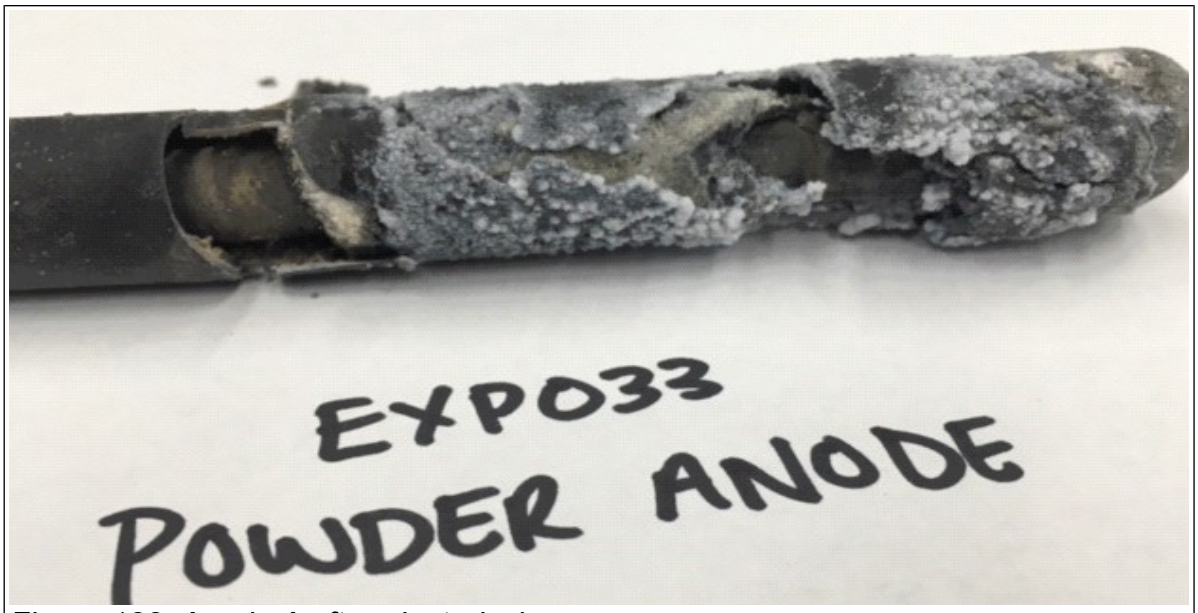


Figure 188: Anode A after electrolysis

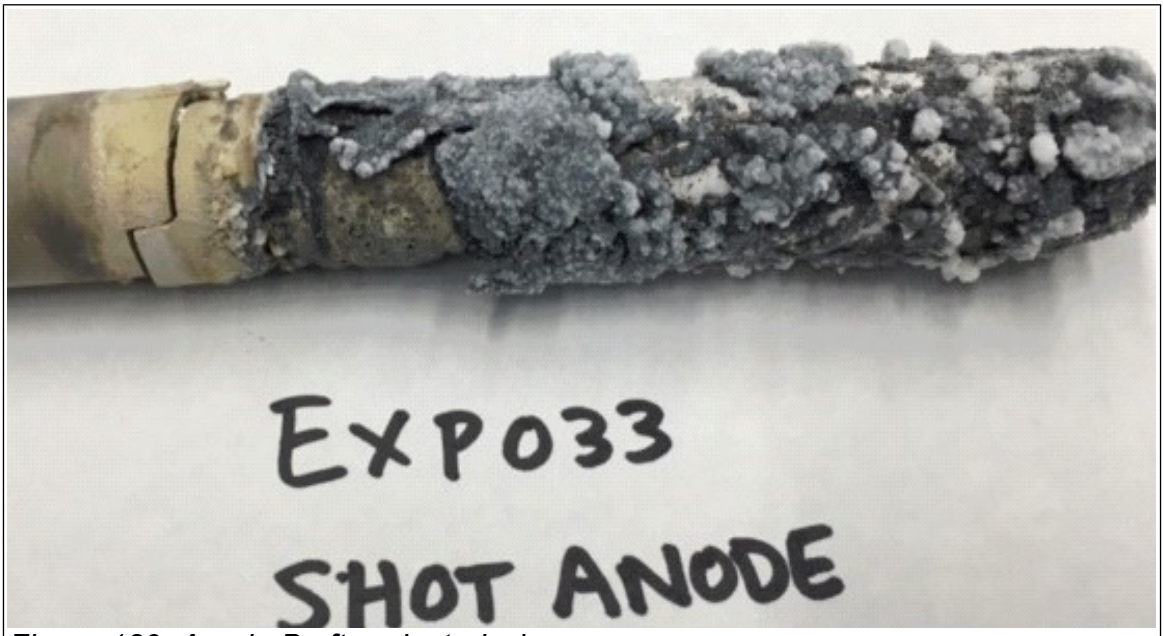


Figure 189: Anode B after electrolysis

SEM analysis was done on the membranes to quantify the degradation compared to a standard piece cut from the top of the tubes that were outside the electrolysis zone. Figure 190 highlights the cross-section of the Shot anode post-electrolysis, while Table 26 gives composition data for the three probe points.

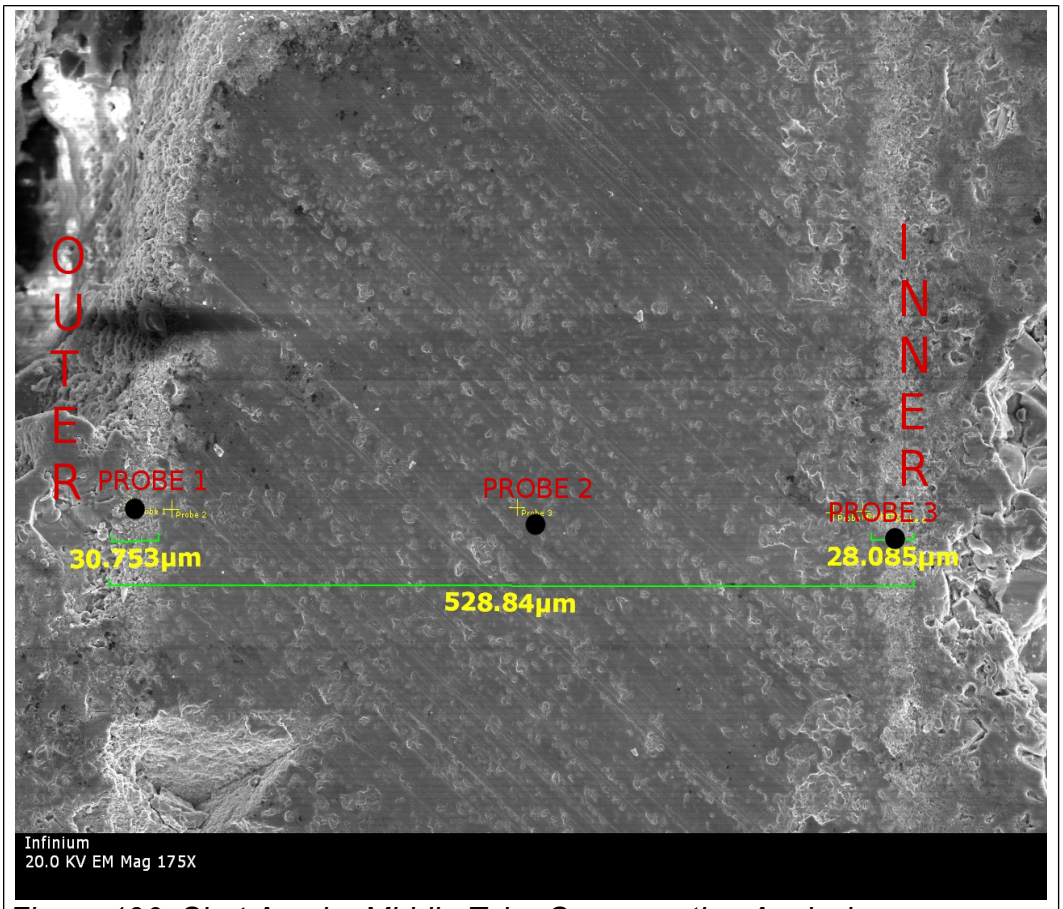


Figure 190: Shot Anode: Middle Tube Cross-section Analysis

	Band Width (µm)	Y (wt%)	Zr (wt%)	O (wt%)	F (wt%)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL		11.8	66.5	21.7	-	-	-	-
PROBE 1	31	8.75	70.18	14.57	5.23	0.87	0.36	-
PROBE 2	529	13.37	72.34	11.04	3.21	-	-	-
PROBE 3	28	5.4	70.87	13.02	1.61	1.41	0.36	-

Table 25: Shot Anode Membrane Analysis

The outer edge (Figure 190, Probe 1) shows a 31 micron band with a 26% decrease in Y, 5.5% increase in Zr, and 33% decrease in O. The decrease in Y can be explained by the dissolution of yttrium into the bath, while the oxygen and zirconium concentrations account for the makeup. Flux components were also seen throughout the 31 micron band, confirming slight migration into the outer membrane.

Probe 2 highlights the bulk concentration of the tube post-electrolysis. The biggest

change was the presence of 3.21wt% Fluorine and 49% decrease in O concentration. This supports the conversion of yttrium oxide to yttrium fluoride in the membrane. Finally, Probe 3 highlights the inner membrane concentration. There was a distinct 28 micron band, with severe degradation. Similarly to the outer membrane, there was a very high (54%) decrease in Y, with a notable decrease in oxygen (40%). Since the membrane cracked, the presence of flux on the inner membrane as well makes sense.

The powder anode was put through the same analytical procedure via SEM. Figure 191 highlights the cross-section with the same three probe areas to be analyzed.

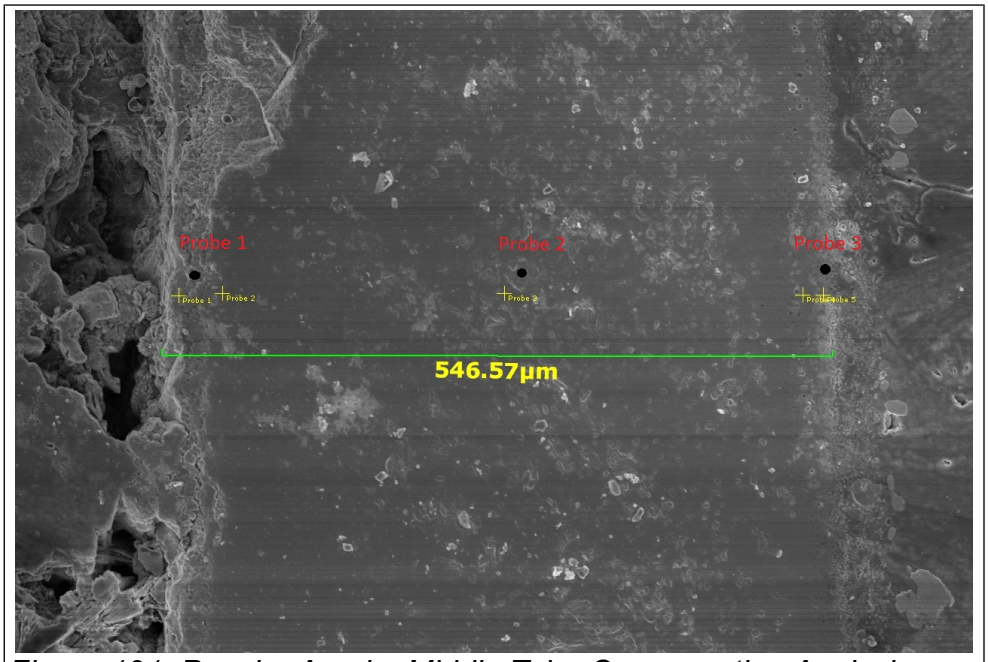


Figure 191: Powder Anode: Middle Tube Cross-section Analysis

Table 26 gives an overview of the banding and concentrations across the membrane post-electrolysis.

	Band Width (µm)	Y (wt%)	Zr (wt%)	O (wt%)	F (wt%)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL		11.8	66.5	21.7	-	-	-	-
PROBE 1	20	2.2	55.07	12.54	14.09	5.96	10.15	-
PROBE 2	800	16.23	69.16	13.51	1.4	-	-	-
PROBE 3	20	14.62	69.66	14.59	1.14	-	-	-

Table 26: Powder Anode Membrane Analysis

The outer membrane (Figure 191, Probe 1) showed a 20 micron band with a significant (81%) decrease in yttrium concentration. Also, this band had a high concentration of the flux components, confirming fast dissolution of the membrane into the flux at this

interface. Probe 2, the bulk concentration, revealed a 38% increase in Y and 38% decrease in O, alongside a 1.4% concentration of F. This supports the conversion of yttrium oxide to yttrium fluoride in the membrane. Finally, Probe 3 (inner membrane) revealed similar results as Probe 2, only indicating yttrium oxide conversion, but no flux constituents present. This indicates that the powder membrane was more stable in the bath then the shot anode, most likely due to the amount of contact points from the powder, minimizing stress to the membrane.

Cathode Characterization

Throughout the experiment metal was able to be collected by pulling the cathode out of the cell and scraping molten metal off the sides. Moving forward it would be beneficial to have a collection cup below the cathode in an effort to contain the produced metal and keep it from mixing into the flux or reaching the anodes.

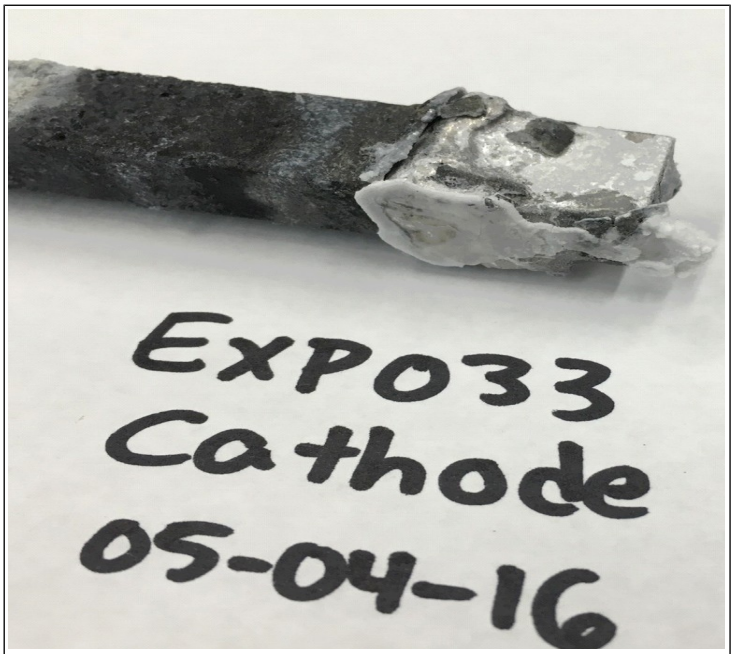


Figure 192: Cathode with metal after extraction.

Summary

The electrolysis made metal, but eventually the metal migrated to the anodes, terminating the experiment. The cell shorting could have had an effect on degree of degradation seen to both anodes post-experiment. The powder and shot anodes performed similarly during electrolysis (342.9 and 340 Ahrs respectively), however the powder anode appeared less degraded via SEM analysis. No flux components were seen at the inner edge, indicating that the membrane stayed partially stable. This is most likely due to the increased number of contact points from the powder, putting less stress on the membrane. Going forward, powder anodes will be tested further to confirm these findings.

Exp 034: Thin-membrane Electrolysis with Cryolite Ratio (CR) 1.5 Flux

Overview

The goal of this experiment was to test the effectiveness of a CR 1.5 flux with yttria and alumina flux additives. Both anodes were filled with Ag powder (versus Ag shot) to optimize contact area inside thin anodes. The adjustments made to the flux should help minimize anode corrosion and extend the length of time electrolysis can be applied. The experiment ran for 66 hours and a total of 236.9Ahrs were passed between the two anodes. This experiment yielded no recoverable metal.

Experimental Parameters

Quad cell: 3

Anode(s): 2 3/4" OD 8YSZ with 4" Ag powder in between alumina sheath and anode

Cathode: TiB_2 with 1" immersion

Reference: Thick 8YSZ tube

Flux: CR 1.5 with 5wt% yttria and 2.5wt% alumina

Crucible: 5" OD isomolded graphite

Bubbling Tube: Steel 3/8" OD

Crucible temperature: 770C

Duration: 67 hours (236.9 Ahrs)

Procedure

1. Make sure vessel is not grounded,
2. No yellow alumina inside flux
3. Bubble for 10 minutes using steel bubbling tube through thermocouple port.
4. After 10 mins, carefully lower both anodes and reference electrode until 0.25" from the bottom inside open ended alumina exhausts
5. Connect each anode to a mass flow meter to correlate the flow to current
6. Connect EI303C between Anode A and ref, EI303D between Anode B and ref, EI303E between ref and crucible
7. Pulse 5A between two anodes at 2 minute intervals

Results and Discussion

Figures 193, 194 and 195 show anode A current and oxygen flow rate, currents in both anodes and the cathode, and voltages in both anodes and the cathode throughout the 67-hour experiment.

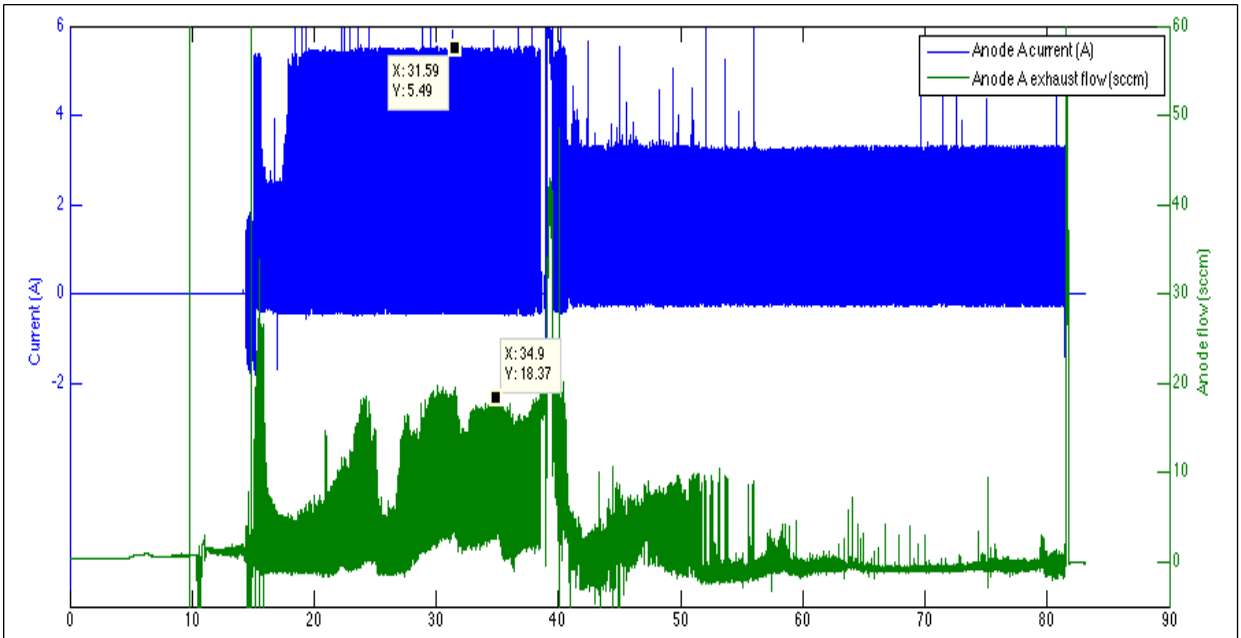


Figure 193: Anode A current and oxygen flow rate vs. time.

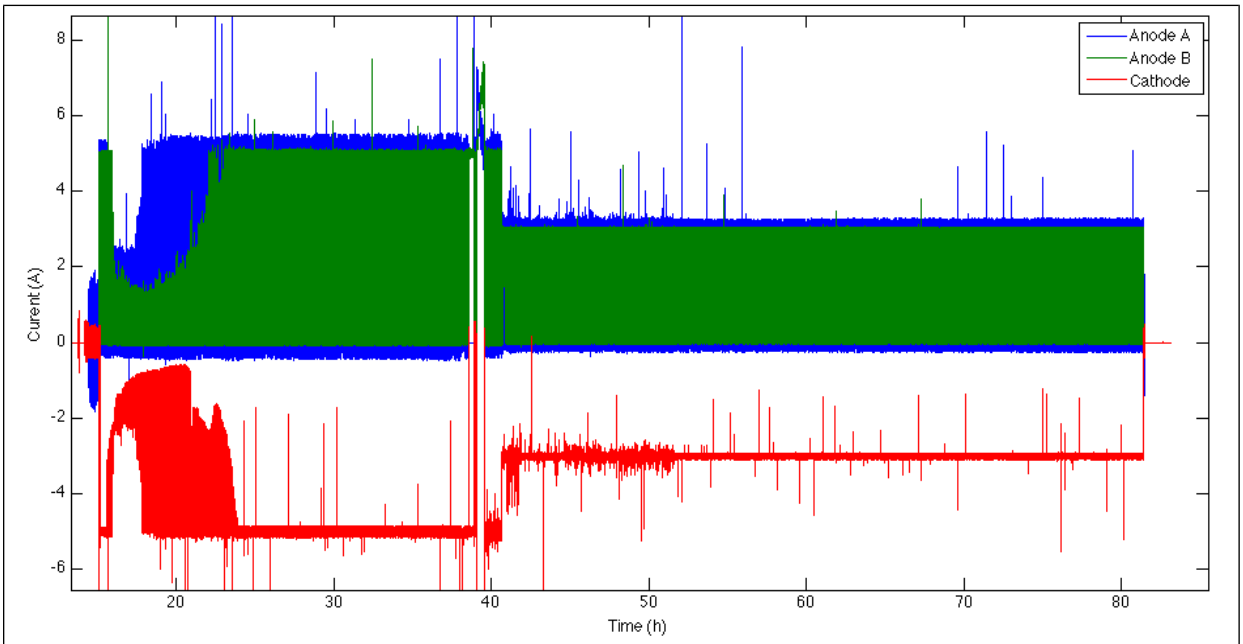


Figure 194: Current in both anodes and cathode vs. time.

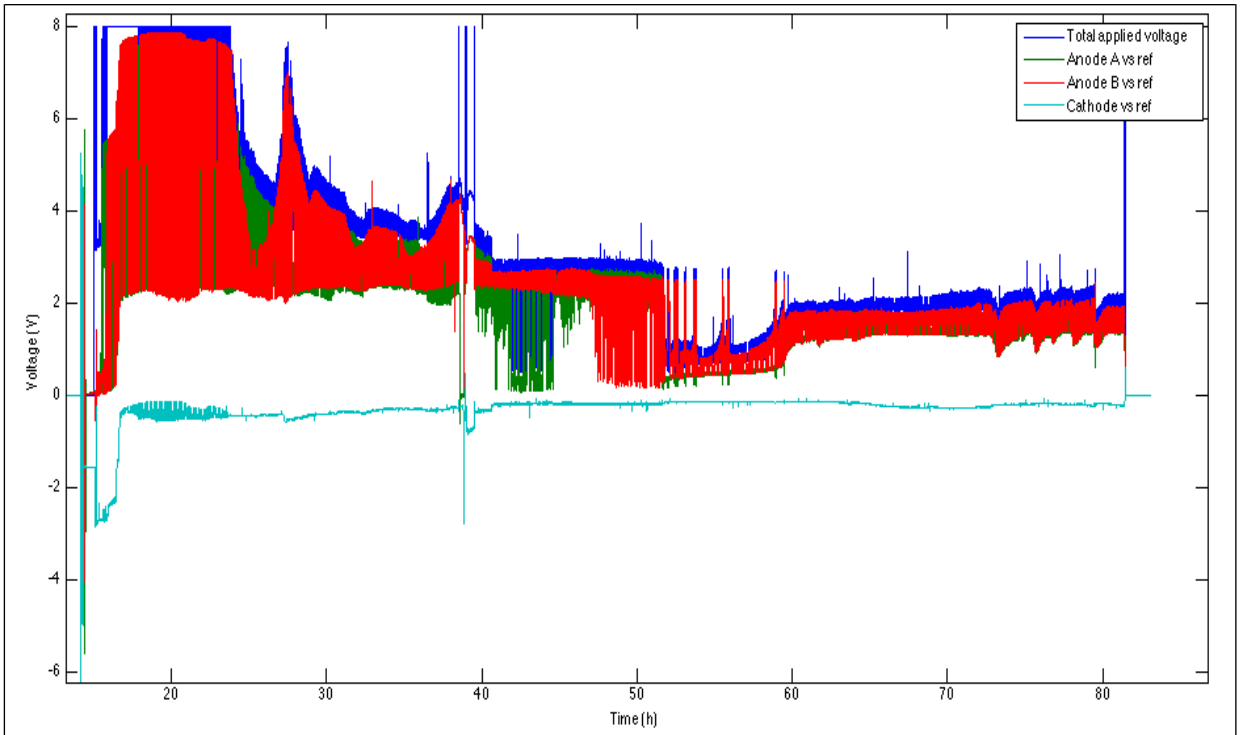


Figure 195: Anode and cathode voltages vs. time.

Anode Characterization

Anodes A and B both accumulated a dark yellow crud layer on the outside (Figure 196). High concentrations of silver point to a conversion of silver to silver fluoride. This means that the silver was directly exposed to the flux from a broken membrane. The zirconium concentration is a direct correlation to the solubility into the flux/silver mixture.

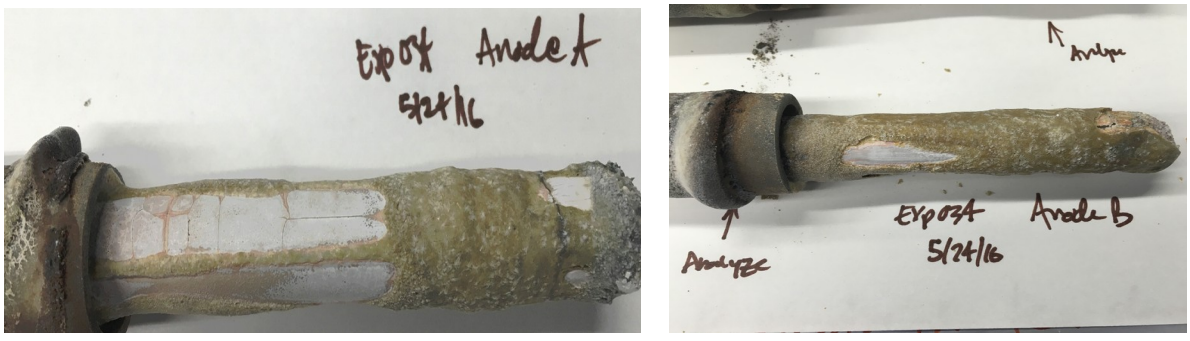


Figure 196: Anodes A (left) and B (right) post experiment. Yellow scaling was seen on both anodes, which SEM verified to be regions of flux with low Zr (2-6wt%) and Ag (8-9wt%)concentrations.

Further examination of the membrane was done by cross-sectioning the tube and probing the inner, outer and bulk concentrations of the tube via SEM using EDS to

calculate the relative concentrations at these points (Figure 5).

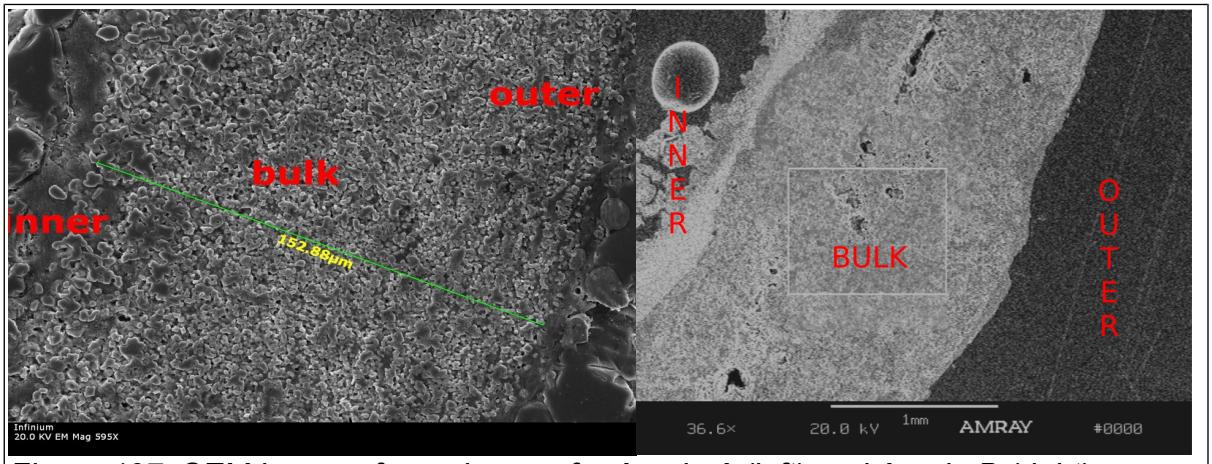


Figure 197: SEM image of membranes for Anode A (left) and Anode B (right).

Concentrations were taken at the inner, outer and bulk locations. The compositions for each tube were compared against a control taken from the tube outside the electrolysis area. Table 27 gives an overview of the composition across anode A and Table 28 across anode B.

	Band Width (µm)	Y (wt%)	Zr (wt%)	O (wt%)	F (wt%)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL		13.22	61.6	25.18	-	-	-	-
PROBE 1		3.91	55.72	13.79	11.25	8.06	7.27	-
PROBE 2		3.62	60.22	14.65	11.4	4.57	3.08	2.46
PROBE 3		3.74	59.89	14.11	11.74	5.68	3.53	1.53

Table 27: Anode A Membrane Analysis

	Band Width (µm)	Y (wt %)	Zr (wt%)	O (wt%)	F (wt %)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL		13.85	61.01	25.14	-	-	-	-
PROBE 1		8.8	62.88	14.99	9.62	2.9	0.81	-
PROBE 2		14.31	62.98	11.91	8.89	1.91	-	-
PROBE 3		10.54	55.05	14.51	13.59	4	1.72	0.58

Table 28: Anode B Membrane Analysis

Both anodes were found to be leeched of yttrium oxide. The YSZ was cracked and held together by flux solidified in the cracks. This meant that there was a lot of flux leached into the anode and mixed with the silver powder. Anode A noticed an average of 72% decrease in Y concentration in the bulk and on tboth the inner and outer edge. This is analogous to severe degradation as the Y is dissolving in the bath and the flux is

leaching its way into the membrane. Anode B revealed similar findings, however the degree of degradation was suppressed. The decrease in Y concentration reached a maximum of 36% on the outer edge, however the bulk concentration of Y actually increased. This has been seen before and explained by the conversion of yttrium oxide to yttrium fluoride. The 53% decrease in O confirms this explanation.

Figure 198: Alumina sheath cross section. Notice purplish tint.

Figure 198 shows a cross section example of the anodes filled with flux and reacting with the alumina spacer inside the anodes. Note that the alumina spacer changed to purple. SEM analysis shows 0.18wt % K added in the alumina spacer but otherwise no significant change in its composition.

Cathode Characterization

The cathode sheath post experiment was coated in a black crud that is rich in aluminum, potassium, and flourine and has smaller amounts of oxygen and zirconia (Figure 199). The cathode appeared to have metal on it, but it was not possible to gather a pure sample off of the TiB2 (Figure 200).

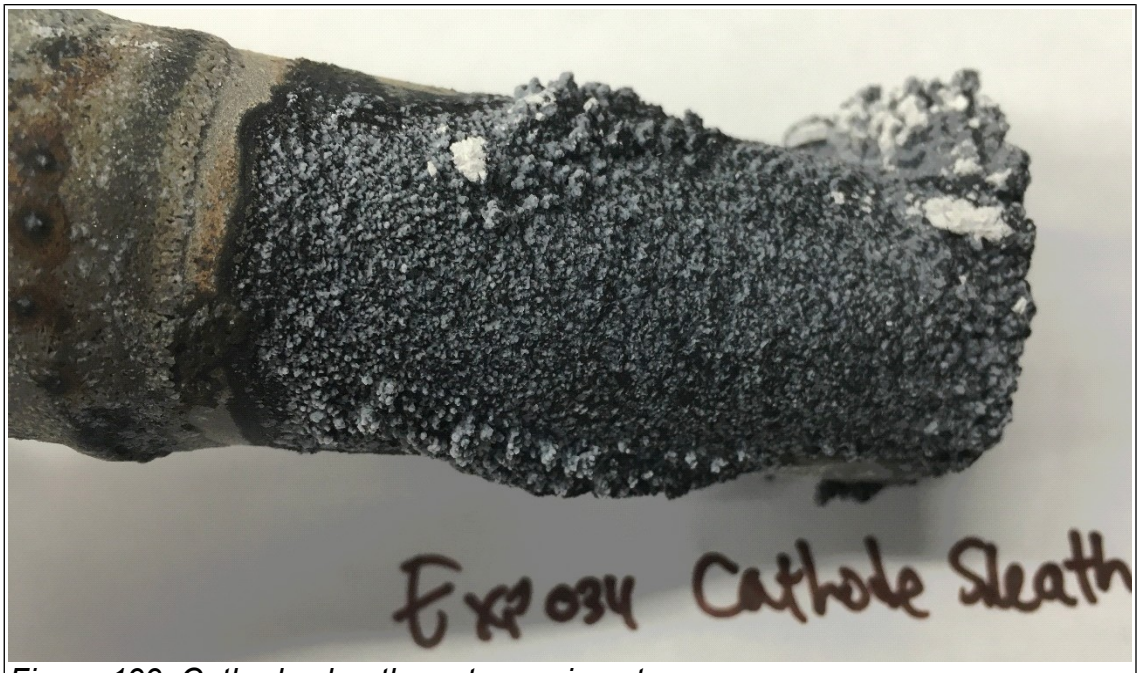


Figure 199: Cathode sheath post-experiment.

Summary

The powders anodes performed as expected under electrolysis, however the metal didn't seem to wet the cathode as much as it usually did. This is believed to happen due to the higher crucible temperature of 770C measured. Further investigation supports a eutectic of yttrium fluoride with zirconium fluoride above 750C, which would accelerate dissolution of the tube and ultimately cause the experiment to fail. Tighter engineering controls will be put in place for future experiments, preventing the crucible temperature from exceeding 750C.

Figure 200: Cathode from Exp034. Metal was seen on the cathode, but unable to recover a measurable quantity.

Exp 035: Thin-membrane Electrolysis with Yttrium Oxide and Yttrium Fluoride

Overview

The goal of this experiment was to test the effectiveness of a CR 1.5 flux with both yttrium oxide and yttrium fluoride as additives. The oxide and fluoride should work together to better stabilize the membrane and prevent degradation. Both anodes were filled with Ag powder (versus Ag shot) to optimize contact area inside thin anodes. The adjustments made to the flux should help minimize anode corrosion and extend the length of time electrolysis can be applied. The experiment was run for 66 hours at 720C crucible operating temperature and yielded 8g of recoverable metal.

Experimental Parameters

Quad cell: 3

Anode(s): 2 3/4" OD 8YSZ with 4" Ag powder in between alumina sheath and anode

Cathode: TiB_2 with 1" immersion

Reference: Thick 8YSZ tube

Flux: CR 1.5 with 5wt% yttria, 4wt% YF3 and 10wt% alumina

Crucible: 5" OD isomolded graphite

Bubbling Tube: Steel 3/8" OD

Crucible temperature: 720C

Duration: 66 hours (303.1 Ahrs)

Procedure

1. Make sure vessel is not grounded,
2. No yellow alumina inside flux
3. Bubble for 10 minutes using steel bubbling tube through thermocouple port.
4. After 10 mins, carefully lower both anodes and reference electrode until 0.25" from the bottom inside open ended alumina exhausts
5. Connect each anode to a mass flow meter to correlate the flow to current
6. Connect EI303C between Anode A and ref, EI303D between Anode B and ref, EI303E between ref and crucible
7. Pulse 5A between two anodes at 2 minute intervals

Results and Discussion

Figures 201, 202 and 203 show voltage and current in each anode.

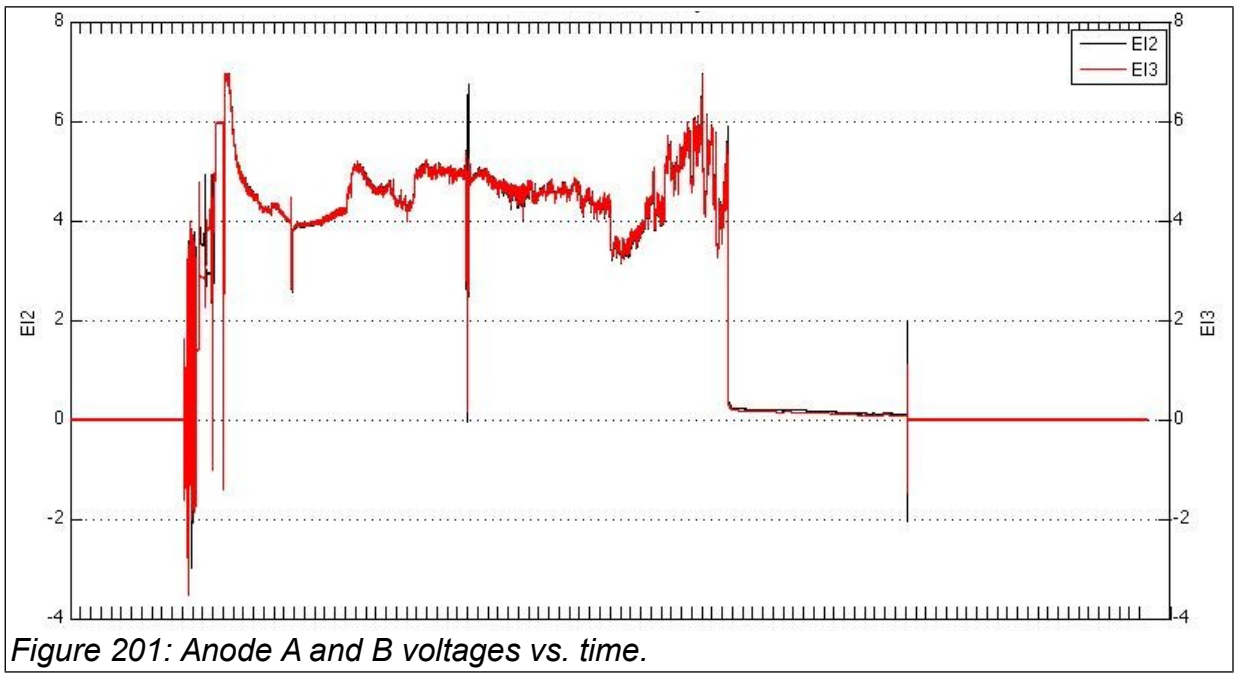


Figure 201: Anode A and B voltages vs. time.

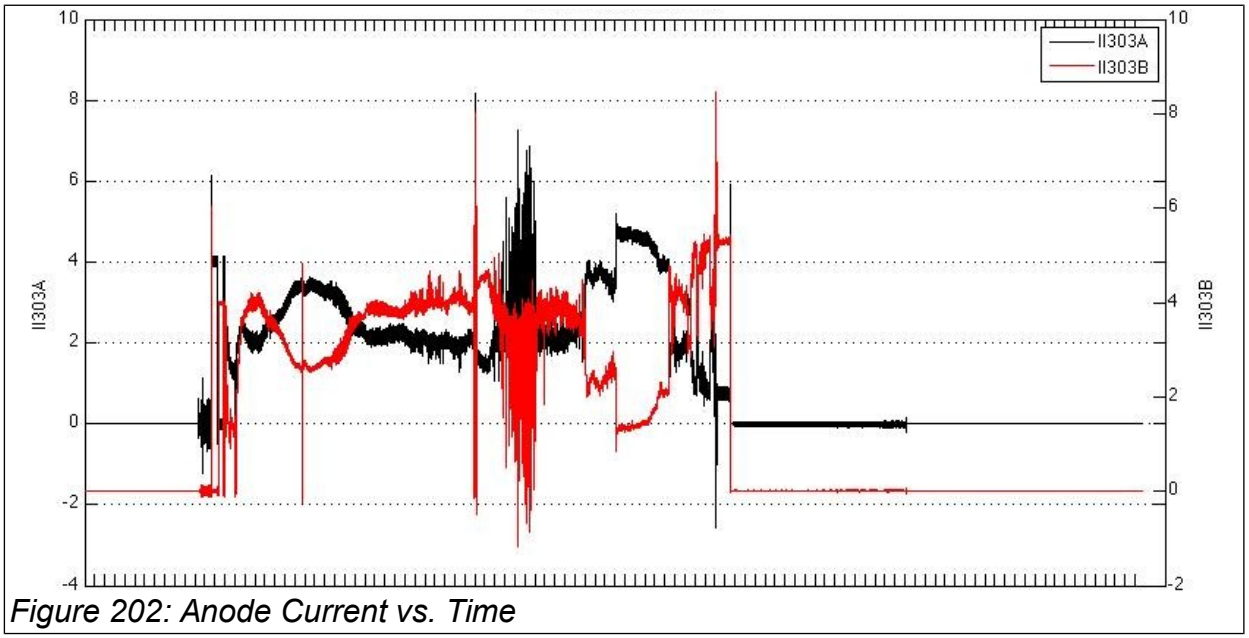


Figure 202: Anode Current vs. Time

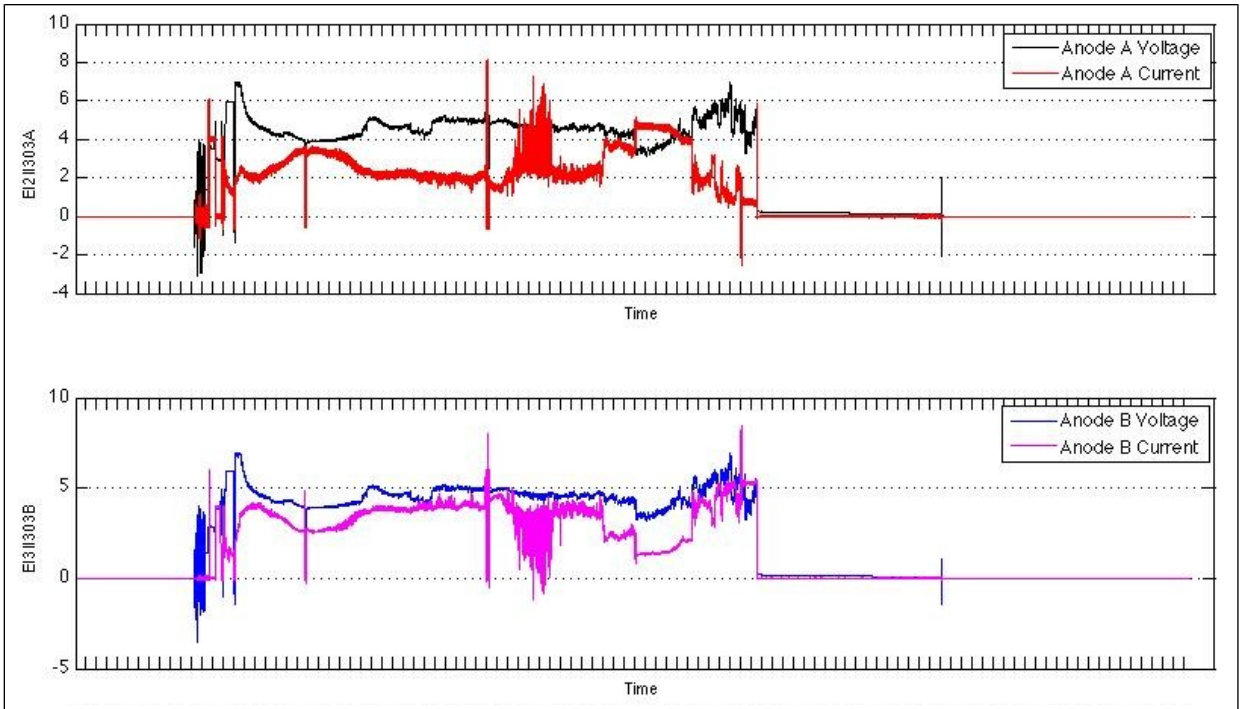


Figure 203: Voltage and current vs. time for each anode.

Anode Characterization

Anode A ran for 130.1 Ahrs and was stressed up to a maximum of 7V. Anode B ran for 173.0 Ahrs and was also stressed to a maximum of 7V. The stress to the anodes were reflective of the visual degradation post-experiment Figure 204).

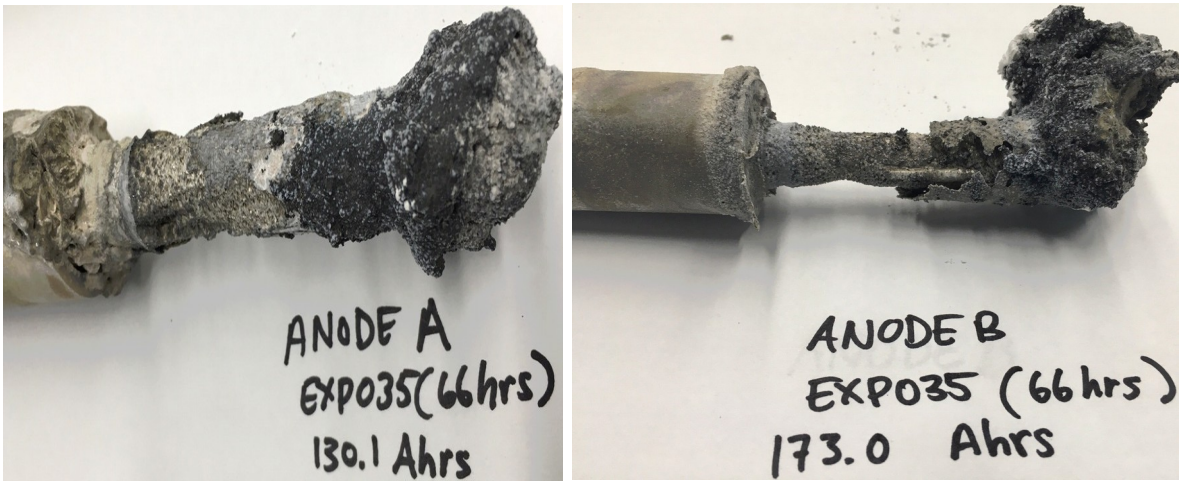


Figure 204: Anodes A (left) and B (right) post experiment. Anode A was partially consumed while complete degradation of Anode B was observed.

The anodes were cut and cross-sectioned for SEM analysis. Probe points were taken across the membrane at the inner and outer edges along with a bulk composition.

These results can be seen below in Table 29 and Table 30.

	Y (wt%)	Zr (wt%)	O (wt%)	F (wt%)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL	13.23	63.37	23.4	-	-	-	-
PROBE 1	6.48	73.31	13.03	4.83	2.17	0.18	-
PROBE 2	14.51	65.2	14.35	5.95	-	-	-
PROBE 3	13.92	65	15.46	5	0.61	-	-

Table 29: Anode A Membrane Analysis

	Y (wt%)	Zr (wt%)	O (wt%)	F (wt%)	K (wt%)	Al (wt%)	Ag (wt%)
CONTROL	16.99	68.49	14.52	-	-	-	-
PROBE 1	6.44	60.29	11.09	11.69	5.35	4.91	0.22
PROBE 2	14.9	65.78	19.33	-	-	-	-
PROBE 3	11.83	64.77	16.72	5.46	0.89	0.25	0.07

Table 30: Anode B Membrane Analysis

Yttrium leaching was seen in both anodes on the outer edge. The outer edge of Anode B also showed a small amount of Ag, probably due to silver powder migration. Flux penetration was seen 20-30 microns on both anodes. The bulk concentration of tube A revealed fluorine and reduction in oxygen, supporting the conversion of yttrium oxide to yttrium fluoride. Anode B, however, did not experience this same conversion. This is a good sign because the additives to the flux may have suppressed this conversion.

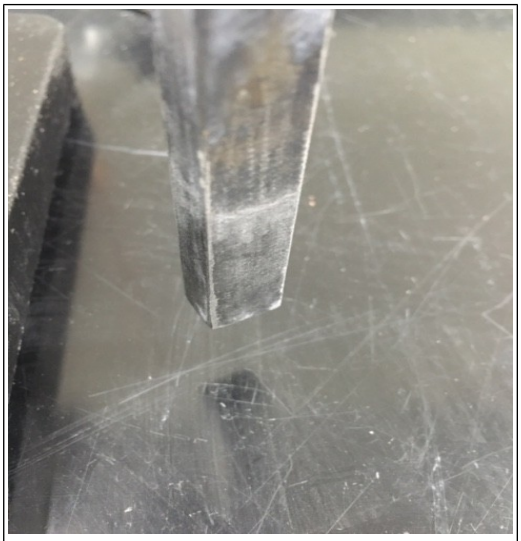


Figure 205: Exp035 cathode post-electrolysis. Metal was recovered by dipping the cathode into the metal pool on the bottom, allowing it to wet the TiB2, pulling it out and scraping it off.

Cathode Characterization

The cathode post-electrolysis came out very clean (Figure 205). Multiple metal samples were taken throughout the experiment to see how the composition changed over time. Figure 206 Shows the change in metal quality over time.

Comp.	Wt%
Al	99.95
Zr	0
Y	0
K	0.05
Ag	0

2 Ahrs
Max 5V

Comp.	Wt%
Al	100
Zr	0
Y	0
K	0
Ag	0

8 Ahrs
Max 6V

Comp.	Wt%
Al	97.08
Zr	0
Y	0
K	0
Ag	2.92

150 Ahrs
Max 7V

Figure 206: Al metal compositions as a function of time (Ahrs). Samples were taken at 2, 8 and 150 Ahrs. The operating voltage was increased to see if the metal quality changed. The metal quality stayed consistent until 150Ahrs and above 6V operating potential.

Summary

The additions of yttrium fluoride and yttrium oxide helped stabilize the membrane during electrolysis. Visual evidence showed a severely degraded membrane. The cell produced 8g of recoverable aluminum metal, and EDS results show lower impurity concentration than prior experiments. Further investigation is needed into the flux additions in order to better stabilize the membrane for longer electrolysis runs.

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Aluminum-Scandium Alloys

Introduction

Motivation

Scandium (Sc) is the best strengthener for aluminum alloys per mole of alloy addition.[55] Al_3Sc coherent precipitates are very fine and stable at high temperature, making these alloys particularly suited to welding or sintering, such as in 3-D printing. Indeed, with yield stress of 525 MPa[56]—twice that of leading powder alloy AlSi10Mg[57]—the strength-density ratio σ_y/ρ of sintered Scalmalloy (3.6% Mg 1% Sc) powder at 1.94×10^5 m²/s² is 20% higher than that of Ti-6-4.[58] Tensile and bending stiffness/density E/ρ and $E^{1/3}/\rho$ are 3% and 40% higher respectively than titanium alloys. At $3300/kg, Sc is expensive, but many alloys benefit from just 0.2 wt% Sc. Low Sc solubility in Al leads to a commercial master alloy composition of Al-2 wt% Sc with a market price of about $115/kg.[59]

Though more abundant than lead in the Earth's crust,[60] Sc is very dispersed and hard to isolate, such that worldwide production of scandium oxide Sc_2O_3 is only 10 tonnes per year (TPY). Sc_2O_3 is the lowest-cost form of scandium available in the market today. With a market price of $1200/kg for 98-99% pure material,[61] the metals basis cost is $1840/kg, *cf.* scandium fluoride ScF_3 at $1300/kg[62] with metals basis cost of $2947/kg. That said, multiple new mines are under construction or in planning stages, as described below, leading to up to 250 TPY Sc_2O_3 production by 2022. This will likely lead to substantial price reduction, and could enable up to 80,000 TPY of Al-0.2% Sc alloy production, which is about half of world titanium production.[63]

Though there are a number of potential methods for producing Al-2%Sc master alloy in the literature, the simplest methods either use expensive ScF_3, or require powder Al handling or additional process steps. Thus, a new low-cost method for direct production of Al-2%Sc at high yield from Sc_2O_3, with material cost around $37/kg, could potentially lead to higher margins than today's methods.

This Introduction describes new scandium resources or production facilities, followed by a literature survey of Al-Sc production methods, a description of INFINIUM approaches, and an outline of work described here and potential follow on beyond the project.

New Scandium Resources

Multiple new scandium oxide resources are likely to come online in 2-5 years, listed below in order of when they will likely start production.

Sumitomo Metal Mining (SMM)

SMM is building a scandium oxide add-on to its existing Ni-Co mine and High Pressure Acid Leach (HPAL) processsing facility at Taganito, Philippines. They started building a pilot operation in 2013,[64] and announced construction of 7.5 TPY Sc_2O_3 capacity in 2016[65] with plans to begin operation in late 2017. This is the most mature new scandia

source, and sets a rough five-year timeline from pilot plant announcement to production.

Rusal

The Russian aluminum producer Rusal is building a Sc_2O_3 production plant at its UAZ smelter in the Ural Mountains.[66] The plant will use a "carbonization process" to isolate scandium from red mud, and has produced >99% pure Sc_2O_3. Rusal is also developing a method for producing Al-2%Sc alloy, as described below.

NioCorp

A new Nb-Sc-Ti mine and production facility in Elk Creek, Nebraska is based on a resource with 72 ppm Sc and 0.56% Nb_2O_5. NioCorp announced completion of pilot plants for scandium recovery in April 2016[67] and whole ore pre-leach in June 2016,[68] and plans to produce 97 TPY Sc_2O_3. We estimate production will start in 2019-2020.

Australian Mines

Australian Mines has purchased options on two new scandium projects in Australia.[69] The Flemington Ni-Co-Sc mine in NSW is estimated to have 2085 tonnes of economically recoverable Sc_2O_3 at 434 ppm. The Sconi mine in QLD was optioned from Metallica Materials[70] also for Ni-Co-Sc production. The company plans 50 TPY Sc_2O_3 production at Sconi by 2020, plus an unknown amount at Flemington.

Scandium International

A new primary scandium mine in Nyngan, NSW, Australia is based on a 235-500 ppm Sc deposit. Scandium International plans to produce 37.7 TPY Sc_2O_3,[71] and has finalized their Environmental Impact Statement, flowsheet, and models.[72] We estimate production will start in 2020-2021.

CleanTeq

A new primary scandium mine in Syerston, NSW, Australia is based on a 386-686 ppm Sc deposit.[73] Though CleanTeq built a pilot plant,[74] production timeline and final volume are unclear, we estimate 30 TPY Sc_2O_3 production starting around 2020-2022.

Platina Resources

A new platinum-scandium mine at Owendale, NSW, Australia includes 380+ ppm Sc, 500 ppm Co, and 0.5 ppm Pt. Platina Resources has completed its pre-feasibility study proposal and a path to its Environmental Impact Statement with production rate estimate of 30 TPY Sc_2O_3.[75] We estimate a production start date in 2021-2023.

Texas Mineral Resources

A new project would produce scandium from coal ash and overburden.[76]

Current Production and Literature

INFINIUM believes that today's Al-2%Sc master alloy likely comes from dissolving scandium metal in aluminum. Such dissolution is slow, and yield is irregular, due to the high melting point of Sc, and formation of a set of high-melting intermetallic compounds at the interface. Figure 207 shows the Al-Sc phase diagram illustrating these points. Scandium metal is also expensive, due to costly processes operating at small scale, such as hydrofluorination at 700°C and calcium metallothermic reduction in welded tantalum retorts.

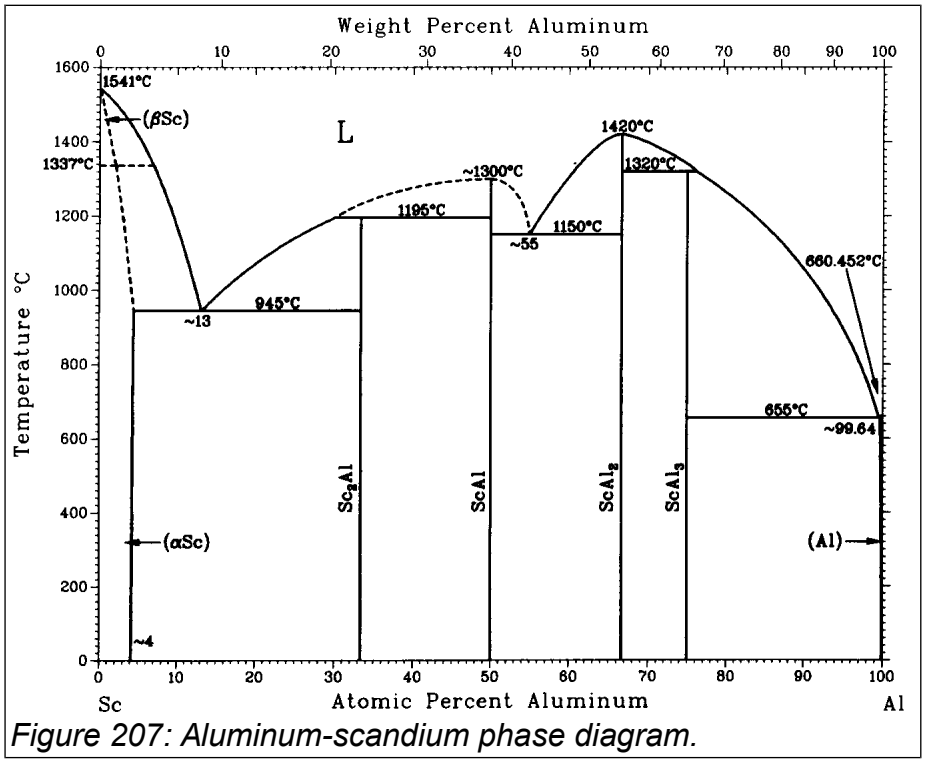

Figure 207: Aluminum-scandium phase diagram.

Despite limited availability of scandium, the high performance of Al-Sc alloys has motivated work on variations around two approaches to direct alloy production approaches: electrolysis and metallothermic reduction.

Electrolysis

Papers and patents on electrolysis generally reduce mixtures of Al_2O_3 and Sc_2O_3 dissolved in sodium or potassium cryolite AlF_3-NaF or AlF_3-KF. They produce Al alloys with 0.5-1.5 wt% Sc.[77-80] One group used electrolysis of Sc_2O_3 in $CaCl_2$ with an Al cathode which produced Al-2%Sc, but with up to 0.65 wt% calcium, making the alloys unusable.[81] On the other end of the spectrum, a 1963 patent by Nuclear Corp. of America produced solid Sc metal granules by electrolysis of Sc_2O_3 dissolved in ScF_3-NaF.[82] Because this requires separation of the granules from solid salt afterward, this apparently has not been commercialized.

Metallothermic

It is well known in literature that ScF_3 reacts with Al to produce AlF_3 and Al-Sc alloy.[83,84] The high Sc-basis cost of ScF_3 has motivated multiple efforts at promoting reaction of Sc_2O_3 with Al as follows:

Powder mixture: Alcoa patented a method for mixing Al powder with Sc_2O_3, pressing the result into pellets, and then immersing those pellets into liquid Al to produce Al-2%Sc.[85] Sc reduction yield was inconsistent (reported at up to 114% in some cases, average of 98.5%), and depended on pellet forming conditions.

Reaction with calcium or magnesium metal: Several papers use Ca vapor or in alloy to reduce Sc_2O_3 into an Al alloy.[86,87] The resulting alloy generally has at least 1% Ca, making it unusable. Magnesium can be a suitable reducing agent,[88,89] and one Russian paper described reacting Sc_2O_3 with an Al alloy with 15-17% Mg, producing an alloy with up to 2.5% Sc,[90] but this paper omits important process details (e.g. "Preliminarily, an Al-Mg alloy carefully mixed with a flux was charged into a reaction vessel" with no details on the flux or mixing method). It also results in Mg:Sc ratio unsuitable for alloys such as Scalmalloy without distilling out Mg.

Note it is possible that the Chinese Al-2%Sc master alloy purchased by INFINIUM mentioned above was produced by Sc_2O_3 or ScF_3 reaction with Al-Ca alloy, rather than dissolution of Sc with retained Ca from a retort reduction process.

In situ fluoride conversion: One patent uses NH_4HF_2 to convert Sc_2O_3 to fluoride in situ and react it with Al_2O_3.[91] The thermodynamics of this process are unclear, as are the disposition of ammonia and HF during and after the reaction. It is likely not practical for use due to capital cost of safe operation. Another dissolves ScF_3 or Sc_2O_3 in chloride-fluoride salts, such as with AlF_3-NaF-KCl,[92] but with low conversion yield from the oxide.

Mechanical alloying: One study reported reduction of Sc_2O_3 by Al to produce Al-Sc by ball milling followed by remelting,[93] but this is very slow and expensive.

Summary

The above methods generally fall into four categories: (1) use of ScF_3 (or its formation from Sc_2O_3 *in situ*), (2) reduction by Ca or Mg which remain in the alloy, (3) preparation of pressed metal-Sc_2O_3 powder pllets, or (4) electrolysis of Al_2O_3-Sc_2O_3 producing Al with a small amount of Sc. The latter option results in high yield using low-cost Sc_2O_3, but at small scale is a relatively expensive method for producing the large amount of aluminum in the alloy.

Our Approaches

Based on the above considerations, INFINIUM proposes—and has made significant progress toward—electrolytic reduction of Sc_2O_3 dissolved in a ScF_3-AlF_3-NaF bath. This either reduces Sc_2O_3 to inject Sc into an Al cathode and produce Al with 2-10% Sc, or co-reduces Al_2O_3 and Sc_2O_3 to produce Al-20%Sc or Al-10%Sc. The co-reduction approach can use an Al cathode to dilute the alloy to 2% Sc *in situ* as it is produced, or use separate Al dilution after reduction.

Figure 208 compares flow sheets and costs for today's production of Al-2%Sc master alloy *vs.* the proposed processes. These would use Sc_2O_3 at high yield, minimizing raw material costs. They would also require much lower current (fewer and/or smaller cells) than complete reduction of all of the aluminum for Al-2%Sc.

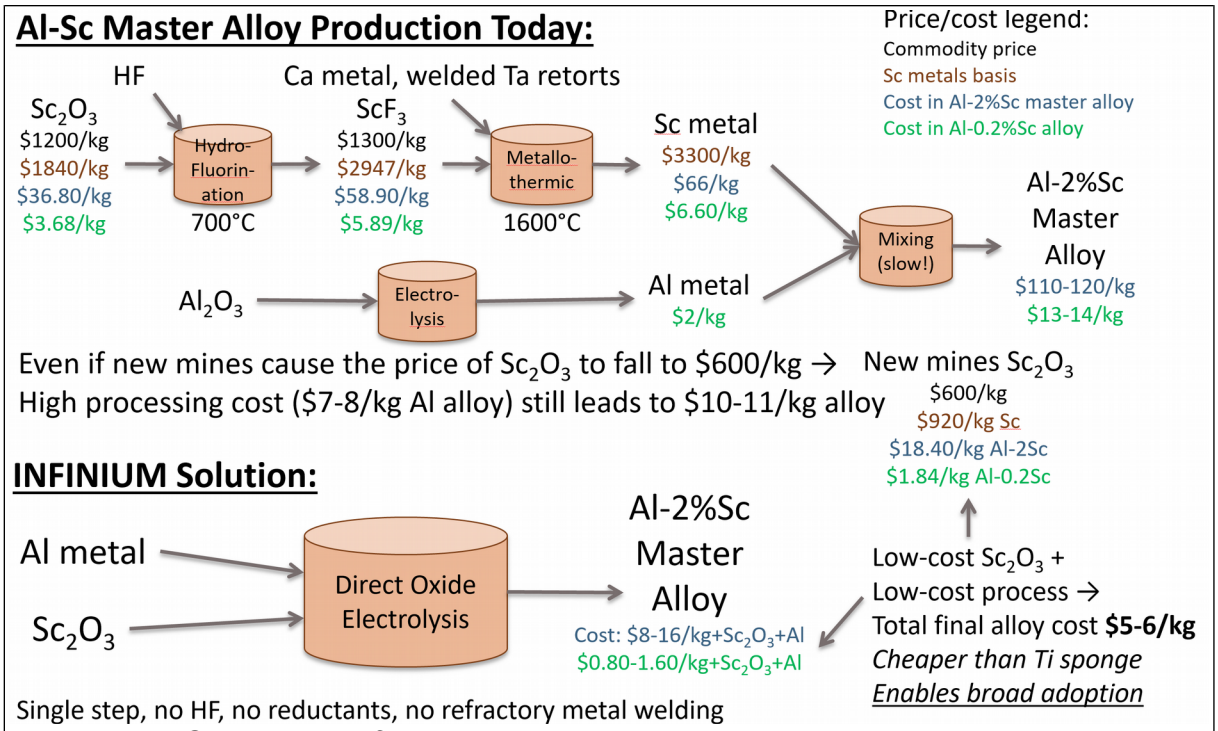

Figure 208: Comparative flow sheets and costs illustrating INFINIUM cost advantage in Al-2%Sc master alloy production and impact on final alloy cost.

The INFINIUM team began this Al-Sc alloy journey with a goal of reducing all of the 98% Al and 2% Sc in a manner similar to the electrolysis papers listed above, and achieved a similar result, as described in Exp040-042 (p. 233). The team concluded that we do not want to expensively make inexpensive aluminum.

The first experiment on Sc_2O_3 reduction from a ScF_3-LiF bath into an aluminum cathode resulted in considerable metallothermic reaction, and the higher bath density than liquid aluminum led to bath-metal mixing (Exp044 p. 239). We then focused on aluminothermic reduction of ScF_3, because of its simplicity, with an attempt at aluminothermic reduction of Sc_2O_3. But we could not reduce Sc_2O_3 using Al, observed ScF_3 reduction yield did not exceed 27%, and scandium fluoride cost is about 60% higher than oxide on a metals basis, leading to expectation of higher raw material cost when using this route. At the end of the project, we returned to electrolysis with a new approach as described starting on the next page.

Aluminum-Scandium Alloy Electrolysis Approach

INFINIUM completed bath density measurements (p. 246), finalized bath composition, and conducted electrolysis trials, first with Sc_2O_3 dissolved in the bath, and then with additional feeding of Sc_2O_3.

The AlF_3 in the bath has two purposes: it reduces density, and it balances out the metallothermic reaction between ScF_3 in the bath and the Al cathode by providing Al ions for reduction. As described in Figure 209, this cell will have both metallothermic and electrolytic reactions, the former constantly and the latter under applied potential. Indeed, here in Exp 064-067, we see metallothermic reaction transferring Sc from bath to metal, and electrolysis increasing the metal cathode's concentration of either Al, as demonstrated in Exp 065-066, or Sc, as demonstrated in Exp 067.

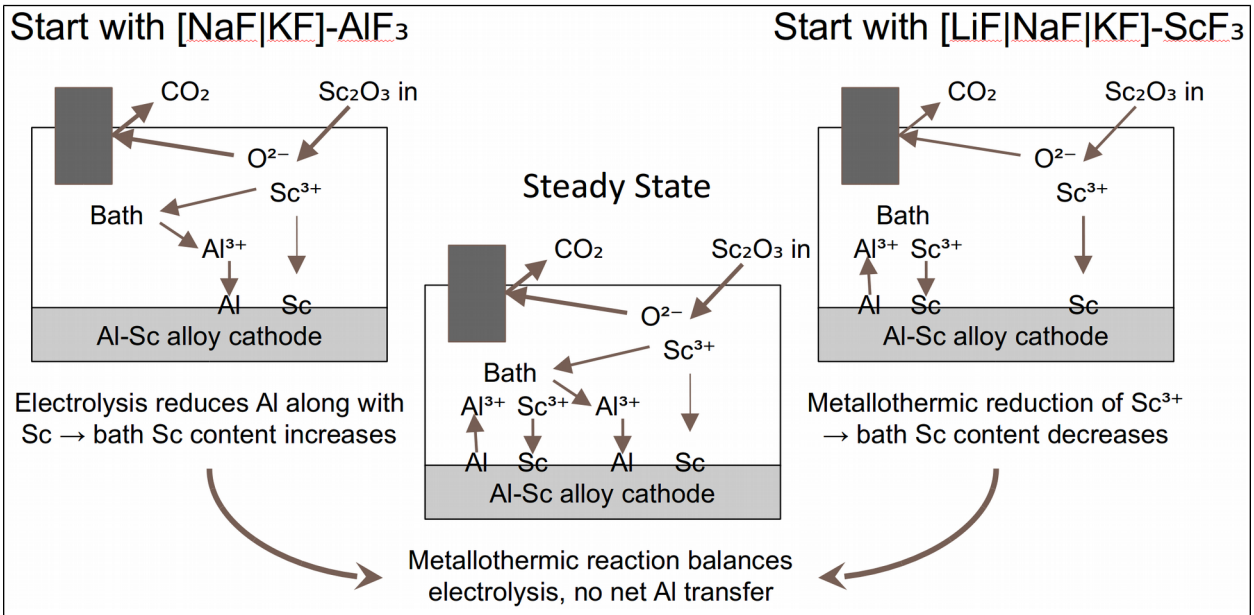

Figure 209: Electrolytic & metallothermic reactions leading to bath-metal steady-state.

In these electrolysis experiments, the concentration of Sc in bath far exceeded that reduced into the cathode. This makes sense due to the lower electronegativity of Sc than Al. But this makes it difficult to precisely characterize the ratio of metallothermic to electrolytic reaction, nor to establish or measure a long-term steady-state bath composition. However, by feeding only Al metal and Sc_2O_3, these experiments demonstrated the ability to make Al-2%Sc using very low-cost input materials as described in the cost model on p. 216. This demonstrates an electrolysis intermediate between co-reduction of both oxides to make all of the alloy, and reduction of Sc_2O_3 to pure scandium for later mixing with Al. This intermediate path is not only efficient with low-cost raw materials, but also novel and likely patentable, and thus can form the foundation for scale-up and commercialization of low-cost production of Al-Sc alloy.

Aluminum-Scandium Scale-Up Electrolysis Cell Narrative

Overview

INFINIUM uses industrial scale-up and deployment vision based on Techno-Economic Analysis (TEA, p. 216) to guide experiment decisions. The TEA shows that of the two approaches considered in this project, electrolysis is most likely lower-cost than metallothermic production, such that electrolysis scale-up is outlined here.

Detailed Design Narrative

Producing Al-2Sc master alloy by electrolysis using scandium oxide can proceed at some point between two extremes. On the one hand, one can co-reduce both metals from their respective oxides at the appropriate ratio; this is the INFINIUM approach for producing Mg-Nd master alloy.[94] This is discussed on p. 27, with techno-economic analysis for a 500 TPY plant having total cell current of 200 kA (*e.g.* 10×20kA cells) and showing production cost just over $8.00/kg plus the scandium oxide input.

On the other hand, rather than produce $8/kg aluminum, it should be much less expensive to start with commodity aluminum at close to $1.50/kg, and use it as a reactive liquid cathode to electrowin scandium metal into it from ScF_3-based bath such as ScF_3-AlF_3-NaF-Sc_2O_3. For this approach, a 500 TPY plant would only need to actually electrolyze 10 TPY of scandium, which would require about 3000 A current at high current efficiency and utilization. (In practice, with lower current efficiency and electrolysis time it could require as much as 7000 A.) However, adding pure Sc to the surface of the liquid aluminum cathode at high current density could produce solid Al_3Sc at the bath-metal interface, possibly creating dendrites and shorting the cell.

Between these two there are intermediates, such as purchasing 90% of the alloy as aluminum metal and electrowinning the other 8% Al and 2% Sc, or purchasing 80% as Al and producing the other 18% Al and 2% Sc as described below. This should avoid precipitation of Al_3Sc, but will require much more cell current, and more production cells, than electrowinning pure Sc into an Al cathode.

Electrowinning 2 wt% Sc Into Liquid Aluminum

As mentioned above, this method would use a reactive cathode of liquid aluminum and electrolyze Sc_2O_3 dissolved in ScF_3-AlF_3-NaF bath. Figure 210 shows the Al-Sc phase diagram,[95] and indicates that at 2 wt% Sc, the liquidus temperature is about 800°C. Based on experiments, particularly Exp054, Exp055 and Exp056 (p. 221), our new target temperature is 1000°C. This and new density measurements led to determination of electrolyte bath composition as described in Exp058 (p. 246).

A 500-800A cell would reduce 200-250 g/hr Sc depending on current efficiency, and produce 10-12 kg/hr Al-2%Sc. 10-12 kg batches would need to be tapped every hour, or 100 kg batches could be tapped twice/day.

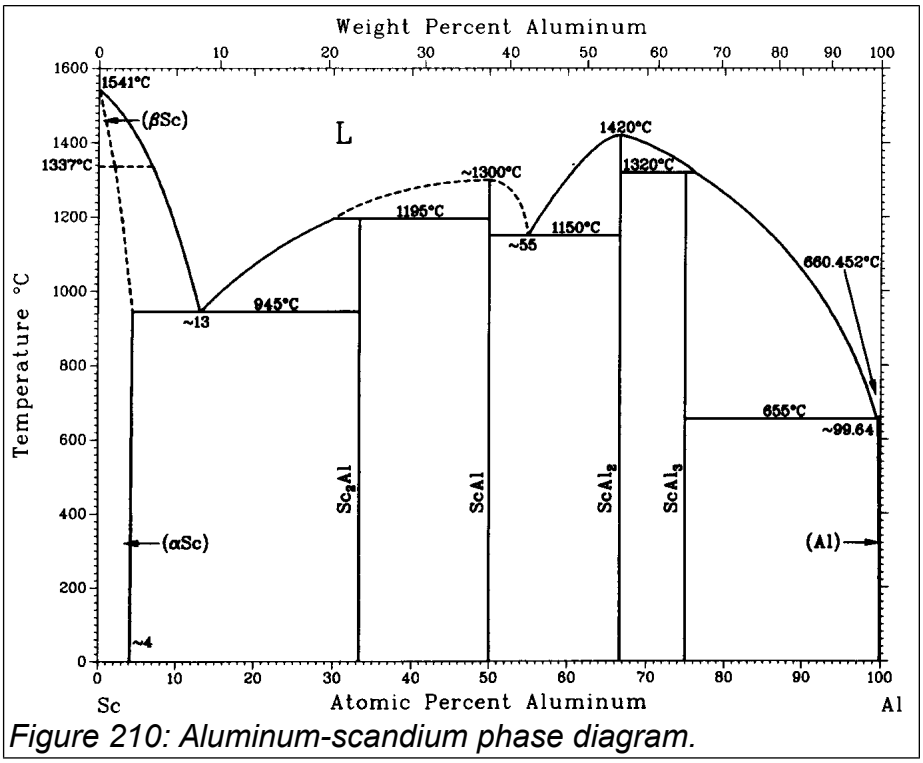

Figure 210: Aluminum-scandium phase diagram.

For example, a 40 cm ID crucible with 120 kg metal (53 liters at density 2.3) would have 42 cm metal depth. An additional 20 kg molten salt bath (density 2.173) would have 7.3 cm depth. Rougly optimal bath composition is described in Exp 058 (starting p. 246) as 26.8 wt% ScF_3, 24.6% AlF_3, 46.6% NaF, 2% Sc_2O_3. Operation would proceed as follows:

- Melt, heat to 1000°C and charge 98 kg Al to the crucible, adding to the 20 kg Al-2%Sc already present, taking care to minimize mixing of bath into the metal.
- Run electrolysis at 500-800 A (0.4-0.65 A/cm² on the bath-metal interface) for 8-10 hours, which would produce 2 kg Sc at roughly 45-90% current efficiency. Observed current efficiency will determine the electrolysis current and duration required for this batch size and tapping frequency. Note that the 20 kg bath contains just 400 g Sc_2O_3, so this would require a continuous feed of the 3 kg Sc_2O_3 needed to produce this alloy over the course of the run.
- Remove (*e.g.* siphon) 100 kg Al-2%Sc, leaving 20 kg Al-2Sc behind in the cell.

A production rate of 200 kg/day over 250 days/year corresponds to about 50 tonnes/year of production from a single cell. A 500 TPY plant would have about ten of these cells, *i.e.* total current of 5000-8000 A, *cf.* 200 kA for co-reduction of all of the Al and Sc as described above.

Next Steps

The next step will be to build and operate this type of cell for pilot production of Al-2%Sc master alloy at 1 kg and then 10 kg batches.

The first priority would be to manage the carbon dust issue found in Exp 065-067 (p.

249). This can be managed by one of several methods including:

- Use controlled localized introduction of air to burn off carbon dust;
- Insert an insulating dam between the anode and cathode connection (which could be the crucible) to prevent carbon dust from forming a complete bridge and shorting the cell;
- Develop more advanced cell design, such as INFINIUM Pure Oxygen Anodes™ to eliminate anode production of carbon dust, and a TiB_2-lined crucible.

Second, in the ScF_3-AlF_3-NaF bath with dissolved Sc_2O_3, there are four variables which need to remain in balance in order to achieve constant bath composition, toward steady-state electrolytic production of Al-Sc alloy and consistent product composition.

- O:F anion ratio Fluoride content in the bath is relatively constant, and oxygen content changes according to the balance between Sc_2O_3 feed rate and oxygen consumption at the anode. As a first cut, current decay at constant voltage, or voltage increase at constant current, can serve as control signals to indicate when and how much it is necessary to change feed rate; INFINIUM has experience with this and similar feed control methods.
- Sc:Al cation ratio Sc_2O_3 feed rate and the relative rates of metallothermic and electrolytic reactions change this ratio in a complex way. As shown in Figure 1 (p. 11) the metallothermic reaction moves Al from metal to bath, and Sc from bath to metal. The electrolytic reaction moves both from bath to metal. Therefore, increasing current results in more electrolysis reaction relative to metallothermic, and less Sc in the bath relative to Al. In addition, because Al is more electronegative than Sc, increasing cathode current density leads to higher ratio of reduced Sc^{3+} to Al^{3+} at the cathode, leading to less Sc in the bath than Al. So current can be used to regulate this parameter. That said, controlling it requires measurement of either bath or metal composition, such as by X-Ray Fluorescence (XRF) or Laser-Induced Breakdown Spectroscopy (LIBS).
- Cryolite ratio Na:(Sc+Al) Like fluoride, sodium is a relatively passive bystander, but it leaves the bath by evaporation more quickly than ScF_3 or AlF_3. Keeping it in control will require feeding excess NaF.
- Bath content Evaporation and metal removal both decrease total bath content, and level in the cell. For cost modeling purposes, we assume bath loss is roughly 1 kg per 100 kg of production, such that one must feed that much ScF_3-AlF_3-NaF mixture into the cell. This is a much higher loss rate than INFINIUM has experienced with other electrolysis processes, but a lower rate on the basis of metal product. It may be possible to recover some of the bath taken out of the cell in product removal, though maintaining purity could be a challenge. And it may be possible to feed just AlF_3 and Sc_2O_3, or NaF and Sc_2O_3, generating ScF_3 *in situ,* though we have yet to demonstrate this capability.

Aluminum-Scandium Electrolysis Techno-Economic Analysis

Overview

This project considered three methods for Al-Sc master alloy production: co-electrolysis of Al_2O_3 and Sc_2O_3, electrolysis of Sc_2O_3 into a reactive aluminum cathode, and metallothermic reduction of ScF_3 by aluminum. This section compares a comprehensive model of co-electrolysis costs, with a cost model of electrolysis raw materials and utilities, and raw material costs for metallothermic reduction.

Al-Sc Co-Reduction Cost Model

The pure aluminum technical cost model section beginning on p. 56 includes a result for a small plant producing 250-1000 tonnes/year aluminum on page 59. Producing Al-2%Sc using this method should have very similar operating cost to producing pure Al, with the notable exception of high Sc_2O_3 input cost. Results showed operating cost of $8.15/kg plus Sc_2O_3 for a 500 TPY plant making Al-2%Sc from oxides using Pure Oxygen Anodes™. This is much higher than for a large aluminum plant, but much lower than the price premium over raw material costs (Al-2%Sc price $115/kg, scandium basis cost of 2% Sc_2O_3 is $38/kg, so price premium is $77/kg).

Reactive Cathode Electrolysis Cost Model

Mass Balance

The overall reaction for this process using the range of approaches described above is:

$$Al + Al_2O_3 + Sc_2O_3 + C \rightarrow \text{Al-Sc alloy} + CO + CO_2 \quad (37)$$

Target product alloy composition is Al-2 wt%Sc for commercial master alloy. The model can accept any range of Al metal and oxide input fractions. It assumes 2:1 CO:CO_2 anode production mole ratio, which roughly doubles carbon consumption and CO_2 emissions vs. 100% CO_2. For this study, we consider three cases, written using mass coefficients relative to product output (numbers may not total exactly due to rounding):

1. $0.98\ Al + 0.031\ Sc_2O_3 + 0.006\ C \rightarrow 1.0\ (Al\text{-}2\%Sc) + 0.009\ CO + 0.007\ CO_2$
2. $0.9\ Al + 0.152\ Al_2O_3 + 0.031\ Sc_2O_3 + 0.046\ C \rightarrow 1.0\ (Al\text{-}2\%Sc) + 0.072\ CO + 0.056\ CO_2$
3. $0.8\ Al + 0.340\ Al_2O_3 + 0.031\ Sc_2O_3 + 0.096\ C \rightarrow 1.0\ (Al\text{-}2\%Sc) + 0.149\ CO + 0.117\ CO_2$

The model estimates bath loss rate at 20% of Sc_2O_3 feed rate, which is a high loss rate to be conservative. Bath composition is 50-33-17 ScF_3-NaF-AlF_3 by mass, plus dissolves 5 wt% total oxide, with Al:Sc oxide concentration ratio in the bath assumed to be equal to the oxide feed ratio. This leads to 94.0% of the scandium input from Sc_2O_3 and 6.0% from ScF_3, with 93-94% overall scandium yield.

There is an additional inert gas stream set at 80 mol% of the exit gas, *i.e.* four times the molar flow rate of $CO+CO_2$. The gas defaults to argon, with an option to use nitrogen.

Default plant output is 100 tonnes/year (TPY) at 80% capacity factor, *i.e.* 14.3 kg/hour. At this scale the plant uses about 3 TPY Sc_2O_3, which is a significant fraction of current world production, but a small fraction of likely future production.

Raw Materials Costs

In July 2016, INFINIUM was able to purchase Sc_2O_3 for $1200/kg and ScF_3 for $1300/kg, as described in the Introduction. Here we consider this as the base case, as well as a future scenario in which high production reduces the Sc_2O_3 cost to $600/kg.

Energy Balance and Utilities

The energy balance begins with theoretical current required to reduce the oxide feed, which is 510 A, 3910 A, and 8160 A for the three mass balance cases listed above. This is divided by assumed current yield of 70% to give total plant current (summed over all cells) of 730 A, 5590 A and 11.7 kA respectively. Maximum cell current is 2000 A, resulting in 1, 3 and 6 cells in the plant respectively. Cell voltage is 4 V, and with low current, cells require four times as much power for heating – particularly aluminum metal heating – *vs.* for electrolysis.

Consumables

The model only explicitly includes fluoride bath (assumes ScF_3 input at ~10% of Sc_2O_3 input rate), carbon anodes and argon gas as consumables. It assumes consumption of 60% of the carbon, with 40% wasted in stubs, again to be conservative. There is an additional "Miscellaneous Materials" line item set to 3% of raw materials to account for other items.

Cost Model Results

Table 31 shows the cost of raw materials and utilities for the three cases described above, using a price of $1200/kg for Sc_2O_3. Labor and overhead cost estimates are not mature at this point.

Cost Category	Case 1 98% Al	Case 2 90% Al	Case 3 80% Al
Raw Materials	$45.41/kg	45.89	46.86
Utilities	.09	0.67	1.41

Table 31: Raw material and utility costs in $/kg Al-2%Sc product for the three production cases with cost of Sc_2O_3 at $1200/kg.

Case 1 has the raw material and utilities costs. Though not discussed here, because of its much lower current, that case also has the fewest cells and lowest capital cost. Across the board, raw material costs are much higher than utilities costs.

With a 50% reduction in Sc_2O_3 cost to $600/kg, prices fall considerably, as shown in Table 32. In this scenario, raw material cost is much lower, but other costs such as utilities remain the same.

Cost Category	Case 1 98% Al	Case 2 90% Al	Case 3 80% Al
Raw Materials	$24.31/kg	24.95	25.94
Utilities	.09	.67	1.41

Table 32: Raw material and utility costs in $/kg Al-2%Sc product for the three production cases with cost of Sc_2O_3 at $600/kg.

Metallothermic Raw Material Costs

The metallothermic reaction is:

$$Al + ScF_3 \rightarrow Al\text{-}Sc\ alloy + AlF_3 \quad (38)$$

At $1300/kg ScF_3, the scandium metal basis cost is $2946/kg, leading to $58.93/kg for the scandium in Al-2%Sc alloy assuming perfect material yield. The best experimental scandium material yield achieved in Task 2 was 27%, although others have reported considerably higher yields. Reaching $100/kg requires over 60% material yield. And achieving the $65/kg upper bound for electrolysis cost requires 94% material yield.

Aluminum-Scandium Alloy Experiments

Metallothermic Reduction

Exp045-053 Metallothermic Reduction of Al-2%Sc Alloy and Scandia Solubility

In an attempt to understand the solubility of scandium oxide in the ScF3-LiF matrix with the presence of Al metal, it was found that Sc migrated into the metal statically. To understand this observation, a series of experiments were run to see how long the reaction takes and how the metal and flux interact. Two procedures were used simultaneously to maximize experiment throughput and data collection. To characterize the bath/metal interaction, one procedure used an induction coil to quickly melt the sample and visually observe the interaction. For temperature control and stable atmosphere, another procedure used the quad cell.

Controlled Quad Cell Procedure

1. Load metal on bottom of isomolded graphite crucible.
2. Pre-mix flux with oxide and pack on top of the metal inside the crucible.
3. Load the crucible into QC3
4. Dry the flux at 90C for 4 hours and 400C for 12 hours.
5. Heat up graphite at 10C/min to desired operating temperature (800,900,1000C).
6. Once molten, mix the metal using a 3/8” isomolded graphite rod.
7. Allow bath and metal to stay molten for 8 hours until cooling down.

   (NOTE: the bath was quenched when repeating this experiment by adding a tapped hole into the graphite crucible and pulling it out of the hot zone of the furnace when the 8 hours had elapsed).

High-throughput Induction Coil Procedure

1. Load metal on bottom of isomolded graphite crucible.
2. Pre-mix flux with oxide and pack on top of the metal inside the crucible.
3. Load the crucible into the induction coil and cover with a graphite lid.
4. Adjust the current to be constant at 400A.
5. Once the graphite is orange, examine every 2 minutes until the mixture is molten.
6. Once molten, mix the metal using a 3/8” isomolded graphite rod.
7. Keep hot for 5 minutes. Then stop applying current.
8. Observe the bath as it begins to freeze. The metal should freeze first, allowing the flux to be poured off the top.

9. Slowly quench the flux in a steel cup, leaving the frozen metal behind.
10. Cap the crucible again and heat up the metal.
11. Once the metal is molten, quench onto a separate steel cup.

Metal that was quenched in the induction coil onto a steel cup was digested and analyzed via ICP-OES. Below is an example of the metal concentration and minor impurities found:

Exp045 ICP-OES Analysis Report

Metal: Al-Sc

Element	Concentration (wt. %)
Sc	5.756
Fe	0.017
Li	<0.005
Zr	<0.005
Mg	<0.005
Al (balance)	94.21

Notes: Re-melted via induction coil

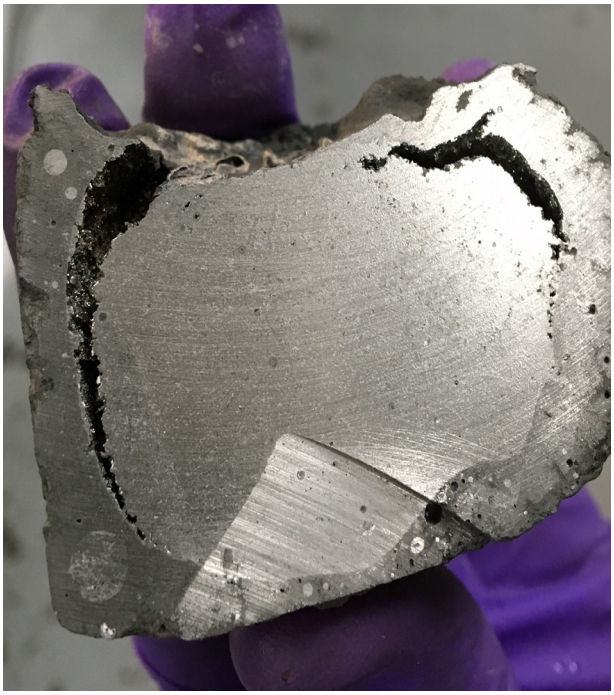

Figure 211: Visual cross-section of Al-Sc after melting in induction coil

Exp048 ICP-OES Analysis Report

This experiment reacted 50 g AlF_3-KF-Sc_2O_3 with 10 g Al metal. The flux phase had 5 wt% Sc_2O_3, or 2.5 g. Following the reaction, the metal had 0.5 wt% Sc, or 0.05 g Sc, indicating 3% conversion of Sc_2O_3.

Exp054-056 Equilibrium Studies and Flux Characterization

Optimizing the flux chemistry alongside crucible temperature and metal:bath ratio will directly impact the Sc compound to metal conversion rate and quality. There are switching reactions that complicate the bath chemistry, and as a result, makes it difficult characterize the flux. A matrix of experiments were designed to target a bath chemistry with high oxide solubility, low volatility and low cost. Table 33 below highlights the experimental conditions and resulting scandium concentrations measured.

Experiment and Well Designation	Flux Composition	Temp (C)	Soak Time (hrs)	Oxide Concentration (wt%)	Metal:Bath Ratio	Resulting Scandium Concentration (%)
Exp054-W1	NaF-AlF3	800	4	5	0.2	0.5 - EDS
Exp054-W2	NaF-AlF3	800	4	5	5	0.5 - EDS
Exp054-W3	ScF3-LiF	800	4	2	5	0.7 – EDS
Exp054-W4	NaF-AlF3	800	24	5	0.2	0.6 – EDS
Exp054-W5	NaF-AlF3	800	24	5	5	0.5 - EDS
Exp054-W6	LiF-ScF3	800	24	2	5	0.8 – EDS
Exp055-W1	LiF-ScF3	1000	8	23 (excess)	0.5	4.9 – EDS
Exp055-W2	LiF-ScF3	1000	8	23 (excess)	1	3.1 – EDS
Exp055-W3	LiF-ScF3	1000	8	0	0.29	1.1 – EDS
Exp055-W4	LiF-ScF3	1000	8	0	0.57	0.7 – EDS
Exp055-W5	NaF-ScF3	1000	8	10	0.29	6.11 - ICP
Exp056-W1	NaF-ScF3	1000	8	10	0.29	4.55 - ICP
Exp055-W6	NaF-ScF3	1000	8	0	0.2	5.94 - ICP
Exp056-W2	NaF-ScF3	1000	8	0	0.2	5.85 - ICP
Exp055-W7	NaF-ScF3	1000	8	10	0.57	6.03 - ICP
Exp056-W3	NaF-ScF3	1000	8	10	0.57	4.37 - ICP
Exp055-W8	NaF-ScF3	1000	8	0	0.4	4.15 - ICP
Exp056-W4	NaF-ScF3	1000	8	0	0.4	3.34 - ICP
Exp055-W9	NaF-ScF3	1000	8	10	0.91	3.3 - ICP
Exp056-W5	NaF-ScF3	1000	8	10	0.91	3.97 - ICP
Exp055-W10	NaF-ScF3	1000	8	0	0.81	3.59 - ICP
Exp056-W6	NaF-ScF3	1000	8	0	0.81	2.77 - ICP

Table 33: Bath-metal equilibrium overview: experimental parameters and resulting scandium concentration in the aluminum alloy.

Statistical analysis

In this designed experiment the amount of aluminum and scandium oxide were varied, the scandium fluoride and sodium fluoride were held fixed. It was found the ICP-OES technique used could show differences in the treatment of the sample as shown in the figure below. The values are the grams of oxide, fluoride or aluminum metal used in the experiment. The designed experiment is shown in Table 34.

Treatment	ScF_3 (g)	NaF (g)	Sc_2O_3 (g)	Al(g)
T1	11.76	19.2	3.1	8.8
T2	11.76	19.2	0	6.215
T3	11.76	19.2	3.1	17.6
T4	11.76	19.2	0	12.45
T5	11.76	19.2	3.1	34.6
T6	11.76	19.2	0	24.9

Table 34: Metallothermic reduction designed experiment.

Where:

Treatment	ScF_3	NaF	Sc_2O_3	Al
T1	11.76	19.2	3.1	8.8
T2	11.76	19.2	0	6.215
T3	11.76	19.2	3.1	17.6
T4	11.76	19.2	0	12.45
T5	11.76	19.2	3.1	34.6
T6	11.76	19.2	0	24.9

One-way ANOVA: T1, T2, T3, T4, T5, T6

```
Sourc DF     SS     MS      F      P
Factor 5   35.60   7.121  20.9   0.000
Error 29   9.879   0.341
Total 3    45.484

S = 0.5836   R-Sq = 78.28%   R-Sq(adj) = 74.54%


Individual 95% CIs For Mean Based on
Pooled StDev
Level  N   Mean    StDev  ---+---------+---------+---------+------
T1     6  5.3343  0.8552                       (----*----)
T2     6  5.8923  0.1053                             (----*----)
T3     5  5.0350  0.9261                     (----*-----)
T4     6  3.7459  0.4471        (---*----)
T5     6  3.6328  0.3653      (----*----)
T6     6  3.1759  0.4623  (----*----)
                          ---+---------+---------+---------+----
```

This shows that this technique can be used to analyze the samples with very little

variance, and also shows that the response in terms of concentration are statically significant since the P value was zero. Treatment 2 produced the highest concentration of scandium in the final aluminum alloy.

To make an estimate of the conversion, the total amount of scandium in the flux, both scandium oxide and fluoride were summed and the concentration of scandium in the final alloy used, it was assumed that the final weight of alloy was the same as initial charge of aluminum, this does add an error since some of the aluminum was consumed in the reaction, increasing the concentration in final alloy.

One-way ANOVA: t11, t12, t13, t14, t15, t16

```
Source  DF        SS        MS      F      P
Factor   5  0.060365  0.012073  48.59  0.000
Error   29  0.007206  0.000248
Total   34  0.067572

S = 0.01576   R-Sq = 89.34%   R-Sq(adj) = 87.50%
                  Individual 95% CIs For Mean Based on
                Pooled StDev
Level  N     Mean    StDev  -----+---------+---------+---------+----
T1     6  0.06529  0.01047  (---*--)
T2     6  0.07097  0.00118    (--*---)
T3     5  0.12326  0.02267                  (---*---)
T4     6  0.09016  0.01072         (---*---)
T5     6  0.17484  0.01758                              (---*---)
t6     6  0.15283  0.02225                           (---*--)
                              -----+---------+---------+---------+----
                                  0.070     0.105     0.140     0.175

Pooled StDev = 0.01576
```

Treatment five, the largest excess of aluminum produced the greatest conversion.

General Regression Analysis: Convert versus Gr Al, Sc_2O_3

```
Regression Equation
Convert  =  0.0431816 + 0.00423096 Gr Al - 0.00286684 Sc2O3
Coefficients

Term            Coef    SE Coef        T      P
Constant   0.0431816  0.0058821   7.3411  0.000
Gr Al      0.0042310  0.0002996  14.1238  0.000
Sc2O3     -0.0028668  0.0019200  -1.4931  0.145

Summary of Model
S = 0.0167872      R-Sq = 86.65%        R-Sq(adj) = 85.82%
PRESS = 0.0107831 R-Sq(pred) = 84.04%
```

This work shows that conversion is not improved by addition of scandium oxide in the flux as the P value is above 0.05. The most significant factor is the amount of excess

aluminum present to convert the scandium fluoride to scandium metal by aluminum. This can be illustrated by plotting the recovery of scandium values from the flux as function of scandium oxide addition.

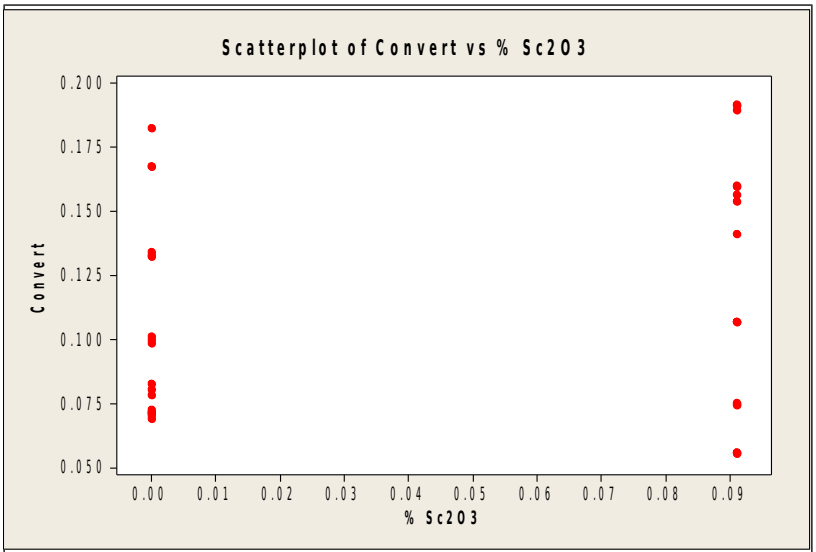

Figure 212: Conversion as a function of scandium oxide additions. There is no improvement of conversion with oxide present

Engineering hurdles

One challenge was the high volatility of the potassium cryolite flux. The more volatile the flux, the quicker the composition can shift, decreasing cell efficiency. Composition change can also result in a higher viscosity melt, which can slow mass transport and kinetics in the bath.

Solubility of scandium oxide in ScF_3-LiF was also low (at most 2wt%, see Exp043 p. 220). In electrolysis, low oxide solubility limits the production rate and can also lead to fluoride reduction. Fluoride reduction is unwanted due to the production of CF_4 gas and also a shift in flux ratio, leading to higher viscosity and slower kinetics as mentioned before.

Exp059-063: Vacuum Metallothermic Conversion of Al-Sc in ScF_3-NaF Matrix

Overview

The goal was to test the validity of a thermodynamics paper that claims the potential for upwards of 90% conversion of scandium fluoride to Sc metal dissolved in Al under 100 Pa pressure at 865 C, by a proposed mechanism of AlF evaporation from the bath.[96] The challenge is to achieve this pressure with a mechanical pump which will require tight control of the sealing surfaces. The vessel is 310 stainless steel, which can hold vacuum at the operating temperature, the crucible is isomolded graphite and the flux is a ScF_3-NaF matrix.

Experimental Parameters

Furnace: Quad Cell 3 (resistance heating)

Flux: 38.4gNaF- 23.5gScF_3 – 49.8g Al metal

Crucible: Isomolded graphite - 3” OD x 2.75” ID x 2.5” Depth

Crucible Temperature: 1000^{o}C.

Procedure

1. Premix flux components in the chemical fume hood.
2. Load all of the components into the isomolded graphite crucible.
3. Ensure that the inside surface of the 310 stainless steel vessel is clean and free of any debris or water absorbing insulation.
4. Use isopropyl alcohol to clean each port and the gasket and then cover with a light layer of vacuum grease.
5. Load the crucible into the furnace and attach vacuum line.
6. Slowly pull vacuum to the vessel to prevent the powders from becoming airborne.
7. Plug in the K-type thermocouple and begin heating at 10C/min until 1000C is reached.
8. Keep furnace hot for 24 hours and then cool the furnace down to room temperature at 10C/min.
9. Once cool, slowly back-fill the vessel with air until atmospheric pressure is reached.
10. Remove the assembly, break apart and analyze the Al metal for Sc content via ICP-OES.

Results and Discussion

The thermodynamic calculations mentioned above predict that AlF gas evaporates from

the bath, and it should condenses up in the cold zone of the cell. Figure 213 shows the vessel post-experiment and likely confirms this theory. Table 35 shows the results measured via ICP-OES for scandium concentration and other impurities. The concept paper showed higher yields correlated to higher vacuum, with a noticable jump at 100 Pa. However, due to the fact that the vessel used in this experiment was not designed to hold such a seal, 1/25 atm was the measured pressure for this experiment.

Figure 213: Post Experiment teardown. Notice the condensation accumulating on the inside of the vessel and on the bottom of the top plate setup.

Quantitative Analysis Results (RIR)

General information

Analysis date	2016/10/28 09:49:35		
Sample name	Exp058_AlF_161021	Measurement date	2016/10/21 10:12:13
File name	Exp058_AlF_161021_Theta_2-Theta.raw	Operator	
Comment			

Qualitative analysis results

Phase name	Formula	Figure of merit	Phase reg. detail	DB card number
Cryolite	Al F6 Na3	0.322	User (COD)	9012281
Chiolite, syn	Na5 Al3 F14	0.750	ICDD	00-030-1144

Weight ratio

Phase name	Content(%)
Cryolite	95.5(5)
Chiolite, syn	4.54(7)

QAGraphs.emf

Measurement profile

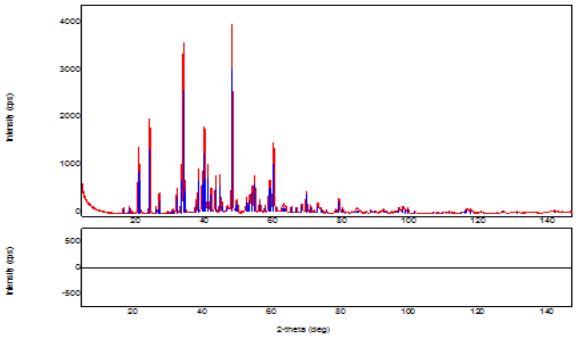

Figure 214: XRD report from condensed flux found inside the vessel after Exp 058.

ICP-OES Analysis Report - Exp058 Vacuum Conversion						
	Fe	Na	Sc	Th	Zr	Al
058 - L	465.1	42.44	47961.67	17.74	8.75	952502.1
058 - M	449.8	315.73	54284.73	16.94	11.75	945923.5
058 - R	458.76	58.76	48911.47	17.43	9.65	951542.7
Average (ppn	457.8867	138.9767	50385.96	17.37	10.05	949989.4
Average (%)	0.0458	0.0139	5.039	0.0017	0.0010	95.00

Table 35: ICP-OES results for relative concentrations in the 51.2g AlSc alloy recovered from the vacuum experiment rig. Note: Sc content is 5.039wt%

Calculated Sc yield based on AlSc conversion:

23.5g ScF_3 × 1mol ScF_3/101.95g ScF_3 × 44.96g Sc/1mol Sc = **10.36g Sc available**

51.2g Al-Sc alloy × 5.039wt% Sc concentration = **2.58g Sc converted**

2.58g converted ÷ 10.36g available = **25% yield**

Though this yield is higher than in previous experiments, the low yield relative to predictions is most likely due to lower vacuum then anticipated. The next step was to run a replicate experiment in a furnace with the capability to support such a vacuum and the results will be compared.

Experiments 060-063 were run to further test and improve on the vacuum conversion method. Table 36 highlights the results from these experiments as well as the changes made.

Name	Major Change	Vacuum	Temp (C)	Al Metal Added	Final Metal Weight	Sc %	Conversion	Notes / Failure Mode
Exp060	Changed from QC3 to small furnace for better vacuum	15 torr	1000	49.8g	32g	18	26.20%	1:1 metal to bath ratio
Exp061	3:1 metal small furnace	20 mtorr	1000	150g	132g	2.2	10.00%	Temp control failed and Al partially melted
Exp062	Back to QC3 with new 310 vessel with welded bottom. Higher metal:bath	200 mtorr	1000	150g	37g	3.87	5.00%	3:1 metal to bath ratio
Exp063	Large batch 1:1 metal to bath ratio	200 mtorr	1000	2000g	1960	4.76	27.00%	

Table 36: Summary of vacuum metallothermic reduction experiments.

Exp068 Produce 10 kg Al-Sc Alloy

NFINIUM produced just over 10 kg of Al-2wt% Sc master alloy from Al metal and rich Al-Sc alloy from Exp 063, in partial fulfillment of Milestone M3.5 (since it was made using ScF_3). This was done by remelting the Al-Sc alloy together with additional aluminum into two 5.44 kg batches in two Quad Cells.

Parameters

Furnaces: Quad Cells 1 and 3 (resistance heating)

Crucibles: Isomolded graphite - 6” OD x 5.5” ID x 11.5” Depth

Crucible Temperature: 1000ºC.

Results and Discussion

At the end, the crucibles contained two metal ingots weighing 5.44 kg and 5.439 kg and measuring roughly 14 cm OD × 25 cm tall (Figure 215). Those were sectioned and analyzed using ICP-OES to examine the Al and Sc concentrations in the top, middle and bottom as shown in Table 6.

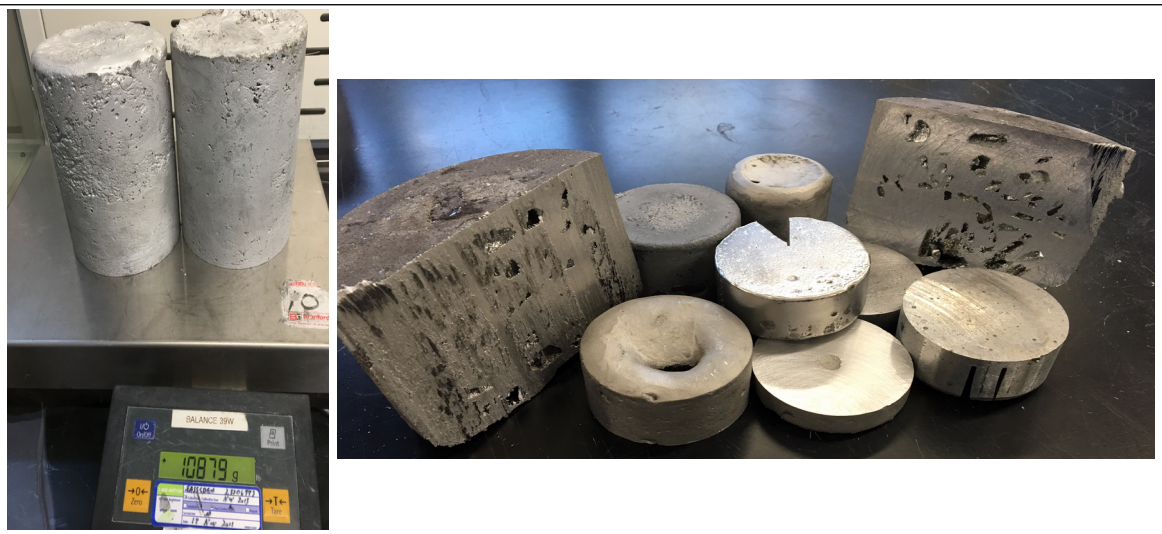

Figure 215: Final ingots of Al-Sc alloy, sectioned pieces of the ingots.

Al-Sc Ingot #	Sample Location	Sc Concentration
1	Top	1.91%
1	Middle	2.41%
1	Bottom	3.80%
2	Top	1.04%
2	Middle	0.52%
2	Bottom	8.69%

Table 6: Final Al-Sc Ingots and cross-section analysis via ICP-OES.

Different parts of the ingots exhibited very different scandium concentrations. This indicated that significant macrosegregation developed during solidification.

We note that commercial Al-2%Sc master alloy is shipped in plates with <1.5 cm thickness or high-area flat “waffle ingots” rather than this type of tall cylindrical ingot. This form likely allows for much more rapid solidification, resulting in less macrosegregation. For this reason, if INFINIUM uses this method to produce large (>1kg) samples for prospective customers, we will need to consider remelting it and making plate in this form.

Electrolysis Experiments

Exp036-040: Carbon Anode Electrolysis

Experiments 36-40 served two purposes: as a baseline test of production of Al metal with carbon anodes in potassium cryolite flux, and to remove the aluminum oxide present in our smelter-grade aluminum fluoride raw material (8wt% measured by LECO-O).

Experimental Parameters

Quad cell: 3

Anode(s): 2x 1.25”OD x 6” graphite rods with alumina sheathed steel current collectors.

Cathode: TiB_2 (0.5” L x 0.2” W x 3” H) with alumina sheathed steel current collector.

Crucible: 5” OD isomolded graphite.

Procedure

1. Make sure vessel is not grounded.
2. Attach mass spectrometer to anode A fitting.
3. Ensure no off-the-shelf alumina is inside the flux.
4. Bubble for 10-20 minutes using steel bubbling tube through thermocouple port.
5. While bubbling, lower both anodes 5.5” into the flux.
6. Position cathode at center of cup.
7. Lower cup until touching bottom, then raise 0.5”.
8. Carefully lower cathode until 5” into flux. Cathode should be 0.5”-1” above cup.
9. Attach C and D both to the cathode (relay switch limited to 20A each).
10. Attach EI303C between cathode (black) and cup (red).
11. Attach EI303D between anode A (red) and cup (black).
12. Attach EI303E between anode B (red) and cup (black).
13. Stop bubbling.
14. Set sequence to have a current limit to 40 A and step the voltage to desired potential.

Results

The crucible temperature, operating potential and cryolite ratio were the three parameters changed over the course of these experiments to optimize cell efficiency. Table 37 shows the parameters and results during these experiments.

Experiment	Cell Potential (V)	Resulting Current (A)	Charge Passed (A·hr)	Crucible Temperature (C)	Cryolite Ratio	Metal Recovery Rate
Exp036	5	18	41.9	700	1.3	5-10%
Exp038	3.5	15	56.6	760	1.3	10-15%
Exp039	3.5	10	224	760	1.3	20-25%
Exp040	4.8	40	152	800	1.3	20-25%

Table 37: Summary of carbon anode electrolysis experiments. Operating parameters highlighted alongside experimental results.

Exp041-042: Electrolysis to Al-2%Sc Alloy

Once carbon anode electrolysis proved to be successful and repeatable, the goal was to add scandium oxide in situ and electrowin the scandium into the aluminum. Electrolysis in Exp041 and Exp042 both worked to force Sc into the aluminum metal. The challenge was collecting the metal without it back-reacting in the cell. Therefore, a graphite containment (Figure 216) was made to isolate the metal.

The goal of these experiments was to make pure Aluminum metal via carbon anode electrolysis in a higher melting, lower volatility CR1.6 flux, then add Sc_2O_3 to the bath and make Al-Sc alloy. If metal cannot be made, the cryolite ratio would be reduced back to 1.3 where metal has been successfully made in prior experiments.

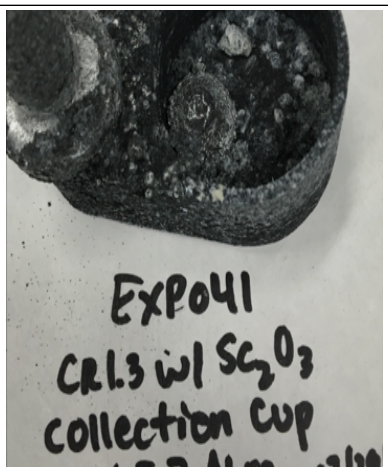

Figure 216: Graphite collection cup for aluminum-scandium alloy

Once Aluminum metal was confirmed, Scandium Oxide would be added until a 1:1 ratio of Al_2O_3-Sc_2O_3 was reached. Electrolysis would continue and the resulting metal would be analyzed for Scandium content and other impurities.

Exp041 ran electrolysis for 30 min and was unable to make aluminum metal. AlF_3 (100g) was added in situ to alter the cryolite ratio down to 1.3. Electrolysis resumed and aluminum metal was successfully made. The experiment ran for 7 hours, 65.3Ahrs and produced 17g of recoverable metal. The bulk concentration of scandium in the resulting product was 1.4wt%.

Exp042 ran for 6.5 hours and produced 10g of recoverable metal over 133 Ahrs. ICP was used to accurately measure the scandium concentration in the aluminum sample to 1.31wt% (see results section for more details).

Experimental Parameters

Quad cell: 3

Anode(s): 2 1.25”OD x 6” graphite rods with alumina sheathed steel current collectors.

Cathode: TiB_2 (0.5” L x 0.2” W x 3” H) with alumina sheathed steel current collector.

Flux: CR 1.3 – 8” (20.3 cm) height no alumina.

Crucible: 5” OD (12.7 cm) isomolded graphite.

Crucible Temperature: 700°C.

Procedure

1. Confirm vessel is electrically isolated from frame.

2. Attach mass spectrometer to anode A fitting.
3. Do not use off-the-shelf alumina inside flux.
4. Bubble for 10-20 minutes using steel bubbling tube through thermocouple port.
5. While bubbling, lower both anodes 5.5” into the flux.
6. Position cathode at center of cup.
7. Lower cup until touching bottom, then raise 0.5”.
8. Carefully lower cathode until 5” into flux. Cathode should be 0.5”-1” above cup.
9. Attach C and D both to the cathode (relay switch limited to 20A each).
10. Attach EI303C between cathode (black) and cup (red).
11. Attach EI303D between anode A (red) and cup (black).
12. Attach EI303E between anode B (red) and cup (black).
13. Stop bubbling.
14. Set sequence to have a current limit to 40 A and step the voltage limit to from 3-5.5V, holding for five minutes at each step.
15. Run current controlled at 40A for 100A·hr.

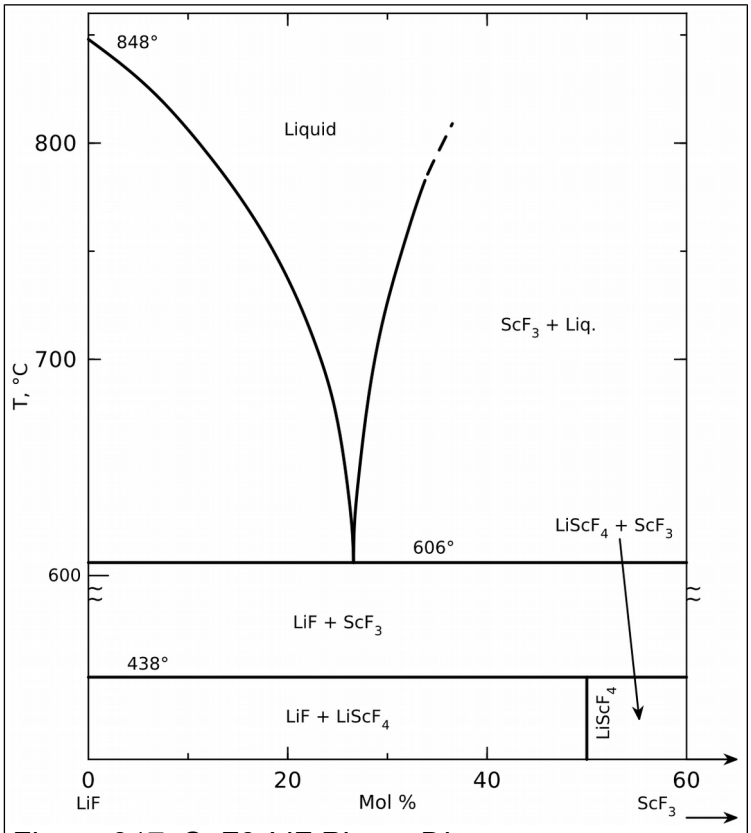

Figure 217: ScF3-LiF Phase Diagram

Exp043: Measure Scandium Oxide Solubility in ScF_3-LiF

Overview

In Exp043 INFINIUM tested two bath types for scandium oxide solubility: potassium cryolite, and ScF_3-LiF. The goal was to estimate the Sc_2O_3 solubility in a ScF_3-LiF bath at electrolysis temperature between 800° and 1000°C. Figure 217 shows the ScF_3-LiF phase diagram, which indicates a deep eutectic at 606°C and 27 mol% ScF_3 (59 wt% ScF_3). With good Sc_2O_3 solubility it should be possible to directly electrowin Sc into an Al cathode.

Experimental Parameters

The following weights were recorded for each well inside an argon-purged glove box:

Melt 1 (0wt% Sc_2O_3 control): 9.852g ScF_3 (58.85wt%)
6.888g LiF (41.14wt%)
16.740g total

Melt 2 (4.55wt% Sc_2O_3): 19.825g ScF_3 (58.8wt%)
13.890g LiF (41.2wt%)
33.700g total
1.532g Sc_2O_3 pre-loaded to bottom of well

Melt 3 (10wt% Sc2O3): 19.539g ScF_3 (58.607wt%)
13.800g LiF (41.393wt%)
33.339g total
3.334g Sc_2O_3 pre-loaded to bottom of well

Figure 218 highlights the design features for the solubility rig. The rig has 3 wells spaced evenly apart, each 2.54cm in diameter x 5.08cm deep. This corresponds to a well volume of 25.74cm³. A melting crucible was made to pre-melt the flux and pour into the wells. The melting crucible was 2.38cm in diameter x 10.16cm depth. This corresponded to a volume of 45.22cm³. The pre-mixed flux for melt 2 (33.7g total) was packed all the way to the top of the well indicating a packing density of 0.745 g/cm³. This will be used for future experiments to calculate the amount of flux that can be prepared for a known crucible volume.

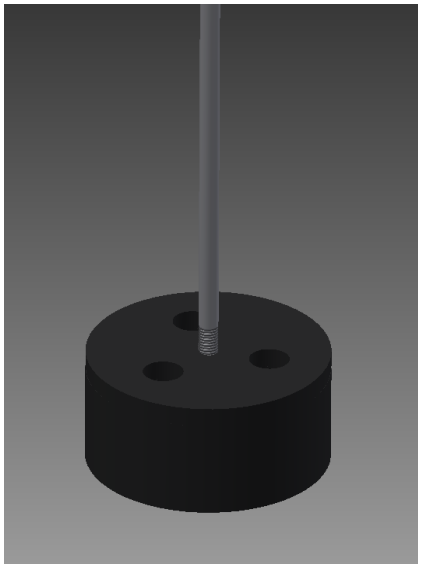

Figure 218: Graphite Solubility Rig for Sc_2O_3 in ScF_3-LiF and potassium cryolite.

Procedure

ScF_3-LiF Bath

The following procedure was designed to optimize solubility measurements while keeping precise

temperature control.

1. Dry thoroughly, weigh out and mix well the fluoride salt and oxide powders, three samples each with: 19.8 g ScF_3, 13.6 g LiF, and 0, 1.65 and 3.3 g Sc_2O_3 respectively (0, 5, 10 wt%).
2. Induction melt each sample in a graphite crucible and pour into testing container.
3. Place the mixtures into small open-top machined wells in a graphite block.
4. Insert the graphite block into a Quad Cell furnace with a thermocouple within 3 cm of the salt mixtures, seal the cell and purge with argon.
5. Heat the furnace and apparatus to the experiment temperature, and let it soak at that temperature for at least three hours, or until water concentration in the exit gas line has fallen below 0.01 mol% (as measured by mass spectroscopy).
6. Observe the liquid wells through the viewport, mix them with a rod, and look for undissolved oxide, which should be there in suspension.
7. Let any undissolved oxide settle for at least ten minutes.
8. Dip in a sample retrieval device and withdraw a quenched liquid sample free of solid precipitate from each well with at least 0.05 g of salt from each well, preferably 0.1 g.
9. Use LECO oxygen to analyze multiple samples from each well.

Figure 219: Sample retrieval device. 1/4" OD 304 S.S. tube x 1/2" L x 0.035" Wall. Crimped and seam welded at bottom, tack-welded to 1/8" dia. 309 S.S. rod.

Results and Discussion

After 21 hours at 800°C, an attempt was made to collect samples following the procedure above. However, the flux did not quench to the wire as expected, so a sample retrieval device was made as shown in Figure 219.

The furnace setpoint was then changed to 900C to collect three more samples. Two and a half hours later samples were taken from wells 2 and 3 at a crucible temperature of 910C. The furnace was then ramped up to 1000C to let it equilibrate over three days. After 65 hours, samples were again taken from wells 2 and 3. Finally, the set point was brought back down to 900C for the last sample. The results can be seen in Table 38 below.

Sample				Oxygen (%)	Weight (g)
Temp (°C)	*Elapsed Time (hr)*	*Well*	*Sc2O3 (wt. %)*		
21	*0*				
800	21	1	0	0.218	0.0954
				0.330	0.1023
800	21	2	5	0.362	0.1005
				0.403	0.1065
800	21	3	10	0.383	0.1046
				0.229	0.0959
900	23.5	2	5	0.287	0.1016
				0.320	0.0984
900	23.5	3	10	0.409	0.1051
				0.432	0.0899
1000	90	2	5	0.707	0.1073
				0.583	0.0943
1000	90	3	10	0.424	0.0746
				0.461	0.0651
900 ↓	92	2	5	0.644	0.0366
				0.604	0.0449
900 ↓	92	3	10	0.442	0.0359
				0.490	0.0414
Re-test w/ less weight					
800		2	5	0.445	0.0527
				0.562	0.0296

Table 38: LECO Oxygen Results showing solubility at 800, 900 and 1000C. Samples were taken as temperature was increased and again as temperature was decreased on the back end of the experiment.

The data showed that solubility increased significantly when taking measurements coming back down in temperature. This could be because there was not enough time given for the oxide to fall back out of solution. The solubility reached a peak of about 2 wt% Sc_2O_3 basis at 1000°C.

The solubility rig was removed from the furnace and can be seen in Figure 220. The rig showed no signs of wear or air burn.

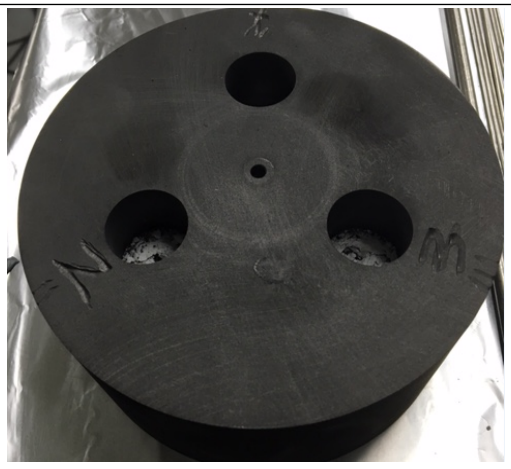

Figure 220: Post-experiment Graphite Solubility Rig with ScF_3-LiF-Sc_2O_3 Flux shown in well.

KF-AlF_3 Bath

Using the same experimental setup and parameters from the previous solubility test, a potassium cryolite bath was analyzed for oxide solubility. Samples were taken at 800, 900 and 1000C to see how the scandium oxide solubility

changes with temperature. These results can be seen below in Table 39. Oxide ion solubility for this system is below that for ScF_3-LiF, so this is not a good system for continuous feeding, dissolution, and electrolysis of Sc_2O_3.

Temperature (C)	Oxide Added (%)	Resulting Oxide Solubility (%)
800	5 (excess)	0.2
900	5 (excess)	0.4
1000	5 (excess)	0.5

Table 39: Scandium oxide solubility in the potassium cryolite 1.3 flux

Exp044: Sc_2O_3 Electrowinning Using a Liquid Al Cathode in ScF_3-LiF Bath

Overview

The goal of this experiment was to test the ability to electrowin scandium metal into a reactive liquid aluminum cathode in order to produce Al-2Sc master alloy at low cost. Due to the high cost of ScF_3, this experiment operated in a small tapered isomolded graphite crucible using just 100 g total bath. It used a carbon (graphite) anode centered in the crucible, with the crucible itself connected to the anode, and an Al-10 wt% Sc (7 mol%) liquid reference electrode.

Figure 221 shows the experimental apparatus which includes anode, reference electrode lead and thermocouple, all of which are in insulating alumina tube sheaths.

Experimental Parameters

Quad cell: 3

Anode(s): 0.5” diameter graphite rod

Cathode: Graphite crucible with liquid aluminum

Reference: Graphite in 90%Al-10%Sc

Flux: (58.8 wt%ScF_3-41.2 wt%LiF)- 2 wt% Sc_2O_3

Crucible: 3” OD isomolded graphite

Bubbling Tube: Steel 3/8” OD

Crucible temperature: 850°C

Duration: 4 hours (1.2 A·h)

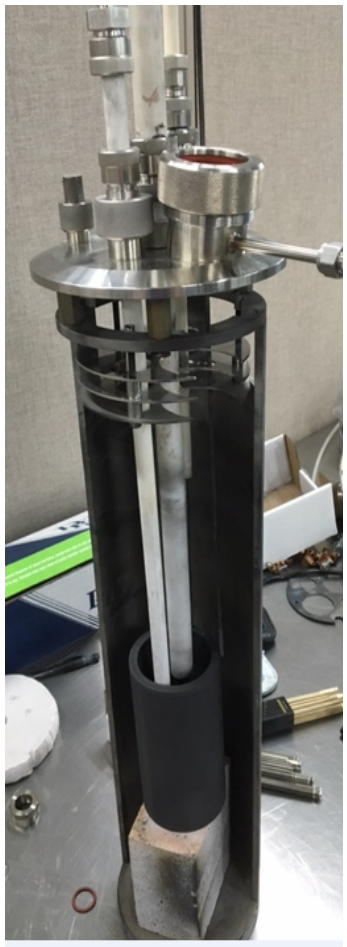

Figure 221: Electrolysis configuration for Exp044

Procedure

1. Using a Quad Cell furnace purged with argon, melt 14 g pure aluminum into the bottom of the crucible shown in Figure 221. Confirm good wetting to the crucible by both visual observation, and by measuring electrical conduction between the crucible and a graphite electrode inserted into the liquid aluminum. Allow the crucible and aluminum to cool to room temperature before opening the cell.
2. Using an induction heater in a glovebox, melt 5.4 g pure aluminum and 0.6 g pure scandium metals into a 1.9 cm outer diameter boron nitride crucible for use as a reference electrode. Allow the crucible and metal to cool to room temperature before removing from the glovebox.

3. Assemble the apparatus in a Quad Cell “cage” as follows:
    a) Insert the reference electrode into the isomolded graphite crucible in the location shown in Figure 7.
    b) Insert the tungsten reference electrode lead through the furnace lid, radiation baffles and crucible lid, and into the boron nitride crucible holding the Al-Sc alloy reference electrode.
    c) Rotate the lid away from the top of the crucible, and add the mixed ScF_3-LiF-Sc_2O_3 bath powders to the crucible, while holding the tungsten reference electrode lead in place.
    d) Secure the crucible lid to the top of the crucible, and add the anode, thermocouple, and sheath tubes to the apparatus – so they are about 1 cm inside the lid.
    e) Screw the steel anode lead into to the crucible.
4. Insert the apparatus/cage into the Quad Cell.
5. Purge the cell with argon and heat the apparatus to a crucible temperature of 850°C. It need not heat up at a slow controlled rate, a quick ramp-up is fine. Let the apparatus stand at this temperature for one hour.
6. Insert a wire through the thermocouple port to take a flux sample, then re-insert the thermocouple.
7. Lower the thermocouple so it descends 2-3 cm below the base of the lid.
8. Set the operating voltage to 1 V DC between the anode and cathode, and lower the anode slowly until the current increases, indicating the level of the top of the bath. Turn of DC power and lower it another 1 cm into the bath.
9. Run Electrochemical Impedance Spectroscopy (EIS) between the anode and cathode from 10 Hz to 10 kHz.
10. Run Linear Sweep Voltammetry (LSV) from open circuit potential to -5 V at 10 mV/sec.
11. Run electrolysis at 0.6 A current for two hours, measuring voltages of the anode, cathode and reference electrode. If there is evidence of CF_4 formation, reduce the current by 0.1 A at a time until this goes away. Disconnect every half hour to measure these voltages, then run EIS and LSV, and take a flux sample on a wire through the thermocouple port, including final measurements and sample at the

Figure 222: Section view of Exp044 electrolysis crucible. Tapered geometry to minimize flux, bottom well for liquid aluminum, side well to hold reference electrode and tapped hole to make connection to crucible.

end.

12. Withdraw the anode, thermocouple lead, and reference electrode lead, and allow the apparatus to cool to room temperature. It need not cool at a controlled rate, shutting off the furnace is fine.
13. Cut the crucible open, remove the reference electrode intact if possible, and take samples of the bath and aluminum anode for analysis by ICP, SEM, and LECO-oxygen (bath) and carbon (metal).

Results and Discussion

Figure 223 highlights the impedance measurement taken to prior to electrolysis.

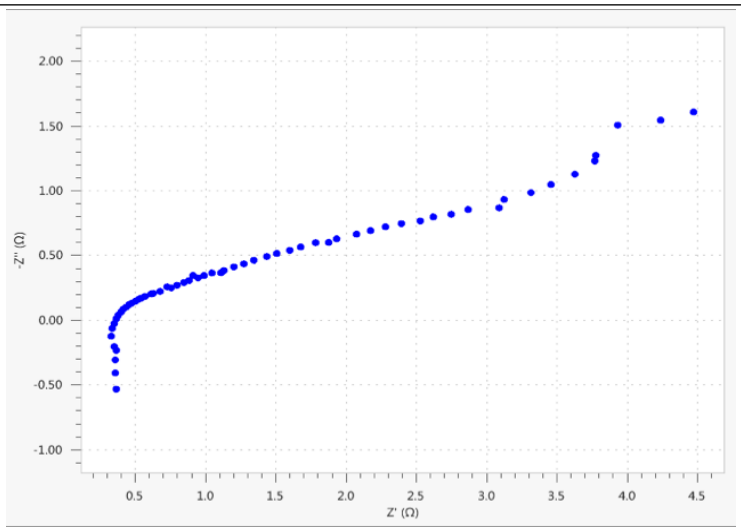

Figure 223: Impedance measurement

Figures 224 and 225 show anode-cathode current and anode-cathode and reference-cathode voltages throughout the 4-hour experiment. These sweeps were used to target

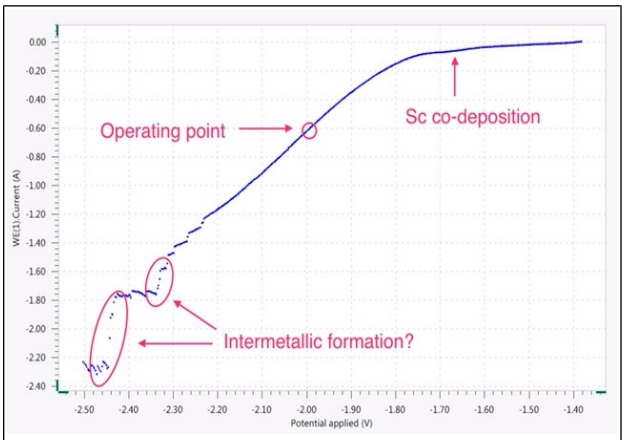

Figure 224: Working Electrode Current vs Voltage. Shows inter-metallic formation as well as the potential for depositing Sc

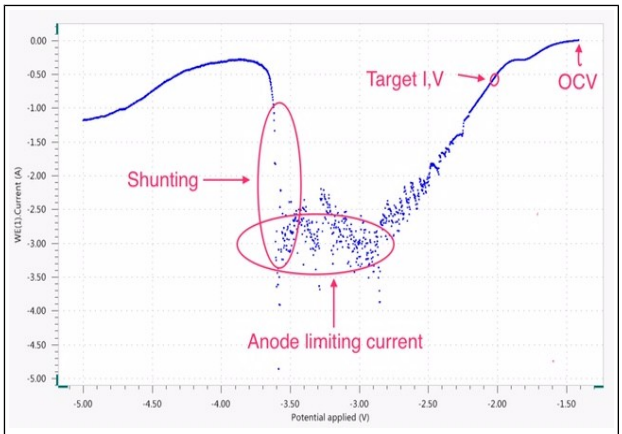

Figure 225: Shunting effects seen after 2V. This voltage sweep identified the optimal operating potential

the ideal operating voltage, which turned out to be 2.0V.

Bath and Alloy Characterization

Upon teardown of the experiment, the metal at the bottom appeared to disperse completely in the bath (Figure 226). As a result, a two-phase mixture of bath and metal made it difficult to decipher the scandium concentration electrowon into the aluminum. Further investigation is necessary to prevent metal dispersion in this configuration.

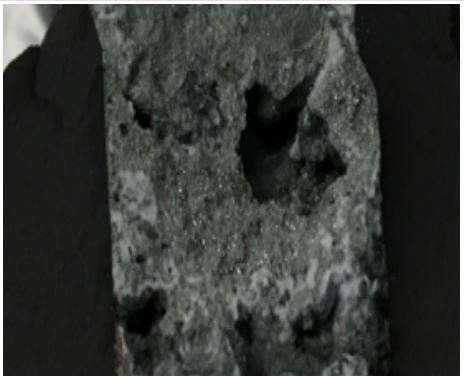

Figure 226: Well consisted mostly of disperse and dissolved metal. No metal without flux inclusions could be recovered.

Exp057: Scandium Oxide Solubility in ScF_3-AlF_3-NaF Bath

Due to relatively low solubility of Sc_2O_3 in ScF_3-LiF bath, and the metallothermic reaction between ScF_3 and Al without AlF_3 present, the goal of this experiment is to estimate the Sc_2O_3 solubility in a ScF_3-AlF_3-NaF bath at 1000°C. The AlF_3-NaF system also has higher oxide solubility than AlF_3-LiF, though with higher volatility. Figure 227 shows the ScF_3-NaF and AlF_3-NaF phase diagrams. Both show eutectic regions in cryolite ratio ranges of 1.5-2.0 (60-67 mol% NaF). We could not find a published ScF_3-AlF_3 phase diagram.

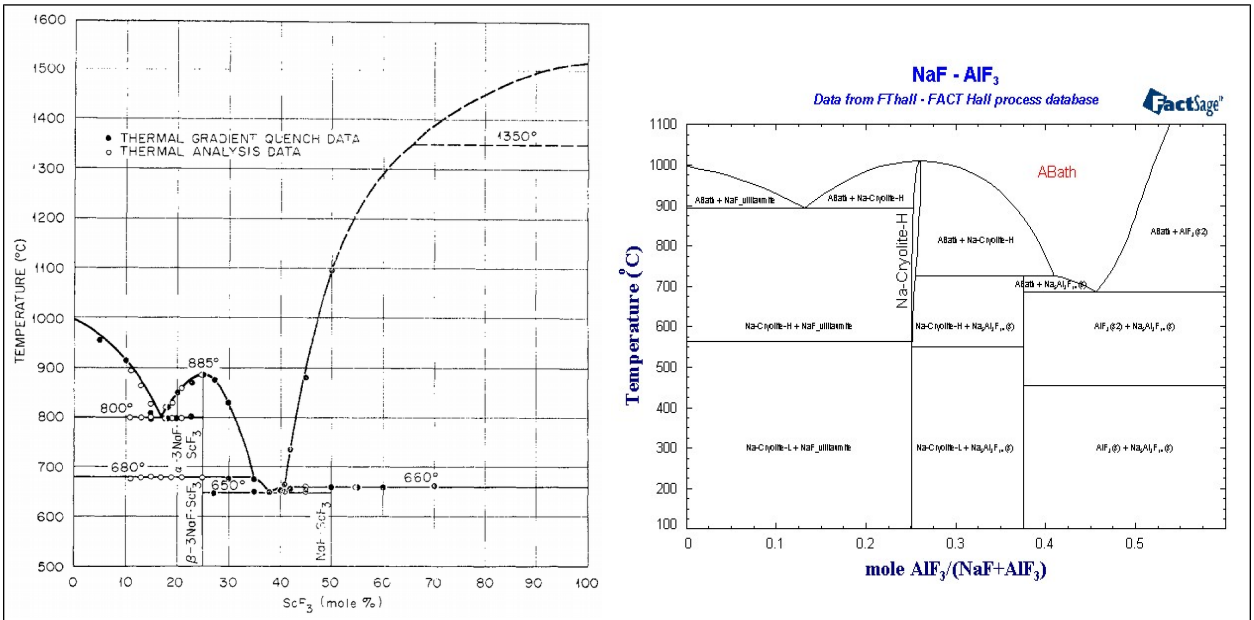

Figure 227: ScF_3-NaF and AlF_3-NaF phase diagrams.

This experiment assumes that there is a stable liquid phase for mixtures of AlF_3 and ScF_3 in the cryolite ratio range of 1.5-2.0, where that ratio is defined as the molar concentration ratio of NaF to (ScF_3+AlF_3). Test bath compositions are in Table 40. The first six samples represent three ScF_3:AlF_3 ratios from 8:1 to 2:1, and two cryolite ratios (CR) 1.5 and 2.0. The last four tested the similarity in oxide solubilities between use of aluminum oxide and scandium oxide for roughly 1:1 ScF_3:AlF_3 ratio, and the same two cryolite ratios.

Bath constituent powders were mixed well and loaded into the graphite block with drilled wells shown in Figure 228. Each well has a tapered design that allows the powders to be packed and when the mixture melts, most but not all of the bath will fill the bottom channel. This allows the excess insoluble oxide to settle out and the bath sample to be taken off the top.

Oxide Solubility Experiment 1

T=1000°C, argon atmosphere in quad cell ten-well system

Stir each when the block reaches temperature, let sit ~4 hrs, then cool rapidly

Sample	Moles $ScO_{1.5}$	Moles ScF_3	Moles AlF_3	Moles $AlO_{1.5}$	CR	Moles NaF
1	0.1	0.8	0.1		1.5	1.35
2	0.1	0.8	0.1		2	1.8
3	0.1	0.7	0.2		1.5	1.35
4	0.1	0.7	0.2		2	1.8
5	0.1	0.6	0.3		1.5	1.35
6	0.1	0.6	0.3		2	1.8
7	0.05	0.45	0.5		1.5	1.425
8	0.05	0.45	0.5		2	1.9
9		0.5	0.45	0.05	1.5	1.425
10		0.5	0.45	0.05	2	1.9
Molar mass	69	102	84	51		42

Mass Sc_2O_3	Mass ScF_3	Mass AlF_3	Mass Al_2O_3	Mass NaF	Total mass, g
1.35	15.94	1.64	0.00	11.07	30.00
1.20	14.19	1.46	0.00	13.15	30.00
1.36	14.11	3.32	0.00	11.21	30.00
1.21	12.55	2.95	0.00	13.29	30.00
1.38	12.24	5.04	0.00	11.34	30.00
1.23	10.87	4.48	0.00	13.43	30.00
0.68	9.11	8.33	0.00	11.88	30.00
0.60	8.05	7.36	0.00	13.99	30.00
0.00	10.12	7.50	0.51	11.88	30.00
0.00	8.94	6.63	0.45	13.99	30.00

Table 40: ScF_3-AlF_3-NaF-Sc_2O_3 bath compositions in oxide solubility test.

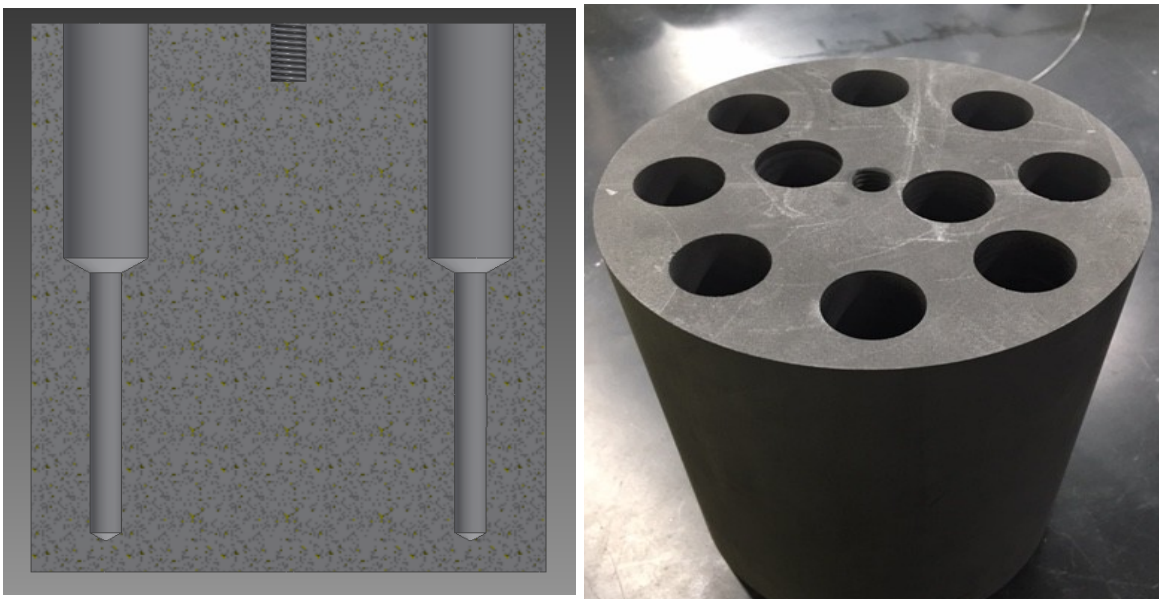

Figure 228: Graphite block with drilled wells: CAD cutaway (left), photograph (right).

Experimental Procedure

1. Dry thoroughly, weigh out and mix well the fluoride salt and oxide powders.
2. Place the mixtures into small open-top machined wells in the graphite block.
3. Insert the graphite block into a Quad Cell furnace with a thermocouple within 3 cm of the salt mixtures, seal the cell and purge it with argon.
4. Heat the furnace and apparatus to the experiment temperature, and let it soak at that temperature for at least three hours, or until water concentration in the exit gas line has fallen below 0.01 mol% (as measured by mass spectroscopy).
5. Observe the liquid wells through the viewport, mix them with a wire, and look for undissolved oxide, which should be there in suspension.
6. Let any undissolved oxide settle for at least three hours.
7. Dip in a welding wire and withdraw a liquid sample free of solid precipitate from each well with at least 0.5 g of bath from each well, preferably 1.0 g or more.
8. Run LECO oxygen analysis on multiple samples from each wire.

Experimental Results

The results from LECO oxygen analysis can be seen below in Table 41. For scandium oxide solubility, the result would be multiplied by a factor of 138÷48 = 2.875, showing that wells 5 and 7 through 10 all yielded compositions with scandium oxide solubility over 3wt%. Well 5 has the highest solubility at 3.60wt%, which has a cryolite ratio of 1.5 and ScF_3:AlF_3 molar ratio of 2:1.

Replicate	1	2	3	4	5	6	7	8	9	10
1	0.687	0.471	0.944	0.732	1.26	1.08	1.13	1.13	1.17	1.25
2	0.717	0.493	0.987	0.718	1.27	1.01	1.23	1.18	1.27	1.11
3	0.657	0.497	0.859	0.658	1.22	0.947	1.22	1.11	1.21	1.22
Average	0.687	0.487	0.930	0.703	1.250	1.012	1.193	1.140	1.217	1.193

Table 41: LECO-Oxygen results for each sample well. Numbers are reported as an average oxygen content in wt %.

This experiment showed that oxide solubility is lower than this at both higher and lower ScF_3:AlF_3 ratio. And oxide solubility is higher for cryolite ratio of 1.5 than 2.0 across the range of ScF_3:AlF_3 ratios. Finally, based on similarities among samples 7-10, oxide ion solubility appears independent of whether those oxide ions come from Sc_2O_3 or Al_2O_3.

It is worth noting that oxide solubility at the highest ScF_3:AlF_3 ratios in samples 1-2 is similar to that of ScF_3-LiF system at this temperature from the prior experiment. Due to the lower volatility of NaF, it could be beneficial to measure oxide solubility in the ScF_3-AlF_3-LiF system.

Exp 058: Density Measurements in the $NaF\text{-}ScF_3\text{-}AlF_3\text{-}Sc_2O_3$ Bath System

Overview

The goal was to measure the density of the four candidate $NaF\text{-}ScF_3\text{-}Sc2O_3$ bath compositions at 1000°C. Density separation between the bath and molten aluminum metal is very important to avoid mixing of the two phases, which was likely the cause of failure of Exp 044 (p. 239). This experiment was conducted in an inert atmosphere inside a glovebox with an induction coil as the heat source. An isomolded graphite crucible was used with an insulation sleeve around it to maintain desired operating temperature. An 1/8" 304L stainless steel welding wire was used as a cold finger that was dipped to the bottom of the crucible well, then the height was measured using calipers to calculate an accurate density.

Experimental Parameters

Furnace: Induction Melter inside argon-purged glove box

Flux: CR 1.5 and 2.0 $NaF\text{-}AlF_3\text{-}ScF_3\text{-}Sc_2O_3$ with 2:1 and 0.9:1 ScF_3:AlF_3 ratios

Crucible: 2" OD, 2X 3/4" bored holes, 3½" depth with flat bottom.

Crucible Temperature: 1000°C.

Procedure

1. Premix flux components in the chemical fume hood.
2. Cover components and load into the anti-chamber of the glove box.
3. Slowly pull vacuum to the chamber and wait until a vacuum of 10^{-4} torr is achieved.
4. Close the vacuum valve and slowly purge the anti-chamber with argon from the pressurized glove-box.
5. Bring flux mixtures into the glove box and load into 3" OD isomolded graphite crucible.
6. Weigh both of the 2-well graphite crucibles and record the weights in the lab notebook.
7. Cover crucible with a cap and heat up until molten.
8. Once molten, grab the crucible with stainless steel tongs, slowly agitate the crucible by hand to mix and pour the contents into the marked 2-well isomolded graphite crucible.
9. Re-weigh the graphite crucible and record the weight difference in the lab notebook to identify the weight of the flux added.
10. Repeat this procedure until all 4 wells contain the melted flux components.

11. Take one of the 2-welled graphite pieces and contain it in the firebrick sleeve made (Figure 229). Put the cap on and load the K-type thermocouple down the designated port located in the cap and aligned to the crucible block.
12. Plug in the K-type thermocouple and begin heating until 1020°C is reached.
13. Allow the temperature to slowly fall to 1000°C
14. Remove the cap and dip an 1/8” stainless steel welding wire to the bottom of the crucible well and remove quickly.
15. Using calipers, measure the quenched salt height on the welding wire.
16. Put the cap back on and repeat steps 12-14 for the second well.
17. Once both wells are samples, remove the isomolded graphite rig and repeat steps 11-14

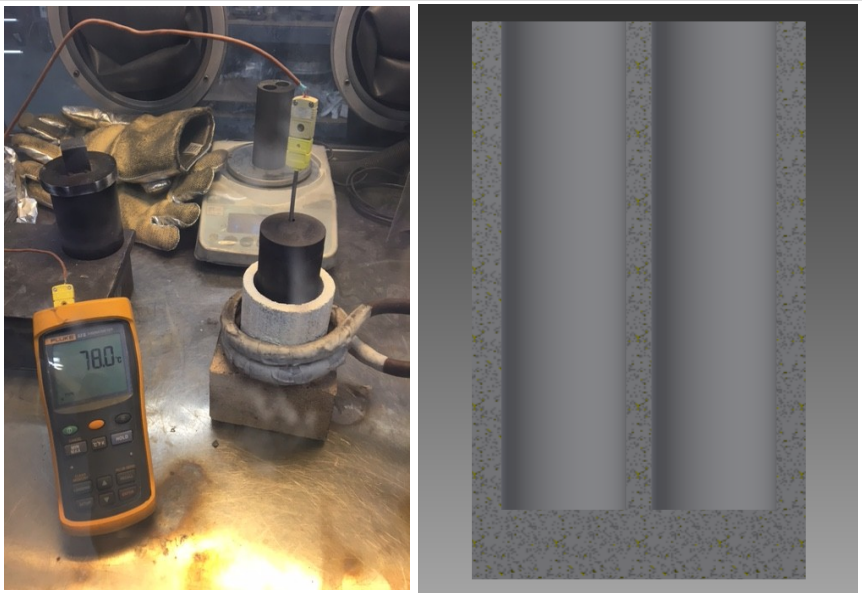

Figure 229: Left: Experimental setup in the induction coil; Right: cross-section of the 2-well isomolded graphite crucible rig.

Results and Discussion

Table 1 below highlights the compositional breakdown for each of the 4-wells tested in this experiment. Table 2 shows the results from the experiment taken according to the written procedure outlined above.

Sample	Weight% Sc2O3	Moles ScF_3	Moles AlF_3	Moles NaF	Moles NaF
Well 1	2	0.61	0.30	1.35	1.5
Well 2	2	0.61	0.30	1.84	2
Well 3	2	0.46	0.51	1.47	1.5
Well 4	2	0.46	0.51	1.94	2
Molar mass	69	102	84		42

Table 42: Compositions for each of the 4 wells tested in the experimental rig.

	Containment Weight (g)	Sample Weight (g)	Well Diameter (cm)	Measured Height (cm)	Calculated Density (g/cm^3)
Well 1	285.05	28.02	1.905	4.256	2.310
Well 2	285.05	29.77	1.905	4.674	2.235
Well 3	284.81	32.867	1.905	5.024	2.295
Well 4	284.81	29.414	1.905	4.749	2.173

Table 43: Density measurements calculated from quenched salt heights inside controlled induction furnace.

The molten densities in each well varied within 6% of each other. The molten density of Aluminum metal is 2.377 – 0.000311 × (T – 660.3°C),[97] at 1000°C this is 2.27 g/cm^3. Well 4 reported the lowest density of 2.173, which is the furthest away from molten aluminum. Therefore, well 4, with ScF_3:AlF_3:NaF molar ratio 0.9:1.0:3.8, would be the best candidate to avoid molten metal and flux interactions. With 2 wt% Sc_2O_3, i.e. about 2/3 of saturation solubility (see oxide solubility p. 243), bath mass fractions are roughly: 26.8% ScF_3 24.6% AlF_3 46.6% NaF 2% Sc_2O_3. This is the flux composition which will be used in final electrolysis experiments.

Exp064: Static Baseline Test for Electrolysis to Al-2%Sc Alloy

Goal of this experiment was to find the static baseline Sc concentration in the CR 2.0 $NaF\text{-}AlF_3\text{-}ScF_3\text{-}Sc_2O_3$ flux selected for electrolysis and given below. The bath was preloaded with 2wt% scandium oxide and mixed well. The experiment ran for 8 hours and the final Sc concentration in the Al was 5.0wt% analyzed via ICP-OES.

Exp065-067: Electrolysis to Al-2%Sc Alloy

Overview

The goal is to electrowin Sc into Al metal using electrolysis in $NaF\text{-}AlF_3\text{-}ScF_3\text{-}Sc_2O_3$ flux at 1000°C. This method of Al-Sc master alloy production has potential for very low raw material cost based on Sc_2O_3 consumption and high Sc yield. If successful, this should be the basis of parameter optimization experiments.

Experimental Parameters

Furnace: Quad Cell

Bath: $NaF\text{-}AlF_3\text{-}ScF_3\text{-}Sc_2O_3$ bath with CR 2.0, 10:9 AlF_3:ScF_3 mole ratio, 2 wt% Sc_2O_3.

Reference electrode: W wire inserted into a small BN cup containing Al-20 wt% Sc.

Crucible Temperature: 1000°C.

Procedure

Exp 065-066 stopped at step 8.

1. Place Al metal into the lower side of the crucible with slanted insert, and melt in the glove box using the induction melter.
2. Premix 300 g flux components in the chemical fume hood.
3. Prepare TiB_2 cathode, 0.375” OD graphite anode.
4. Place the crucible in the cage.
5. Add pre-mixed bath powder to the crucible.
6. Place cage with crucible into the Quad Cell, and heat up to 1000°C.
7. Withdraw small (1-3 g) samples of both bath and metal for analysis: LECO oxygen (bath) and ICP-OES (metal).
8. Run electrolysis at up to 8 V and up to 10 A, to approximately 15 A·h total charge, i.e. enough to reduce 6 g Sc_2O_3 at about 60% current yield.
9. Withdraw approximately 80 g Al-Sc alloy using a steel ladle.
10. Withdraw a small (1-3 g) sample of bath for LECO oxygen analysis using.
11. Feed 6 g Sc_2O_3, and stir the bath until dissolved.

12. Add Al metal.
13. Repeat from step 8 2-4 times.
14. Cool to room temperature and disassemble.

To insert a long break (>1 hour), pause after step 10.

Results and Discussion

Exp 065 and 066 used a custom crucible with stepped interior, creating a deep well for Al metal wrapped around the TiB_2 cathode, and a shallow ledge for the anode and reference electrode. Those experiments began with 15 g Al metal and 100 g bath.

In Exp 065, the metal grew to 21 g with 2.0 wt% Sc. This is considerably lower than the 5 wt% Sc from Exp 064 with no electrolysis, indicating considerable Al electrowinning into from the bath into the metal. However, The metal also had 5 wt% Fe, likely because there was originally not sufficient bath to immerse the anode, and adding more bath resulted in bath contact with the steel lead attached to the TiB_2 cathode.

In Exp 066, a change in crucible geometry eliminated contact between the bath and steel lead. In this case, the metal grew to 17.5 g with 2.31 wt% Sc.

Exp 067 used an isomolded graphite 5" OD, 4" depth with flat bottom, 20° isomolded graphite slanted insert to pool metal on lower side. This held 300 g bath, with 100 g Al initially placed in the cell. Though the original plan was for three electrolysis runs, carbon buildup on the bath surface shorted the cell after just two runs of the loop from steps 7-12, preventing further runs. Observations and measurements were as follows:

- Run 1 Step 7 bath and metal samples: Metal was Al-3.14 wt% Sc indicating some metallothermic reaction but less than the 8-hour soak in Exp 064. Bath had 1.87% oxygen, which would correspond to 5.37 wt% Sc_2O_3, or rather, the 2 wt% Sc_2O_3 added plus 2.50 wt% Al_2O_3 which came in as an impurity of the fluorides.
- Run 1 Step 8 electrolysis: Unlike prior experiments, it was tricky to get the 100 g aluminum to attach to the TiB_2 cathode. Ran to 17.5 A·h charge, saw a small amount of bubbling.
- Run 1 Step 9 Al-Sc alloy withdrawal: Successful, pulled out 80 g Al-3.44 wt% Sc alloy, an 0.30 wt% increase in Sc vs. the initial concentration, corresponding to about 0.3 g Sc added by electrolysis. This differed from both Exp 045 and 046 which both showed less Sc after electrolysis than in the 8-hour static case.
- Run 1 Step 10 bath sample: Bath had 1.47 wt% oxygen, a reduction of 0.40 wt% i.e. 1.2 g oxide ions, corresponding to 4.0 A·h, indicating 23% current yield on the 17.5 A·h passed. The 0.3 g Sc added to the metal by electrolysis would consume 0.46 g Sc_2O_3 from the bath, corresponding to 0.05 wt% reduction in bath oxygen. Therefore, 0.35 wt% of the reduction in oxygen corresponds to 2.23 g Al_2O_3 reduction, making 1.18 g aluminum.

- Run 1 Step 11 feeding 6 g Sc_2O_3: Appeared to dissolve well.
- Run 1 step 12: Added 80 g Al metal, stirred with a graphite rod to agglomerate it with ~20 g existing metal remaining in the cell.
- Run 2 Step 8 electrolysis: Ran to 14.7 A·h charge, saw significant growth in carbon dust on the bath surface which we believed would lead to low current efficiency.
- Run 2 Step 9 Al-Sc alloy withdrawal: Successful, pulled out 60 g Al-3.42 wt% Sc alloy, very close to that of Run 1 Step 9. About 2.7 g Sc entered the newly-added Al metal by some combination of metallothermic and electrolytic reactions.
- Run 2 Step 10 bath sample: Bath had 1.23 wt% oxygen, indicating more oxide reduction than feed. Reduction of 0.24 wt% oxygen vs. Run 1 Step 10 corresponds to 0.72 g oxygen consumed from the bath, plus 2.09 g oxygen consumed from the 6 g Sc_2O_3 added in Run 1 Step 11, leading to 2.81 g oxygen consumed by electrolysis. This corresponds to 9.43 A·h charge, indicating 64% current efficiency on the 14.7 A·h passed.

As mentioned above, at this point we chose to end the experiment due to the quantity of carbon dust on the surface, which we believed would lead to low current efficiency, though the oxide consumption data indicate otherwise.

In all, this experiment produced about 140 g metal in the two runs with 3.42-3.44% Sc. Based on the amount fed and decline in bath oxygen concentration, 1.9 g oxygen from the original bath was consumed, corresponding to 5.4 g 13.9 g scandium oxide was consumed to produce the scandium in the alloy. With 3:1 bath:metal ratio, this did not approach steady-state, but consumed 116% of the oxide fed in. This indicated reduction of oxide from the original bath, and good prospects for high process material yield.

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Computer Modeling

INFINIUM used COMSOL[98] to model electrochemical reactions.

Model Description

Conservation of charge: $\nabla \cdot (\sigma \nabla \varphi) = 0$, φ is electrical potential. Assumes steady state (negligible local rate of charge accumulation) in good conductors. Used to model current density in molten salts, solid electrolytes, and metals.

Documentation

See the COMSOL manual.[98]

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